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**Report prepared for the WHO annual consultation on the
composition of influenza vaccines for the Southern
Hemisphere 2020**

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WHO Vaccine Recommendation Meeting for the 2020 Southern Hemisphere influenza season

WHO CC, London – Final Report for the meeting: 23-26 September 2019

Epidemiology

Influenza activity in the WHO European Region, October 2018 – August 2019

Timing and spread

Compared to the previous four years, the 2018-2019 influenza season (weeks 40/2018 to 20/2019) in the WHO European Region started several weeks later than the 2016-2017 and 2017-2018 seasons, and several weeks earlier than the 2014-2015 and 2015-2016 seasons (**Figure E1-A**). As measured by the proportion of respiratory specimens, from patients presenting to sentinel primary care sites with influenza-like illness (ILI) or acute respiratory infection (ARI) symptoms, that tested positive for influenza virus the initial increase in activity (the first week with more than 10% of specimens from the WHO European Region testing positive for influenza viruses) occurred from week 49/2018 (**Figure E1-B**). The proportion of specimens positive for influenza peaked at 60% in week 5/2019, slightly higher than the peaks of 53-55% in the previous four seasons (**Figure E1-A**). Peak activity, in which $\geq 40\%$ of sentinel specimens tested positive for influenza, lasted for 10 weeks, compared to an average of 11 weeks (range: 9-14 weeks) for the previous four seasons, before declining to less than 10% in week 17/2019 (**Figure E1-B**). Based on the intensity of influenza activity reported weekly by each country/territory/region, activity increased in the east of the WHO European Region –notably in Kyrgyzstan, Armenia, Georgia, Turkey and Ukraine – prior to other countries (**Figure E2**).

Distribution of viruses

Between week 40/2018 and week 20/2019, just over 205,000 influenza detections from sentinel and non-sentinel primary care sources were reported to The European Surveillance System (TESSy) (**Table E1-A**), a slight decrease (approximately 1,000) compared to the 2017-2018 season. Combined sentinel and non-sentinel source data from the Region, showed an overwhelming majority (99%) of type A virus detected with just 1% being type B viruses. This trend was generally consistent when stratified by sub-region, but with slightly higher proportions of type B virus detections of 7% and 5% reported for Central Asia and West Asia, respectively (**Table E1-A**). Combined sentinel and non-sentinel data for type A viruses showed a slight majority (57%) to be A(H1N1)pdm09 viruses, with 43% being A(H3N2) viruses. However, there was much greater variation in these proportions when stratified by sub-region, whereby the majority of A(H1N1)pdm09 viruses ranging from 54% to 71% detected in Southern Europe, Eastern Europe, Northern Europe and Central Asia, while A(H3N2) viruses predominated (58% and 62%) in West Asia and Western Europe, respectively (**Table E1-A**). Only 6% of type B viruses reported were ascribed to a lineage, being primarily from Southern and Northern Europe, of which 60% were B/Yamagata and 40% were B/Victoria.

In general, the regional distribution of influenza by type and subtype remained consistent when analysis was restricted to sentinel source data only, although there was higher predominance of A(H3N2) viruses in West Asia (68% from 58%) and A(H1N1)pdm09 viruses in Eastern Europe (70% from 59%) compared to combined sentinel and non-sentinel data sources (**Table E1-B**). Nearly all influenza B virus detections from sentinel source data were from Southern Europe, in which a majority (90%) were of the B/Yamagata-lineage. The subtype distribution proportions for influenza type A viruses detected, from both sentinel and non-sentinel sources, were relatively consistent for most weeks during the season (**Figure E1-B/C**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections) could be assigned by 43 countries, territories or regions within the WHO European Region, with only seven having insufficient numbers (ten or fewer) of detections. All 43 countries, territories or regions reported a predominantly influenza type A virus season (**Figure E3**). Two countries reported dominance of type A without specifying a subtype, and 41 were able to ascribe dominance (the proportion of detections of an influenza subtype was 60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes to the season: 23 in which A(H1N1)pdm09 was dominant, six in which A(H3N2) was dominant, and 12 in which A(H1N1)pdm09 and A(H3N2) were co-dominant.

Since the end of the 2018-2019 influenza season (weeks 21-34/2019), a total of 877 influenza detections from non-sentinel sources across the Region have been reported. Of these, 756 (86%) were type A viruses and 121 (14%) were type B. Of the 407 type A viruses subtyped (54%), 70 (17%) were A(H1N1)pdm09 and 337 (83%) were A(H3N2). Only 15 (12%) of the type B viruses detected were ascribed to a lineage, of which 12 (80%) were B/Victoria and three (20%) were B/Yamagata. Over the same post-season period, there have been just eight detections of influenza virus from sentinel sources, of which six were type A (five subtype A(H3N2) and one unknown) and two were type B.

Influenza virus characterisation

Genetic characterisations of 4,102 influenza viruses, collected between weeks 40/2018 to 20/2019 and ascribed to genetic clades as defined at the start of the 2018-2019 season, were reported to TESSy:

- All but three of the 1,885 A(H1N1)pdm09 viruses characterised fell in the A/Michigan/45/2015 vaccine component clade (6B.1);
- Among 2,163 A(H3N2) viruses attributed to a clade, 1,435 (66%) fell in the 3C.2a1b subgroup, 548 (25%) in clade 3C.3a, 57 (3%) in the northern hemisphere 2018-2019 vaccine virus component 3C.2a1 subclade, 70 (3%) in subclade 3C.2a2, 33 in 3C.2a3 subclade, 5 (<1%) in clade 3C.2a, with 6 (<1%) being attributed to a clade, subclade or subgroup not listed;
- All 25 B/Victoria-lineage viruses characterised belonged to clade 1A (represented by B/Brisbane/60/2008), but 5 of these fell in a subclade with a two amino acid deletion in HA (1A.Δ2; represented by the vaccine component B/Colorado/06/2017) and 15 fell in a subclade with a three amino acid deletion in HA (1A.Δ3; represented by B/Hong Kong/269/2017);

- All 29 influenza B/Yamagata-lineage viruses belonged to clade 3, represented by the vaccine component B/Phuket/3073/2013.

Neuraminidase inhibitor susceptibility was assessed for 2,824 influenza viruses with collection dates since week 40/2018: 1,668 A(H1N1)pdm09, 1,121 A(H3N2) and 35 type B. Very few viruses showed any reduction in susceptibility. Eight A(H1N1)pdm09 viruses carried an H275Y amino acid substitution in NA indicative of highly reduced inhibition (HRI) by oseltamivir which was confirmed for three of the viruses by phenotypic assay. Two A(H3N2) viruses showed reduced inhibition (RI) by oseltamivir only in phenotypic assays and one type B virus showed evidence of RI by oseltamivir and zanamivir.

Two reassortant A(H1N2) influenza viruses were detected – one each in December 2018 and April 2019 in Sweden (A/Ystad/1/2018) and Denmark (A/Denmark/3176/2019), respectively - in specimens collected from persons aged over 65 years. Whole genome sequencing showed both reassortants to contain seven genes (PB2, PB1, PA, HA, NP, M and NS) and one (NA-N2) from circulating human seasonal A(H1N1)pdm09 and A(H3N2) viruses, respectively. Both reassortants were susceptible to neuraminidase inhibitors.

Hospitalised cases

Between week 40/2018 and week 20/2019, 15 countries (Belgium, Czech Republic, Denmark, Estonia, Finland, France, Ireland, Netherlands, Romania, Russian Federation, Slovakia, Spain, Sweden, Ukraine and the United Kingdom) reported a total of 17,314 hospitalised laboratory-confirmed influenza cases. Fourteen of these countries reported a total of 7,345 cases admitted to Intensive Care Units (ICUs) and ten countries reported 9,969 cases from other hospital wards. Among cases admitted to ICUs, a majority (99%) of influenza type A viruses were detected and of the 3,357 viruses subtyped, 66% were A(H1N1)pdm09 and 34% were A(H3N2). Approximately half (47%) of the 4,092 influenza-associated ICU admissions with a recorded age were in patients aged 65 years or older, and 45% were aged from 15-64 years (**Figure E4-A**).

Eighteen countries, territories or areas (Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Georgia, Kazakhstan, Kosovo*, Kyrgyzstan, Montenegro, North Macedonia, Republic of Moldova, Russian Federation, Serbia, Tajikistan, Turkmenistan, Ukraine and Uzbekistan) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 43,126 cases between week 40/2018 and week 20/2019. The majority of SARI cases were among children in most countries, although higher proportions were observed in older adults in Albania, Belarus, Bosnia and Herzegovina, Montenegro, Serbia and Kosovo (**Figure E4-B**). Of 9,683 SARI cases tested for influenza virus over the period, 2,848 (29%) were positive. The great majority (99%) were infected with influenza type A viruses and, of the 90% subtyped, 79% were A(H1N1)pdm09 and 21% were A(H3N2). Only 32 (1%) of the influenza positive cases were infected with type B viruses, with the one ascribed to a

* in accordance with. United Nations Security Council. Resolution 1244 (1999).

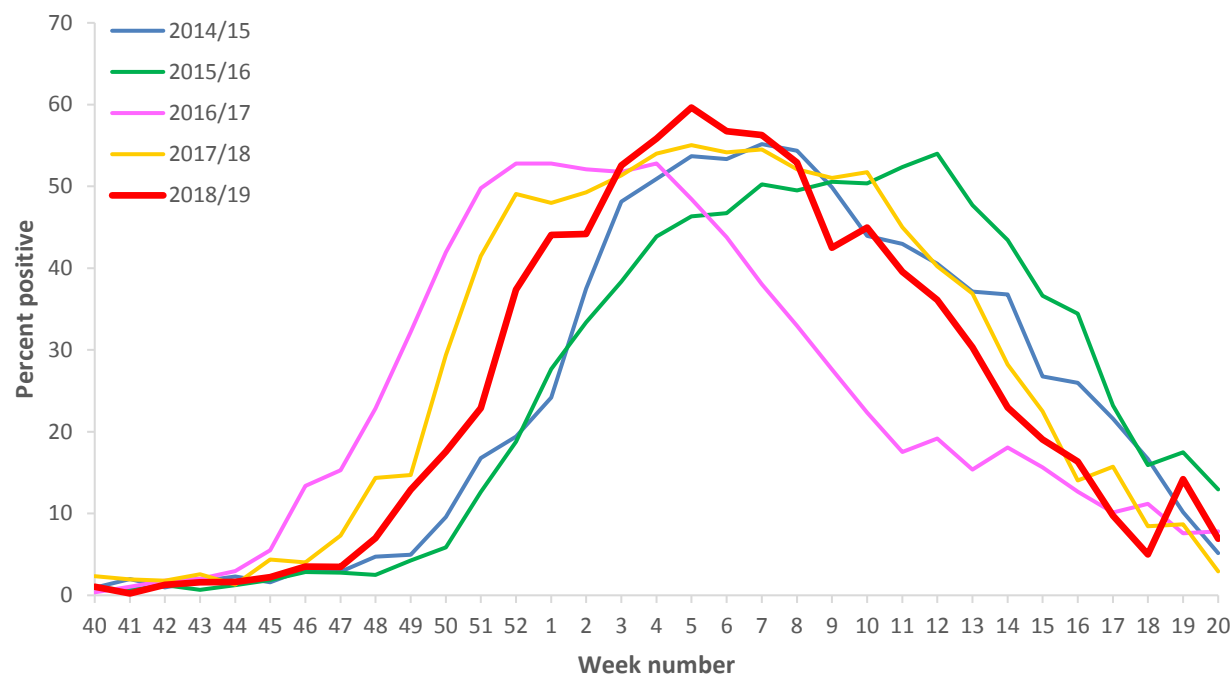
lineage being B/Yamagata. Information on distributions of influenza types and subtypes or lineages by age group are not available for these data.

Mortality data

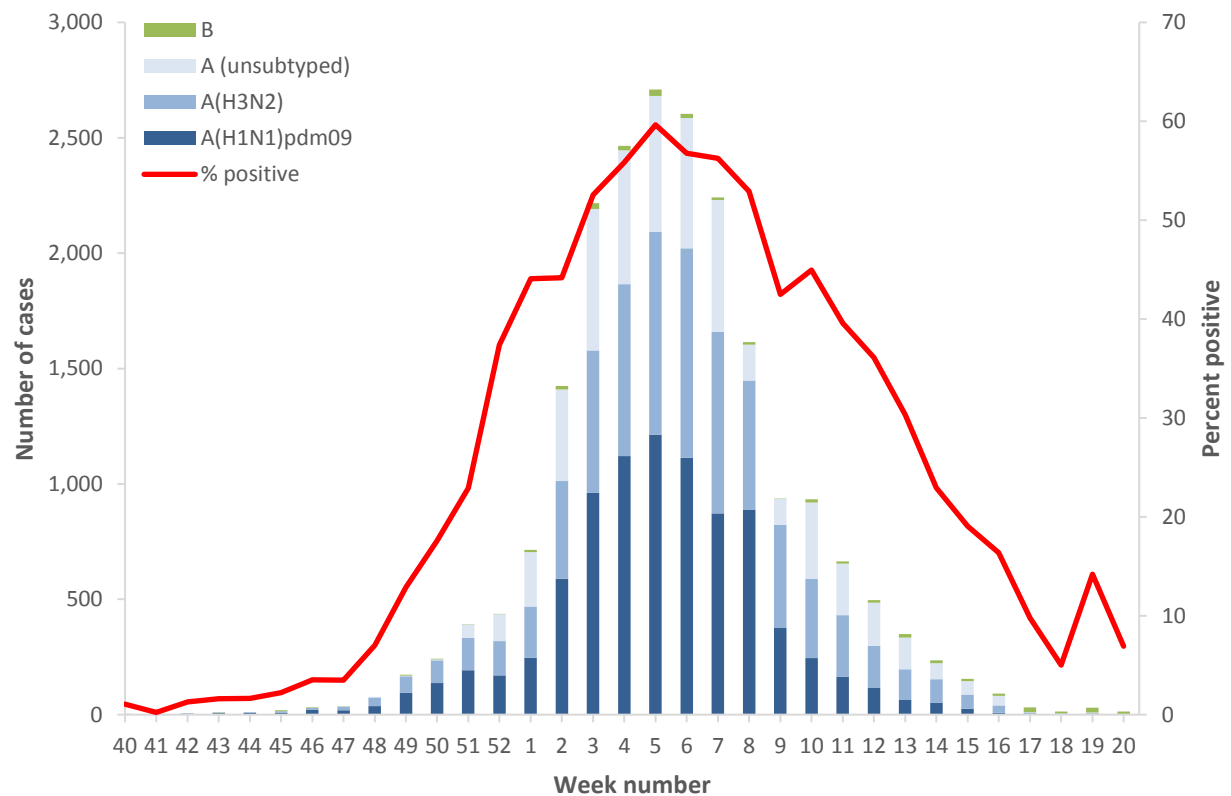
Pooled analyses of all-cause excess and influenza-attributable mortality from 24 European countries or sub-national regions participating in the EuroMOMO network [Austria, Belgium, Berlin (Germany), Denmark, England (United Kingdom (UK)), Estonia, Finland, France, Greece, Hessen (Germany), Hungary, Ireland, Italy, Luxemburg, Malta, the Netherlands, Northern Ireland (UK), Norway, Portugal, Scotland (UK), Spain, Sweden, Switzerland and Wales (UK)] showed increased excess mortality in Southern and South-Eastern Europe during the winter season of 2018-2019. Based on pooled analysis, an increase in all-cause excess mortality was observed from mid-January 2019 and, by the end of January, the pooled estimates confirmed a marked elevated mortality from the beginning of 2019 notably among adults and elderly. Excess mortality began to decline around mid-March and by the end of March it had returned to normal levels. The all-cause mortality observed in the 2018-2019 winter season was moderate. Overall, across the participating countries, there was an excess mortality rate around 20 per 100,000 population with a mortality rate of 10 per 100,000 population being attributable to influenza (EuroMOMO preliminary estimates).

Figure E1. Seasonal (weeks 40–20) influenza activity in the WHO European Region

A. Percentage of sentinel ILI and ARI specimens testing positive for influenza by season and week, 2014–2019 (five seasons)



B. Sentinel ILI and ARI specimens testing positive for influenza by week, (sub)type and percentage positive, 2018–2019 season



C. Non-sentinel ILI and ARI specimens testing positive for influenza by week and (sub)type, 2018–2019 season

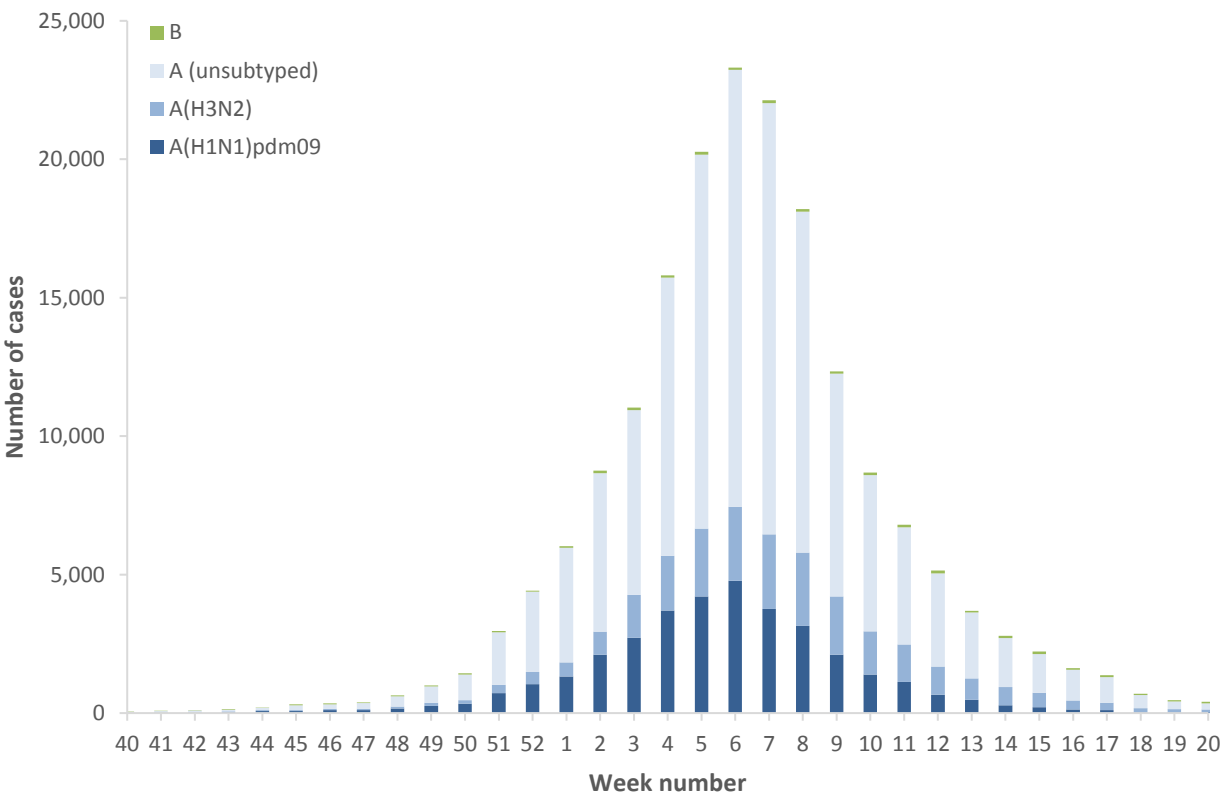
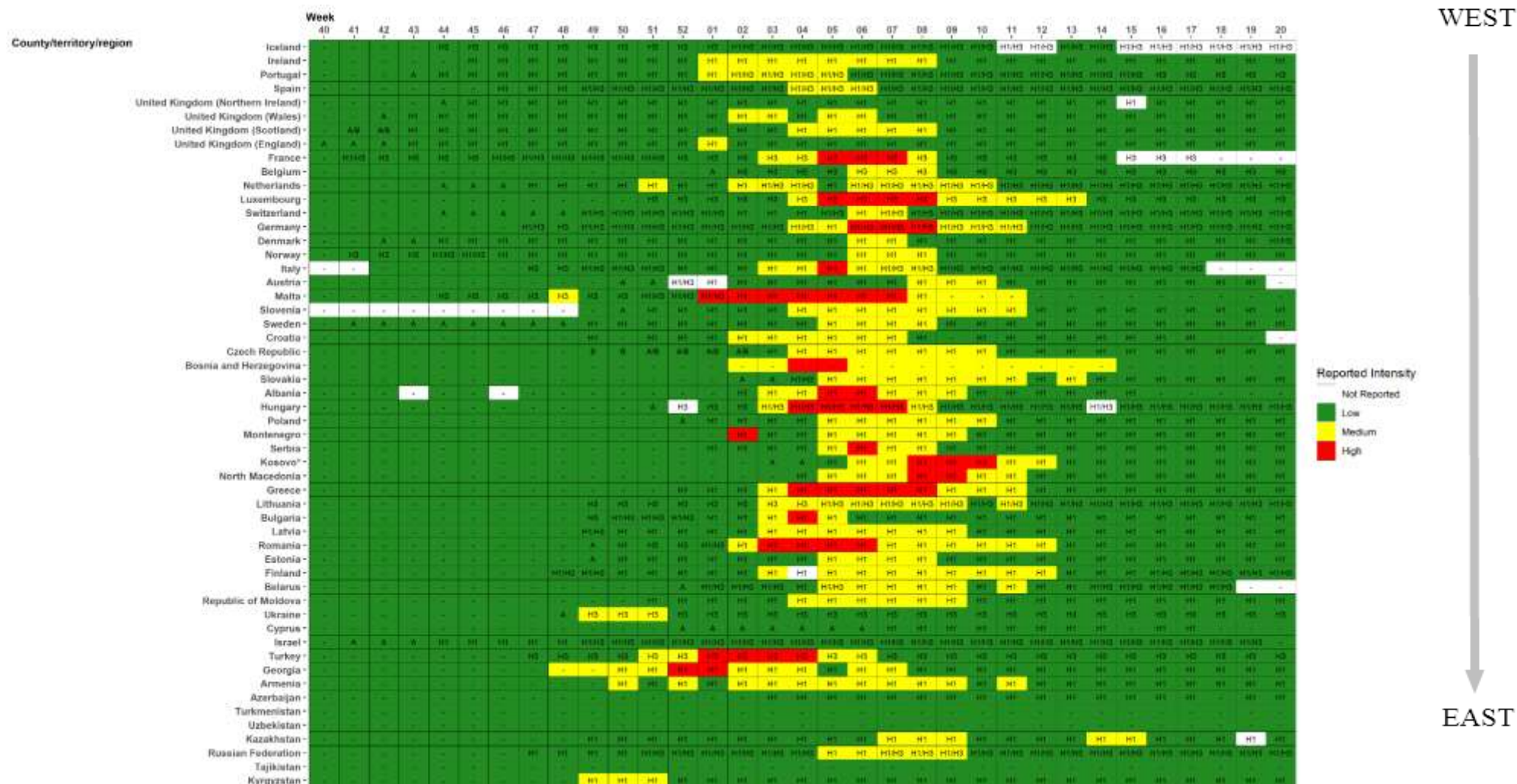


Figure E2. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or region, 2018–2019 season (weeks 40–20)

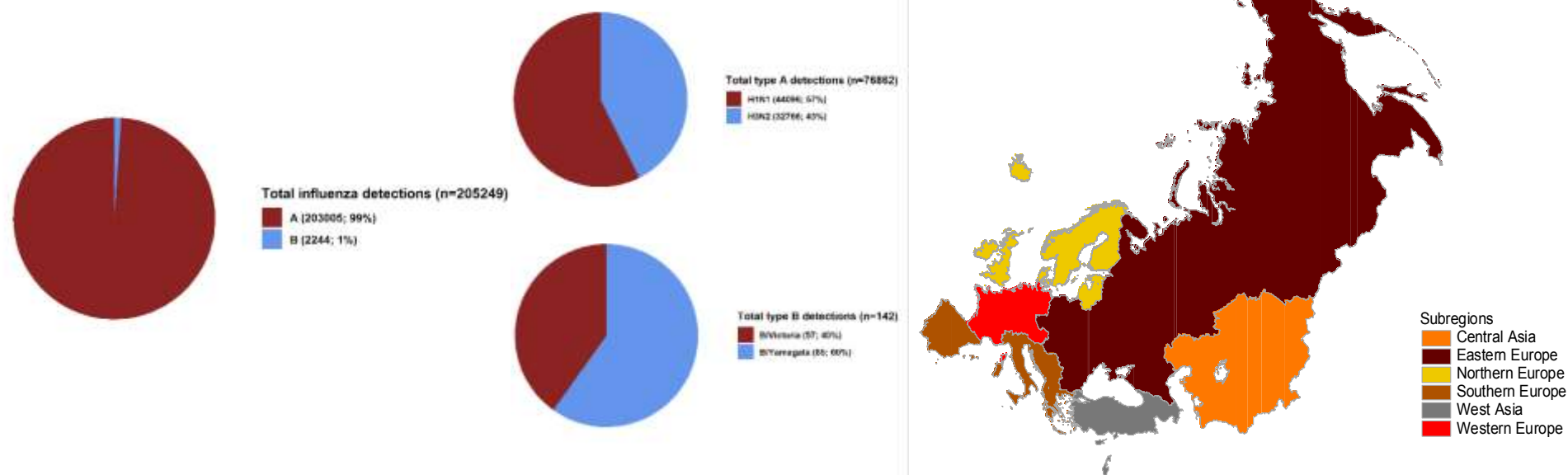


* In accordance with. United Nations Security Council. Resolution 1244 (1999).

Low intensity influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.
Dominant virus (sub)type is defined as at least 60% of detections.

Table E1. Distribution of Influenza Detections in the WHO European Region², 2018–2019 season (weeks 40–20)

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

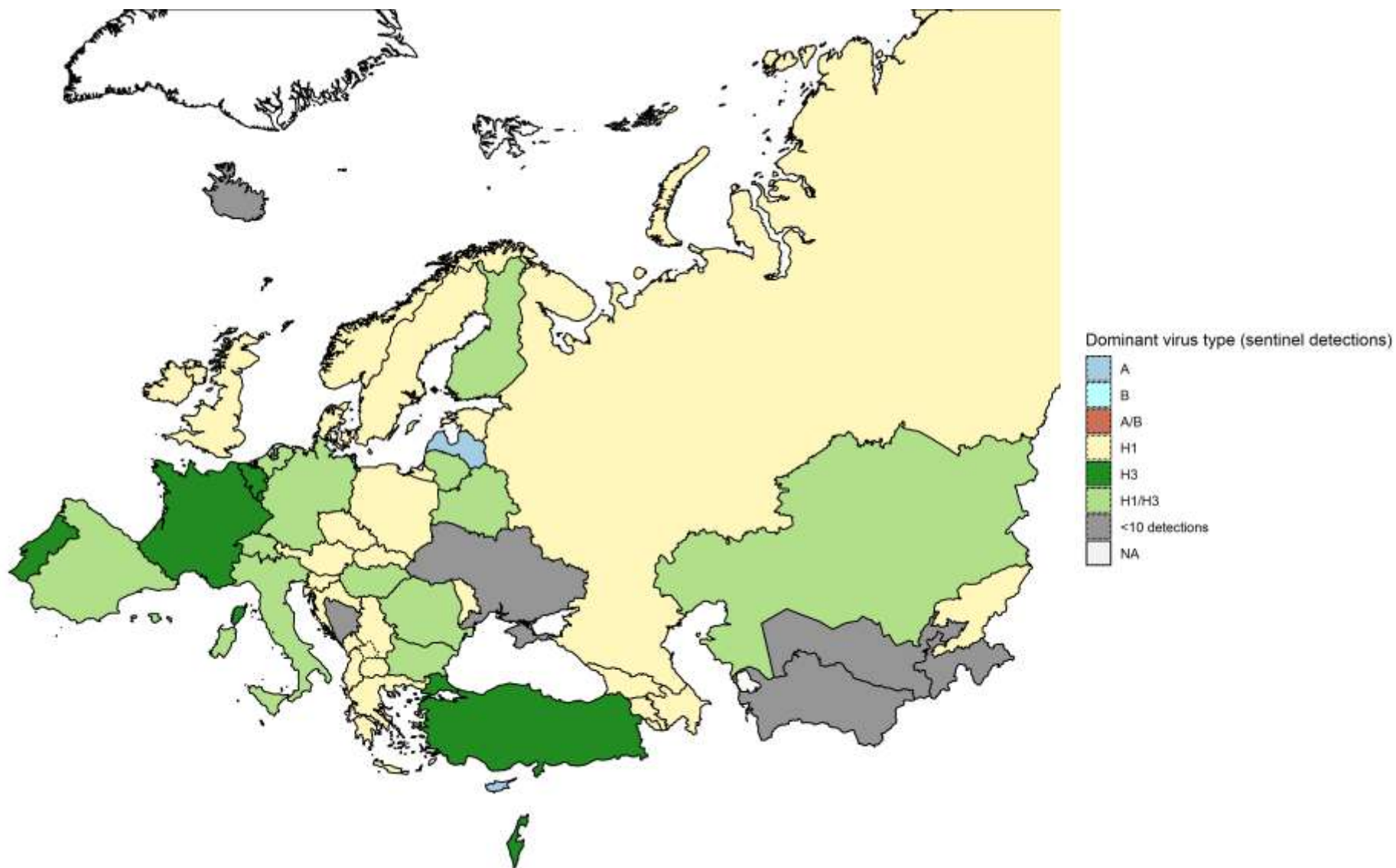
Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	32,820 (99%)	49,215 (99%)	25,337 (98%)	91,147 (99%)	3,898 (95%)	588 (93%)	203,005 (99%)
Influenza A subtyped	20,670	9,052	22,075	21,788	2,689	588	76,862
A(H1N1)pdm09	11,142 (54%)	3,476 (38%)	13,127 (59%)	14,796 (68%)	1,140 (42%)	415 (71%)	44,096 (57%)
A(H3N2)	9,528 (46%)	5,576 (62%)	8,948 (41%)	6,992 (32%)	1,549 (58%)	173 (29%)	32,766 (43%)
Influenza B	204 (1%)	306 (1%)	434 (2%)	1,068 (1%)	188 (5%)	44 (7%)	2,244 (1%)
Influenza B lineage determined	63	5	4	66	3	1	142
B/Yamagata-lineage	49 (78%)	2 (40%)	1 (25%)	32 (48%)	0 (0%)	1 (100%)	85 (60%)
B/Victoria-lineage	14 (22%)	3 (60%)	3 (75%)	34 (52%)	3 (100%)	0 (0%)	57 (40%)
Total	33,024	49,521	25,771	92,215	4,086	632	205,249

² WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

B. Laboratory confirmed influenza from sentinel site ILI and ARI specimens by geographic sub-region and influenza type and subtype/lineage

Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	10,470 (99%)	3,874 (>99%)	1,736 (99%)	2,761 (99%)	1,984 (94%)	253 (95%)	21,078 (99%)
Influenza A subtyped	5,988	3,766	1,632	2,523	1,861	253	16,023
A(H1N1)pdm09	3,467 (58%)	1,561 (41%)	1,139 (70%)	1,822 (72%)	603 (32%)	171 (68%)	8,763 (55%)
A(H3N2)	2,521 (42%)	2,205 (59%)	493 (30%)	701 (28%)	1,258 (68%)	82 (32%)	7,260 (45%)
Influenza B	117 (1%)	9 (<1%)	19 (1%)	17 (1%)	133 (6%)	12 (5%)	307 (1%)
Influenza B lineage determined	52	5	4	2	0	0	63
B/Yamagata-lineage	47 (90%)	2 (40%)	1 (25%)	0 (0%)	-	-	50 (79%)
B/Victoria-lineage	5 (10%)	3 (60%)	3 (75%)	2 (100%)	-	-	13 (21%)
Total	10,587	3,883	1,755	2,778	2,117	265	21,385

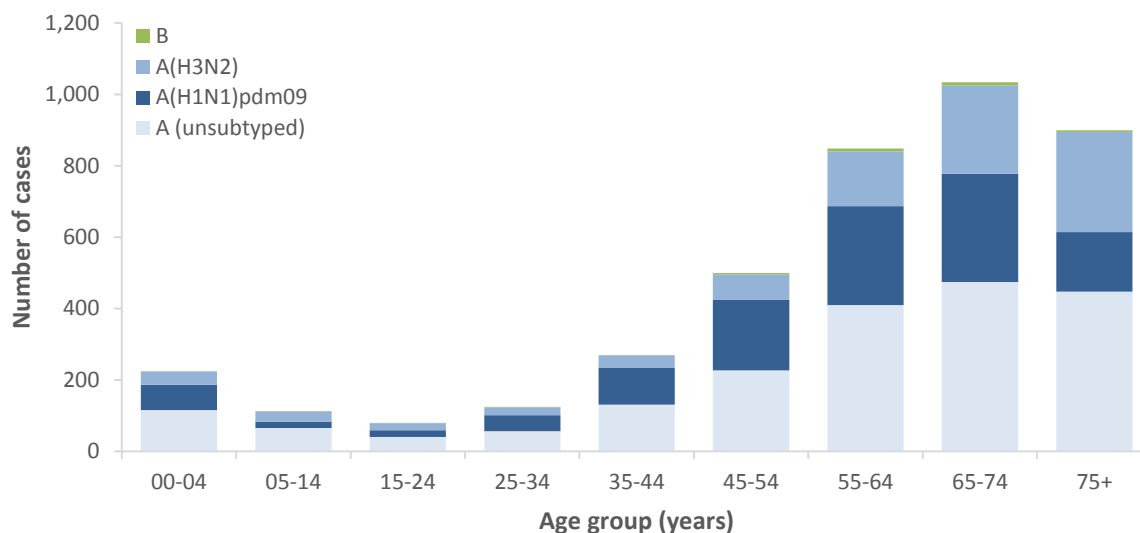
Figure E3. Influenza circulation³ in the WHO European Region by (sub)type and country, territory or region, 2018–2019 season (weeks 40–20)



³ Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Figure E4. Hospitalised influenza cases in the WHO European Region, 2018–2019 season (weeks 40–20)

A. Distribution of virus (sub)type in laboratory-confirmed hospitalised cases in intensive care units^a by age group



^a Reporting countries: Czech Republic, Denmark, Estonia, Finland, France, Ireland, Netherlands, Romania, Russian Federation, Slovakia, Spain, Sweden, Ukraine and the United Kingdom.

B. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or region, 2018-2019 season (weeks 40-20)

Country, territory or area	SARI cases with age data	Percentage of total cases				
		0-<5 years	5-<15 years	15-<30 years	30-<65 years	≥65 years
Albania	983	10	7	15	49	20
Armenia	4,189	58	21	7	10	4
Azerbaijan	147	23	6	38	32	1
Belarus	490	19	9	15	39	17
Bosnia and Herzegovina	572	8	4	5	44	40
Georgia	3,000	55	15	11	12	7
Kazakhstan	20,809	61	13	13	9	3
Kyrgyzstan	3,490	89	4	4	3	0
Moldova, Republic of	4,334	63	10	2	14	12
Montenegro	79	11	0	1	47	41
North Macedonia	861	45	10	7	29	10
Russian Federation	1,321	41	13	18	20	8
Serbia	1,121	13	5	7	43	32
Tajikistan	201	89	11	0	0	0
Turkmenistan	325	73	19	2	2	4
Ukraine	807	45	25	17	11	2
Uzbekistan	277	74	13	6	7	0
Kosovo ^b	120	13	4	5	55	23

^b In accordance with. United Nations Security Council. Resolution 1244 (1999).

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2019-01-31, totalling 1153, have been received from 53 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Influenza A viruses represented 88% of the samples, of which A(H1N1)pdm09 viruses and A(H3N2) viruses were received in similar numbers but A(H1N1)pdm09 viruses predominated over A(H3N2) viruses at a ratio of ~1.2:1. Influenza B viruses represented 12% of the samples received with influenza B viruses of the B/Victoria lineage outnumbering those of the B/Yamagata lineage at a ratio of ~3:1 (**Table 1**).

Of the influenza-positive samples derived from individual patients (n = 1153) received, 490 (42%) were virus (V) isolates, 450 (39%) were clinical specimens (CS) and 213 were sent as CS/V pairs.

Table 2 shows the recovery/isolation rates from samples received with collection dates after 2019-01-31. Viruses were recovered efficiently. For samples sent as clinical samples, that were taken for culture, virus recoveries of between 73% and 94% were achieved, depending on virus subtype or lineage. Recoveries of samples sent as viruses were $\geq 92\%$. For A(H3N2) viruses, efficient recovery was achieved as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but only approximately a half (53%) of these could be assessed by HI assay as the other viruses had low HA titres (<4 HA units in the presence of 20nM oseltamivir) or were unable to agglutinate red blood cells.

Table 3 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses or reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown. For A(H1N1)pdm09 test viruses the HI titres obtained with antisera raised against the vaccine viruses, egg-propagated A/Michigan/45/2015 and egg-propagated A/Brisbane/02/2018, were compared to the respective homologous titres. For A(H3N2) test viruses the HI titres of antisera raised against cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a virus), the vaccine virus egg-propagated A/Singapore/INFIMH-16-0019/2016 (a 3C.2a1 virus), the vaccine virus egg-propagated A/Switzerland/8060/2017 (a 3C.2a2 virus) and a High Yield/Growth Reassortant (X-327) derived from egg-propagated A/Kansas/14/2017 were compared to the titres of the antisera with their homologous viruses. For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres of antisera raised against both tissue culture-propagated and egg-propagated cultivars of A/Norway/3275/2018, egg-propagated A/Switzerland/8060/2017 and X-327 derived from egg-propagated A/Kansas/14/2017 (clade 3C.3a) were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Victoria lineage the HI titres of test viruses with antisera raised against one of three tissue culture-propagated B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016 – clade 1A viruses), tissue-culture propagated B/Colorado/06/2017 (a 1A- Δ 162-163 virus) and

against B/Cote d'Ivoire/1662/2018 (a 1A- Δ 162-164 virus with the additional HA1 amino acid substitution K136E) were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Yamagata lineage the HI titres of test viruses with an antiserum raised against the egg-propagated vaccine virus, B/Phuket/3073/2013, were compared to the titre of the antiserum with the homologous virus.

Table 1a. Summary of clinical samples and virus isolates received, with collection dates after 2019-01-31, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2019													
FEBRUARY													
Algeria	4			1	0	3	0	3					
Armenia	15			5	in process	10	1	1					
Austria	5			4	4	1	1	0					
Belarus	1					1	1						
Bosnia	29			18	in process	11	in process						
Bulgaria	42			23	22	19	12	6					
Cameroon	2			2	0								
Cyprus	15			14	13	1	0	1					
Czech Republic	10			10	10								
Denmark	6			3	3	3	0	3					
Estonia	8			6	4	2	1	1					
Finland	4			4	4								
Georgia	6			3	3	1	0	1		2	2		
Germany	26			9	9	17	9	8					
Ghana	1			1	in process								
Greece	17			11	10	5	1	4				1	1
Iran	5			2	2	2	2			1	1		
Iraq	4			2	0	2	in process						
Ireland	7			3	3	3	2	1		1	1		
Italy	12			4	4	8	3	5					
Jordan	9			2	1	7	0	6					
Kazakhstan	20			18	17	2	2						
Kosovo	10			9	4	1	0						
Latvia	7			1	1	6	5	1					
Macedonia	22			17	17	4	0	4					
Malta	8	1	0	5	2	3	0	1					
Moldova	2			2	0	2	0	1					
Morocco	27			24	22	3	0	3					
Palestine	1			1	0								
Poland	28	1	0	22	7	5	0	3					
Portugal	13			6	6	7	6	1					
Russia	16			8	8	6	5	1				2	2
Senegal	4			6	4	2	in process			1	1	1	0
Serbia	10			4	4	4	0	4					
Slovakia	13			10	10	3	3						
Slovenia	7			3	3	3	1	2		1	1		
Sudan	1			1	1								
Sweden	5			1	1	2	2			1	1	1	1
Switzerland	8			2	2	6	0	6					
Tajikistan	3	1	in process						2	in process			
Turkmenistan	1											1	1
Ukraine	30			21	19	9	8						
United Kingdom	14			4	4	10	10						
MARCH													
Armenia	4			2	in process	2	in process						
Austria	2												
Belarus	12			5	4	7	4	3				2	2
Bosnia	12			7	in process	5	in process						
Bulgaria	1			1	1								
Cameroon	1					1	1						
Cyprus	1					1	0	0					
Czech Republic	2					2	1	1					
Denmark	16			7	7	9	1	8					
Estonia	7			5	5	2	0	2					
Finland	7			2	2	5	2	3					
France	9			4	4	4	3	1				1	1
Georgia	5			2	2	2	1	1		1	1		
Germany	23			7	7	16	12	4					
Ghana	0												
Greece	15			8	3	6	1	4				1	1
Hong Kong	2									1	1	1	1
Iceland	7			4	4	3	2	1					
Iran	4					3	3					1	1
Iraq	1			1	0								
Ireland	17			5	5	11	6	4		1	1		
Italy	15			6	6	9	8	1					
Jordan	7			3	2	2	1	1				2	1
Kazakhstan	3			1	1	2	2						
Kosovo	14			13	12	1	1						
Latvia	2			1	1	1	1						
Macedonia	16			9	9	7	3	3					
Moldova	1			1	0								
Norway	4					1	in process			1	in process	2	in process
Poland	7			7	5								
Portugal	5			3	3	2	1	1					
Russia	40			23	23	14	2	12		2	2	1	1
Senegal	6					4	in process			2	2		
Serbia	10			6	4	4	0	3					
Seychelles	2					2	in process						
Slovakia	4			2	2	2	2						
Slovenia	8			4	4	3	0	3		1	1		
South Africa	2			2	2								
Sweden	5			2	2	2	0	2		1	1		
Switzerland	11			5	3	6	1	5					
Tajikistan	5	1	in process						4	in process			
Ukraine	7			6	6	1	1						
United Kingdom	20			13	2	5	0					2	2

Table 1b. Summary of clinical samples and virus isolates received, with collection dates after 2019-01-31, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage		
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹	
2019														
APRIL														
Angola	5	4	0	1	0									
Armenia	4			2	in process	2	0							
Bosnia	3					3	2							
Cameroon	18			17	in process	1	0							
Cyprus	1			1	1									
Czech Republic	2					2	1	1						
Denmark	1			1	1									
Finland	6			2	2	4	3	1						
France	15			3	3	12	7	5						
Georgia	4			1	1	3	0	2						
Ghana	2					2	2							
Hong Kong	1											1	1	
Iceland	7			1	1	4	1	3		2	2			
Iran	1					1	1							
Iraq	2			2	0									
Ireland	3			1	0	2	1	1						
Jordan	6			4	3	2	1	1						
Kosovo	8			2	1	2	1	1						
Macedonia	7	4	0	2	2	4	1	3						
Madagascar	5									1	1			
Mauritania	1					4	in process			1	0			
Moldova	1									1	1			
Norway	5					1	1			1	in process	3	in process	
Russia	10			3	3	4	3	1		2	2	1	in process	
Senegal	2					2	in process							
Seychelles	2					1	in process		1	in process				
Slovakia	1			1	1									
Slovenia	2			2	2									
South Africa	2					2	2							
Switzerland	2			2	1									
Tajikistan	1	1	in process											
Turkmenistan	7									7	7			
Ukraine	1			1	1									
United Kingdom	15					15	0							
MAY														
Armenia	1					1	1							
Cameroon	5			5	in process									
Finland	1					1	0	1						
France	1									1	1			
Georgia	1					1	0	1						
Hong Kong	2									2	2			
Iceland	7			2	2	5	1	4						
Iraq	1			1	0									
Jordan	1											1	1	
Madagascar	11					2	in process			9	3			
Norway	9					3	in process			3	in process	3	in process	
Portugal	1									1	0			
Russia	5			3	3	1	1			1	1			
Senegal	1					1	1							
Seychelles	3					3	in process							
South Africa	10					10	8	2						
JUNE														
Cameroon	1			1	0									
Ghana	2					2	1							
Hong Kong	35			8	8	12	8	4		15	15			
Iceland	2					2	1	1						
Madagascar	14									14	11			
Norway	8			2	in process	2	1	1		3	in process	1	in process	
Oman	23			12	12	1	1		1	8	in process	1	in process	
Seychelles	1					1	in process							
South Africa	6					6	3	3						
JULY														
Norway	11			4	in process	3	in process			4	in process			
Oman	10			3	in process	2	2		3	2	2			
AUGUST														
Norway	7			3	in process	4	in process							
	1153	13	0	546	399	458	191	166	11	0	95	64	30	18
		1.13%		47.35%		39.72%			0.95%		8.24%		2.60%	
53 Countries				88.2%							11.8%			

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

One virus from Denmark was A(H1N2)pdm09 reassortant

* As of 2019-09-16

Table 2. Isolation rates for specimens collected after 2019-01-31

Seasonal Virus	Clinical sample			Virus isolate		
	Number processed	Number propagated¹	Percent recovered	Number processed	Number propagated¹	Percent recovered
H1	158	116	73%	310	286	92%
H3	102	24²	24%	291	175²	60%
		67³	89% (26%)⁴ 66%		112³	99% (61%)⁴ 39%
B Victoria	16	15	94%	58	57	98%
B Yamagata	5	4	80%	15	14	93%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2019-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ^{1a}	401	307 (76.6%)	62 (15.5%)	32 (7.9%)
H1 ^{1b}	401	315 (78.6%)	68 (17.0%)	18 (4.4%)
H3 ^{2a}	199	62 (31.2%)	91 (45.7%)	46 (23.1%)
H3 ^{2b}	199	3 (1.5%)	27 (13.6%)	169 (84.9%)
H3 ^{2c}	199	1 (0.5%)	5 (2.5%)	193 (97.0%)
H3 ^{2d}	179	69 (38.5%)	14 (7.9%)	96 (53.6%)
H3 ^{3a}	147	71 (48.3%)	32 (21.8%)	44 (29.9%)
H3 ^{3b}	115	0	6 (5.2%)	109 (94.8%)
H3 ^{3c}	36	2 (5.6%)	0	34 (94.4)
H3 ^{3d}	147	0	0	147 (100.0%)
B/Victoria ^{4a}	72	10 (13.9%)	2 (2.8%)	60 (83.3%)
B/Victoria ^{4b}	72	29 (40.3%)	22 (30.6%)	21 (29.1%)
B/Victoria ^{4c}	72	31 (43.1%)	0	41 (56.9%)
B/Yamagata ⁵	18	17 (94.4%)	1 (5.6%)	0

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1a. Compared to homologous titres for ferret antiserum raised against vaccine virus A/Michigan/45/2015 (genetic subgroup 6B.1)

1b. Compared to homologous titres for ferret antiserum raised against vaccine virus A/Brisbane/02/2018 (genetic subgroup 6B.1A1)

2a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

2b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (genetic subgroup 3C.2a1)

2c. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Switzerland/8060/2017 (genetic subgroup 3C.2a2)

2d. Compared to homologous titres for ferret antiserum raised against cell-propagated A/Kansas/14/2017 (genetic subgroup 3C.3a)

3a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Norway/3275/2018 (genetic subgroup 3C.2a1b + T131K) in PRNA (2-fold)

3b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Norway/3275/2018 (genetic subgroup 3C.2a1b + T131K) in PRNA (2-fold)

3c. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Switzerland/8060/2017 (genetic subgroup 3C.2a2) in PRNA (2-fold)

3d. Compared to homologous titres for ferret antiserum raised against egg-propagated reassortant virus NYMC X-327, derived from A/Kansas/14/2017 (genetic subgroup 3C.3a) in PRNA (2-fold)

4a. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture propagated B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016) - Clade 1A

4b. Compared to homologous titres for ferret antisera raised against tissue-culture propagated B/Colorado/06/2017- Clade 1A(Δ2)

4c. Compared to homologous titres for ferret antisera raised against tissue-culture propagated B/Cote d'Ivoire/1662/2018 - Clade 1A(Δ3) 'African'

5. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 47 countries or administrative regions in four WHO Regions: African (5: Algeria, Angola, Cameroon, Ghana and South Africa), Eastern Mediterranean (7: Iran, Iraq, Jordan, Morocco, Oman, Palestine and Sudan), European (34) and Western Pacific (1: Hong Kong SAR) (**Table 1**). **Table 4** summarises the antigenic (HI) results for viruses recovered from clinical specimens with collection dates after 2019-01-31.

Tables 5-1 to 5-26 show the results of HI assays of H1N1 viruses carried out since the February 2019 report. Test viruses in each table are sorted by both date of collection and by genetic group and viruses with collection dates after 2019-01-31, sorted by genetic group, are shown below red lines in **Tables 5-8 to 5-26**. A summary of the HI results for viruses with collection dates after 2019-01-31 is shown in **Table 5-27** including a breakdown by the genetic group of the HA where known.

The vast majority (92%) of viruses collected after 2019-01-31 were antigenically similar to the egg-propagated former vaccine virus A/Michigan/45/2015, being recognised at titres within 4-fold of the titre of the antiserum with the homologous virus (**Table 4** and **Table 5-27**) with 77% being recognised at titres within 2-fold of the homologous titre of the antiserum. The majority (95%) were also recognised by the antiserum raised against the vaccine virus recommended for use in the Northern Hemisphere 2019-2020, egg-propagated A/Brisbane/02/2018, at titres within 4-fold of the homologous titre of the antiserum, with 79% being recognised at titres within 2-fold of the homologous titre.

Antisera raised against seven reference viruses (A/Bayern/69/2009, A/Slovenia/2903/2015, A/Paris/1447/2017, A/Switzerland/2656/2017, A/Switzerland/3330/2017, A/Norway/3433/2018 and A/Ireland/84630/2018) recognised ≥92% of test viruses with collection dates after 2019-01-31 at titres within 4-fold of the titres of the antisera with their respective homologous viruses, with over 80% being recognised at titres within 2-fold of the respective homologous titres (**Table 5-27**). The antiserum raised against cell culture-propagated A/Lviv/N6/2009 recognised only 48% of test viruses at titres within 4-fold of the homologous titres. The antiserum raised against A/Lviv/N6/2009 is an unusual virus/antiserum combination with A/Lviv/N6/2009 encoding **HA1** amino acid substitutions of **G155G/E**, with E predominating, and **D222G**.

All test viruses with collection dates after 2019-01-31, for which HA gene sequences were known, fell into subclade 6B.1A which is defined by the amino acid substitutions **S74R**, **S164T** (altering a glycosylation motif introduced at residues 162-164 in clade 6B.1 viruses) and **I295V** in **HA1**, compared with the clade 6B.1 prototype A/Michigan/45/2019 virus. **Figure 1** shows a phylogenetic tree for HA genes of representative A(H1N1)pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses.

The HA of most test viruses carry the **S183P** amino acid substitution in **HA1**, but this is not retained in all viruses. Within subclade 6B.1A genetic clusters of viruses encoding a range of **HA1** amino acid substitutions have emerged with the major groups being:

- 6B.1A2, with HAs defined by the substitutions **L233I** in **HA1** and **V192A** in **HA2** (**V520A**),
- 6B.1A5, with HAs defined by **N129D**, **T185I** and **N260D** in **HA1**,
- 6B.1A6, with HAs defined by **T120A** in **HA1**,
- 6B.1A7, with HAs defined by **K302T** in **HA1** and **I76M**, **N169S** and **E179D** in **HA2** (**I404M**, **N496S** and **E506D**).

Of the viruses with collection dates after 2019-01-31 analysed at the Francis Crick Institute: 53.3% fell into genetic group 6B.1A5, 16.7% into group 6B.1A7, 14.4% into group 6B.1A6, 6.9% into group 6B.1A2, and 8.5% fell into the original 6B.1A subclade. **Table 5-27** shows that viruses in subclade 6B.1A and each of the four genetic groups identified above show similar patterns of recognition by the panel of post-infection ferret antisera.

An amino acid sequence alignment of the HAs for representative H1N1pdm09 viruses is shown in **Figure 2**; the vaccine viruses are shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 shows a H1pdm09-HA structure with the amino acid substitutions that differentiate subclade 6B.1A viruses (**S74R**, **S164T** and **I295V** in **HA1**) from clade 6B.1 viruses indicated in red and those that define two main groups of subclade 6B.1A viruses (**T120A**, **S183P/S** and **N129D**, **T185I** and **N260D** in **HA1**) indicated in blue and black.

Figure 4 shows a phylogenetic tree for the NA genes of representative H1N1pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses. The vast majority encoded the amino acid substitutions **G77R**, **V81A**, **I188T** and **N449D**, compared with the NA of A/Michigan/45/2015. Those with HA genes in genetic groups 6B.1A5 and 6B.1A6 also carry the NA substitution **D416N**, and viruses with HA genes in genetic group 6B.1A5 carry additional NA substitutions of **Q51K** and **T452I** many with **F74S** and **I389K** also. Those with HA genes in group 6B.1A7 have the NA substitution **M314I**.

Figure 5 shows a NA amino acid alignment for the same representative set of H1N1pdm09 viruses as used in **Figure 2**, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 404 viruses with collection dates after 2019-01-31. All but three viruses showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 14**). The exceptions were A/Cyprus/919/2019 with highly reduced inhibition (HRI) by oseltamivir (NA N295S), A/Trnava/535/2019 in Slovakia with HRI by oseltamivir (NA H275Y) and A/Denmark/2697/2019 with reduced inhibition (RI) by zanamivir (NA Q136X and D151X).

Conclusion.

All A(H1N1)pdm09 viruses received fell within subclade 6B.1A encoding the **HA1** amino acid substitutions **S74R**, **S164T** (which alters the glycosylation motif at residues **162 to 164**) and **I295V**. The majority of viruses in this subclade carry **HA1 S183P** amino acid substitution and a number of genetic groups defined by additional amino acid substitutions have emerged. The predominant genetic group was 6B.1A5 representing 53% of the viruses tested with collection dates after 2019-01-31. The majority of test viruses in each genetic group remained antigenically similar to the former vaccine virus, A/Michigan/45/2015, and the currently recommended vaccine virus, A/Brisbane/02/2018, as determined by assays using panels of post-infection ferret antisera. The vast majority of test viruses (>99%) retained susceptibility to oseltamivir and zanamivir.

Table 4a. Summary of influenza A(H1N1)pdm09 viruses collected after 2019-01-31, analysed by HI

MONTH	H1N1pdm09							
	Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against				
				Clade 6B.1 virus			Clade 6B.1A1 virus	
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
2019								
FEBRUARY								
Algeria	1	0						
Armenia	5	0						
Austria	4	4	1	2	1	2	2	
Bosnia	18	0						
Bulgaria	23	22	18	4		22		
Cameroon	2	0						
Cyprus	14	13	11	2		6	6	1
Czech Republic	10	10	10			10		
Denmark	3	3		2	1	3		
Estonia	6	4	4			4		
Finland	4	4	1	1	2	2	2	
Georgia	3	3	3			3		
Germany	9	9	9			9		
Ghana	1	0						
Greece	11	10	10			9	1	
Iran	2	2	2			2		
Iraq	2	0						
Ireland	3	3	3			2	1	
Italy	4	4	4			4		
Jordan	2	1	1			1		
Kazakhstan	18	17	13	4		8	8	1
Kosovo	9	4	4			4		
Latvia	1	1	1			1		
Macedonia	17	17	6	10	1	7	9	1
Malta	5	2	2			1	1	
Morocco	24	22	22			16	6	
Palestine	1	0						
Poland	22	7	7			4	3	
Portugal	6	6	5		1	5		1
Russia	8	8	6	2		8		
Serbia	6	4		1	3	3	1	
Slovakia	10	10	10			10		
Slovenia	3	3	2	1		3		
Sudan	1	1	1			1		
Sweden	1	1			1			1
Switzerland	2	2	2			2		
Ukraine	21	19	6	9	4	12	6	1
United Kingdom	4	4	4			4		
MARCH								
Armenia	2	0						
Belarus	5	4	3	1			2	2
Bosnia	7	0						
Bulgaria	1	1	1			1		
Denmark	7	7	1	2	4	3	4	
Estonia	5	5	5			5		
Finland	2	2			2		2	
France	4	4	3	1		4		
Georgia	2	2	2			2		
Germany	7	7	7			7		
Greece	8	3	3			3		
Iceland	4	4	3	1			3	1
Iraq	1	0						
Ireland	5	5	4		1	4		1
Italy	6	6		6		6		
Jordan	3	2	2			2		
Kazakhstan	1	1	1			1		
Kosovo	13	12	12			12		
Latvia	1	1	1			1		
Macedonia	9	9	2	1	6	4		5
Moldova	1	0						
Poland	7	5	5			4	1	
Portugal	3	3	3			1	2	
Russia	23	23	22	1		23		
Serbia	6	4		3	1	4		
Slovakia	2	2	2			2		
Slovenia	4	4	4			4		
South Africa	2	2	2			2		
Sweden	2	2		1	1	1		1
Switzerland	5	3	3			3		
Ukraine	6	6	3	2	1	3	1	2
United Kingdom	13	2	2			2		

Table 4b. Summary of influenza A(H1N1)pdm09 viruses collected after 2019-01-31, analysed by HI

MONTH	H1N1pdm09							
			Post infection ferret antisera raised against					
			Clade 6B.1 virus			Clade 6B.1A1 virus		
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
2019								
APRIL								
Angola	1	0						
Armenia	2	0						
Cameroon	17	0						
Cyprus	1	1	1			1		
Denmark	1	1		1			1	
Finland	2	2	1	1			2	
France	3	3	3			3		
Georgia	1	1	1			1		
Iceland	1	1	1				1	
Iraq	2	0						
Ireland	1	0						
Jordan	4	3	3			3		
Kosovo	2	1	1			1		
Macedonia	2	2	1	1		2		
Russia	3	3	3			3		
Slovakia	1	1	1			1		
Slovenia	2	2	1		1	2		
Switzerland	2	1	1				1	
Ukraine	1	1			1		1	
MAY								
Cameroon	5	0						
Iceland	2	2	2			1	1	
Iraq	1	0						
Russia	3	3	3			3		
JUNE								
Cameroon	1	0						
Hong Kong	8	8	8			8		
Norway	2	0						
Oman	12	12	10	2		12		
JULY								
Norway	4	0						
Oman	3	2	2			2		
AUGUST								
Norway	3	0						
47 Countries	546	401	307	62	32	315	68	18
			76.6%	15.5%	8.0%	78.6%	17.0%	44.5%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg-propagated vaccine virus)

4 Reduction in HI titre with post-infection ferret antiserum raised against A/Brisbane/02/2018 (egg-propagated vaccine virus)

H1N2 reassortant

Numbers highlighted in blue show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Haemagglutination inhibition titre																
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera											
					A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/Slov	A/Paris	A/Swit	A/Swit	A/Norway		
					45/15	7/09	69/09	N6/09	1/11	5659/12	2903/15	1447/17	2656/17	3330/17	3433/18	
					Egg	Egg	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	Egg	MDCK	
					NIB	F07/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F22/13 ⁻¹	F17/15 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	
					F42/16 ⁻¹											
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	
REFERENCE VIRUSES																
A/Michigan/45/2015			2015-09-07	ES/IE3	640	640	320	320	320	640	1280	2560	1280	320	1280	
A/California/7/2009	clone 38-32		2009-04-09	ES/IE3	320	640	320	320	320	640	1280	640	320	1280		
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	<	320	320	<	40	320	<	40	320		
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	1280	80	160	640	320	320	1280		
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	640	640	640	640	320	640	1280	2560	640	640	2560	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	160	160	80	160	320	320	640	320	160	1280	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	640	320	320	640	640	1280	2560	1280	640	2560	
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	320	80	320	320	1280	2560	640	320	2560	
A/Switzerland/2656/2017		6B.1A	2017-12-21	ES/IE3	640	640	640	320	640	640	1280	2560	1280	640	2560	
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	640	320	160	320	320	640	2560	1280	640	2560	
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	160	160	80	160	160	640	1280	640	320	2560	
TEST VIRUSES																
A/Moscow/205/2018		6B.1A	2018-12-10	MDCK1/MDCK1	640	640	640	80	320	320	1280	2560	1280	640	2560	
A/Moscow/207/2018		6B.1A	2018-12-13	MDCK1/MDCK1	640	640	320	80	320	320	1280	2560	640	320	2560	
A/Moscow/217/2018		6B.1A	2018-12-18	MDCK1/MDCK1	640	640	640	160	320	640	1280	2560	1280	640	2560	
A/Moscow/215/2018		6B.1A	2018-12-18	MDCK1/MDCK1	160	80	80	<	80	160	320	640	320	160	1280	
A/Norway/4114/2018		6B.1A	2018-12-18	MDCK1/MDCK1	1280	1280	1280	320	640	640	2560	5120	2560	1280	5120	
A/Moscow/225/2018		6B.1A	2018-12-23	MDCK1/MDCK1	1280	320	320	160	320	320	1280	2560	1280	320	2560	
A/Moscow/228/2018		6B.1A	2018-12-27	MDCK1/MDCK1	1280	160	160	320	320	320	1280	2560	1280	640	2560	
A/Hong Kong/108/2019		6B.1A	2018-12-28	MDCK1/MDCK1	640	640	320	160	640	640	1280	1280	1280	640	2560	
A/Hong Kong/103/2019		6B.1A	2018-12-31	MDCK1/MDCK1	1280	640	640	160	320	320	1280	2560	1280	640	2560	
A/Croatia/5357/2018		6B.1A1	2018-12-04	MDCK1/MDCK3	640	80	320	80	160	160	640	1280	640	320	1280	
A/Hong Kong/2956/2018	K154X, N156K	6B.1A2	2018-11-25	MDCK1/MDCK1	40	<	80	40	<	<	40	<	<	<	80	
A/Cyprus/FR866/2018	N156K	6B.1A2	2018-12													

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-02-19)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Mich 45/15 Egg NIB F42/16 ⁻¹	A/Cal 7/09 Egg F07/16 ⁻¹	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F13/18 ⁻¹	A/Astrak 1/11 MDCK F22/13 ⁻¹	A/HK 5659/12 MDCK F17/15 ⁻¹	A/Slov 2903/15 Egg NIB F48/16 ⁻¹	A/Paris 1447/17 MDCK F03/18 ⁻²	A/Swit 2656/17 Egg F20/18 ⁻¹	A/Swit 3330/17 Egg F23/18 ⁻¹	A/Norway 3433/18 MDCK F04/19 ^{*1}
					6B.1					5	6A	6B.1	6B.1A	6B.1A	6B.1A5
REFERENCE VIRUSES															
A/Michigan/45/2015			2015-09-07	E3/E3	640	640	320	320	640	320	1280	1280	640	320	1280
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	640	320	320	320	320	320	640	1280	640	160	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	<	<	320	320	<	<	<	320	40	40	320
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	1280	80	80	160	640	160	160	1280
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	640	320	640	640	640	320	1280	2560	640	320	2560
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	640	320	160	160	320	320	640	1280	320	320	1280
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	320	320	320	320	640	1280	640	320	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	320	160	320	320	640	2560	640	320	2560
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	320	640	640	320	320	1280	1280	1280	320	2560
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	160	160	160	160	320	1280	320	320	1280
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	160	160	80	160	160	640	1280	320	160	1280
TEST VIRUSES															
A/Saint-Etienne/1883/2018			2018-08-31	MDCK3/MDCK1	640	320	320	80	320	320	1280	1280	640	320	2560
A/Georgia/664/2018			2018-11-13	MDCK1	320	320	320	160	320	160	640	1280	640	320	1280
A/Ireland/80216/2018		6B.1A	2018-11-12	MDCK1/MDCK1	640	320	320	160	320	320	640	1280	640	320	1280
A/Ireland/78012/2018		6B.1A1	2018-11-05	MDCK1/MDCK1	320	160	160	<	160	160	640	1280	320	160	1280
A/Georgia/675/2018		6B.1A2	2018-11-20	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Georgia/736/2018		6B.1A2	2018-12-06	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Georgia/864/2018		6B.1A2	2018-12-24	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Georgia/912/2018		6B.1A2	2018-12-26	MDCK1	640	320	640	160	640	320	1280	2560	1280	640	2560
A/Georgia/927/2018		6B.1A2	2018-12-28	MDCK1	640	640	320	320	320	320	1280	2560	640	640	2560
A/Georgia/91/2019		6B.1A2	2019-01-07	MDCK1	1280	640	640	320	640	640	2560	2560	1280	640	2560
A/Marseille/1989/2018		6B.1A5	2018-10-05	MDCK2/MDCK1	1280	320	320	160	320	320	1280	2560	640	320	2560
A/Ireland/70370/2018		6B.1A5	2018-10-07	MDCK2/MDCK1	640	160	160	160	160	160	640	1280	320	160	1280
A/Ireland/78268/2018		6B.1A5	2018-11-06	MDCK1/MDCK1	640	160	160	320	160	320	640	1280	320	320	2560
A/Lyon/2065/2018		6B.1A5	2018-11-08	MDCK3/MDCK1	320	80	320	320	160	160	640	1280	320	320	2560
A/Lyon/2088/2018		6B.1A5	2018-11-12	MDCK2/MDCK1	1280	320	640	320	320	320	1280	2560	640	640	2560
A/Ireland/81752/2018		6B.1A5	2018-11-20	MDCK2/MDCK1	640	320	320	320	320	640	1280	2560	640	640	2560
A/Netherlands/10000/2019		6B.1A5	2019-01-02	MDCK-MIX2/MDCK1	640	160	160	320	160	160	640	1280	320	160	2560
A/Bosnia and Herzegovina/1244/2019		6B.1A5	2019-01-04	MDCK1	640	320	320	320	320	320	640	2560	640	320	2560
A/Switzerland/4553/2019		6B.1A5	2019-01-07	MDCK2	640	320	320	320	320	320	640	1280	640	320	2560
A/Switzerland/4217/2019		6B.1A5	2019-01-08	MDCK2	640	320	640	320	320	320	1280	2560	640	320	2560
A/Bosnia and Herzegovina/1782/2019		6B.1A5	2019-01-08	MDCK1	640	320	320	320	320	320	640	2560	640	320	2560
A/Netherlands/10003/2019		6B.1A5	2019-01-10	MDCK-MIX2/MDCK1	640	320	320	320	320	320	640	2560	640	320	2560
A/Netherlands/10004/2019		6B.1A5	2019-01-11	MDCK-MIX2/MDCK1	640	640	640	640	640	640	1280	2560	640	640	2560
A/Switzerland/5126/2018		6B.1A6	2018-12-31	MDCK2	1280	640	640	160	640	640	1280	2560	640	320	2560
A/Netherlands/10001/2019		6B.1A6	2019-01-07	MDCK-MIX2/MDCK1	640	320	320	160	320	160	1280	2560	640	320	2560
A/Ireland/79897/2018		6B.1A7	2018-11-12	MDCK1/MDCK1	640	160	160	80	160	160	640	2560	640	160	2560
A/Netherlands/10615/2018		6B.1A7	2018-12-28	MDCK-MIX2/MDCK1	640	320	320	320	320	320	1280	2560	640	320	2560
A/Netherlands/10614/2018		6B.1A7	2018-12-28	MDCK-MIX2/MDCK1	640	320	320	160	320	320	640	2560	640	320	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-02-26)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												NEW
				Post-infection ferret antisera												
				A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/HK 5659/12	A/Slov 2903/15	A/Paris 1447/17	A/Swit 2656/17	A/Swit 3330/17	A/Norway 3433/18	A/Ireland 84630/18	
				Egg	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	MDCK	
				F42/16 ⁻¹	F07/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F22/13 ⁻¹	F17/15 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	
Genetic group	6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6				
REFERENCE VIRUSES																
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	640	640	320	640	640	1280	2560	1280	320	2560	1280
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	640	640	640	320	640	640	1280	2560	1280	320	2560	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	<	320	320	<	<	40	320	40	40	320	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	1280	80	160	160	1280	320	160	1280	320
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	640	640	640	640	640	640	640	2560	640	320	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	640	320	320	160	320	640	1280	2560	640	320	2560	1280
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	640	320	320	640	640	1280	2560	1280	320	2560	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	320	160	320	320	1280	2560	1280	640	2560	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	640	320	320	640	640	1280	2560	1280	640	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	320	320	160	320	320	640	1280	640	640	1280	1280
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	160	80	160	320	640	1280	320	320	2560	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK2	640	320	320	160	320	320	640	2560	640	320	2560	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	1280	640	640	640	1280	2560	5120	2560	640	5120	2560
TEST VIRUSES																
A/Slovenia/2615/2018		6B.1A	2018-10-15	MDCKx/MDCK1	1280	1280	640	320	640	1280	2560	5120	2560	1280	5120	2560
A/Illinois/08/2018		6B.1A1	2018-01-16	C2/MDCK1	640	640	320	160	640	640	1280	2560	1280	640	5120	2560
A/Illinois/37/2018		6B.1A1	2018-02-08	C2/MDCK1	1280	1280	640	320	640	1280	2560	5120	1280	640	5120	2560
A/Baden-Wuerttemberg/161/2018		6B.1A2	2018-08-16	C1/MDCK1	1280	640	640	320	640	1280	2560	5120	1280	640	5120	2560
A/Schleswig-Holstein/3/2018		6B.1A4		C2/MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560	1280
A/Lisboa/25/2018		6B.1A5	2018-10-25	MDCK1/MDCK1	640	160	320	160	320	160	640	2560	640	320	2560	1280
A/Serbia/17140/2018		6B.1A5	2018-12-29	C3/MDCK1	640	160	320	320	160	320	640	2560	640	320	2560	1280
A/Slovenia/82/2019		6B.1A5	2019-01-03	MDCKx/MDCK1	640	320	320	160	320	320	640	2560	640	320	2560	1280
A/Slovenia/80/2019		6B.1A5	2019-01-03	SIATx/MDCK1	640	320	320	160	320	640	640	1280	640	320	2560	1280
A/Avila/33/2019		6B.1A5	2019-01-15	MDCK1/MDCK1	640	320	320	320	640	320	1280	2560	640	320	2560	1280
A/Valladolid/28/2019		6B.1A5	2019-01-15	MDCK1/MDCK1	1280	640	640	320	640	320	1280	2560	1280	640	2560	2560
A/Valladolid/559/2018		6B.1A6	2018-09-24	MDCK1/MDCK1	640	320	160	80	160	320	640	1280	640	320	2560	1280
A/Switzerland/4642/2018		6B.1A6	2018-12-27	MDCK3	640	320	320	160	320	640	1280	5120	640	320	2560	1280
A/Lisboa/26/2018		6B.1A7	2018-10-25	MDCK3/MDCK1	1280	640	640	160	640	320	1280	5120	1280	640	5120	2560
A/Ireland/83112/2018		6B.1A7	2018-11-22	MDCK2	1280	640	640	320	640	640	2560	5120	1280	640	2560	2560
A/Ireland/82546/2018		6B.1A7	2018-11-22	MDCK2	1280	640	640	320	640	640	2560	5120	1280	640	5120	2560
A/Lisboa/29/2018		6B.1A7	2018-12-06	MDCK1/MDCK1	1280	320	640	160	640	320	1280	2560	1280	640	2560	1280
A/Lisboa/27/2018		6B.1A7	2018-12-08	MDCK1/MDCK1	640	320	320	160	320	160	1280	5120	640	320	2560	1280
A/Lisboa/28/2018		6B.1A7	2018-12-10	MDCK2/MDCK1	640	320	160	80	320	160	640	2560	640	320	2560	1280
A/Lisboa/30/2018		6B.1A7	2018-12-14	MDCK2/MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560	1280
A/Slovenia/3088/2018		6B.1A7	2018-12-26	SIATx/MDCK1	1280	640	640	320	640	640	2560	5120	2560	640	2560	2560
A/Burgos/561/2018		6B.1A7	2018-12-28	MDCK1/MDCK1	2560	1280	1280	640	640	1280	2560	5120	2560	1280	5120	5120
A/Portugal/119/2018		6B.1A7	2018-12-29	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560	1280
A/Valladolid/1/2019		6B.1A7	2019-01-02	MDCK1/MDCK1	1280	640	640	320	640	1280	2560	5120	2560	1280	5120	5120
A/Valladolid/8/2019		6B.1A7	2019-01-08	MDCK1/MDCK1	1280	640	640	320	640	640	1280	5120	1280	640	5120	2560
A/Burgos/6/2019		6B.1A7	2019-01-08	MDCK1/MDCK1	1280	640	640	160	320	640	1280	5120	1280	640	5120	2560
A/Valladolid/10/2019		6B.1A7	2019-01-09	MDCK1/MDCK1	640	320	320	160	320	320	1280	2560	1280	640	2560	1280
A/Valladolid/16/2019		6B.1A7	2019-01-11	MDCK1/MDCK1	1280	640	640	320	640	640	2560	5120	2560	1280	5120	5120
A/Palencia/19/2019		6B.1A7	2019-01-12	MDCK1/MDCK1	640	320	320	160	320	320	1280	2560	1280	640	2560	2560
A/Valladolid/22/2019		6B.1A7	2019-01-13	MDCK1/MDCK1	1280	640	320	160	640	320	1280	5120	1280	640	5120	2560
A/Burgos/35/2019		6B.1A7	2019-01-15	MDCK1/MDCK1	640	320	320	160	640	320	1280	2560	1280	640	2560	2560
A/Valladolid/29/2019		6B.1A7	2019-01-15	MDCK1/MDCK1	1280	640	640	320	640	640	2560	5120	1280	640	5120	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-03-05)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/Mich 45/15 Egg NIB F42/16 ⁻¹ 6B.1	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F13/18 ⁻¹	A/Slov 2903/2015 Egg NIB F48/16 ⁻¹ 6B.1	A/Paris 1447/17 MDCK F03/18 ⁻²	A/Swit 2656/17 Egg F20/18 ⁻¹	A/Swit 3330/17 Egg F23/18 ⁻¹	A/Norway 3433/18 MDCK F04/19 ⁻¹	A/Ire 84630/18 MDCK F08/19 ⁻¹
REFERENCE VIRUSES													
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	640	320	2560	2560	1280	640	2560	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	320	320	40	320	80	80	320	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	160	1280	320	160	1280	160
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	320	320	160	1280	1280	640	320	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	80	1280	2560	640	320	2560	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	320	320	1280	2560	1280	640	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	320	160	640	1280	640	320	1280	1280
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	80	640	1280	320	320	2560	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK2	640	320	160	1280	5120	640	320	5120	2560
TEST VIRUSES													
A/Ghana/3724/2018		6B.1A	2018-11-02	C1/MDCK1	1280	640	160	1280	2560	1280	640	2560	2560
A/Bulgaria/1464/2018		6B.1A	2018-12-11	MDCK1	640	320	160	1280	2560	640	640	2560	2560
A/Estonia/118253/2018		6B.1A	2018-12-17	MDCK2	640	320	160	640	1280	640	320	2560	1280
A/Bulgaria/1526/2018		6B.1A	2018-12-27	MDCK1	640	320	80	1280	2560	640	320	2560	1280
A/Bulgaria/1533/2018		6B.1A	2018-12-28	MDCK1	640	320	80	1280	2560	640	320	2560	1280
A/Norway/3737/2018		6B.1A3	2018-11-27	MDCK1	1280	640	640	1280	2560	1280	640	2560	2560
A/Stockholm/21/2018		6B.1A5	2018-09-30	MDCK0/MDCK1	640	160	160	320	1280	320	320	2560	640
A/Norway/3379/2018		6B.1A5	2018-10-23	MDCK1/MDCK1	320	160	80	640	1280	320	320	2560	640
A/Ghana/3709/2018		6B.1A5	2018-11-01	C1/MDCK1	640	320	80	1280	2560	640	320	5120	2560
A/Ghana/3699/2018		6B.1A5	2018-11-01	C1/MDCK1	1280	640	160	1280	5120	640	640	5120	2560
A/Ghana/3693/2018		6B.1A5	2018-11-01	C1/MDCK1	640	640	160	2560	2560	1280	640	5120	2560
A/Ghana/748/2018		6B.1A5	2018-11-06	C1/MDCK1	1280	640	160	1280	5120	640	640	2560	2560
A/Norway/3488/2018		6B.1A5	2018-11-07	MDCK1/MDCK1	320	80	40	320	640	320	160	1280	640
A/Norway/3473/2018		6B.1A5	2018-11-07	MDCK1	640	320	320	640	2560	640	320	2560	1280
A/Norway/3524/2018		6B.1A5	2018-11-10	MDCK1/MDCK1	640	640	320	1280	2560	640	640	5120	1280
A/Norway/3585/2018		6B.1A5	2018-11-14	MDCK1	1280	640	640	1280	2560	1280	640	5120	2560
A/Norway/3513/2018		6B.1A5	2018-11-15	MDCK1/MDCK1	640	320	320	1280	2560	1280	320	2560	1280
A/Norway/3637/2018		6B.1A5	2018-11-26	MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Norway/3672/2018		6B.1A5	2018-11-28	MDCK1	640	160	160	640	1280	640	320	2560	1280
A/Norway/3679/2018		6B.1A5	2018-11-29	MDCK1	640	320	160	640	1280	640	320	2560	640
A/Norway/3652/2018		6B.1A5	2018-11-29	MDCK1/MDCK1	640	160	160	640	1280	640	320	2560	640
A/Estonia/117956/2018		6B.1A5	2018-11-29	MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Lulea/1/2018		6B.1A5	2018-12-04	MDCK0/MDCK1	320	160	160	640	1280	320	320	1280	640
A/Eskilstuna/2/2018		6B.1A5	2018-12-06	MDCK0/MDCK1	320	320	160	640	1280	320	320	2560	1280
A/Bulgaria/1441/2018		6B.1A5	2018-12-11	MDCK1	1280	640	640	1280	2560	640	640	5120	1280
A/Norway/4138-2/2018		6B.1A5	2018-12-24	MDCK1	640	320	160	640	2560	320	320	2560	1280
A/Norway/90/2019		6B.1A5	2019-01-06	MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Norway/95/2019		6B.1A5	2019-01-07	MDCK1	640	320	160	1280	2560	320	320	2560	1280
A/Norway/140/2019		6B.1A5	2019-01-07	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Norway/223/2019		6B.1A5	2019-01-14	MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Halmstad/3/2018		6B.1A7	2018-11-12	MDCK1/MDCK1	1280	640	160	1280	5120	1280	640	5120	2560
A/Norway/3478/2018		6B.1A7	2018-11-14	MDCK1	320	160	80	640	1280	320	160	1280	640
A/Norway/82/2019		6B.1A7	2019-01-05	MDCK1	1280	320	160	2560	5120	640	640	5120	2560
A/Norway/103/2019		6B.1A7	2019-01-07	MDCK1	640	320	160	1280	2560	640	320	5120	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-03-12)

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Mich	A/Bayern	A/Lviv	A/Slov	A/Paris	A/Swit	A/Swit	A/Norway	A/Ire	
				45/15	69/09	N6/09	2903/2015	1447/17	2656/17	3330/17	3433/18	84630/18	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	MDCK	
Passage history				NIB			NIB			NIB			NIB
Ferret number				F42/16 ¹	F09/15 ¹	F13/18 ²	F48/16 ¹	F03/18 ²	F20/18 ¹	F23/18 ¹	F04/19 ¹	F08/19 ¹	
Genetic group				6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	
REFERENCE VIRUSES													
A/Michigan/45/2015	G155E, D222G clone 37	6B.1	2015-09-07	E3/E2	640	320	320	1280	2560	640	320	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	40	320	320	40	320	40	80	320	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	640	640	80	640	160	160	1280	160
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	640	320	160	640	1280	640	320	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	160	<	640	1280	320	160	1280	640
A/Switzerland/2656/2017	clone 35	6B.1A	2017-12-21	E5/E3	640	320	320	1280	2560	640	320	2560	1280
A/Switzerland/3330/2017		6B.1A5	2017-12-20	E6/E2	320	160	80	320	1280	320	320	1280	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK1	320	80	<	320	640	160	160	1280	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK2	640	160	80	640	2560	320	320	2560	1280
TEST VIRUSES													
A/Mahajanga/4593/2018			2018-10-02	MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Oman/5690/2018			2018-10-31	MDCK1	320	160	160	320	1280	320	160	1280	640
A/Oman/6184/2018			2018-11-16	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Niigata/796/2018		6B.1A	2018-08-20	MDCK2/MDCK2/MDCK1	640	320	160	640	2560	640	320	2560	1280
A/Tochigi/18139/2018		6B.1A	2018-09-05	MDCK1/MDCK2/MDCK1	320	80	<	640	640	160	160	1280	640
A/Antsirabe/4203/2018		6B.1A	2018-09-17	MDCK1	640	160	80	640	1280	320	160	2560	640
A/Antsirabe/4399/2018		6B.1A	2018-09-27	MDCK1	320	160	80	640	1280	320	160	2560	1280
A/Antsirabe/4579/2018		6B.1A	2018-10-01	MDCK1	640	320	320	1280	2560	640	320	2560	1280
A/Antananarivo/4575/2018		6B.1A	2018-10-02	MDCK1	640	320	160	640	1280	640	320	2560	1280
A/Kyrgyzstan/25/2018		6B.1A	2018-11-12	MDCK1	1280	640	320	1280	2560	1280	640	5120	2560
A/Kyrgyzstan/24/2018		6B.1A	2018-11-12	MDCK1	640	320	160	640	1280	640	320	2560	1280
A/Kyrgyzstan/29/2018		6B.1A	2018-11-19	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Kyrgyzstan/28/2018		6B.1A	2018-11-19	MDCK1	640	320	80	640	1280	320	320	2560	1280
A/Kyrgyzstan/30/2018		6B.1A	2018-11-21	MDCK1	1280	640	160	1280	2560	640	640	5120	1280
A/Kyrgyzstan/32/2018		6B.1A	2018-11-27	MDCK1	1280	320	320	1280	2560	640	640	5120	2560
A/Kyrgyzstan/35/2018		6B.1A	2018-11-28	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Kyrgyzstan/42/2018		6B.1A	2018-12-03	MDCK2	640	320	160	640	2560	640	320	2560	1280
A/Kyrgyzstan/88/2018		6B.1A	2018-12-04	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Kyrgyzstan/87/2018		6B.1A	2018-12-04	MDCK1	640	320	160	1280	1280	640	320	2560	1280
A/Kyrgyzstan/127/2018		6B.1A	2018-12-12	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Kyrgyzstan/131/2018		6B.1A	2018-12-14	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Kyrgyzstan/130/2018		6B.1A	2018-12-14	MDCK1	1280	640	320	1280	5120	640	640	5120	2560
A/Kyrgyzstan/134/2018		6B.1A	2018-12-17	MDCK1	1280	320	160	1280	2560	640	320	2560	2560
A/Yokohama/99/2018		6B.1A2	2018-09-22	AX-4 1/MDCK2/MDCK1	320	160	80	640	1280	320	160	1280	640
A/Oman/4604/2018		6B.1A4	2018-09-30	MDCK1	640	160	160	1280	2560	640	320	2560	1280
A/Oman/4657/2018		6B.1A4	2018-10-01	MDCK1	320	160	<	320	1280	160	160	1280	640
A/Oman/5458/2018		6B.1A4	2018-10-28	MDCK1	640	160	80	640	1280	160	160	1280	640
A/Oman/4868/2018		6B.1A5	2018-10-10	MDCK1	320	160	80	320	1280	320	160	1280	640
A/Oman/4954/2018		6B.1A5	2018-10-13	MDCK1	320	160	80	320	640	320	160	1280	640
A/Oman/5355/2018		6B.1A5	2018-10-24	MDCK1	640	160	160	640	1280	320	160	2560	640
A/Oman/5346/2018		6B.1A5	2018-10-24	MDCK1	320	160	160	640	1280	320	160	1280	640
A/Oman/5582/2018		6B.1A5	2018-11-01	MDCK1	640	160	160	640	1280	320	320	2560	640
A/Oman/6038/2018		6B.1A5	2018-11-12	MDCK1	640	320	160	640	1280	320	320	2560	1280
A/Oman/6141/2018		6B.1A5	2018-11-15	MDCK1	320	160	160	640	1280	320	160	2560	640
A/Oman/6109/2018		6B.1A5	2018-11-15	MDCK1	320	160	80	320	1280	320	160	1280	640
A/Oman/6070/2018		6B.1A5	2018-11-15	MDCK1	640	160	160	640	1280	320	320	2560	1280
A/Oman/6305/2018		6B.1A5	2018-11-23	MDCK1	1280	320	320	1280	2560	640	320	2560	1280
A/Oman/4903/2018		6B.1A6	2018-10-09	MDCK1	320	160	80	320	1280	320	160	1280	640
A/Oman/5179/2018		6B.1A6	2018-10-19	MDCK1	640	320	160	640	1280	320	320	1280	1280
A/Oman/5442/2018		6B.1A6	2018-10-23	MDCK1	320	160	160	320	1280	160	160	1280	640
A/Oman/5437/2018		6B.1A6	2018-10-28	MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Oman/5537/2018		6B.1A6	2018-10-31	MDCK1	640	160	160	640	1280	320	320	2560	640
A/Oman/5753/2018		6B.1A6	2018-11-05	MDCK1	320	160	160	640	1280	320	160	1280	640
A/Oman/6247/2018		6B.1A6	2018-11-21	MDCK1	640	320	160	1280	2560	640	640	2560	1280
A/Luxembourg/1621/2019		6B.1A6	2019-01-07	MDCK1	1280	320	160	1280	2560	640	320	5120	1280
A/Oman/5679/2018		6B.1A7	2018-11-03	MDCK1	640	320	160	1280	2560	640	640	2560	1280
A/Montenegro/82/2018		6B.1A7	2018-12-31	MDCKx/MDCK1	640	320	160	1280	2560	640	320	2560	1280
A/Luxembourg/1616/2019		6B.1A7	2019-01-09	MDCK1	640	320	160	640	1280	640	320	2560	1280
A/Montenegro/146/2019		6B.1A7	2019-01-10	MDCK1	640	160	80	640	1280	320	160	1280	640
A/Luxembourg/1423/2019		6B.1A7	2019-01-10	MDCK1	640	160	80	640	1280	320	320	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-03-20)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Mich 45/15 Egg NIB F42/16 ^{*1} 6B.1	A/Bayern 69/09 MDCK F09/15 ^{*1}	A/Lviv N6/09 MDCK F13/18 ^{*2}	A/Slov 2903/2015 Egg NIB F48/16 ^{*1} 6B.1	A/Paris 1447/17 MDCK F03/18 ^{*2} 6B.1A	A/Swit 2656/17 Egg F20/18 ^{*1} 6B.1A	A/Swit 3330/17 Egg F23/18 ^{*1} 6B.1A5	A/Norway 3433/18 MDCK F04/19 ^{*1} 6B.1A5	A/Ire 84630/18 MDCK F08/19 ^{*1} 6B.1A6	A/Bris 02/18 Egg F09/19 ^{*1} 6B.1A1	
				Passage history	Ferret number	Genetic group								
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E2	640	320	320	1280	2560	640	320	1280	1280	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	<	320	160	<	160	40	40	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	160	640	1280	160	1280	320	320	1280	320	640
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	320	1280	2560	640	320	2560	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	160	40	640	1280	320	160	1280	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	320	320	1280	2560	640	320	2560	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	160	160	80	320	640	320	160	640	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	<	320	1280	160	160	1280	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK2	320	160	80	640	1280	320	160	1280	640	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	640	320	320	1280	2560	640	320	1280	1280	640
IVR-190(A/Brisbane/02/2018)		6B.1A1	2018-01-04	E3/D8/E1	1280	640	320	2560	5120	1280	640	2560	2560	1280
TEST VIRUSES														
A/Saudi Arabia/464601/2018			2018-08-30	MDCK1	320	160	80	640	1280	320	160	1280	640	640
A/Saudi Arabia/462678/2018		6B.1A	2018-08-24	MDCK1	320	160	80	640	1280	320	320	2560	1280	640
A/Saudi Arabia/462832/2018		6B.1A	2018-08-25	MDCK1	640	320	160	1280	1280	640	320	2560	640	640
A/Saudi Arabia/463062/2018		6B.1A	2018-08-26	MDCK2	640	320	160	1280	2560	640	320	2560	1280	1280
A/Saudi Arabia/460066/2018		6B.1A5	2018-08-14	MDCK1	320	160	80	640	1280	320	160	1280	640	640
A/Saudi Arabia/465645/2018		6B.1A5	2018-09-02	SIAT1	320	160	80	640	1280	320	160	2560	640	640
A/Saudi Arabia/466143/2018		6B.1A5	2018-09-04	MDCK1	640	320	80	1280	2560	640	320	2560	1280	1280
A/Belgium/G0024/2019		6B.1A5	2019-01-07	MDCK2	320	160	80	320	1280	160	160	2560	640	640
A/Saudi Arabia/458398/2018		6B.1A6	2018-08-07	MDCK1	640	160	80	640	1280	320	320	1280	640	640
A/Luxembourg/2575/2019		6B.1A6	2019-01-13	MDCK2	640	160	80	640	1280	320	320	2560	640	1280
A/Belgium/S0270/2019		6B.1A7	2019-01-15	MDCK2	320	160	80	640	1280	320	160	1280	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-03-26)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
					6B.1			6B.1	6B.1	6B.1	6B.1	6B.1	6B.1	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	320	160	640	1280	640	320	1280	1280	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	320	40	320	40	40	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	320	1280	1280	160	640	160	160	1280	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	160	1280	1280	640	320	1280	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	80	1280	1280	640	320	2560	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	640	320	1280	2560	1280	640	2560	1280	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	80	640	1280	640	320	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	80	40	320	640	160	160	1280	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	80	640	1280	320	160	1280	640	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	640	320	1280	5120	1280	640	2560	2560	640
TEST VIRUSES														
A/England/708/2018			2018-12-27	SIAT1/MDCK1	640	320	160	640	1280	320	320	1280	1280	640
A/Dakar/26/2018		6B.1A	2018-01-08	P0/MDCK1	640	320	80	1280	1280	640	320	2560	1280	640
A/Dakar/16/2018		6B.1A	2018-01-22	P0/MDCK1	1280	640	160	1280	2560	640	320	2560	1280	1280
A/Dakar/22/2018		6B.1A	2018-03-19	P0/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280
A/Kazakhstan/8332/2018		6B.1A	2018-12-13	MDCK1	1280	320	160	1280	2560	640	320	2560	1280	640
A/Kazakhstan/8335/2018		6B.1A	2018-12-19	MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280
A/Kazakhstan/19/2018		6B.1A	2018-12-25	MDCK1	1280	320	160	2560	2560	640	640	2560	2560	1280
A/Niedersachsen/1/2019		6B.1A	2019-01-02	P1/MDCK1	1280	320	160	1280	2560	1280	320	2560	2560	1280
A/Kazakhstan/8363/2019		6B.1A	2019-01-04	MDCK1	640	320	160	640	1280	320	160	2560	1280	640
A/Saint-Etienne/137/2019		6B.1A1	2019-01-11	MDCK2/MDCK1	2560	640	160	1280	5120	1280	640	5120	2560	2560
A/Grenoble/43/2019		6B.1A2	2018-12-28	MDCK2/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	640
A/Lyon/10/2019		6B.1A2	2018-12-31	MDCK2/MDCK1	1280	320	160	1280	2560	640	640	2560	2560	1280
A/Lyon/CHU/R19.01.51/2019		6B.1A2	2019-01-01	MDCK2/MDCK1	1280	320	320	2560	2560	1280	640	2560	2560	2560
A/Dakar/15/2018		6B.1A5	2018-09-10	P1/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280
A/Dakar/31/2018		6B.1A5	2018-09-24	P1/MDCK1	1280	160	80	1280	1280	640	320	1280	1280	640
A/Dakar/30/2018		6B.1A5	2018-09-24	P1/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280
A/Dakar/33/2018		6B.1A5	2018-10-31	P0/MDCK1	640	160	80	640	1280	320	160	1280	1280	640
A/Dakar/34/2018		6B.1A5	2018-11-14	P1/MDCK1	1280	320	160	1280	1280	640	320	2560	1280	1280
A/Hamburg/5/2018		6B.1A5	2018-11-28	P2/MDCK1	320	80	80	320	640	160	80	1280	640	320
A/Berlin/58/2018		6B.1A5	2018-12-07	P1/MDCK1	640	320	160	640	1280	320	320	2560	1280	640
A/Hamburg/6/2018		6B.1A5	2018-12-10	P1/MDCK1	320	160	160	320	640	160	160	1280	640	320
A/Hamburg/7/2018		6B.1A5	2018-12-17	P1/MDCK1	640	320	160	640	1280	320	320	2560	1280	640
A/England/733/2018	S157L	6B.1A5	2018-12-17	SIAT2/MDCK1	160	80	80	80	320	40	80	640	80	160
A/England/700/2018		6B.1A5	2018-12-21	SIAT1/MDCK1	320	160	160	320	640	160	160	1280	640	320
A/Saint-Etienne/2322/2018		6B.1A5	2018-12-23	MDCK2/MDCK1	320	80	80	320	640	160	160	1280	320	320
A/England/732/2018	N156S	6B.1A5	2018-12-26	SIAT1/MDCK1	80	160	40	40	160	40	40	1280	40	160
A/Lyon/4/2019		6B.1A5	2018-12-27	MDCK2/MDCK1	640	160	160	640	1280	320	160	2560	640	640
A/Lyon/CHU/R18.134.99/2018		6B.1A5	2018-12-29	MDCK2/MDCK1	640	160	160	640	1280	320	160	1280	640	640
A/Lyon/CHU/R19.03.40/2019		6B.1A5	2019-01-03	MDCK2/MDCK1	640	320	160	1280	1280	640	320	2560	640	640
A/England/1/2019		6B.1A5	2019-01-03	SIAT1/MDCK1	640	320	160	640	1280	320	320	2560	640	640
A/Lyon/CHU/R19.04.49/2019		6B.1A5	2019-01-05	MDCK2/MDCK1	640	160	40	320	1280	320	160	1280	640	320
A/Annecy/35/2019		6B.1A5	2019-01-06	MDCK2/MDCK1	640	320	160	1280	1280	640	320	2560	1280	1280
A/Gueret/64/2019		6B.1A5	2019-01-07	MDCK2/MDCK1	640	160	80	640	1280	320	160	2560	640	640
A/Lyon/HFME/47/2019		6B.1A5	2019-01-07	MDCK2/MDCK1	640	160	80	640	1280	320	320	2560	640	640
A/England/2/2019		6B.1A5	2019-01-07	SIAT1/MDCK1	640	320	320	640	1280	640	320	2560	1280	1280
A/England/620/2018		6B.1A6	2018-11-12	SIAT2/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Toulon/2307/2018		6B.1A6	2018-12-18	MDCK2/MDCK1	640	160	80	640	1280	320	160	1280	640	640
A/England/731/2018	N156D	6B.1A6	2018-12-24	SIAT1/MDCK1	80	160	80	40	320	80	<	320	40	160
A/Mecklenburg-Vorpommern/5/2018		6B.1A6	2018-12-27	P1/MDCK1	1280	320	160	2560	2560	1280	640	2560	2560	1280
A/Lyon/6/2019		6B.1A6	2018-12-29	MDCK2/MDCK1	640	160	160	1280	2560	640	320	2560	1280	640
A/Lyon/96/2019		6B.1A6	2019-01-08	MDCK2/MDCK1	640	320	80	1280	1280	640	320	2560	1280	1280
A/Bayern/50/2018		6B.1A7	2018-12-06	P1/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280
A/Lyon/128/2019		6B.1A7	2019-01-09	MDCK2/MDCK1	1280	320	160	1280	2560	640	320	2560	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-03-27)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻²	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
	Passage history	Genetic group		6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES													
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	320	320	1280	2560	640	320	1280	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	<	320	160	<	160	40	40	320	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	160	640	1280	160	1280	320	320	1280	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	320	1280	2560	640	320	2560	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	160	40	640	1280	320	160	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	320	320	1280	2560	640	320	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	160	160	80	320	640	320	320	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	<	320	1280	160	160	1280	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	80	640	1280	320	160	1280	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	640	320	320	1280	2560	640	320	1280	640
TEST VIRUSES													
A/Thuringen/8/2019			2019-01-28	C1/MDCK1	640	160	160	640	1280	320	320	1280	640
A/Croatia/5331/2018		6B.1A	2018-12-03	MDCKx/MDCK1	2560	640	320	2560	5120	1280	1280	2560	2560
A/Antalya/2/2018		6B.1A2	2018-12-06	SIAT1/MDCK1	1280	320	160	1280	2560	640	640	2560	1280
A/Antalya/1/2018		6B.1A2	2018-12-06	SIAT1/MDCK1	1280	320	160	1280	1280	640	640	1280	640
A/Greece/1585/2018		6B.1A2	2018-12-18	MDCK1/MDCK1	2560	640	320	1280	2560	1280	640	2560	1280
A/Saarland/1/2019		6B.1A2	2019-01-07	C1/MDCK1	640	160	80	640	640	320	320	2560	1280
A/Nordrhein-Westfalen/9/2019		6B.1A2	2019-01-17	C1/MDCK1	1280	320	160	1280	1280	640	640	2560	1280
A/Greece/199/2019		6B.1A2	2019-01-18	MDCK1	1280	320	160	1280	2560	640	640	2560	1280
A/Greece/1589/2018		6B.1A5		MDCK1/MDCK1	1280	640	320	1280	2560	640	640	2560	1280
A/Bucuresti/238999/2018		6B.1A5	2018-12-07	SIAT2/MDCK1	640	160	160	640	1280	320	640	1280	640
A/Bucuresti/239045/2018		6B.1A5	2018-12-10	SIAT1/MDCK1	320	160	80	320	640	160	320	1280	640
A/Bursa/1874/2018		6B.1A5	2018-12-12	Cx/MDCK1	640	320	320	1280	1280	640	640	2560	1280
A/Greece/1607/2018		6B.1A5	2018-12-27	MDCK1/MDCK1	640	320	320	640	1280	640	640	2560	640
A/Greece/1619/2018		6B.1A5	2018-12-28	MDCK1/MDCK1	1280	640	320	1280	2560	640	640	2560	1280
A/Greece/1617/2018		6B.1A5	2018-12-28	MDCK1/MDCK1	1280	320	320	640	1280	320	640	2560	1280
A/Greece/1616/2018		6B.1A5	2018-12-31	MDCK1/MDCK1	1280	320	320	1280	1280	640	640	2560	1280
A/Rheinland-Pfalz/1/2019		6B.1A5	2019-01-10	C1/MDCK1	640	320	320	640	640	320	640	2560	1280
A/Bucuresti/239839/2019		6B.1A5	2019-01-11	MDCK2	640	320	160	1280	2560	640	640	2560	1280
A/Thuringen/1/2019		6B.1A5	2019-01-11	C1/MDCK1	1280	320	320	640	640	320	640	2560	1280
A/Greece/144/2019		6B.1A5	2019-01-14	MDCK1	640	160	40	640	1280	320	320	2560	640
A/Berlin/7/2019		6B.1A5	2019-01-14	C1/MDCK1	640	320	320	1280	640	320	640	2560	1280
A/Brandenburg/2/2019		6B.1A5	2019-01-15	C1/MDCK1	1280	320	80	1280	2560	640	1280	2560	1280
A/Rheinland-Pfalz/3/2019		6B.1A5	2019-01-17	C1/MDCK1	640	320	160	640	1280	320	640	2560	640
A/Greece/232/2019		6B.1A5	2019-01-21	MDCK1	640	320	160	1280	1280	640	320	2560	1280
A/Sachsen/5/2019		6B.1A5	2019-01-21	C1/MDCK1	640	320	160	640	1280	320	640	2560	640
A/Bremen/7/2019		6B.1A5	2019-01-31	C1/MDCK1	1280	320	160	640	1280	640	640	2560	1280
A/Greece/1519/2018		6B.1A6		MDCK1/MDCK1	640	160	80	640	1280	320	320	2560	640
A/Palestine/10N/2018		6B.1A6	2018-12-10	MDCK1	2560	640	320	1280	2560	640	1280	5120	2560
A/Greece/168/2019		6B.1A6	2019-01-17	MDCK1	1280	320	160	640	1280	640	640	1280	640
A/Istanbul/1683/2018		6B.1A7	2018-11-21	Cx/MDCK1	1280	320	160	1280	2560	640	1280	2560	1280
A/Croatia/5299/2018		6B.1A7	2018-11-28	MDCKx/MDCK1	640	320	80	640	2560	640	640	1280	640
A/Croatia/5327/2018		6B.1A7	2018-12-03	MDCKx/MDCK1	640	160	80	640	2560	320	640	1280	640
A/Bucuresti/239594/2019		6B.1A7	2019-01-07	MDCK1	1280	320	160	1280	2560	640	640	2560	1280
A/Baden-Wuerttemberg/7/2019		6B.1A7	2019-01-21	C1/MDCK1	640	160	80	640	1280	320	640	1280	640
A/Berlin/26/2019		6B.1A	2019-02-04	C1/MDCK1	2560	640	160	1280	2560	1280	1280	2560	2560
A/Rheinland-Pfalz/27/2019		6B.1A2	2019-02-21	C1/MDCK1	640	160	160	1280	640	640	640	2560	1280
A/Brandenburg/18/2019		6B.1A2	2019-02-28	C1/MDCK1	640	80	80	640	640	320	320	1280	640
A/Niedersachsen/31/2019		6B.1A5	2019-02-05	C1/MDCK1	1280	320	160	640	1280	320	640	2560	640
A/Hamburg/1/2019		6B.1A5	2019-02-11	C1/MDCK1	640	320	320	640	640	320	640	2560	1280
A/Nordrhein-Westfalen/63/2019		6B.1A5	2019-02-18	C1/MDCK1	640	320	160	640	640	320	640	1280	640
A/Bayern/68/2019		6B.1A5	2019-02-19	C1/MDCK1	640	160	80	320	320	160	160	1280	320
A/Sachsen/28/2019		6B.1A7	2019-02-11	C1/MDCK1	640	160	160	640	640	320	640	2560	1280
A/Thuringen/16/2019		6B.1A7	2019-02-11	C1/MDCK1	640	160	80	640	640	320	320	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-04-02)

					Haemagglutination inhibition titre									
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				A/Mich	A/Bayern	A/Lviv	A/Slov	A/Paris	A/Swit	A/Swit	A/Norway	A/Ire	A/Bris	
				45/15	69/09	N6/09	2903/2015	1447/17	2656/17	3330/17	3433/18	84630/18	02/18	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	MDCK	Egg	
				F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻²	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹	
Passage history														
Ferret number														
Genetic group														
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	160	160	640	1280	640	320	1280	640	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	160	320	40	160	80	80	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	640	320	160	1280	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	160	160	640	1280	640	320	1280	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	80	640	2560	640	320	1280	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	640	640	1280	2560	1280	640	2560	1280	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	160	640	1280	640	640	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	40	320	640	320	160	1280	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	320	160	1280	2560	640	320	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	320	1280	2560	1280	640	2560	1280	1280
TEST VIRUSES														
A/Tanger/346/2018			2018-12-31	MDCK1/SIAT1	640	160	160	640	1280	640	320	1280	640	640
A/Armenia/1647/2018		6B.1A	2018-12-28	MDCK1	320	80	80	320	640	320	320	640	320	320
A/Cyprus/FA13/2019		6B.1A1	2019-01-11	MDCK1	320	80	40	320	640	640	320	640	640	320
A/Armenia/1640/2018		6B.1A2	2018-12-27	MDCK1	640	160	80	640	1280	1280	320	1280	640	640
A/Jordan/30003/2019		6B.1A2	2019-01-06	MDCK1	640	320	160	640	1280	1280	640	1280	1280	640
A/Albania/9168/2019		6B.1A2	2019-01-20	MDCK1	320	160	80	320	640	640	320	640	320	320
A/Valladolid/12/2019		6B.1A5	2019-01-09	SIAT1/SIAT1	1280	640	320	1280	2560	1280	1280	2560	1280	1280
A/Cyprus/F68/2019		6B.1A5	2019-01-12	MDCK2	320	80	<	320	640	320	160	640	320	320
A/Turkey/55/2019		6B.1A5	2019-01-14	MDCK1	320	80	80	320	640	320	320	1280	640	320
A/Albania/9164/2019		6B.1A5	2019-01-18	MDCK1	640	320	160	640	1280	1280	640	2560	640	640
A/Albania/9161/2019		6B.1A5	2019-01-18	MDCK1	640	160	160	320	640	640	320	1280	640	320
A/Jordan/10134/2019		6B.1A5	2019-01-23	MDCK1	640	320	320	640	1280	640	640	1280	1280	1280
A/Armenia/1550/2018		6B.1A6	2018-12-18	MDCK1	320	80	40	320	640	640	320	640	640	320
A/Jordan/30002/2018		6B.1A6	2018-12-21	MDCK1	640	160	80	640	1280	640	320	1280	640	320
A/Armenia/26/2019		6B.1A6	2019-01-08	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Latvia/01-042924/2019		6B.1A6	2019-01-14	MDCK2/MDCK1	1280	320	160	640	1280	1280	640	1280	1280	640
A/Latvia/01-035913/2019		6B.1A6	2019-01-14	MDCK2/MDCK1	640	160	80	640	1280	640	320	1280	1280	640
A/Latvia/01-078881/2019		6B.1A6	2019-01-28	MDCK2/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Latvia/01-075097/2019		6B.1A6	2019-01-28	MDCKx/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Latvia/19-090472/2019		6B.1A6	2019-01-31	MDCKx/MDCK1	640	320	160	1280	1280	1280	640	1280	1280	640
A/Cyprus/F33/2019		6B.1A7	2019-01-04	MDCK1	640	320	160	640	1280	1280	640	1280	1280	640
A/Turkey/46/2019		6B.1A7	2019-01-10	MDCK1	640	160	160	640	1280	1280	640	1280	1280	640
A/Cyprus/F96/2019		6B.1A7	2019-01-14	MDCK1	640	320	160	1280	1280	640	640	2560	1280	640
A/Latvia/01-040279/2019		6B.1A7	2019-01-14	MDCK2/MDCK1	640	160	80	320	640	640	320	640	640	320
A/Valladolid/23/2019		6B.1A7	2019-01-14	SIAT1/SIAT1	640	160	80	640	1280	640	640	1280	640	640
A/Latvia/02-033557/2019		6B.1A6	2019-02-11	MDCK1/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	640
A/Latvia/03-012374/2019		6B.1A6	2019-03-04	MDCK1/MDCK1	640	320	160	640	1280	1280	640	1280	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-04-16)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1		6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1	
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	320	160	1280	1280	1280	640	1280	640	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	160	160	40	160	80	40	160	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1J/MDCK3	160	640	640	160	640	320	160	640	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	160	160	640	1280	640	320	1280	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1J/MDCK3	640	160	80	640	1280	640	320	1280	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	640	320	1280	1280	1280	640	1280	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	80	640	1280	640	640	640	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	160	80	40	320	640	320	160	640	320	160
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1J/MDCK2	320	80	40	320	640	320	320	640	640	320
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	320	1280	2560	1280	640	2560	1280	1280
TEST VIRUSES														
A/Andalucia/190266/2018			2018-12-13	SIAT1J/MDCK1	640	320	320	640	1280	640	640	2560	1280	640
A/Macedonia/727/2019			2019-01-25	MDCK1	640	320	160	1280	1280	1280	640	1280	640	640
A/Andalucia/190267/2018		6B.1A	2018-12-14	SIAT1J/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	640
A/Macedonia/711/2019		6B.1A	2019-01-19	MDCK1	640	320	80	640	1280	1280	640	1280	640	640
A/Malta/53336/2019		6B.1A	2019-01-21	MDCK1	320	160	80	320	640	320	320	640	320	640
A/Malta/53480/2019		6B.1A	2019-01-23	MDCK1	640	160	80	640	640	640	640	1280	1280	640
A/Malta/53465/2019		6B.1A	2019-01-23	MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Mauritius/1-1077/2018		6B.1A5	2018-12-12	MDCK1J/MDCK1	1280	320	160	640	1280	1280	1280	1280	1280	1280
A/Greece/237/2019		6B.1A2	2019-01-30	MDCK2	640	160	160	640	2560	1280	640	2560	1280	640
A/Mauritius/1-1001/2018		6B.1A5	2018-11-06	MDCK1J/MDCK1	320	80	<	160	320	320	160	640	640	320
A/Mauritius/1-1011/2018		6B.1A5	2018-11-10	MDCK1J/MDCK1	320	80	<	160	320	320	160	640	320	320
A/Navarra/2907/2018		6B.1A5	2018-11-15	MDCK1	640	160	80	640	1280	640	320	1280	640	320
A/Mauritius/1-1047/2018		6B.1A5	2018-11-27	MDCKx/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Baleares/3014/2018		6B.1A5	2018-12-07	MDCK1	320	160	80	320	640	320	160	1280	320	1280
A/Mauritius/1-1092/2018		6B.1A5	2018-12-16	MDCK1J/MDCK1	640	160	80	640	2560	640	640	2560	640	640
A/Palestine/1/2019		6B.1A5	2019-01-02	MDCK1	640	320	80	1280	1280	1280	640	1280	1280	640
A/Mauritius/1-02/2019		6B.1A5	2019-01-03	MDCK1J/MDCK1	1280	160	160	640	1280	640	640	2560	1280	640
A/Galicia/296/2019		6B.1A5	2019-01-04	MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Extremadura/280/2019		6B.1A5	2019-01-04	SIAT1J/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Andalucia/268/2019		6B.1A5	2019-01-05	SIAT1J/MDCK1	640	160	80	640	1280	640	320	1280	640	320
A/Mauritius/1-11/2019		6B.1A5	2019-01-05	MDCK1J/MDCK1	320	160	80	320	640	320	320	1280	640	320
A/Andalucia/271/2019		6B.1A5	2019-01-07	SIAT1J/MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Galicia/174/2019		6B.1A5	2019-01-07	MDCK1	640	320	160	640	1280	640	320	1280	1280	640
A/Galicia/173/2019		6B.1A5	2019-01-07	MDCK1	640	320	160	640	1280	640	640	2560	640	640
A/Mauritius/1-14/2019		6B.1A5	2019-01-10	MDCKx/MDCK1	320	80	<	160	320	80	640	320	320	320
A/Andalucia/276/2019		6B.1A5	2019-01-10	SIAT1J/MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Mauritius/1-30/2019		6B.1A5	2019-01-10	MDCKx/MDCK1	1280	320	80	640	1280	640	640	1280	1280	640
A/Mauritius/1-28/2019		6B.1A5	2019-01-10	MDCKx/MDCK1	640	160	80	320	640	320	640	1280	640	640
A/Mauritius/1-48/2019		6B.1A5	2019-01-14	MDCKx/MDCK1	1280	320	160	1280	1280	640	640	1280	1280	640
A/Mauritius/1-47/2019		6B.1A5	2019-01-15	MDCKx/MDCK1	640	160	160	320	640	640	320	1280	640	640
A/Macedonia/712/2019		6B.1A5	2019-01-19	MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Macedonia/704/2019		6B.1A5	2019-01-19	MDCK1	640	320	80	1280	1280	640	320	1280	1280	640
A/Macedonia/719/2019		6B.1A5	2019-01-22	MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Malta/53463/2019		6B.1A5	2019-01-23	MDCK1	320	160	80	320	640	320	320	1280	1280	640
A/Macedonia/714/2019		6B.1A5	2019-01-25	MDCK1	640	160	80	640	640	640	320	1280	640	320
A/Mauritius/1-147/2019		6B.1A5	2019-01-28	MDCK2	320	160	320	640	640	320	320	1280	320	320
A/Greece/261/2019		6B.1A5	2019-01-30	MDCK1	1280	320	320	1280	2560	1280	640	2560	1280	1280
A/Palestine/3/2018		6B.1A6	2018-11-11	MDCK1	640	160	80	1280	1280	640	320	1280	640	640
A/Aragon/190085/2018		6B.1A6	2018-12-30	SIAT1J/MDCK2	1280	160	160	1280	2560	1280	1280	>5120	2560	1280
A/Palestine/2/2019		6B.1A6	2019-01-03	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Palestine/7/2019		6B.1A6	2019-01-10	MDCK1	640	320	80	640	1280	1280	640	2560	1280	640
A/Palestine/8/2019		6B.1A6	2019-01-13	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/PaisVasco/3139/2018		6B.1A7	2018-12-11	SIAT1J/MDCK2	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/PaisVasco/3135/2018		6B.1A7	2018-12-13	MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/PaisVasco/3130/2018		6B.1A7	2018-12-17	SIAT1J/MDCK2	640	160	80	640	1280	640	320	1280	640	640
A/Extremadura/282/2019		6B.1A7	2019-01-03	SIAT1J/MDCK1	640	320	160	640	1280	640	320	1280	1280	640
A/Extremadura/278/2019		6B.1A7	2019-01-03	SIAT1J/MDCK1	640	320	160	640	1280	640	320	1280	640	640
A/Palestine/5/2019		6B.1A7	2019-01-05	MDCK1	640	320	80	640	1280	1280	640	2560	640	640
A/Madrid/209/2019		6B.1A7	2019-01-13	MDCK1	1280	320	160	640	1280	640	640	1280	1280	640
A/Greece/341/2019		6B.1A	2019-02-04	MDCK2	1280	320	160	1280	2560	1280	640	2560	2560	1280
A/Greece/324/2019		6B.1A2	2019-02-03	MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	1280
A/Greece/351/2019		6B.1A2	2019-02-05	MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Greece/1063-19/2019		6B.1A2	2019-02-21	MDCK1	2560	320	160	1280	2560	2560	1280	2560	1280	2560
A/Nordrhein-Westfalen/90/2019		6B.1A2	2019-03-14	C1J/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Malta/54659/2019		6B.1A5	2019-02-02	MDCK1	640	160	80	640	1280	640	320	1280	640	320
A/Malta/516155/2019		6B.1A5	2019-02-03	MDCK1	1280	160	160	1280	2560	640	640	2560	1280	640
A/Greece/1046-19/2019		6B.1A5	2019-02-20	MDCK1	320	80	40	320	640	320	320	1280	640	320
A/Greece/1078-19/2019		6B.1A5	2019-02-21	MDCK1	640	320	160	640	1280	640	640	1280	1280	640
A/Greece/1125-19/2019		6B.1A5	2019-02-25	MDCK1	1280	640	320	1280	2560	1280	640	2560	2560	640
A/Greece/1383-19/2019		6B.1A5	2019-03-07	MDCK1	1280	640	160	1280	2560	1280	640	5120	1280	640
A/Niedersachsen/116/2019		6B.1A5	2019-03-07	C1J/MDCK1	640	320	160	640	1280	1280	320	1280	640	640
A/Greece/1387-19/2019		6B.1A5	2019-03-12	MDCK1	640	160	80	1280	1280	640	320	1280	640	640
A/Sachsen-Anhalt/43/2019		6B.1A5	2019-03-12	C1J/MDCK1	640	160	160	640	640	640	320	1280	640	640
A/Bremen/19/2019		6B.1A5	2019-03-18	C1J/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Niedersachsen/150/2019		6B.1A5	2019-03-19	C1J/MDCK1	640	160	160	640	640	640	320	1280	640	640
A/Bayern/104/2019		6B.1A5	2019-03-26	C1J/MDCK1	640	160	160	640	1280	640	320	1280	640	640

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-04-26)

					Haemagglutination inhibition titre									
					Post-infection ferret antisera									
Viruses	Other information		Collection date	Passage history	A/Mich 45/15 Egg F31/16 ⁻¹ 6B.1	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F13/18 ⁻¹ NIB	A/Slov 2903/2015 Egg F48/16 ⁻¹ 6B.1	A/Paris 1447/17 MDCK F03/18 ⁻² 6B.1A	A/Swit 2656/17 Egg F20/18 ⁻¹ 6B.1A	A/Swit 3330/17 Egg F23/18 ⁻¹ 6B.1A5	A/Norway 3433/18 MDCK F04/19 ⁻¹ 6B.1A5	A/Ire 84630/18 MDCK F08/19 ⁻¹ 6B.1A6	A/Bris 02/18 Egg F09/19 ⁻¹ 6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	160	160	640	1280	640	640	1280	640	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	160	40	160	80	80	160	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	640	320	320	1280	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	320	1280	1280	1280	640	1280	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	80	640	1280	640	320	1280	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	320	320	1280	1280	1280	640	2560	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	80	640	1280	640	640	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	80	640	640	320	320	1280	640	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	320	160	640	1280	1280	640	1280	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	320	1280	1280	1280	1280	2560	1280	1280
TEST VIRUSES														
A/Fujian-Gulou/SWL1884/2018			2018-11-05	E3/E2	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Marrakech/358/2019			2019-01-09	MDCK1/MDCK2	320	<	160	640	640	640	640	1280	1280	640
A/Sale/404/2019			2019-01-14	MDCK1/MDCK1	320	160	640	640	1280	640	640	640	640	640
A/Marrakech/427/2019			2019-01-15	MDCK1/MDCK1	2560	640	640	2560	2560	2560	2560	5120	5120	2560
A/Meknes/386/2019			2019-01-16	MDCK1/MDCK1	320	320	160	160	320	320	320	1280	160	320
A/Meknes/217/2018		6B.1A	2018-12-13	MDCK1/MDCK1	<	160	80	640	640	640	320	1280	640	640
A/Meknes/387/2019		6B.1A	2019-01-14	MDCK1/MDCK1	2560	640	320	2560	2560	2560	1280	5120	2560	2560
A/Rabat/454/2019		6B.1A	2019-01-21	MDCK1/MDCK1	640	160	80	640	1280	640	1280	1280	640	640
A/Marrakech/357/2019		6B.1A5	2019-01-08	MDCK1/MDCK1	40	<	80	40	40	80	80	80	160	80
A/Marrakech/355/2019		6B.1A5	2019-01-08	MDCK1/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Oujda/337/2019		6B.1A5	2019-01-08	MDCK1/MDCK1	640	160	160	640	640	640	640	2560	640	640
A/Skhirat/405/2019		6B.1A5	2019-01-14	MDCK1/MDCK1	160	<	80	320	160	320	160	320	160	320
A/Marakech/586/2019		6B.1A5	2019-01-23	MDCK1/MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	640
A/Rabat/423/2019		6B.1A6	2019-01-07	MDCK1/MDCK1	640	160	80	640	1280	640	640	1280	640	640
A/Rabat/367/2019		6B.1A6	2019-01-09	MDCK1/MDCK1	320	80	160	640	640	640	320	1280	1280	640
A/Fes/350/2019		6B.1A6	2019-01-09	MDCK1/MDCK1	1280	160	80	640	640	640	640	2560	1280	640
A/Fes/416/2019		6B.1A6	2019-01-15	MDCK1/MDCK1	2560	320	160	2560	1280	2560	1280	5120	2560	1280
A/Sale/424/2019		6B.1A6	2019-01-16	MDCK1/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Fes/506/2019		6B.1A6	2019-01-17	MDCK1/MDCK1	640	160	80	640	1280	640	1280	1280	640	640
A/Meknes/401/2019		6B.1A6	2019-01-17	MDCK1/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Beni Mellal/413/2019		6B.1A7	2019-01-14	MDCK1/MDCK1	640	160	80	640	320	640	320	1280	640	640
A/Temara/408/2019		6B.1A7	2019-01-17	MDCK1/MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	1280
A/Fes/407/2019		6B.1A7	2019-01-17	MDCK1/MDCK1	640	320	160	1280	1280	1280	1280	2560	1280	640
A/Beni Mellal/1405/2019		6B.1A5	2019-02-01	MDCK2	640	320	160	1280	1280	640	1280	2560	1280	640
A/Oujda/11116/2019		6B.1A5	2019-02-04	MDCK1	640	320	320	640	1280	1280	640	2560	1280	640
A/Greece/1092-19/2019		6B.1A5	2019-02-22	MDCK3	1280	640	320	1280	1280	1280	640	2560	1280	640
A/Fes/1380/2019		6B.1A6	2019-02-05	MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Agadir/1414/2019		6B.1A6	2019-02-08	MDCK1	640	160	80	640	1280	1280	640	2560	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-05-24)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ^{*1}	F09/15 ^{*1}	F13/18 ^{*1}	F48/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F23/18 ^{*1}	F04/19 ^{*1}	F08/19 ^{*1}	F09/19 ^{*1}
	Passage history			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Michigan/45/2015		2015-09-07	E3/E3	640	160	160	640	1280	640	640	1280	640	640
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	80	320	160	40	160	80	80	320	40	80
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK2	320	640	640	160	640	320	160	1280	160	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	640	160	160	640	1280	640	640	1280	640	640
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	640	160	40	640	640	640	320	1280	640	320
A/Switzerland/2656/2017		2017-12-21	E5/E2	640	320	160	640	1280	640	640	1280	640	640
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	640	160	80	640	640	320	640	1280	640	320
A/Norway/3433/2018		2018-10-30	MDCK3	160	80	<	160	320	160	160	640	320	160
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	1280	160	160	1280	1280	1280	640	2560	1280	640
A/Brisbane/02/2018		2018-01-04	E3/E1	1280	320	160	1280	1280	1280	640	1280	1280	1280
TEST VIRUSES													
A/Switzerland/8251/2019		2019-02-12	MDCK1	640	160	160	640	1280	640	320	1280	320	640
A/Switzerland/6005/2019		2019-03-04	MDCK1	640	320	160	640	1280	640	640	2560	640	640
A/Switzerland/3847/2019		2019-04-01	MDCK1	320	80	40	320	640	320	320	1280	320	320
A/Switzerland/4025/2019		2019-02-28	MDCK1	2560	640	160	1280	2560	2560	1280	2560	1280	1280
A/Switzerland/7387/2019		2019-03-12	MDCK1	1280	320	160	1280	2560	1280	1280	2560	2560	1280
A/Switzerland/8332/2019		2019-03-18	MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-05-30)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
	Passage history			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	2560		320	320	1280	1280	1280	640	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	320	160	<	160	80	160	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	160	640	640	160	640	320	320	1280	160
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	1280	320	160	640	1280	1280	640	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	40	640	640	320	1280	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	2560	640	320	1280	2560	1280	2560	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	160	80	640	640	640	640	1280	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	80	40	160	320	320	320	1280	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	2560	320	160	1280	2560	2560	1280	2560	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	2560	320	320	1280	2560	1280	1280	2560	1280
TEST VIRUSES													
A/Kosovo/38/2019			2019-01-16	MDCK1	2560	320	160	1280	1280	2560	1280	2560	1280
A/Irkutsk/5/2019		6B.1A5	2018-12-30	MDCK1/MDCK1	640	320	160	640	640	640	640	2560	640
A/Kosovo/19/2019		6B.1A5	2019-01-11	MDCK1	640	160	160	640	640	640	640	2560	640
A/Kosovo/31/2019		6B.1A5	2019-01-15	MDCK1	1280	320	320	1280	640	1280	640	2560	1280
A/Kosovo/45/2019		6B.1A5	2019-01-17	MDCK1	640	320	160	640	640	640	640	2560	640
A/Kosovo/54/2019		6B.1A5	2019-01-18	MDCK1	640	320	160	640	640	640	640	2560	640
A/Kosovo/75/2019		6B.1A5	2019-01-21	MDCK1	1280	320	320	640	640	1280	640	2560	1280
A/Kosovo/85/2019		6B.1A5	2019-01-23	MDCK1	640	160	80	640	1280	640	640	2560	640
A/Slovenia/382/2019		6B.1A7	2019-01-18	MDCKx/MDCK1	1280	320	160	1280	640	1280	640	2560	1280
A/Kosovo/76/2019		6B.1A7	2019-01-21	MDCK1	1280	320	160	1280	1280	1280	1280	2560	1280
A/Kosovo/81/2019		6B.1A7	2019-01-22	MDCK1	2560	640	320	1280	1280	2560	1280	2560	1280
A/Kosovo/134/2019		6B.1A7	2019-01-25	MDCK1	1280	160	80	640	1280	1280	640	1280	1280
A/Samara/RII-2/2019		6B.1A	2019-02-11	MDCK1/MDCK1	640	160	160	640	640	640	320	1280	640
A/Kaliningrad/RII-3/2019		6B.1A	2019-02-12	MDCK1/MDCK1	2560	640	640	2560	1280	2560	1280	5120	2560
A/Grozny/RII-1/2019		6B.1A	2019-02-18	MDCK1/MDCK1	1280	320	160	1280	640	1280	640	2560	1280
A/Grozny/RII-6/2019		6B.1A	2019-02-19	MDCK1/MDCK1	1280	320	160	1280	640	1280	640	2560	1280
A/Slovenia/1377/2019		6B.1A2	2019-03-02	MDCKx/MDCK1	1280	320	160	1280	1280	2560	1280	2560	1280
A/Voronezh/2/2019		6B.1A5	2019-02-06	MDCK1/MDCK1	2560	640	320	2560	1280	2560	1280	5120	2560
A/Slovenia/1170/2019		6B.1A5	2019-02-18	MDCKx/MDCK1	640	320	160	640	640	640	640	2560	640
A/S. Peteresburg/RII-232/2019		6B.1A5	2019-02-19	MDCK1/MDCK1	1280	320	320	1280	640	1280	640	2560	1280
A/S. Peteresburg/RII-228/2019		6B.1A5	2019-02-20	MDCK1/MDCK1	640	160	80	640	640	640	640	2560	640
A/S. Peteresburg/RII-260/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	1280	320	320	1280	640	1280	640	2560	1280
A/S. Peteresburg/RII-257/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	640	320	160	640	640	1280	640	2560	640
A/Slovenia/888/2019		6B.1A7	2019-02-05	MDCKx/MDCK1	1280	320	160	640	640	1280	640	1280	640
A/Slovenia/999/2019		6B.1A7	2019-02-09	SIATx/SIAT1	1280	320	160	1280	1280	1280	640	2560	1280
A/Makhachkala/RII-6/2019		6B.1A7	2019-02-19	MDCK1/MDCK1	5120	640	320	2560	1280	2560	1280	5120	2560
A/Slovenia/1424/2019		6B.1A7	2019-03-05	MDCKx/MDCK1	1280	320	160	1280	640	1280	640	2560	1280
A/Krasnodar/5/2019		6B.1A7	2019-03-06	MDCK1/MDCK1	2560	640	320	2560	1280	2560	1280	2560	1280
A/Slovenia/1489/2019		6B.1A7	2019-03-10	MDCKx/MDCK1	2560	640	320	2560	2560	2560	2560	2560	2560
A/Slovenia/1512/2019		6B.1A7	2019-03-12	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280
A/Slovenia/1772/2019		6B.1A7	2019-04-10	MDCKx/MDCK1	2560	320	320	1280	1280	1280	1280	2560	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-06-12)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	2560	320	320	1280	1280	2560	640	2560	1280	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	160	320	160	40	160	160	80	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK2	320	640	1280	160	640	640	320	1280	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	1280	320	160	1280	1280	1280	640	1280	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	40	320	640	640	320	640	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	320	320	1280	1280	1280	640	2560	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	160	80	640	640	640	640	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	80	40	320	640	320	320	1280	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	2560	320	160	1280	2560	2560	1280	2560	1280	640
TEST VIRUSES														
A/Bulgaria/863/2019			2019-01-28	SIAT2/MDCK1	1280	640	320	1280	2560	2560	1280	5120	2560	1280
A/Zambia/4-173/2018		6B.1A	2018-05-07	MDCK1	1280	640	160	1280	2560	1280	1280	2560	1280	1280
A/Zambia/4-206/2018		6B.1A	2018-05-09	MDCK1	1280	160	160	1280	2560	1280	640	1280	1280	1280
A/Zambia/4-221/2018		6B.1A	2018-05-23	MDCK1	2560	640	320	2560	2560	2560	1280	2560	2560	1280
A/Zambia/4-334/2018		6B.1A	2018-08-14	MDCK1	1280	640	320	1280	2560	1280	1280	2560	5120	1280
A/Zambia/1-364/2018		6B.1A	2018-09-04	MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Zambia/1-321/2018		6B.1A5	2018-08-04	MDCK1	640	320	160	640	1280	640	640	1280	1280	640
A/Serbia/3992/2019		6B.1A5	2019-01-10	MDCK1	640	160	160	640	1280	640	640	1280	640	640
A/Bulgaria/856/2019		6B.1A6	2019-01-29	SIAT2/MDCK1	1280	320	160	1280	2560	1280	1280	2560	2560	1280
A/Serbia/4035/2019		6B.1A7	2019-01-18	MDCK1	1280	320	160	1280	2560	1280	640	1280	1280	1280
A/Kosovo/73/2019		6B.1A7	2019-01-21	MDCK2	640	160	160	640	2560	1280	320	1280	640	320
A/Bulgaria/869/2019		6B.1A7	2019-01-31	SIAT2/MDCK1	1280	320	160	640	1280	1280	640	2560	1280	1280
A/Bulgaria/905/2019			2019-02-01	SIAT2/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Bulgaria/1044/2019			2019-02-08	SIAT2/MDCK1	1280	640	160	1280	2560	2560	1280	2560	2560	1280
A/Bulgaria/1128/2019			2019-02-19	SIAT2/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Bulgaria/904/2019		6B.1A	2019-02-01	SIAT2/MDCK1	640	320	160	1280	1280	1280	640	1280	1280	640
A/Bulgaria/936/2019		6B.1A	2019-02-04	SIAT2/MDCK1	1280	320	80	1280	2560	1280	640	2560	1280	1280
A/Bulgaria/923/2019		6B.1A	2019-02-04	SIAT2/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/Bulgaria/913/2019		6B.1A	2019-02-04	SIAT2/MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	1280
A/Bulgaria/1143/2019		6B.1A	2019-02-18	SIAT2/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Bulgaria/1041/2019		6B.1A2	2019-02-07	SIAT2/MDCK1	640	320	160	1280	2560	1280	640	1280	1280	640
A/Bulgaria/1145/2019		6B.1A2	2019-02-18	SIAT2/MDCK1	1280	320	160	640	1280	1280	640	1280	1280	640
A/Bulgaria/1154/2019		6B.1A2	2019-02-19	SIAT2/MDCK1	1280	640	160	1280	2560	2560	1280	2560	2560	1280
A/Bulgaria/1149/2019		6B.1A2	2019-02-20	SIAT2/MDCK1	1280	320	160	1280	1280	1280	640	1280	1280	640
A/Bulgaria/1197/2019		6B.1A2	2019-02-26	SIAT2/MDCK1	2560	640	320	1280	2560	2560	1280	2560	2560	1280
A/Serbia/4287/2019		6B.1A5	2019-02-01	MDCK1	160	40	40	160	320	160	160	640	320	160
A/Serbia/4525/2019		6B.1A5	2019-02-08	MDCK1	640	320	320	640	1280	1280	640	2560	1280	640
A/Bulgaria/1163/2019		6B.1A5	2019-02-25	SIAT2/MDCK1	1280	320	160	640	1280	640	640	2560	1280	640
A/Bulgaria/1200/2019		6B.1A5	2019-02-26	SIAT2/MDCK1	1280	640	320	1280	2560	1280	1280	2560	2560	1280
A/Sudan/1/2018		6B.1A5	2019-02-27	MDCK1	1280	320	320	1280	1280	640	640	2560	1280	640
A/Serbia/5147/2019		6B.1A5	2019-03-04	MDCK1	320	80	40	160	640	320	160	640	320	320
A/Serbia/5188/2019		6B.1A5	2019-03-06	MDCK1	640	160	160	640	640	640	320	1280	640	640
A/Serbia/5237/2019		6B.1A5	2019-03-09	MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Slovenia/1719/2019		6B.1A5	2019-04-01	SIATx/MDCK3	320	160	80	320	640	320	320	1280	320	320
A/Bulgaria/865/2019		6B.1A6	2019-02-01	SIAT2/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Bulgaria/981/2019		6B.1A6	2019-02-04	SIAT2/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Serbia/4296/2019		6B.1A7	2019-02-01	MDCK1	320	160	80	320	640	640	320	640	640	320
A/Serbia/4504/2019		6B.1A7	2019-02-07	MDCK1	320	80	40	320	640	320	320	640	320	320
A/Bulgaria/1158/2019		6B.1A7	2019-02-22	SIAT2/MDCK1	2560	640	320	2560	2560	2560	1280	2560	2560	1280
A/Bulgaria/1174/2019		6B.1A7	2019-02-25	SIAT2/MDCK1	2560	640	320	2560	2560	2560	1280	2560	2560	1280
A/Bulgaria/1172/2019		6B.1A7	2019-02-25	SIAT2/MDCK1	640	320	80	640	1280	1280	640	1280	640	640
A/Bulgaria/1216/2019		6B.1A7	2019-02-28	SIAT2/MDCK1	1280	320	160	1280	5120	2560	640	2560	2560	1280
A/Serbia/5144/2019		6B.1A7	2019-03-04	MDCK1	640	160	160	640	1280	640	640	2560	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-06-14)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	1280				1280	1280	640	1280	2560	320
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	160	40	160	80	80	320	20	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK2	320	640	640	160	640	320	320	1280	320	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	160	160	640	1280	1280	640	1280	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	40	640	1280	1280	640	1280	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	320	320	1280	2560	1280	640	2560	2560	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	80	640	640	640	640	1280	1280	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	80	40	640	320	320	320	1280	640	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	320	160	1280	2560	1280	640	2560	2560	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	320	1280	2560	1280	640	2560	1280	1280
TEST VIRUSES														
A/Kyrgyzstan/28/2019		6B.1A	2018-12-04	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	1280	2560	1280
A/Kyrgyzstan/19/2019		6B.1A	2018-12-04	MDCKx/MDCK1	640	160	80	640	1280	640	640	1280	2560	640
A/Kyrgyzstan/10/2019		6B.1A	2018-12-04	MDCKx/MDCK1	1280	640	160	1280	2560	1280	1280	2560	2560	1280
A/Kyrgyzstan/6/2019		6B.1A	2018-12-04	MDCKx/MDCK1	640	320	160	640	1280	1280	640	1280	2560	640
A/Kyrgyzstan/2/2019		6B.1A	2018-12-04	MDCKx/MDCK1	320	160	160	320	640	320	320	640	2560	640
A/Kyrgyzstan/21/2019		6B.1A	2018-12-06	MDCKx/MDCK1	640	160	80	640	1280	640	640	1280	2560	640
A/Kyrgyzstan/14/2019		6B.1A	2018-12-06	MDCKx/MDCK1	1280	160	80	640	1280	1280	640	1280	2560	640
A/Kyrgyzstan/11/2019		6B.1A	2018-12-06	MDCKx/MDCK1	640	320	160	640	1280	1280	640	1280	2560	640
A/Kyrgyzstan/29/2019		6B.1A	2018-12-07	MDCKx/MDCK1	640	160	80	640	640	640	320	640	2560	640
A/Kyrgyzstan/18/2019		6B.1A	2018-12-07	MDCKx/MDCK1	320	80	80	320	320	320	160	640	640	320
A/Kyrgyzstan/17/2019		6B.1A	2018-12-07	MDCKx/MDCK1	640	320	160	640	640	640	320	1280	2560	640
A/Kyrgyzstan/37/2019		6B.1A	2018-12-10	MDCKx/MDCK1	320	160	80	320	640	320	320	640	1280	320
A/Kyrgyzstan/20/2019		6B.1A	2018-12-10	MDCKx/MDCK1	640	160	80	640	640	640	320	640	1280	320
A/Kyrgyzstan/8/2019		6B.1A	2018-12-10	MDCKx/MDCK1	320	80	40	320	640	320	320	640	1280	320
A/Kyrgyzstan/4/2019		6B.1A	2018-12-10	MDCKx/MDCK1	640	80	80	320	320	320	320	640	2560	640
A/Kyrgyzstan/34/2019		6B.1A	2018-12-11	MDCKx/MDCK1	640	160	80	320	640	640	320	640	1280	320
A/Kyrgyzstan/30/2019		6B.1A	2018-12-11	MDCKx/MDCK1	320	80	80	320	320	320	160	640	1280	320
A/Kyrgyzstan/25/2019		6B.1A	2018-12-11	MDCKx/MDCK1	320	160	80	320	640	320	320	640	2560	320
A/Kyrgyzstan/24/2019		6B.1A	2018-12-11	MDCKx/MDCK1	640	160	80	320	640	320	320	640	1280	320
A/Kyrgyzstan/5/2019		6B.1A	2018-12-11	MDCKx/MDCK1	320	160	80	320	640	320	320	640	1280	320
A/Kyrgyzstan/40/2019		6B.1A	2018-12-12	MDCKx/MDCK1	320	80	80	320	320	320	160	640	640	640
A/Kyrgyzstan/39/2019		6B.1A	2018-12-12	MDCKx/MDCK1	320	160	160	640	640	320	320	1280	640	640
A/Kyrgyzstan/13/2019		6B.1A	2018-12-12	MDCKx/MDCK1	1280	320	80	1280	1280	1280	640	2560	2560	640
A/Kyrgyzstan Osh/6/2019		6B.1A	2018-12-13	MDCKx/MDCK1	640	160	160	640	1280	1280	640	2560	1280	640
A/Kyrgyzstan Osh/2/2019		6B.1A	2018-12-18	MDCKx/MDCK1	320	160	80	320	640	320	320	640	640	320
A/Kyrgyzstan Osh/3/2019		6B.1A	2018-12-20	MDCKx/MDCK1	320	80	40	320	640	320	320	640	640	320
A/Kyrgyzstan/22/2019		6B.1A	2018-12-24	MDCKx/MDCK1	320	160	80	320	640	320	320	640	640	320
A/Kyrgyzstan Osh/4/2019		6B.1A	2018-12-26	MDCKx/MDCK1	160	80	<	160	320	160	160	320	320	320
A/Kyrgyzstan/15/2019		6B.1A	2018-12-26	MDCKx/MDCK1	320	160	80	320	320	320	320	640	640	640
A/Kyrgyzstan/3/2019		6B.1A	2018-12-27	MDCKx/MDCK1	640	160	80	320	640	640	320	1280	640	640
A/Kyrgyzstan/16/2019		6B.1A	2019-01-04	MDCKx/MDCK1	640	160	160	640	640	640	320	1280	1280	640
A/Kyrgyzstan/12/2019		6B.1A	2019-01-09	MDCKx/MDCK1	320	160	40	320	640	640	320	1280	1280	320
A/Kyrgyzstan/36/2019		6B.1A7	2018-12-21	MDCKx/MDCK1	320	80	40	160	320	320	160	320	320	320
A/Kyrgyzstan/1/2019		6B.1A7	2018-12-26	MDCKx/MDCK1	640	160	80	640	640	640	320	640	640	320
A/Kyrgyzstan/41/2019		6B.1A7	2019-01-22	MDCKx/MDCK1	640	160	80	640	640	640	320	1280	2560	320
A/Iran/164026/2019		6B.1A5	2019-02-03	SIAT1/SIAT1	640	320	160	1280	1280	1280	640	1280	2560	1280
A/Iran/162976/2019		6B.1A5	2019-02-03	SIAT1/SIAT1	640	160	160	640	1280	1280	640	2560	2560	640
A/Denmark/3176/2019	H1N2	6B.1A5	2019-04-15	SIAT2/SIAT1	320	160	160	320	640	640	320	1280	1280	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-16. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-06-18)

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg	
				F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	NIB F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹	
				6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1	
REFERENCE VIRUSES														
A/Michigan/45/2015	G155E	6B.1	2015-09-07	E3/E3	2560	320	320	1280	1280	2560	640	2560	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	80	320	160	40	160	80	160	40	80	
A/Lviv/N6/2009		G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	640	640	320	640	160	320
A/Slovenia/2903/2015		clone 37	2015-10-26	E4/E2	1280	320	160	1280	1280	1280	640	1280	640	640
A/Paris/1447/2017	clone 35	6B.1A	2017-10-20	MDCK1/MDCK3	640	320	40	640	640	640	320	640	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	640	320	1280	1280	1280	640	2560	1280	640
A/Switzerland/3330/2017		6B.1A5	2017-12-20	E6/E2	320	160	80	320	640	640	640	640	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	80	40	320	640	320	320	1280	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	160	1280	1280	1280	1280	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	160	1280	2560	2560	1280	2560	1280	1280
TEST VIRUSES														
A/Moldova/863.51/2019		6B.1A5	2018-12-17	MDCK3/MDCK1	1280	320	320	1280	2560	1280	1280	2560	1280	1280
A/Moldova/237.05/2019		6B.1A5	2019-01-25	MDCK1	640	320	160	640	640	640	640	1280	640	640
A/Ukraine/8317/2019		6B.1A	2019-02-05	MDCK3/MDCK1	320	80	<	320	640	640	320	640	320	320
A/Denmark/2276/2019		6B.1A2	2019-02-26	MDCK4/MDCK1	320	80	40	320	640	640	320	1280	640	640
A/Denmark/2728/2019		6B.1A2	2019-03-19	MDCK3/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Ukraine/8305/2019		6B.1A5	2019-02-12	MDCK2/MDCK1	640	80	40	320	640	640	320	1280	640	320
A/Ukraine/8304/2019		6B.1A5	2019-02-12	MDCK2/MDCK1	320	80	40	320	640	320	320	1280	320	320
A/Ukraine/8310/2019		6B.1A5	2019-02-20	MDCK2/MDCK1	320	80	40	320	640	320	320	1280	640	320
A/Denmark/2373/2019		6B.1A5	2019-03-01	MDCK2/MDCK1	320	80	80	320	640	640	320	1280	640	320
A/Denmark/2398/2019		6B.1A5	2019-03-05	MDCK3/MDCK1	320	80	80	320	640	320	320	1280	320	320
A/Ukraine/8298/2019		6B.1A5	2019-03-06	MDCK1/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Denmark/2697/2019		6B.1A5	2019-03-14	MDCK3/MDCK1	640	320	320	640	1280	1280	640	2560	1280	640
A/Denmark/2729/2019		6B.1A5	2019-03-18	MDCK2/MDCK1	320	80	80	320	640	320	320	1280	320	320
A/Ukraine/8274/2019		6B.1A6	2019-02-20	MDCK2/MDCK1	640	320	160	640	1280	1280	640	1280	1280	640
A/Ukraine/8278/2019		6B.1A6	2019-02-27	MDCK3/MDCK1	640	160	160	640	1280	1280	640	1280	1280	640
A/Denmark/2336/2019		6B.1A6	2019-02-28	MDCK4/MDCK1	640	320	80	1280	1280	1280	640	2560	1280	1280
A/Denmark/2434/2019		6B.1A6	2019-03-06	MDCK3/MDCK1	640	160	80	640	1280	1280	640	1280	1280	640
A/Denmark/2225/2019		6B.1A7	2019-02-26	MDCK3/MDCK1	640	160	160	640	1280	640	640	1280	640	640
A/Denmark/2653/2019		6B.1A7	2019-03-12	MDCK4/MDCK1	320	80	40	320	640	640	320	640	640	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-17. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-06-25)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
					6B.1		6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1	
REFERENCE VIRUSES														
A/Michigan/45/2015	G155E	6B.1	2015-09-07	E3/E3	2560	320	320	1280	1280	1280	1280	2560	1280	1280
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	80	320	160	40	160	160	80	160	40	160
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	320	640	640	160	320	320	320	1280	160	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	640	320	160	1280	640	1280	640	1280	1280	640
A/Paris/1447/2017	clone 37	6B.1A	2017-10-20	MDCK1/MDCK3	640	320	40	640	640	640	320	1280	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	1280	640	160	1280	1280	1280	640	2560	1280	1280
A/Switzerland/3330/2017		6B.1A5	2017-12-20	E6/E2	320	160	80	320	640	640	640	1280	640	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	320	160	40	640	640	640	320	1280	640	640
A/Ireland/84630/2018	clone 35	6B.1A6	2018-11-28	MDCK1/MDCK3	640	160	160	1280	1280	1280	640	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	160	1280	1280	1280	1280	2560	1280	1280
TEST VIRUSES														
A/Kyrgyzstan/48/2019		6B.1A	2019-01-06	MDCKx/SIAT1	1280	160	80	1280	1280	1280	640	2560	1280	640
A/Austria/1119457/2019		6B.1A	2019-01-23	SIAT1/MDCK1	1280	160	80	640	1280	1280	640	2560	1280	640
A/Austria/1120563/2019		6B.1A	2019-01-28	SIAT1/MDCK1	640	160	80	640	1280	1280	640	2560	640	640
A/Austria/1113414/2018		6B.1A5	2018-12-26	SIAT1/MDCK1	640	160	80	320	640	640	320	1280	640	320
A/Umea/3/2018		6B.1A5	2018-12-26	MDCK0/MDCK1	640	160	160	640	1280	640	640	2560	640	640
A/Austria/1113546/2018		6B.1A5	2018-12-27	SIAT1/MDCK1	640	160	160	640	1280	640	640	2560	1280	640
A/Austria/1113762/2019		6B.1A5		SIAT1/MDCK1	640	160	80	320	640	640	640	1280	640	640
A/Umea/4/2018		6B.1A5	2018-12-28	MDCK0/MDCK1	320	160	80	320	640	320	320	1280	640	640
A/Linköping/71/2018		6B.1A5	2018-12-31	MDCK1/MDCK1	640	80	80	320	640	320	320	1280	640	640
A/Linköping/1/2019		6B.1A5	2019-01-02	MDCK1/MDCK1	640	160	160	640	640	640	320	1280	640	640
A/Austria/1114707/2019		6B.1A5	2019-01-03	SIAT1/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Moldova/27.01/2019		6B.1A5	2019-01-03	MDCK3/MDCK2	1280	320	160	640	1280	640	640	2560	1280	1280
A/Linköping/3/2019		6B.1A5	2019-01-03	MDCK0/MDCK1	320	80	80	160	320	320	160	640	320	160
A/Sundsvall/2/2019		6B.1A5	2019-01-23	MDCK0/MDCK1	640	160	80	320	640	640	320	1280	320	320
A/Gavle/2/2019		6B.1A5	2019-01-28	MDCK0/MDCK1	320	80	40	320	640	320	320	640	320	320
A/Austria/1121448/2019		6B.1A5	2019-01-31	SIAT1/MDCK1	320	160	80	320	640	640	320	2560	640	320
A/Austria/1115871/2019		6B.1A7	2019-01-08	SIAT1/MDCK1	640	160	80	320	640	640	320	1280	640	640
A/Austria/1119591/2019		6B.1A7	2019-01-23	SIAT1/MDCK1	640	160	80	640	1280	1280	640	1280	640	640
A/Sundsvall/1/2019		6B.1A7	2019-01-23	MDCK0/MDCK1	1280	320	160	1280	2560	2560	1280	2560	2560	1280
A/Finland/90/2019		6B.1A2	2019-02-25	MDCK1/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Khmelnitskyi/154/2019		6B.1A5	2019-02-01	SIAT2/MDCK1	640	160	80	640	640	640	640	1280	640	640
A/Gavle/3/2019		6B.1A5	2019-02-05	MDCK0/MDCK1	160	80	<	160	320	320	160	640	320	160
A/Austria/1123857/2019		6B.1A5	2019-02-11	SIAT1/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Ukraine/8294/2019		6B.1A5	2019-02-11	MDCK0/MDCK1	640	160	80	320	640	320	320	1280	640	320
A/Khmelnitskyi/151/2019		6B.1A5	2019-02-12	SIAT1/MDCK1	1280	640	320	1280	2560	1280	2560	1280	1280	1280
A/Khmelnitskyi/152/2019		6B.1A5	2019-02-13	SIAT1/MDCK1	640	320	160	1280	1280	640	640	2560	1280	640
A/Finland/39/2019		6B.1A5	2019-02-17	MDCK2/MDCK1	320	80	40	320	320	320	160	1280	320	320
A/Austria/1125761/2019		6B.1A5		SIAT1/MDCK1	640	160	80	320	640	640	320	1280	640	640
A/Austria/1126231/2019		6B.1A5	2019-02-18	SIAT1/MDCK1	320	160	80	320	640	640	320	1280	640	320
A/Khmelnitskyi/147/2019		6B.1A5	2019-02-19	SIAT1/MDCK1	1280	320	320	1280	1280	1280	640	2560	1280	640
A/Ukraine/8313/2019		6B.1A5	2019-02-21	MDCK1/MDCK1	640	160	80	640	640	640	640	2560	640	640
A/Kyiv/127/2019	N156X	6B.1A5	2019-02-27	SIAT1/MDCK1	640	320	160	640	1280	640	640	2560	640	640
A/Khmelnitskyi/144/2019		6B.1A5	2019-02-27	SIAT1/MDCK1	160	80	40	40	160	80	80	640	80	160
A/Finland/47/2019		6B.1A5	2019-02-28	MDCK1/MDCK1	320	80	40	320	640	320	320	1280	640	320
A/Halmstad/1/2019		6B.1A5	2019-03-16	MDCK0/MDCK1	160	40	<	160	320	160	160	640	160	160
A/Finland/55/2019		6B.1A5	2019-03-19	MDCK1/MDCK1	320	80	40	320	320	320	320	1280	640	320
A/Finland/54/2019		6B.1A5	2019-03-19	MDCK1/MDCK1	320	80	80	320	640	320	160	1280	320	320
A/Sundsvall/3/2019		6B.1A5	2019-03-24	MDCK0/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Dnipro/269/2019		6B.1A5	2019-03-28	SIAT1/MDCK1	320	80	40	160	320	320	160	640	320	160
A/Ukraine/8319/2019		6B.1A5	2019-04-02	MDCK3/MDCK1	320	160	80	320	640	320	320	1280	640	320
A/Finland/103/2019		6B.1A5	2019-04-02	MDCK1/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Finland/102/2019		6B.1A5	2019-04-02	MDCK1/MDCK1	320	160	80	320	640	320	320	1280	640	320
A/Austria/1122894/2019		6B.1A6	2019-02-05	SIAT1/MDCK1	640	160	80	640	1280	1280	640	2560	1280	640
A/Dnipro/257/2019		6B.1A6	2019-02-14	SIAT1/MDCK1	640	160	80	640	1280	640	640	1280	1280	640
A/Kyiv/134/2019		6B.1A6	2019-02-24	SIAT1/MDCK1	1280	320	80	1280	1280	1280	640	2560	1280	640
A/Finland/45/2019		6B.1A6	2019-02-25	MDCK1/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Khmelnitskyi/146/2019		6B.1A6	2019-02-26	SIAT1/MDCK1	1280	320	160	1280	2560	2560	1280	2560	1280	1280
A/Khmelnitskyi/143/2019		6B.1A6	2019-02-27	SIAT1/MDCK1	1280	160	80	640	1280	1280	640	2560	1280	640
A/Khmelnitskyi/201/2019		6B.1A6	2019-03-05	SIAT2/MDCK1	1280	160	80	640	1280	1280	640	1280	1280	640
A/Kyiv/191/2019		6B.1A6	2019-03-09	SIAT2/MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	1280
A/Ukraine/8297/2019		6B.1A6	2019-03-11	MDCK1/MDCK1	1280	320	160	2560	2560	1280	1280	5120	1280	1280
A/Austria/1124185/2019		6B.1A7	2019-02-11	SIAT1/MDCK1	1280	160	80	640	1280	1280	640	2560	1280	640
A/Bulgaria/1171/2019		6B.1A7	2019-02-24	SIAT2/SIAT1	2560	640	160	2560	2560	2560	1280	5120	1280	2560
A/Bulgaria/1238/2019		6B.1A7	2019-03-05	SIAT2/SIAT1	1280	320	160	1280	2560	1280	1280	5120	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-18. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-07-02)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg F31/16 ¹ 6B.1	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F13/18 ¹ NIB F48/16 ¹ 6B.1	A/Slov 2903/2015 Egg F48/16 ¹ 6B.1	A/Paris 1447/17 MDCK F03/18 ¹ 6B.1A	A/Swit 2656/17 Egg F20/18 ¹ 6B.1A	A/Swit 3330/17 Egg F23/18 ¹ 6B.1A5	A/Norway 3433/18 MDCK F04/19 ¹ 6B.1A5	A/Ire 84630/18 MDCK F08/19 ¹ 6B.1A6	A/Bris 02/18 Egg F09/19 ¹ 6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015	G155E G155E, D222G clone 37	6B.1	2015-09-07	E3/E3	2560	640	320	2560	2560	2560	1280	2560	1280	1280
A/Bavaria/69/2009			2009-07-01	MDCKx/MDCK1	160	640	320	80	320	320	160	320	80	160
A/Lviv/6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	1280	640	640	320	1280	160	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	1280	320	320	1280	1280	1280	640	1280	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	320	80	1280	1280	1280	640	2560	1280	1280
A/Switzerland/2656/2017	clone 35	6B.1A	2017-12-21	E5/E3	1280	640	320	2560	2560	2560	1280	2560	1280	2560
A/Switzerland/3330/2017		6B.1A5	2017-12-20	E6/E2	640	320	160	640	1280	1280	1280	1280	640	1280
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	160	80	640	1280	640	1280	1280	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	320	2560	2560	2560	1280	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	2560	640	320	2560	2560	2560	1280	2560	1280	2560
TEST VIRUSES														
A/Karaganda/351/2019		6B.1A	2019-01-17	MDCKx/MDCK1	1280	320	320	1280	1280	2560	1280	2560	1280	1280
A/Almatinskaya/358/2019		6B.1A	2019-01-22	MDCKx/MDCK1	1280	320	160	1280	1280	1280	2560	1280	640	640
A/Karaganda/355/2019		6B.1A	2019-01-22	MDCKx/MDCK1	1280	320	160	1280	640	1280	640	2560	1280	640
A/Karaganda/364/2019		6B.1A	2019-01-25	MDCKx/MDCK1	1280	320	320	1280	640	1280	2560	1280	1280	1280
A/Karaganda/366/2019		6B.1A2	2019-01-23	MDCKx/MDCK1	1280	320	160	1280	640	1280	2560	1280	1280	1280
A/Karaganda/367/2019	N156K	6B.1A2	2019-01-27	MDCKx/MDCK1	1280	320	160	1280	640	1280	2560	1280	1280	1280
A/Karaganda/365/2019		6B.1A2	2019-01-27	MDCKx/MDCK1	1280	320	160	1280	640	1280	2560	1280	1280	1280
A/Lisboa/NEV1A195/2019		6B.1A2	2019-01-29	MDCK1/MDCK1	80	80	40	<	<	80	40	160	40	40
A/Agadir/675/2019		6B.1A5	2019-01-29	MDCK1/MDCK1	2560	640	320	1280	1280	1280	640	2560	1280	1280
A/Almatinskaya/357/2019		6B.1A7	2019-01-17	MDCKx/MDCK1	640	160	160	640	640	640	1280	640	640	640
A/Cyprus/368/2019	N156K	6B.1A7	2019-01-29	MDCK1	2560	320	160	1280	320	1280	1280	2560	1280	1280
A/Cyprus/367/2019		6B.1A7	2019-01-29	MDCK1	2560	320	160	2560	1280	2560	1280	2560	1280	1280
A/Cyprus/389/2019		6B.1A7	2019-01-31	MDCK1	1280	640	160	1280	1280	2560	1280	2560	640	1280
A/Meknes/1879/2019			2019-02-13	MDCK1/MDCK1	1280	320	160	1280	1280	2560	1280	2560	1280	1280
A/Meknes/1888/2019			2019-02-13	MDCK1/MDCK1	1280	320	160	2560	1280	2560	1280	2560	1280	1280
A/Cyprus/957/2019		6B.1A	2019-02-14	MDCK1	1280	320	80	1280	640	1280	640	2560	1280	640
A/Karaganda/370/2019		6B.1A	2019-02-05	MDCKx/MDCK1	640	320	160	1280	640	1280	640	1280	640	640
A/Karaganda/377/2019		6B.1A	2019-02-06	MDCKx/MDCK1	1280	320	320	1280	640	1280	2560	1280	1280	1280
A/Karaganda/372/2019		6B.1A	2019-02-06	MDCKx/MDCK1	1280	320	160	1280	640	1280	2560	1280	1280	1280
A/Karaganda/371/2019		6B.1A	2019-02-06	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Karaganda/374/2019	N156K	6B.1A	2019-02-07	MDCKx/MDCK1	1280	320	320	1280	640	1280	1280	2560	1280	1280
A/Karaganda/376/2019		6B.1A	2019-02-08	MDCKx/MDCK1	2560	640	320	2560	1280	1280	2560	2560	1280	1280
A/Cyprus/1165/2019		6B.1A2	2019-02-22	MDCK1	2560	320	160	2560	5120	2560	1280	2560	2560	2560
A/Lisboa/NEV1A275/2019		6B.1A2	2019-02-27	MDCK1/MDCK1	160	160	80	40	80	160	80	320	80	80
A/Karaganda/369/2019		6B.1A5	2019-02-03	MDCKx/MDCK1	640	320	320	1280	640	1280	640	2560	1280	640
A/Agadir/1305/2019		6B.1A5	2019-02-04	MDCK1/MDCK1	1280	320	320	1280	640	1280	640	2560	1280	640
A/Cyprus/788/2019		6B.1A5	2019-02-11	MDCK1	640	320	160	640	320	640	640	2560	1280	640
A/Marrakech/1738/2019		6B.1A5	2019-02-11	MDCK1/MDCK1	1280	640	320	1280	1280	1280	1280	2560	1280	640
A/Marrakech/1737/2019		6B.1A5	2019-02-11	MDCK1/MDCK1	2560	640	320	1280	1280	1280	1280	2560	1280	1280
A/Agadir/1881/2019		6B.1A5	2019-02-14	MDCK1/MDCK1	1280	640	640	2560	1280	2560	1280	2560	1280	1280
A/Rabat/1890/2019		6B.1A5	2019-02-16	MDCK1/MDCK1	1280	320	80	1280	640	1280	1280	1280	1280	640
A/Agadir/1929/2019		6B.1A5	2019-02-17	MDCK1/MDCK1	1280	640	320	1280	1280	1280	2560	1280	640	640
A/Rabat/2051/2019		6B.1A5	2019-02-24	MDCK1/MDCK1	1280	320	320	1280	1280	1280	2560	1280	1280	640
A/Lisboa/78/2019		6B.1A5	2019-02-25	MDCK1/MDCK1	1280	320	320	1280	640	1280	1280	2560	1280	1280
A/Lisboa/80/2019		6B.1A5	2019-03-10	MDCK1/MDCK1	1280	320	160	1280	640	1280	1280	2560	1280	640
A/Lisboa/82/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	1280	320	320	640	640	1280	1280	2560	1280	640
A/Cyprus/1619/2019		6B.1A5	2019-04-09	MDCK1	1280	320	160	640	1280	1280	640	1280	1280	1280
A/Chefchaouen/1456/2019		6B.1A6	2019-02-05	MDCK1/MDCK1	1280	320	160	1280	1280	2560	1280	2560	1280	1280
A/Fes/1250/2019		6B.1A6	2019-02-05	MDCK1/MDCK1	2560	320	320	2560	1280	2560	1280	2560	1280	1280
A/Chefchaouen/1454/2019		6B.1A6	2019-02-08	MDCK1/MDCK1	2560	320	160	2560	1280	2560	1280	2560	1280	1280
A/Rabat/1433/2019		6B.1A6	2019-02-08	MDCK1/MDCK1	1280	320	160	1280	1280	1280	1280	2560	1280	1280
A/Karaganda/375/2019		6B.1A6	2019-02-12	MDCKx/MDCK1	1280	320	320	2560	1280	1280	1280	2560	1280	640
A/Karaganda/383/2019		6B.1A6	2019-02-12	MDCKx/MDCK1	1280	160	160	1280	640	1280	640	1280	1280	640
A/Karaganda/381/2019		6B.1A6	2019-02-12	MDCKx/MDCK1	2560	320	160	1280	1280	1280	1280	2560	1280	1280
A/Fes/1927/2019		6B.1A6	2019-02-13	MDCK1/MDCK1	1280	320	160	1280	1280	2560	1280	2560	1280	1280
A/Meknes/1872/2019		6B.1A6	2019-02-14	MDCK1/MDCK1	2560	640	640	2560	1280	2560	1280	2560	1280	1280
A/Tetoun/1865/2019		6B.1A6	2019-02-15	MDCK1/MDCK1	1280	320	160	1280	640	1280	1280	2560	1280	640
A/Fes/1921/2019		6B.1A6	2019-02-17	MDCK1/MDCK1	2560	640	640	2560	1280	2560	5120	2560	1280	1280
A/Rabat/1971/2019		6B.1A6	2019-02-20	MDCK1/MDCK1	1280	640	160	1280	1280	1280	640	2560	1280	1280
A/Lisboa/74/2019		6B.1A6	2019-02-27	MDCK1/MDCK1	2560	640	320	2560	1280	5120	5120	2560	1280	1280
A/Karaganda/368/2019		6B.1A7	2019-02-03	MDCKx/MDCK1	640	320	160	1280	640	1280	1280	1280	640	640
A/Cyprus/738/2019		6B.1A7	2019-02-05	MDCK1	1280	320	160	1280	640	1280	640	2560	1280	1280
A/Cyprus/507/2019		6B.1A7	2019-02-05	MDCK1	2560	640	80	2560	1280	2560	1280	5120	2560	2560
A/Cyprus/505/2019		6B.1A7	2019-02-05	MDCK1	640	160	80	640	320	1280	640	1280	1280	640
A/Cyprus/488/2019		6B.1A7	2019-02-05	MDCK1	1280	320	160	640	1280	1280	640	2560	2560	1280
A/Karaganda/378/2019		6B.1A7	2019-02-07	MDCKx/MDCK1	1280	320	160	1280	640	1280	640	2560	1280	640
A/Karaganda/373/2019		6B.1A7	2019-02-07	MDCKx/MDCK1	1280	320	320	1280	1280	2560	1280	2560	1280	1280
A/Cyprus/774/2019		6B.1A7	2019-02-11	MDCK1	1280	320	160	1280	640	1280	640	2560	1280	640
A/Cyprus/753/2019		6B.1A7	2019-02-11	MDCK1	1280	320	160	1280	640	1280	640	2560	1280	640
A/Cyprus/740/2019		6B.1A7	2019-02-11	MDCK1	1280	160	160	1280	640	1280	640	2560	1280	640
A/Cyprus/739/2019		6B.1A7	2019-02-11	MDCK1	2560	320	160	1280	1280	2560	640	5120	2560	1280
A/Cyprus/873/2019		6B.1A7	2019-02-12	MDCK1	2560	640	160	2560	1280	2560	2560	5120	1280	1280

Table 5-19. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-07-09)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1		6B.1	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	1280		320	2560	2560	2560	2560	2560	2560	2560
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	320	40	320	160	80	320	40	160
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	640	320	1280	160	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	1280	320	320	1280	1280	1280	1280	1280	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	320	160	1280	1280	1280	1280	2560	2560	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	2560	640	640	2560	2560	2560	1280	5120	2560	2560
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	320	160	640	640	1280	1280	1280	1280	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	160	80	640	640	640	640	2560	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	640	160	2560	2560	1280	1280	2560	2560	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	640	320	2560	2560	2560	2560	5120	2560	2560
TEST VIRUSES														
A/West Kazakhstan/360/2019			2019-01-04	MDCKx/MDCK1	2560	640	320	2560	2560	2560	1280	2560	2560	1280
A/East Kazakhstan/389/2019			2019-01-16	MDCKx/MDCK1	640	160	160	1280	1280	1280	640	2560	1280	640
A/West Kazakhstan/359/2018		6B.1A	2018-12-07	MDCKx/MDCK1	1280	320	320	2560	2560	2560	1280	2560	2560	1280
A/Almaty/346/2018		6B.1A	2018-12-21	MDCKx/MDCK1	2560	640	640	5120	5120	2560	1280	5120	5120	5120
A/East Kazakhstan/386/2019		6B.1A2	2019-01-04	MDCKx/MDCK1	1280	320	160	1280	1280	2560	1280	2560	2560	1280
A/East Kazakhstan/388/2019		6B.1A2	2019-01-13	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Georgia/335/2019		6B.1A2	2019-01-17	C2/MDCK1	2560	640	640	2560	1280	2560	1280	5120	2560	2560
A/Poland/6732/2019		6B.1A2	2019-01-21	MDCK2	1280	320	160	2560	1280	2560	1280	2560	2560	1280
A/Georgia/372/2019		6B.1A2	2019-01-22	C2/MDCK1	1280	320	320	2560	1280	2560	1280	2560	2560	1280
A/Georgia/378/2019		6B.1A2	2019-01-23	C2/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Egypt/7667/2018		6B.1A6	2018-12-26	MDCK1	2560	640	160	2560	2560	2560	1280	2560	2560	2560
A/Poland/349/2019		6B.1A6	2019-01-31	MDCK1	1280	320	160	1280	1280	2560	1280	2560	2560	1280
A/Almaty/349/2019		6B.1A7	2019-01-03	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/East Kazakhstan/384/2019		6B.1A7	2019-01-05	MDCKx/MDCK1	640	160	160	640	1280	1280	640	1280	1280	640
A/Georgia/627/2019			2019-03-04	MDCK1	1280	320	320	1280	640	1280	640	2560	1280	1280
A/Iceland/11/2019			2019-03-07	MDCK1/MDCK1	320	160	40	640	640	1280	320	1280	640	320
A/West Kazakhstan/449/2019		6B.1A	2019-02-25	MDCKx/MDCK1	640	160	80	640	1280	640	320	1280	640	640
A/West Kazakhstan/465/2019		6B.1A	2019-03-05	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Poland/16714/2019		6B.1A	2019-03-18	MDCK1	1280	320	80	1280	1280	1280	640	2560	1280	1280
A/Georgia/555/2019		6B.1A2	2019-02-19	C2/MDCK1	1280	320	320	1280	1280	2560	1280	2560	1280	1280
A/Georgia/561/2019		6B.1A2	2019-02-20	C2/MDCK1	2560	320	320	2560	1280	2560	1280	2560	2560	1280
A/Georgia/571/2019		6B.1A2	2019-02-25	C2/MDCK1	1280	320	160	1280	1280	1280	1280	2560	1280	1280
A/Georgia/682/2019		6B.1A2	2019-03-19	MDCK1	2560	640	640	2560	2560	2560	1280	5120	2560	2560
A/Poland/1973/2019		6B.1A5	2019-02-08	MDCK1	640	160	160	640	640	640	640	2560	1280	640
A/Poland/6218/2019		6B.1A5	2019-02-13	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/West Kazakhstan/441/2019	N162K (-CHO)	6B.1A5	2019-02-15	MDCKx/MDCK1	320	160	160	640	640	320	320	1280	640	320
A/Poland/11284/2019		6B.1A5	2019-02-19	MDCK1	640	320	160	640	1280	640	640	2560	640	640
A/Minsk/1507/2019		6B.1A5	2019-03-01	MDCK2/MDCK1	640	160	80	640	640	640	640	2560	1280	640
A/Poland/14120/2019		6B.1A5	2019-03-04	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Baranocihi/1485/2019		6B.1A5	2019-03-04	MDCK2/MDCK1	320	80	40	320	320	320	320	1280	640	320
A/Iceland/17/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	640	160	80	640	640	1280	320	1280	640	640
A/Iceland/16/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	640	80	80	640	640	1280	320	1280	640	640
A/Iceland/20/2019		6B.1A5	2019-03-15	MDCK1/MDCK1	640	160	160	640	1280	1280	320	1280	640	640
A/Poland/866/2019		6B.1A5	2019-03-18	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Poland/149/2019		6B.1A5	2019-03-19	MDCK1	1280	320	160	1280	2560	1280	1280	2560	2560	1280
A/Soligorsk/1776/2019		6B.1A5	2019-03-19	MDCK2/MDCK1	640	160	80	320	640	1280	320	1280	640	320
A/Iceland/44/2019		6B.1A5	2019-04-03	MDCK1/MDCK1	640	160	80	640	640	1280	320	1280	640	640
A/Poland/433/2019		6B.1A6	2019-02-06	MDCK1	1280	320	160	1280	2560	1280	2560	2560	2560	1280
A/Poland/68/2019		6B.1A6	2019-02-11	MDCK1	1280	320	160	1280	2560	2560	1280	2560	2560	2560
A/Ukraine/8232/2019		6B.1A6	2019-02-14	MDCK2/SIAT1	640	160	80	640	1280	640	320	1280	640	640
A/Poland/100/2019		6B.1A6	2019-02-20	MDCK1	2560	640	320	2560	2560	2560	1280	5120	2560	2560
A/Ukraine/8235/2019		6B.1A6	2019-03-12	MDCK1/SIAT1	320	160	80	640	640	320	320	1280	640	320
A/Vitebsk/1858/2019		6B.1A6	2019-03-14	MDCK3/MDCK1	640	320	320	1280	1280	1280	640	1280	1280	640
A/Poland/876/2019		6B.1A6	2019-03-19	MDCK1	2560	640	160	2560	2560	2560	1280	2560	2560	2560
A/Georgia/749/2019		6B.1A6	2019-04-08	MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Iceland/68/2019		6B.1A6	2019-05-06	MDCK1/MDCK1	640	160	160	640	1280	1280	320	1280	1280	1280
A/Iceland/74/2019		6B.1A6	2019-05-29	MDCK1/MDCK1	640	160	160	640	1280	1280	320	1280	640	640
A/Poland/553/2019		6B.1A7	2019-02-14	MDCK1	1280	320	80	1280	1280	1280	640	2560	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-20. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-07-16)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	NIB F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	160	160	640	640	1280	640	1280	640
A/Bavaria/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	160	160	40	80	160	80	320	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	320	320	80	80	320	80	640	80
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	160	160	640	1280	640	2560	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	80	<	320	320	640	320	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	1280	640	640	1280	2560	1280	5120	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	80	80	320	640	640	1280	320	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	160	40	<	160	160	160	640	160	160
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	160	160	640	1280	640	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	160	1280	640	2560	640	2560	1280
TEST VIRUSES													
A/Macedonia/556/2018		6B.1A	2018-03-16	MDCK1/MDCK1	160	160	80	640	320	640	320	1280	640
A/Macedonia/1008/2019			2019-02-18	MDCK1	160	40	<	160	80	160	80	320	160
A/Macedonia/1006/2019			2019-02-19	MDCK1	160	40	40	160	160	320	160	320	160
A/Macedonia/584/2019		6B.1A	2019-04-02	MDCK1/MDCK1	320	160	80	320	320	640	320	1280	640
A/Macedonia/1300/2019		6B.1A1	2019-03-12	MDCK1	80	40	<	80	80	160	80	320	160
A/Macedonia/793/2019		6B.1A2	2019-02-05	MDCK1	640	160	80	640	640	1280	320	1280	640
A/Macedonia/803/2019		6B.1A2	2019-02-06	MDCK1	160	80	40	160	160	320	160	320	160
A/Macedonia/788/2019		6B.1A2	2019-02-06	MDCK1	640	160	80	320	320	640	320	640	320
A/Macedonia/764/2019		6B.1A5	2019-02-01	MDCK1	160	40	40	160	160	160	80	640	160
A/Macedonia/810/2019		6B.1A5	2019-02-06	MDCK1	320	160	80	320	160	320	160	640	320
A/Macedonia/883/2019		6B.1A5	2019-02-12	MDCK1	160	80	40	160	160	160	160	320	160
A/Macedonia/1041/2019		6B.1A5	2019-02-21	MDCK1	160	40	<	160	160	160	160	320	640
A/Macedonia/1154/2019		6B.1A5	2019-02-28	MDCK1	160	40	40	160	80	160	80	320	160
A/Macedonia/1163/2019		6B.1A5	2019-03-01	MDCK1	80	40	<	80	80	80	80	320	160
A/Macedonia/1227/2019		6B.1A5	2019-03-06	MDCK1	80	40	<	80	80	80	80	320	80
A/Macedonia/1219/2019		6B.1A5	2019-03-06	MDCK1	80	40	<	80	80	160	160	320	160
A/Macedonia/1277/2019		6B.1A5	2019-03-09	MDCK1	160	40	80	320	160	320	160	640	320
A/Macedonia/1290/2019		6B.1A5	2019-03-11	MDCK1	80	40	40	80	80	160	80	320	160
A/Macedonia/1359/2019		6B.1A5	2019-03-18	MDCK1	320	40	80	640	320	640	320	1280	640
A/Macedonia/1352/2019		6B.1A5	2019-03-18	MDCK1	640	40	160	640	320	640	640	1280	640
A/Macedonia/1470/2019		6B.1A5	2019-04-24	MDCK1	160	80	40	160	160	320	160	1280	320
A/Macedonia/786/2019		6B.1A6	2019-02-06	MDCK1	640	160	80	640	640	640	320	1280	640
A/Macedonia/826/2019		6B.1A6	2019-02-07	MDCK1	160	40	40	320	160	320	160	320	160
A/Macedonia/1012/2019		6B.1A6	2019-02-20	MDCK1	160	80	40	320	160	320	160	640	320
A/Macedonia/1329/2019		6B.1A6	2019-03-13	MDCK1	80	40	<	160	80	160	80	320	160
A/Macedonia/763/2019		6B.1A7	2019-02-01	MDCK1	160	160	40	640	320	640	320	1280	640
A/Macedonia/1056/2019		6B.1A7	2019-02-18	MDCK1	160	40	40	160	160	320	160	320	160
A/Macedonia/1094/2019		6B.1A7	2019-02-21	MDCK1	640	40	160	640	640	1280	640	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-21. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-07-24)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre												NEW	NEW	NEW	
					Post-infection ferret antisera															
					A/Mich	A/Bayern	A/Lviv	A/Slov	A/Paris	A/Swit	A/Norway	A/Ire	A/Bris	A/HK	A/Greece	A/Swit				
					45/15	69/09	N6/09	2903/2015	1447/17	2656/17	3330/17	3433/18	84630/18	02/18	110/19	144/19				
					Egg	MDCK	MDCK	Egg	Egg	Egg	Egg	MDCK	MDCK	Egg	MDCK	Egg				
					F31/16 ¹	F09/15 ¹	F13/18 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F23/18 ¹	F04/19 ¹	F08/19 ¹	F09/19 ¹	F28/19 ¹	F27/19 ¹	F22/19 ¹			
					6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1	6B.1A2	6B.1A5	6B.1A5			
REFERENCE VIRUSES																				
A/Michigan/45/2015	G155E C155E, D222G clone 37	6B.1	2015-09-07	E3/E3	1280	320	320	1280	2560	2560	1280	2560	2560	1280	40	2560	1280			
A/Bavaria/69/2009			2009-07-01	MDCK4/MDCK1	160	640	320	80	320	160	160	640	80	320	40	80	80			
A/Lviv/N6/2009			2009-10-27	MDCK4/MDCK3	320	640	640	160	640	640	320	1280	320	640	40	80	160			
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	1280	640	320	2560	2560	2560	1280	2560	2560	2560	40	1280	2560			
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	320	160	1280	2560	2560	1280	5120	2560	2560	80	2560	1280			
A/Switzerland/2655/2017		6B.1A	2017-12-21	E5/E3	640	320	320	640	640	1280	640	2560	640	640	40	640	640			
A/Switzerland/3330/2017		6B.1A5	2017-12-26	E5/E2	640	160	160	640	1280	640	1280	1280	1280	640	<	640	640			
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK3	640	160	80	640	1280	640	640	2560	1280	640	40	1280	1280			
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	160	1280	2560	2560	1280	2560	2560	1280	40	1280	2560			
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	2560	640	640	2560	5120	2560	5120	5120	5120	1280	160	2560	2560			
A/Hong Kong/110/2019		6B.1A2	2019-01-01	MDCK1/MDCK2	160	160	80	<	40	160	80	320	80	80	1280	160	80			
A/Greece/144/2019		6B.1A5	2019-01-14	E3	1280	320	160	1280	1280	1280	640	2560	1280	640	40	1280	1280			
A/Switzerland/4217/2019		6B.1A5	2019-01-08	E4	1280	320	320	640	1280	1280	640	2560	1280	1280	80	640	1280			
A/Switzerland/4217/2019		6B.1A5	2019-01-08	E4/E1	1280	320	320	640	1280	640	640	2560	1280	1280	80	640	1280			
TEST VIRUSES																				
A/Macedonia/618/2018		6B.1A5	2019-09-14	MDCK1/MDCK3	2560	1280	640	2560	5120	2560	1280	5120	2560	2560	80	2560	2560			
A/Banska Bystrica/115/2019		6B.1A5	2019-01-15	MDCKx/MDCK1	1280	640	320	1280	2560	1280	640	2560	2560	1280	80	1280	2560			
A/Piestany/80/2019		6B.1A7	2019-01-15	MDCK1/MDCK1	1280	320	160	1280	2560	1280	640	2560	2560	1280	40	1280	1280			
A/Kanagawa/AC1831/2019		6B.1A2	2019-01-15	Ex	80	160	160	40	80	160	80	160	80	320	160	40	80			
A/Kosovo/56/2019		6B.1A5	2019-01-18	MDCK1	1280	320	320	1280	2560	1280	640	2560	2560	1280	40	1280	2560			
A/Bratislava/76/2019		6B.1A5	2019-01-18	MDCK1/MDCK1	640	160	160	640	640	2560	640	5120	2560	1280	40	1280	1280			
A/Kanagawa/IC1848/2019		6B.1A2	2019-01-18	Ex	80	160	160	40	40	160	80	160	80	160	320	40	80			
A/Bratislava/93/2019		6B.1A	2019-01-21	MDCKx/MDCK1	2560	320	320	1280	2560	5120	1280	5120	2560	2560	80	1280	2560			
A/Tmava/85/2019		6B.1A5	2019-01-22	MDCK1/MDCK1	1280	320	160	640	1280	2560	1280	2560	1280	2560	80	1280	1280			
A/Trencin/124/2019		6B.1A5	2019-01-28	MDCKx/MDCK1	1280	320	160	640	1280	2560	1280	5120	2560	1280	40	1280	1280			
A/Poprad/134/2019		6B.1A7	2019-01-29	MDCKx/MDCK1	1280	320	160	1280	2560	2560	1280	2560	2560	2560	80	1280	2560			
A/Padova/22/2019		6B.1A2	2019-02-20	MDCK4/MDCK1	1280	320	160	1280	1280	1280	640	1280	1280	1280	40	640	640			
A/Kosice/422/2019		6B.1A2	2019-02-28	MDCKx/MDCK1	2560	640	320	2560	5120	2560	1280	2560	2560	2560	80	1280	2560			
A/Kosovo/198/2019		6B.1A5	2019-02-01	MDCK2	1280	640	320	1280	2560	1280	1280	2560	1280	1280	80	1280	1280			
A/Malacky/145/2019		6B.1A5	2019-02-01	MDCKx/MDCK1	1280	640	640	1280	1280	2560	2560	5120	2560	2560	80	1280	1280			
A/Lubica/183/2019		6B.1A5	2019-02-05	MDCKx/MDCK1	1280	320	320	1280	1280	2560	640	5120	2560	1280	80	1280	1280			
A/Nitra/197/2019		6B.1A5	2019-02-08	MDCKx/MDCK1	1280	320	320	640	1280	2560	640	2560	2560	1280	80	1280	1280			
A/Macedonia/060/2019		6B.1A5	2019-02-11	MDCK3	2560	640	320	1280	2560	1280	1280	2560	2560	1280	80	1280	2560			
A/Bojnice/225/2019		6B.1A5	2019-02-11	MDCKx/MDCK1	640	160	160	640	1280	1280	640	2560	1280	1280	40	640	1280			
A/Bratislava/216/2019		6B.1A5	2019-02-11	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280	80	1280	2560			
A/Piestany/267/2019		6B.1A5	2019-02-12	MDCKx/MDCK1	1280	320	320	1280	1280	1280	640	2560	2560	1280	80	1280	1280			
A/Senica/305/2019		6B.1A5	2019-02-18	MDCKx/MDCK1	640	160	80	320	640	640	320	1280	640	640	40	640	2560			
A/Torino/617/2019		6B.1A5	2019-02-19	MDCK4/MDCK1	640	160	80	640	1280	640	320	1280	640	640	<	640	640			
A/Friuli Venezia Giulia/184/2019		6B.1A5	2019-02-21	MDCK2/MDCK1	1280	640	640	1280	2560	1280	640	2560	2560	1280	80	1280	1280			
A/Kosovo/575/2019		6B.1A5	2019-02-22	MDCK1	1280	640	640	2560	2560	2560	640	2560	2560	1280	80	1280	5120			
A/Padova/26/2019		6B.1A5	2019-02-25	MDCK3/MDCK1	640	320	160	640	1280	640	320	2560	1280	640	40	640	640			
A/Kosovo/655/2019		6B.1A5	2019-02-26	MDCK1	1280	320	160	640	1280	640	640	1280	640	640	40	640	640			
A/Galanta/387/2019		6B.1A5	2019-02-27	MDCKx/MDCK1	1280	320	320	1280	2560	1280	1280	2560	1280	1280	80	1280	1280			
A/Kosovo/730/2019		6B.1A5	2019-02-28	MDCK1	1280	640	320	1280	2560	2560	640	2560	1280	1280	80	1280	2560			
A/South Africa/R01828/2019		6B.1A5	2019-03-01	MDCK1/MDCK1	1280	640	320	1280	2560	1280	5120	2560	1280	1280	80	1280	2560			
A/Kosovo/841/2019		6B.1A5	2019-03-06	MDCK1	1280	320	320	1280	2560	1280	640	2560	1280	1280	40	1280	1280			
A/Kosovo/832/2019		6B.1A5	2019-03-06	MDCK2	2560	640	640	2560	5120	2560	1280	5120	2560	1280	80	2560	2560			
A/Torino/779/2019		6B.1A5	2019-03-06	MDCK4/MDCK1	2560	640	320	1280	2560	1280	640	2560	2560	1280	80	1280	1280			
A/Kosovo/875/2019		6B.1A5	2019-03-07	MDCK1	2560	640	640	2560	5120	2560	1280	5120	2560	2560	80	2560	2560			
A/Kosovo/894/2019		6B.1A5	2019-03-08	MDCK2	2560	640	320	1280	2560	1280	1280	2560	2560	1280	80	5120	2560			
A/Palermo/23/2019		6B.1A5	2019-03-08	MDCK5/MDCK1	1280	640	160	1280	1280	1280	640	1280	2560	1280	80	1280	2560			
A/Kosovo/909/2019		6B.1A5	2019-03-11	MDCK2	2560	640	320	2560	2560	2560	1280	2560	2560	1280	80	1280	1280			
A/Kosovo/926/2019		6B.1A5	2019-03-12	MDCK2	1280	640	320	1280	2560	1280	1280	2560	1280	1280	80	1280	1280			
A/Palermo/26/2019		6B.1A5	2019-03-12	MDCK3/MDCK1	1280	320	320	640	1280	1280	640	1280	1280	1280	80	1280	1280			
A/Parma/370/2019		6B.1A5	2019-03-12	MDCK4/MDCK1	1280	320	320	640	1280	640	640	1280	1280	640	40	1280	640			
A/Kosovo/982/2019		6B.1A5	2019-03-14	MDCK2	1280	320	320	1280	2560	1280	640	2560	1280	1280	40	1280	1280			
A/Kosovo/1023/2019		6B.1A5	2019-03-19	MDCK2	1280	640	640	2560	2560	2560	1280	5120	2560	2560	160	2560	2560			
A/Kosovo/1025/2019		6B.1A5	2019-03-19	MDCK2	640	160	160	640	1280	640	640	1280	640	640	40	640	1280			
A/Prizren/518/2019		6B.1A5	2019-03-25	MDCKx/MDCK1	2560	640	640	1280	2560	2560	1280	2560	2560	1280	80	1280	2560			
A/Kosovo/1137/2019		6B.1A5	2019-03-25	MDCK1	1280	320														

Table 5-22. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-07-31)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	NIB F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	160	160	640	1280	640	320	640	640	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	320	40	160	160	80	320	40	160
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	160	640	640	80	320	320	160	640	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	1280	640	320	1280	2560	2560	1280	2560	1280	2560
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	80	640	1280	1280	640	1280	1280	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	320	320	640	640	1280	640	2560	640	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	160	160	640	1280	640	640	1280	640	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK4/SIAT1/MDCK3	320	80	40	640	640	320	320	640	640	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	320	1280	2560	2560	640	2560	2560	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	640	320	160	1280	1280	1280	640	1280	1280	640
TEST VIRUSES														
A/Estonia/118748/2019		6B.1A	2019-01-16	SIAT1/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Estonia/118723/2019		6B.1A5	2019-01-16	SIAT1/MDCK1	640	320	320	640	1280	1280	640	1280	640	640
A/Estonia/118840/2019		6B.1A5	2019-01-21	SIAT2/MDCK1	320	160	80	640	640	640	320	1280	640	320
A/Estonia/118896/2019		6B.1A5	2019-01-23	SIAT2/MDCK1	640	320	320	1280	1280	640	640	2560	1280	640
A/Estonia/118882/2019		6B.1A	2019-01-23	SIAT2/MDCK1	1280	320	160	1280	2560	1280	1280	2560	1280	1280
A/Czech Republic/1250/2019		6B.1A5	2019-01-30	MDCK3/MDCK1	1280	320	160	640	1280	640	640	5120	1280	640
A/Estonia/119176/2019		6B.1A	2019-02-06	MDCK1	1280	320	320	1280	1280	2560	1280	2560	1280	640
A/Karaganda/380/2019		6B.1A	2019-02-11	MDCKx/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Estonia/120012/2019		6B.1A	2019-03-18	SIAT2/MDCK2	1280	320	160	1280	1280	2560	640	2560	1280	1280
A/Estonia/120077/2019		6B.1A	2019-03-20	SIAT2/MDCK1	640	160	80	640	1280	640	640	1280	640	640
A/Czech Republic/761/2019		6B.1A5	2019-02-04	MDCK3/MDCK1	640	160	80	320	640	320	320	2560	640	640
A/Jordan/10237/2019		6B.1A5	2019-02-07	MDCK1	640	320	160	640	1280	1280	640	1280	1280	640
A/Estonia/119338/2019		6B.1A5	2019-02-11	SIAT2/MDCK1	1280	640	640	1280	2560	2560	1280	2560	2560	2560
A/Czech Republic/1254/2019		6B.1A5	2019-02-11	MDCK3/MDCK1	1280	320	160	640	1280	1280	640	2560	1280	1280
A/Czech Republic/1253/2019		6B.1A5	2019-02-11	MDCK3/MDCK1	640	160	40	320	640	320	320	1280	640	320
A/Czech Republic/1255/2019		6B.1A5	2019-02-13	MDCK3/MDCK1	640	160	80	640	640	640	320	2560	640	640
A/Czech Republic/1257/2019		6B.1A5	2019-02-14	MDCK3/MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Czech Republic/1256/2019		6B.1A5	2019-02-14	MDCK3/MDCK1	1280	640	640	1280	2560	1280	640	5120	2560	1280
A/Czech Republic/1258/2019		6B.1A5	2019-02-17	MDCK3/MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Estonia/119487/2019		6B.1A5	2019-02-18	SIAT2/SIAT1	640	320	320	640	1280	1280	640	2560	1280	640
A/Estonia/119483/2019		6B.1A5	2019-02-19	SIAT2/SIAT1	640	320	160	640	1280	640	640	2560	1280	640
A/Czech Republic/1259/2019		6B.1A5	2019-02-21	MDCK3/MDCK1	640	160	80	640	640	640	320	1280	640	640
A/Czech Republic/1262/2019		6B.1A5	2019-02-26	MDCK3/MDCK1	640	320	160	1280	1280	640	1280	2560	1280	640
A/Czech Republic/1261/2019		6B.1A5	2019-02-27	MDCK3/MDCK1	640	160	40	640	1280	640	640	1280	640	640
A/Estonia/119738/2019		6B.1A5	2019-03-04	SIAT2/MDCK1	1280	640	320	1280	2560	2560	640	2560	1280	1280
A/Jordan/10457/2019		6B.1A5	2019-03-10	MDCK1	640	160	160	640	1280	640	640	1280	640	640
A/Estonia/119976/2019		6B.1A5	2019-03-18	SIAT2/SIAT1	640	160	80	640	1280	640	320	1280	640	640
A/Jordan/20133/2019		6B.1A5	2019-03-21	MDCK1	1280	640	320	1280	2560	1280	640	2560	1280	1280
A/Estonia/120189/2019		6B.1A5	2019-03-27	SIAT2/SIAT1	1280	640	320	1280	1280	1280	640	2560	1280	640
A/Jordan/20157/2019		6B.1A5	2019-04-04	MDCK1	320	160	160	640	640	640	640	1280	640	640
A/Jordan/10646/2019		6B.1A5	2019-04-14	MDCK3	1280	320	320	640	1280	1280	640	1280	1280	640
A/Jordan/10682/2019		6B.1A5	2019-04-21	MDCK3	1280	320	320	1280	1280	1280	640	2560	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-23. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-08-06)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	NIB F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E4	640	320	320	1280	1280	1280	640	1280	640	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	160	40	160	80	80	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	160	320	640	80	320	320	160	640	80	160
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	320	160	1280	2560	2560	1280	2560	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	80	640	1280	1280	640	1280	640	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	640	640	1280	1280	2560	1280	2560	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	160	160	640	1280	640	640	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK4	320	80	<	320	640	320	320	640	320	320
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	320	160	1280	2560	2560	640	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	640	320	160	640	1280	1280	640	1280	1280	640
TEST VIRUSES														
A/Orenburg/178/2019		6B.1A	2019-03-01	MDCK1/MDCK1	640	320	320	1280	1280	1280	640	1280	1280	320
A/Orenburg/201/2019		6B.1A	2019-03-05	MDCK2/MDCK1	320	160	80	640	1280	640	640	1280	640	320
A/V.Novgorod/197/2019		6B.1A	2019-03-06	MDCK1/MDCK1	1280	320	320	1280	2560	1280	640	1280	1280	1280
A/Orenburg/199/2019		6B.1A	2019-03-07	MDCK2/MDCK1	640	320	160	1280	1280	1280	640	1280	1280	1280
A/V.Novgorod/170/2019		6B.1A	2019-03-12	MDCK1/MDCK1	640	160	160	640	1280	1280	640	1280	1280	640
A/Orenburg/180/2019		6B.1A	2019-03-19	MDCK1/MDCK1	1280	320	80	1280	2560	1280	640	2560	1280	1280
A/Moscow/175/2019		6B.1A	2019-03-24	MDCK1/MDCK1	1280	320	320	1280	640	1280	1280	2560	2560	640
A/Vladivostok/123/2019		6B.1A5	2019-03-02	MDCK1/MDCK2	640	160	160	640	1280	640	320	1280	640	320
A/Vladivostok/4/2019		6B.1A5	2019-03-04	MDCK1/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	1280
A/Vladivostok/110/2019		6B.1A5	2019-03-04	MDCK1/MDCK1	320	160	80	320	640	640	640	1280	640	320
A/Vladivostok/111/2019		6B.1A5	2019-03-05	MDCK1/MDCK1	640	320	320	1280	1280	1280	640	2560	1280	640
A/Vladivostok/117/2019		6B.1A5	2019-03-12	MDCK1/MDCK1	1280	320	320	1280	2560	2560	640	2560	1280	1280
A/Moscow/162/2019		6B.1A5	2019-03-15	MDCK1/MDCK1	1280	320	320	1280	1280	1280	640	2560	1280	640
A/Bretagne/1721/2019		6B.1A5	2019-03-18	MDCK1/MDCK1	160	160	80	80	160	320	160	320	40	160
A/Vladivostok/124/2019		6B.1A5	2019-03-18	MDCK1/MDCK1	1280	320	320	1280	2560	1280	640	2560	1280	1280
A/Paris/1772/2019		6B.1A5	2019-03-22	MDCK1/MDCK1	640	160	320	640	320	640	320	1280	640	320
A/Alsace/1787/2019		6B.1A5	2019-03-26	MDCK1/MDCK1	320	160	40	320	320	640	320	1280	640	640
A/Moscow/177/2019		6B.1A5	2019-03-29	MDCK1/MDCK1	640	320	160	640	320	640	640	1280	1280	320
A/Moscow/195/2019		6B.1A5	2019-04-08	MDCK2/MDCK1	640	320	320	1280	640	1280	640	2560	1280	640
A/Brest/2004/2019		6B.1A5	2019-04-14	MDCK1/MDCK1	640	160	160	640	640	640	320	1280	640	640
A/Cheboksary/143/2019		6B.1A7	2019-03-13	MDCK1/MDCK1	1280	320	80	1280	1280	1280	640	1280	1280	1280
A/Moscow/173/2019		6B.1A7	2019-03-14	MDCK1/MDCK1	640	160	80	640	1280	1280	640	1280	1280	640
A/Moscow/193/2019		6B.1A7	2019-03-15	MDCK1/MDCK1	640	160	80	640	1280	1280	640	1280	1280	640
A/Moscow/174/2019		6B.1A7	2019-03-18	MDCK1/MDCK1	640	320	80	1280	1280	1280	640	1280	1280	640
A/Pays de Loire/1751/2019		6B.1A7	2019-03-20	MDCK1/MDCK1	640	320	160	640	1280	1280	640	2560	1280	1280
A/Moscow/164/2019		6B.1A7	2019-03-20	MDCK1/MDCK1	640	320	80	1280	640	1280	640	2560	1280	640
A/Brest/2003/2019		6B.1A7	2019-04-13	MDCK1/MDCK1	1280	320	1280	1280	1280	1280	640	2560	2560	1280
A/Bretagne/1920/2019		6B.1A7	2019-04-13	MDCK1/MDCK1	640	160	80	1280	640	1280	640	1280	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-24. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-08-14)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
					F31/16 [†]	F09/15 [†]	F13/18 [†]	NIB F48/16 [†]	F03/18 [‡]	F20/18 [†]	F23/18 [†]	F04/19 [†]	F08/19 [†]	F09/19 [†]
	Passage history				6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E4	640	320	320	1280	2560	1280	640	2560	1280	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	80	320	320	40	160	160	80	320	40	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	640	320	320	1280	160	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	1280	320	160	1280	1280	1280	640	1280	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	160	80	640	1280	640	640	1280	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	2560	640	1280	2560	2560	2560	1280	2560	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	160	160	640	1280	640	640	1280	640	640
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK4	320	80	40	320	320	320	320	1280	320	160
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	160	160	1280	1280	1280	640	2560	1280	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	320	320	1280	2560	1280	1280	2560	1280	1280
TEST VIRUSES														
A/England/137/2019	N156K	6B.1A2	2019-01-14	SIAT1/MDCK1	80	160	40	40	40	160	80	160	40	80
A/Bosnia and Herzegovia/123G/2019			2019-01-18	MDCK1/MDCK1	1280	320	160	2560	2560	1280	1280	2560	1280	1280
A/England/228/2019		6B.1A5	2019-02-13	SIAT2/MDCK1	640	320	160	640	1280	1280	640	2560	1280	640
A/England/158/2019		6B.1A5	2019-02-23	SIAT1/MDCK1	1280	320	160	1280	1280	1280	640	2560	1280	640
A/Bosnia and Herzegovia/334G/2019		6B.1A5	2019-03-04	MDCK1/MDCK2	1280	160	160	1280	1280	1280	640	2560	1280	640
A/England/246/2019		6B.1A5	2019-03-15	SIAT1/MDCK1	640	160	160	640	1280	640	320	1280	640	640
A/Tomsk/5606V/2019		6B.1A5	2019-04-19	MDCK1/MDCK1	1280	160	160	640	1280	1280	640	2560	640	640
A/Sakha/3-24V/2019		6B.1A5	2019-04-24	MDCK1/MDCK1	1280	640	320	1280	2560	1280	1280	2560	1280	1280
A/Arkhangelsk/115V/2019		6B.1A5	2019-05-06	MDCK1/MDCK1	1280	160	160	640	1280	640	640	2560	640	640
A/England/190/2019		6B.1A6	2019-02-10	SIAT1/MDCK1	1280	320	80	1280	2560	1280	1280	2560	1280	1280
A/England/214/2019		6B.1A6	2019-02-18	SIAT1/MDCK1	1280	640	160	1280	2560	2560	640	2560	1280	1280
A/England/234/2019		6B.1A7	2019-03-09	SIAT1/MDCK1	1280	320	160	1280	2560	1280	640	2560	1280	1280
A/Chuvashia/3890V/2019		6B.1A7	2019-05-06	MDCK1/MDCK1	1280	640	160	1280	2560	1280	1280	2560	1280	1280
A/Belgorod/509V/2019		6B.1A7	2019-05-14	MDCK1/MDCK1	640	160	80	640	1280	1280	640	1280	1280	640
					Vaccine NH 2018-19 SH 2019									Vaccine NH 2019-20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Table 5-25. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-08-14)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ¹	F09/15 ¹	F13/18 ¹	NIB F48/16 ¹	F03/18 ²	F20/18 ¹	F23/18 ¹	F04/19 ¹	F08/19 ¹	F09/19 ¹
	Passage history			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Michigan/45/2015		2015-09-07	E3/E4	1280	320	320	1280	2560	1280	1280	1280	1280	1280
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	80	160	160	<	160	80	80	160	40	80
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	640	160	640	320	320	640	160	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	2560	640	640	2560	1280	2560	1280	1280	1280	1280
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	640	160	40	640	1280	640	320	640	640	320
A/Switzerland/2656/2017		2017-12-21	E5/E2	640	320	320	1280	2560	1280	640	1280	640	640
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	640	160	160	640	1280	640	640	640	640	320
A/Norway/3433/2018		2018-10-30	MDCK4	320	80	<	320	320	320	320	640	320	320
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	1280	160	160	1280	1280	1280	640	1280	640	320
A/Brisbane/02/2018		2018-01-04	E3/E1	1280	320	320	1280	2560	1280	1280	2560	1280	1280
TEST VIRUSES													
A/Ireland/24181/2019		2019-03-04	MDCK1/MDCK1	640	160	160	1280	640	1280	640	1280	640	640
A/Ireland/22361/2019		2019-02-25	MDCK1/MDCK1	1280	320	160	1280	640	1280	640	1280	1280	640
A/Ireland/23159/2019		2019-02-27	MDCK1/MDCK1	640	160	80	320	320	640	320	1280	640	320
A/Ireland/24488/2019	N156S	2019-03-05	MDCK1/MDCK1	80	80	<	<	<	80	40	320	<	40
A/Ireland/26203/2019		2019-03-07	MDCK1/MDCK1	640	320	160	640	1280	640	320	1280	640	640
A/Ireland/29434/2019		2019-03-22	MDCK1/MDCK1	640	320	160	640	1280	640	320	1280	640	640
A/Ireland/22715/2019		2019-02-26	MDCK1/MDCK1	1280	320	160	1280	1280	1280	640	1280	1280	640
A/Ireland/23810/2019		2019-03-01	MDCK2/MDCK1	640	160	80	640	640	640	320	1280	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20

Table 5-26. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-09-10)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
				F31/16 ^{*1}	F09/15 ^{*1}	F13/18 ^{*1}	NIB F48/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F23/18 ^{*1}	F04/19 ^{*1}	F08/19 ^{*1}	F09/19 ^{*1}
	Passage history			6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Michigan/45/2015		2015-09-07	E3/E4	1280	320	320	1280	1280	1280	640	2560	1280	640
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	80	320	160	<	160	160	80	320	40	160
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	160	640	640	80	320	320	160	640	80	640
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	640	320	320	1280	1280	1280	640	1280	1280	640
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	640	160	80	640	1280	640	640	1280	640	320
A/Switzerland/2656/2017		2017-12-21	E5/E2	1280	640	320	1280	2560	1280	640	2560	1280	640
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	640	160	80	640	1280	640	640	1280	640	320
A/Norway/3433/2018		2018-10-30	MDCK4	320	80	<	320	320	320	160	640	160	160
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	1280	320	160	1280	1280	1280	640	1280	1280	640
A/Brisbane/02/2018		2018-01-04	E3/E1	1280	320	160	1280	1280	1280	640	1280	1280	640
TEST VIRUSES													
IVR-193 (A/Darwin/6/2018)			E3/D7/E1	1280	320	160	1280	2560	1280	640	2560	1280	640
IVR-198 (A/Darwin/122/2018)			E3/D7/E1	1280	320	160	1280	2560	1280	640	2560	640	640
IVR-201 (A/Canberra/13/2019)			E3/D9/E1	40	80	80	<	<	80	40	80	<	40
A/Wisconsin/505/2018		2018-11-20	C2/MDCK1	1280	320	320	1280	1280	640	640	2560	1280	640
A/Darwin/122/2018		2018-11-25	E3/E1	640	160	160	640	1280	640	640	1280	640	640
A/Darwin/123/2018		2018-11-25	E3/E1	640	160	160	640	1280	1280	640	1280	640	640
A/Canberra/13/2019		2019-01-19	E3/E1	40	80	40	<	80	80	40	80	<	40
A/Brunei/5/2019		2019-02-25	E3/E1	40	80	80	40	80	80	40	80	40	40
A/Newcastle/46/2019		2019-03-13	E3/E1	40	40	40	<	80	80	40	80	<	<
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	C2/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	E3/E1	640	160	160	640	1280	640	320	1280	640	320
A/Oman/4015/2019		2019-06-04	SIAT1/MDCK1	640	320	160	640	1280	640	640	2560	640	640
A/Oman/4008/2019		2019-06-06	SIAT1/MDCK1	1280	80	320	1280	1280	1280	640	2560	1280	640
A/Oman/4066/2019		2019-06-09	SIAT1/MDCK1	640	160	160	640	1280	640	640	1280	640	640
A/Oman/4090/2019		2019-06-09	SIAT1/MDCK1	1280	320	320	640	1280	1280	640	1280	1280	640
A/Oman/4125/2019		2019-06-13	SIAT2/MDCK1	640	320	160	640	1280	640	640	1280	1280	640
A/Oman/4161/2019		2019-06-15	SIAT2/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Oman/4196/2019		2019-06-19	SIAT1/MDCK1	640	160	160	640	1280	640	640	1280	640	320
A/Oman/4307/2019		2019-06-27	SIAT1/MDCK1	320	80	80	320	640	320	160	640	320	320
A/Oman/4296/2019		2019-06-29	SIAT1/MDCK1	320	160	80	320	640	320	320	1280	320	320
A/Oman/4419/2019		2019-07-10	MDCK1	640	320	320	1280	1280	1280	640	2560	1280	640
A/Oman/4425/2019		2019-07-09	MDCK1	1280	320	160	640	1280	640	640	2560	640	640
A/Oman/4042/2019		2019-06-09	SIAT1/MDCK1	640	160	160	640	1280	640	640	1280	640	320
A/Oman/4123/2019		2019-06-14	SIAT1/MDCK1	640	160	160	640	640	640	320	1280	640	320
A/Oman/4204/2019		2019-06-19	SIAT1/MDCK1	640	160	80	640	640	640	640	1280	640	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20

Table 5-27. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-06-18) - Summary

			Haemagglutination inhibition titre									
			Post-infection ferret antisera									
Viruses	Other information		A/Mich 45/15 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Slov 2903/2015 Egg NIB F48/16 ⁻¹	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK	A/Ire 84630/18 MDCK	A/Bris 02/18 Egg
		Passage history										
		Ferret number	F31/16 ⁻¹	F09/15 ⁻¹	F13/18 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	F04/19 ⁻¹	F08/19 ⁻¹	F09/19 ⁻¹
		Genetic group	6B.1			6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	6B.1A6	6B.1A1
REFERENCE VIRUSES												
A/Michigan/45/2015		6B.1	2560	320	320	1280	1280	2560	640	2560	1280	640
A/Bayern/69/2009	G155E		80	320	160	40	160	80	80	160	40	80
A/Lviv/N6/2009	G155E, D222G		320	640	1280	160	640	640	320	640	160	320
A/Slovenia/2903/2015	clone 37	6B.1	1280	320	160	1280	1280	1280	640	1280	640	640
A/Paris/1447/2017		6B.1A	640	320	40	640	640	640	320	640	640	640
A/Switzerland/2656/2017		6B.1A	1280	640	320	1280	1280	1280	640	2560	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	320	160	80	320	640	640	640	640	640	320
A/Norway/3433/2018		6B.1A5	320	80	40	320	640	320	320	1280	320	320
A/Ireland/84630/2018		6B.1A6	1280	320	160	1280	1280	1280	1280	2560	1280	1280
A/Brisbane/02/2018		6B.1A1	1280	320	160	1280	2560	2560	1280	2560	1280	1280
TEST VIRUSES												
Genetic group	No tested	No with										
All viruses	403	Titre reduced ≤2-fold	313	351	78	324	380	328	361	400	335	315
		%	77.7	87.1	19.4	80.4	94.3	81.4	89.6	99.3	83.1	78.2
		Titre reduced =4-fold	57	50	114	50	20	47	32	2	52	69
		%	14.1	12.4	28.3	12.4	5.0	11.7	7.9	0.5	12.9	17.1
		Titre reduced ≥8-fold	33	2	211	15	3	28	10	1	16	19
		%	8.2	0.5	52.3	7.2	0.7	6.9	2.5	0.2	4.0	4.7
6B.1A	33	Titre reduced ≤2-fold	28	32	5	32	33	29	32	33	30	30
		%	84.8	97.0	15.2	97.0	100.0	87.9	97.0	100.0	90.9	90.9
		Titre reduced =4-fold	4	1	11	1		4	1		3	3
		%	12.2	3.0	33.3	3.0		12.1	3.0		9.1	9.1
		Titre reduced ≥8-fold	1		17							
		%	3.0		51.5							
6B.1A1	1	Titre reduced ≤2-fold								1		
		Titre reduced =4-fold		1			1					
		Titre reduced ≥8-fold	1		1	1		1	1		1	1
6B.1A2	27	Titre reduced ≤2-fold	23	24	2	24	26	24	25	26	25	25
		%	85.2	88.9	7.4	88.9	96.3	88.9	92.6	96.3	92.6	92.6
		Titre reduced =4-fold	2	3	10	2		1	1	1	1	1
		%	7.4	11.1	37.0	7.4		3.7	3.7	3.7	3.7	3.7
		Titre reduced ≥8-fold	2		15	1	1	2	1		1	1
		%	7.4		55.6	3.7	3.7	7.4	3.7		3.7	3.7
6B.1A5	208	Titre reduced ≤2-fold	152	172	59	155	192	151	182	207	157	149
		%	73.1	82.7	28.4	74.5	92.3	72.6	87.5	99.5	75.5	71.6
		Titre reduced =4-fold	32	34	59	40	14	39	20	1	40	46
		%	15.4	16.3	28.4	19.2	6.7	18.8	9.6	0.5	19.2	22.1
		Titre reduced ≥8-fold	24	2	90	13	2	18	6		11	13
		%	11.5	1.0	43.2	6.3	1.0	8.6	2.9		5.3	6.3
6B.1A6	56	Titre reduced ≤2-fold	45	53	4	55	55	50	49	56	50	47
		%	80.4	94.6	7.1	98.2	98.2	89.3	87.5	100.0	89.3	83.9
		Titre reduced =4-fold	10	3	10	1	1	2	6		5	7
		%	17.9	5.4	17.9	1.8	1.8	3.6	10.8		9.00	12.5
		Titre reduced ≥8-fold	1		42			4	1		1	2
		%	1.7		75.0			6.6	1.7		1.7	3.6
6B.1A7	65	Titre reduced ≤2-fold	55	59	8	59	64	61	63	64	63	55
		%	84.6	90.8	12.3	90.8	98.5	93.8	96.9	98.5	96.9	84.6
		Titre reduced =4-fold	6	6	20	6	1	3	2		1	9
		%	9.2	9.2	30.8	9.2	1.5	4.7	3.1		1.5	13.9
		Titre reduced ≥8-fold	4		37			1		1	1	1
		%						1.5		1.5	1.5	1.5
			Vaccine NH 2018-19 SH 2019									Vaccine NH 2019-20

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Vaccine viruses

Reference viruses

Apr 2019
May 2019
Jun 2019
Jul 2019

* amino acid polymorphism



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**

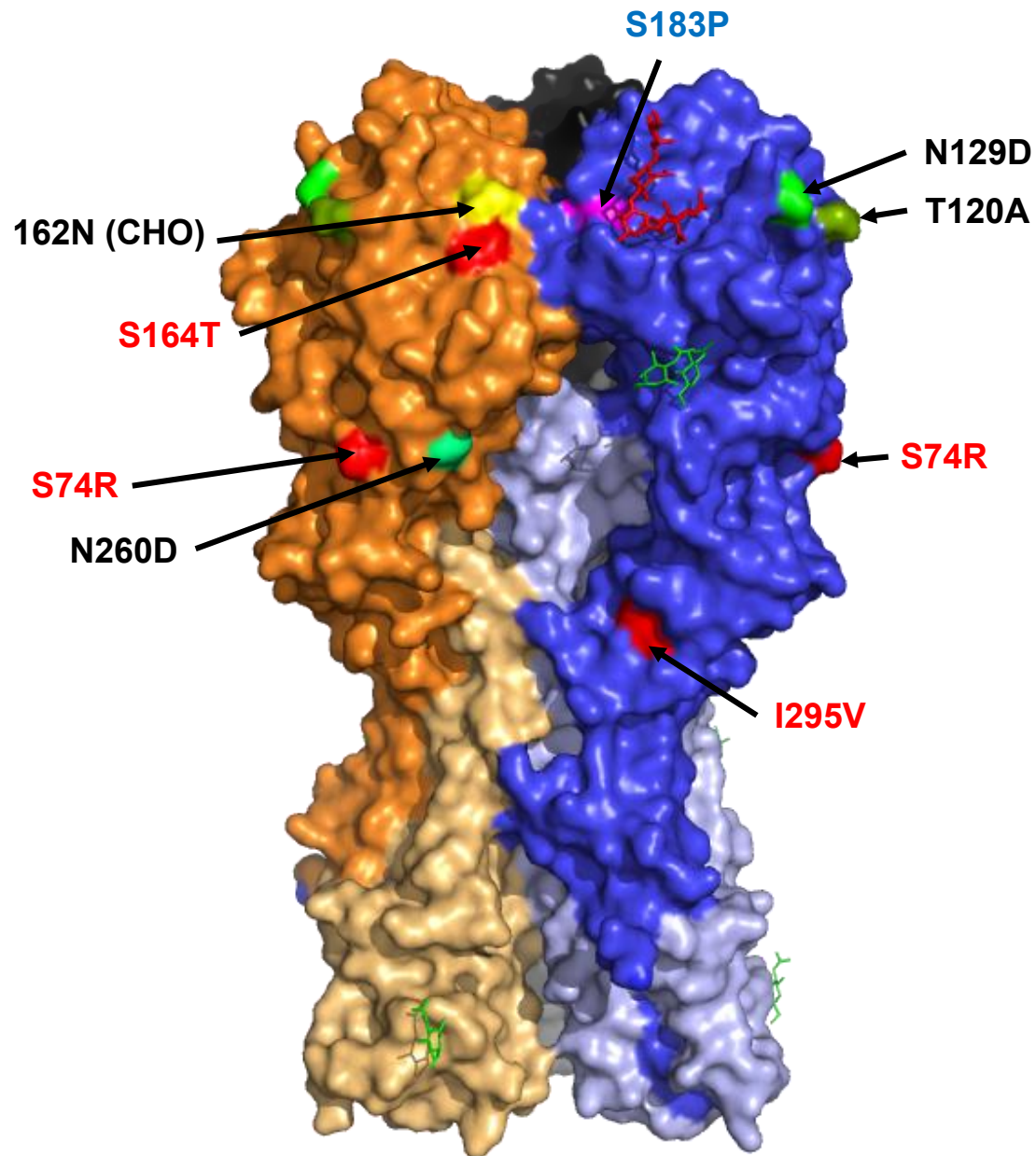


	310	320	330	340	350	360	370	380	390	400	
A/Michigan/45/2015	GKCPKYVKSTKLRLATGLRNVPSIQSRGLFGA	IAGFIEGGWTGMXDGWYGYHHQNEQSGYA	ADLKSTQNAIDKITNKVNSVIEKMNTQFTAVG	KEFNHL							6B.1
A/Slovenia/2903/2015					V						6B.1
A/Paris/1447/2017					V						6B.1A
A/Hong Kong/110/2019					V						6B.1A2
A/Georgia/749/2019					V						6B.1A2
A/Brisbane/02/2018					V						6B.1A1
A/Bretagne/1920/2019	T				V						6B.1A7
A/Slovenia/1772/2019	T	I						G			6B.1A7
A/Buea/19V-3508/2019					V						6B.1A3
A/Switzerland/2656/2017					V						6B.1A6
A/Ireland/84630/2018			X		V						6B.1A6
A/Oman/4425/2019					V						6B.1A6
A/Switzerland/3330/2017					V						6B.1A5
A/Yaounde/19V-2830/2019					V						6B.1A5
A/Vladivostok/111/2019					V						6B.1A5
A/Oman/4015/2019					V						6B.1A5
A/Switzerland/3847/2019					V						6B.1A5
A/Macedonia/1290/2019		R			V						6B.1A5
A/Moscow/195/2019					V						6B.1A5
A/Jordan/20157/2019					V			V			6B.1A5
A/Finland/103/2019					V						6B.1A5
A/Norway/3433/2018					V						6B.1A5
A/Poland/14120/2019					V						6B.1A5
A/Oman/4307/2019					V						6B.1A5
A/Oman/4419/2019					V						6B.1A5
A/Kosovo/1185/2019					V						6B.1A5
A/Ukraine/8319/2019		I			V						6B.1A5
A/Hong Kong/2658/2019					V						6B.1A5
A/Kosovo/832/2019					V				X		6B.1A5
A/Sakha/3-24V/2019					V						6B.1A5

	410	420	430	440	450	460	470	480	490	500	
A/Michigan/45/2015	EKRIENLNKKVDDGFLDIWTYNAELLVLENERFLDYHDSNVKNLYEKVRNQLKNNAKEIGNGC	FEFYHKCDNTCMESVRKNGTYDYPKYSEEAKLNREKI									
A/Slovenia/2903/2015											
A/Paris/1447/2017											
A/Hong Kong/110/2019			N								
A/Georgia/749/2019											
A/Brisbane/02/2018											
A/Bretagne/1920/2019	M									S	
A/Slovenia/1772/2019	M									S	
A/Buea/19V-3508/2019											
A/Switzerland/2656/2017											
A/Ireland/84630/2018											
A/Oman/4425/2019											
A/Switzerland/3330/2017											
A/Yaounde/19V-2830/2019											
A/Vladivostok/111/2019											
A/Oman/4015/2019											
A/Switzerland/3847/2019										S	
A/Macedonia/1290/2019											
A/Moscow/195/2019										T	
A/Jordan/20157/2019											
A/Finland/103/2019											
A/Norway/3433/2018											
A/Poland/14120/2019											
A/Oman/4307/2019											
A/Oman/4419/2019											
A/Kosovo/1185/2019											
A/Ukraine/8319/2019											
A/Hong Kong/2658/2019											
A/Kosovo/832/2019											
A/Sakha/3-24V/2019											

	510	520	530	540	
A/Michigan/45/2015	DGVKLESTRIYQILAIYSTVASSLVLVSLGAISFWMC	NGSLQCRICI			
A/Slovenia/2903/2015					
A/Paris/1447/2017					
A/Hong Kong/110/2019	A				
A/Georgia/749/2019	A				
A/Brisbane/02/2018					
A/Bretagne/1920/2019	D				
A/Slovenia/1772/2019	D				
A/Buea/19V-3508/2019					
A/Switzerland/2656/2017					
A/Ireland/84630/2018					
A/Oman/4425/2019					
A/Switzerland/3330/2017		A			
A/Yaounde/19V-2830/2019		A			
A/Vladivostok/111/2019					
A/Oman/4015/2019					
A/Switzerland/3847/2019					
A/Macedonia/1290/2019					
A/Moscow/195/2019					
A/Jordan/20157/2019					
A/Finland/103/2019					
A/Norway/3433/2018					
A/Poland/14120/2019					
A/Oman/4307/2019					
A/Oman/4419/2019	N				
A/Kosovo/1185/2019					
A/Ukraine/8319/2019		T			
A/Hong Kong/2658/2019					
A/Kosovo/832/2019		X			
A/Sakha/3-24V/2019		R			

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



Signature amino acid substitutions:
Defining subclade 6B.1A
Defining 6B.1A subgroups
 6B.1A subgroup-specific substitutions

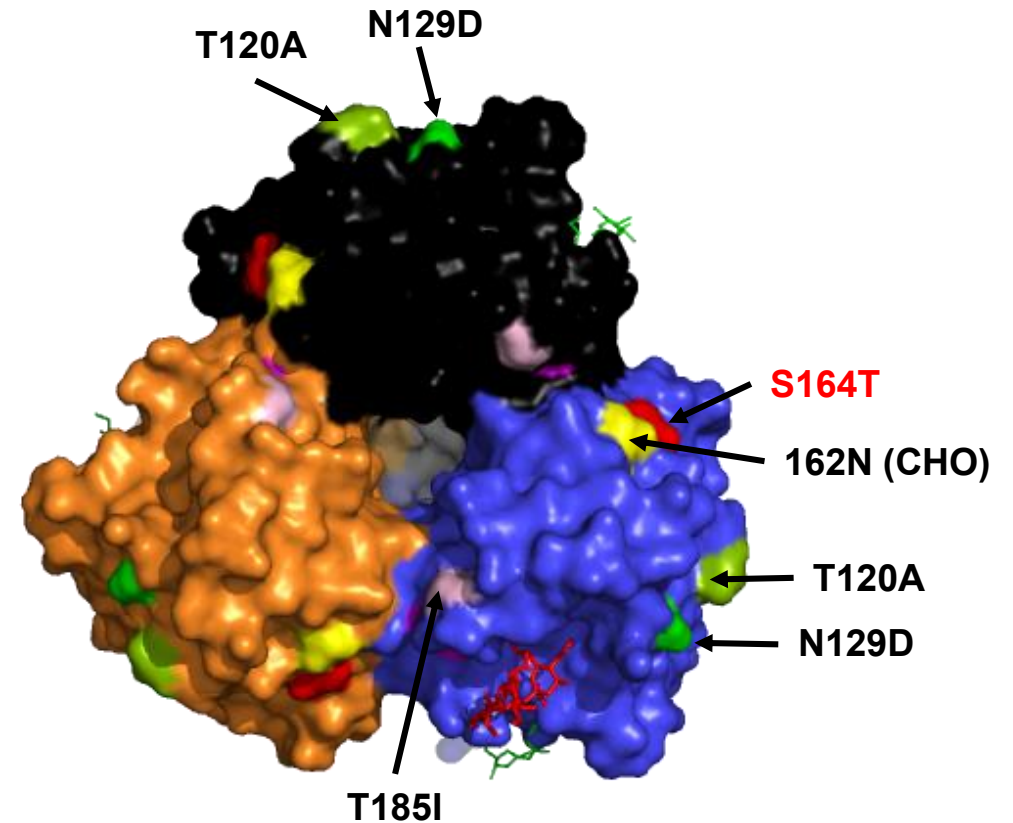


Figure 3. Amino acid substitutions in the HA of recent H1N1pdm09 viruses

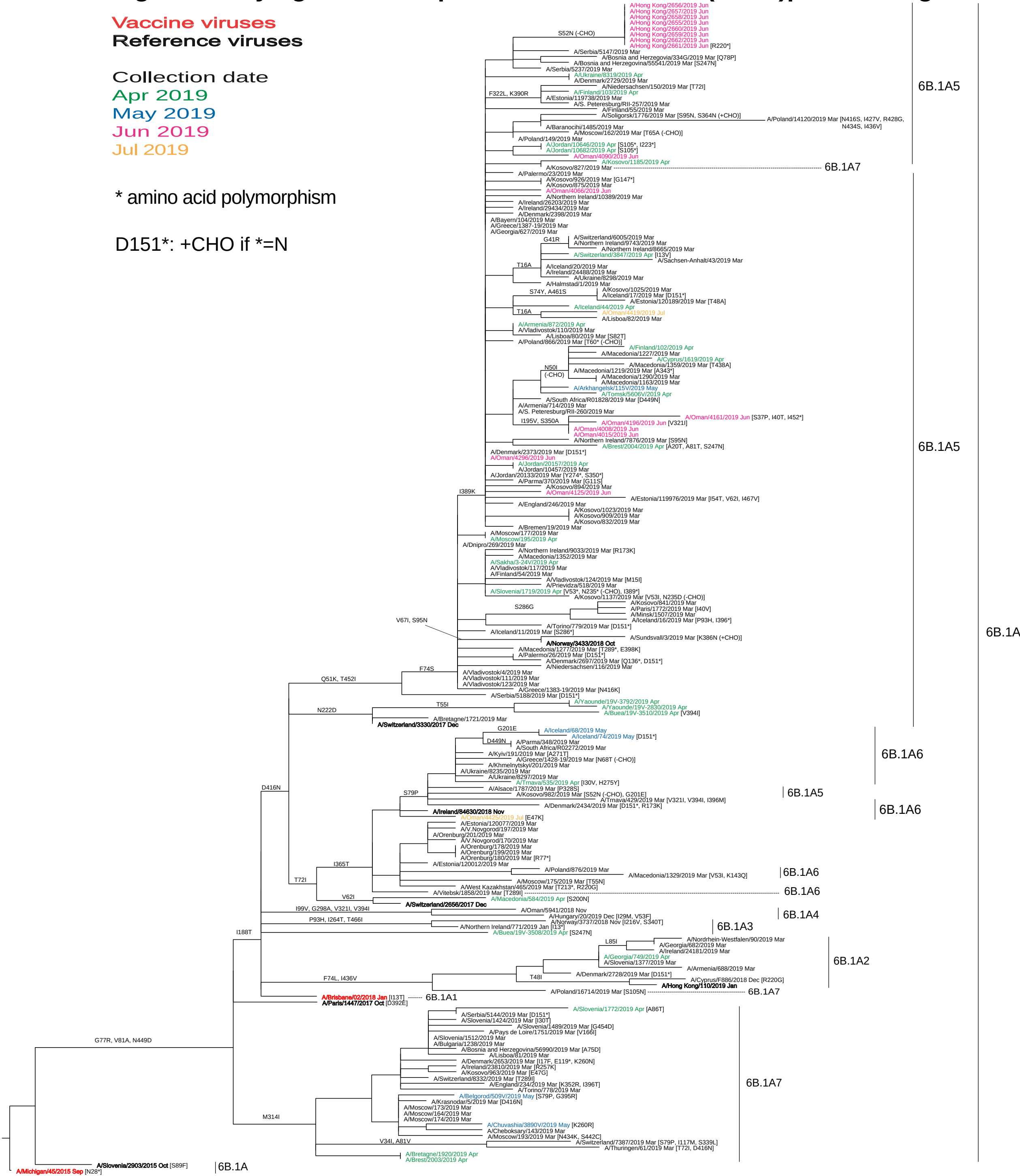
Figure 4. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

Vaccine viruses
Reference viruses

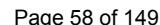
Collection date
Apr 2019
May 2019
Jun 2019
Jul 2019

* amino acid polymorphism

D151*: +CHO if *=N



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400				
A/Michigan/45/2015	RPWVSFNQ	NLEYQMG	YICSGVF	GDNPRPN	DKTGSCGP	VSSNGANG	VKGFSFK	YGNVWIG	RTKSIS	SRKGFEMI	WDPNGWTG	TDNKFSIKQ	DIVGINEWS	6B.1
A/Slovenia/2903/2015														6B.1
A/Paris/1447/2017												E		6B.1A
A/Hong Kong/110/2019														6B.1A2
A/Georgia/749/2019														6B.1A2
A/Brisbane/02/2018														6B.1A1
A/Bretagne/1920/2019			I											6B.1A7
A/Slovenia/1772/2019			I											6B.1A7
A/Buea/19V-3508/2019														6B.1A3
A/Switzerland/2656/2017														6B.1A6
A/Ireland/84630/2018								T						6B.1A6
A/Oman/4425/2019								T						6B.1A6
A/Switzerland/3330/2017														6B.1A5
A/Yaounde/19V-2830/2019														6B.1A5
A/Vladivostok/111/2019														6B.1A5
A/Oman/4015/2019						A						K		6B.1A5
A/Switzerland/3847/2019												K		6B.1A5
A/Macedonia/1290/2019												K		6B.1A5
A/Moscow/195/2019														6B.1A5
A/Jordan/20157/2019												K		6B.1A5
A/Finland/103/2019				L								KR		6B.1A5
A/Norway/3433/2018														6B.1A5
A/Poland/14120/2019												K		6B.1A5
A/Oman/4161/2019												K		6B.1A5
A/Oman/4419/2019												K		6B.1A5
A/Kosovo/1185/2019												K		6B.1A5
A/Ukraine/8319/2019												K		6B.1A5
A/Hong Kong/2658/2019												K		6B.1A5
A/Kosovo/832/2019												K		6B.1A5
A/Sakha/3-24V/2019														6B.1A5

	410	420	430	440	450	460				
A/Michigan/45/2015	GYSGSFVQ	HPELTGL	DCIRPCF	WVLIIRG	PRPEENTI	WTSGSSIS	FCGVNSD	TVGWSWPD	GAELEPFT	IDK
A/Slovenia/2903/2015										
A/Paris/1447/2017					D					
A/Hong Kong/110/2019					V		D			
A/Georgia/749/2019					V		D			
A/Brisbane/02/2018							D			
A/Bretagne/1920/2019			X				D			
A/Slovenia/1772/2019							D			
A/Buea/19V-3508/2019							D			
A/Switzerland/2656/2017			N				D			
A/Ireland/84630/2018			N				D			
A/Oman/4425/2019			N				D			
A/Switzerland/3330/2017			N				D			
A/Yaounde/19V-2830/2019			N				D			
A/Vladivostok/111/2019			N				D	I		
A/Oman/4015/2019			N				D	I		
A/Switzerland/3847/2019			N				D	I		
A/Macedonia/1290/2019			N				D	I		
A/Moscow/195/2019			N				D	I		
A/Jordan/20157/2019			N				D	I		
A/Finland/103/2019			N				D	I		
A/Norway/3433/2018			N				D	I		
A/Poland/14120/2019			S		VG	S.V	D	I		
A/Oman/4161/2019			N				D	X		
A/Oman/4419/2019			N				D	I		
A/Kosovo/1185/2019			N				D	I		
A/Ukraine/8319/2019			N				D	I		
A/Hong Kong/2658/2019			N				D	I		
A/Kosovo/832/2019			N				D	I		
A/Sakha/3-24V/2019			N				D	I		

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2019-01-31 have been received from NICs in 47 countries or administrative regions in four WHO Regions: African (7: Algeria, Cameroon, Ghana, Madagascar, Senegal, the Seychelles and South Africa) Eastern Mediterranean (3: Jordan, Morocco and Oman), Western Pacific (1: Hong Kong SAR) and European (36 countries) (**Table 1**).

Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pigs, turkeys and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses with collection dates since 2019-01-31, 53% had sufficient HA titre to be analysed by HI assays using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs (**Tables 1, 2, 6 & 7**), this being a increase over the proportion (24%) seen in February 2019. The remainder were unable to agglutinate guinea pig RBCs at all (**Tables 2 & 6**), or were unable to agglutinate RBCs in the presence of 20nM oseltamivir. Neutralisation of selected subsets of viruses, a total of 147 test viruses with collection dates after 2019-01-31 and including many for which HI assay could not be performed, were analysed by plaque reduction neutralisation assay (PRNA: **Tables 9-1 to 9-9**).

HI analyses.

A total of 201 viruses with collection dates after 2019-01-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2** and **7**. Individual HI results are shown in **Tables 8-1 to 8-19**. Test viruses in each table are sorted by date of collection and genetic clade/subclade for those viruses that have been sequenced and those below the red horizontal lines in the tables are viruses with collection dates after 2019-01-31 (**Tables 8-5** and **8-7 to 8-19**). The viruses sequenced and analysed by HI with collection dates after 2019-01-31 fell into clade 3C.3a (33%) or subclades 3C.2a1b (54%), 3C.2a2 (1.5%) and 3C.2a3 (0.5%) of clade 3C.2a.

A summary of the HI results for viruses collected since 2019-01-31 with antisera raised against the egg-propagated cultivars of A/Singapore/INFIMH-16-0019/2016 and A/Switzerland/8060/2017 (the most recently recommended vaccine viruses in clade 3C.2a) and cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a clade virus that is A/Hong Kong/4801/2014-like) and A/Kansas/14/2017 (a clade 3C.3a virus) is shown in **Table 7**. A more detailed summary of the whole panel of antisera is shown in **Table 8-20**.

The antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 recognised 77% of the test viruses collected after 2019-01-31 at titres within 4-fold of the titre of the antiserum with the homologous virus, 31% within 2-fold of the

homologous titre. Approximately 80% of viruses in clade 3C.3a were recognised by this antiserum at titres within 4-fold of the homologous titre of the antiserum. Viruses in subclade 3C.2a1b with a T135K substitution were recognised less well with 66% of viruses being recognised at titres within 4-fold of the homologous titre of the antiserum, but viruses in subclade 3C.2a1b with a T131K substitution, and retaining the glycosylation sites at residues 126 and 133 of HA1, were recognised better with 93% of these viruses being recognised at titres within 4-fold of the homologous titre.

Antiserum raised against an egg-propagated cultivar of A/Singapore/INFIMH-16-0019/2016 (with a predominance of glycine at HA1 position 225 [225G]), recommended for use in the Northern Hemisphere 2018/2019 season, recognised only 15% of the test viruses at titres within 4-fold of the homologous titre of the antiserum. This antiserum recognised only 3.9% and 22.6% of subclade 3C.2a1b viruses with the T135K substitution or the T131K substitution, respectively, at titres ≥ 160 within 4-fold of the homologous titre, and it recognised only ~10% of clade 3C.3a viruses at titres within 4-fold of the homologous titre.

Antisera raised against two subclade 3C.2a1b viruses were also used in the tests although no homologous titres were available for the antisera since the viruses were unable to agglutinate RBCs. These antisera, raised against A/La Rioja/2202/2017 (T135K) and A/Norway/3275/2018 (T131K) recognised 44%, and 41%, respectively, of the test viruses collected after 2019-01-31 at titres ≥ 160 , titres similar to the titres of these antisera with most of the reference viruses. These antisera recognised clade 3C.3a viruses markedly less well; but recognised subclade 3C.2a1b viruses at titres ≥ 160 . Within this subclade the antiserum raised against A/La Rioja/2202/2017 recognised 70% and 68% of subclade 3C.2a1b viruses with the T135K substitution or the T131K substitution, respectively, at titres ≥ 160 . The antiserum raised against A/Norway/3275/2018 recognised 51% and 100% of subclade 3C.2a1b viruses with the T135K substitution or the T131K substitution, respectively, at titres ≥ 160 .

Antisera raised against viruses in subclade 3C.2a2 (cell culture-propagated A/Bretagne/1413/2017 and egg-propagated A/Switzerland/8060/2017 [a previously recommended vaccine virus]) recognised the vast majority of test viruses poorly.

An antiserum raised against the 3C.3a reference virus A/England/538/2018 recognised 55% of test viruses at titres within 4-fold of the homologous titre of the antiserum, but recognised 100% of test viruses in clade 3C.3a at titres within 4-fold of the homologous titre and 97% within 2-fold. This antiserum recognised 3C.2a1b viruses less well recognising only 20% of viruses with the T135K substitution and 39% of those with the T131K substitution at titres within 4-fold of the homologous titre. Similarly, the antiserum raised against the cell culture-propagated cultivar of A/Kansas/14/2017 recognised 98% of clade 3C.3a viruses tested with this antiserum at titres within 4-fold of the homologous titre, and the minority (8%) of viruses in subclade 3C.2a1b with the T135K substitution but somewhat more (26%) of those with the T131K substitution. An antiserum raised against X-327, derived from egg-

propagated A/Kansas/14/2017, recognised the large majority of viruses at titres ≥ 8 -fold lower than the titre of the antiserum with the homologous virus, recognising only 24% of test viruses in clade 3C.3a within 4-fold of the homologous titre and those in subclade 3C.2a1b poorly.

Virus Neutralisation results

The results of Plaque Reduction Neutralisation Assay (PRNA) tests are shown in **Tables 9-1 to 9-9**. Summaries of the results are shown in **Table 9-10**.

To summarise, antisera raised against

- cell culture-propagated A/La Rioja/2202/2017 (3C.2a1b + T135K) recognised 76% of the test viruses (96% of the 3C.2a1b viruses with T135K, 98% of the 3C.2a1b viruses with T131K but only 6% of clade 3C.3a viruses) at titres within 4-fold of the homologous titre,
- cell culture-propagated A/Norway/3275/2018 (3C.2a1b + T131K) recognised 70% of the test viruses (81% of the 3C.2a1b viruses with T135K, 100% of the 3C.2a1b viruses with T131K but only a single (3%) clade 3C.3a virus) at titres within 4-fold of the homologous titre,
- cell culture-propagated A/Kansas/14/2017 (3C.3a) recognised 37% of the test viruses (11% of the 3C.2a1b viruses with T135K, 25% of the 3C.2a1b viruses with T131K and all but one (97%) clade 3C.3a test viruses) at titres within 4-fold of the homologous titre,
- egg-propagated A/Norway/3275/2018 (3C.2a1b + T131K) recognised only a small minority (7%) of test viruses in subclade 3C.2a1b well, with 93% being recognised at titres ≥ 8 -fold lower than the titre of the antiserum with the homologous virus, while all test viruses in clade 3C.3a were recognised poorly,
- egg-propagated X-327, derived from A/Kansas/14/2017, recognised all 147 viruses tested poorly at titres ≥ 8 -fold lower than the homologous titre of the antiserum,
- egg-propagated A/Switzerland/8060/2017 (3C.2a2) recognised only two of the 36 test viruses analysed with this antiserum at titres within 4-fold of the homologous titre – both were subclade 3C.2a2 viruses.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating A(H3N2) viruses analysed at the Francis Crick Institute and other representative viruses is shown in **Figure 6**. The HA genes fell within clade 3C.2a, primarily subclade 3C.2a1b, and in clade 3C.3a, but with small numbers falling in subclades 3C.2a2, 3C.2a3 and 3C.2a4.

Major clades/subclades

- Clade **3C.2a** viruses differ from older clade **3C.3a** viruses with substitutions **N128T** (resulting in the gain of a potential glycosylation site), **S138A**, **N144S** (resulting in the loss of a potential glycosylation site), **S159Y**, **K160T** (resulting

in the gain of a potential glycosylation site at residue **158**), and **Q311H** in **HA1** and **D160N** in **HA2**; some of the subclades share the substitution **G142R** in **HA1**.

- Subclade **3C.2a1b**, with the substitutions **E62G**, **K92R**, **N121K**, a reversion **R142G**, **N171K** and **H311Q** in **HA1** and **I77V**, **G155E** in **HA2**, has predominated. Within this subclade there are two main genetic groups defined by
 - **T128A** and **T135K** in **HA1** both resulting in the loss of glycosylation motifs (e.g. A/La Rioja/2202/2018)
 - **T131K** in **HA1** and **V200I** in **HA2** (e.g. A/Norway/3275/2018).
- Clade **3C.3a** viruses have acquired the substitutions **S91N**, **N144K** (resulting in the loss of a glycosylation site) **P193S** and **K326R** in **HA1** with many viruses sharing the substitutions **M17L**, **I149M** and **A201V** in **HA2** (e.g. A/Kansas/14/2017).

Minor subclades

- **T131K**, **R142K** and **R261Q** in **HA1**, define subclade **3C.2a2** (e.g. A/Switzerland/8060/2017).
- **N121K** and **S144K** in **HA1**, define subclade **3C.2a3** (e.g. A/Cote d'Ivoire/544/2016).
- **N31S**, **D53N**, **S144R**, **N171K**, **I192T** and **Q197H** in **HA1**, define subclade **3C.2a4** (e.g. A/Valladolid/182/2017).

Figure 7 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny. The vaccine viruses A/Singapore/INFIMH-16-0019/2016 (3C.2a1), A/Switzerland/8060/2017 (3C.2a2) and A/Kansas/14/2017 (3C.3a) are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade to which each of the HA sequences belongs is also shown.

Figure 8a shows the location on a 2005 H3 HA structure of amino acid substitutions that define recent subclades of viruses compared with the vaccine virus A/Singapore/INFIMH-16-0019/2016. **Figure 8b** shows the location of HA amino acid substitutions found in recently circulating clade 3C.3a viruses compared with A/Switzerland/9715293/2013.

Figure 9 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 10** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue

and the clade/subclade to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of recent viruses, with HA genes in genetic clusters within clade 3C.2a and its subclades, show generally similar subclade clustering. There are several subgroups defined by amino acid substitutions in NA:

- Many recently collected viruses with HA genes in subclade 3C.2a1b share the amino acid substitutions **P126L**, **K220N**, **V303I** and **N329S** (losing a potential glycosylation motif) in the NA and many also have substitutions **S315R**, **E344K** and **G346D**.
- Recent HA clade **3C.3a** viruses have NA genes evolved from clade 3C.2a viruses and have NA substitutions **K75R**, **L140I**, **V149A**, **Y155H**, **N329T** (resulting in the loss of a potential glycosylation site) and many share the additional substitutions **S315R** and **E344K**.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 351 viruses with collection dates after 2019-01-31. All but one showed normal inhibition by oseltamivir and zanamivir (**Table 14**). A/Latvia/03-0738053/2019 showed reduced inhibition (RI) by zanamivir and the NA carried amino acid polymorphism (X) and substitution at two positions (D151X and V165I).

Conclusion.

Viruses in subclade 3C.2a1b were predominant. Of the recovered viruses, 53% were able to agglutinate RBCs in the presence of oseltamivir such that they could be characterised by HI assay.

In HI assays the antiserum raised against a cell culture-propagated clade 3C.2a virus, A/Hong Kong/5738/2014 (genetically similar to the vaccine virus A/Hong Kong/4801/2014), recognised the majority of recently collected test viruses well. However, antisera raised against previous, recently recommended egg-propagated vaccine viruses recognised test viruses poorly; those raised against A/Singapore/INFIMH-16-0019/2016, A/Switzerland/8060/2017 and a high yield reassortant derived from A/Kansas/14/2017 recognised only 15%, 3% and 3% of test viruses, respectively, at titres ≤ 4 -fold reduced compared to the homologous titres of the antisera.

In PRNA, the antisera raised against egg-propagated A/Switzerland/8060/2017 recognised the vast majority (94%) of test viruses poorly while that raised against the high yield reassortant derived from egg-propagated A/Kansas/14/2017 recognised all test viruses poorly. Additionally, antiserum raised against egg-propagated A/Norway/3275/2018 (a subclade 3C.2a1b virus with T131K) also recognised the vast majority (95%) of test viruses poorly. Antisera raised against cell culture-propagated 3C.2a1b viruses recognised the test viruses from subclade 3C.2a1b generally well but viruses from clade 3C.3a much less so.

Table 6. A(H3N2) viruses recovered from specimens collected after 2019-01-31, as assessed by NA activity (MUNANA-based assay)

Situation at the time of the September 2019 VCM (viruses with collection dates Feb-Aug 2019)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a1b	Number in clade 3C.2a2	Number in clade 3C.2a3	Number in clade 3C.2a4
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir								
+	+	+	Yes	90	76 (84%)	56	19	1	0	0
-	+	+	Yes	102	91 (89%)	10	78	2	1	0
-	-	+	Yes	7	7 (100%)	0	7	0	0	0
+	-	+	Yes	0	0	0	0	0	0	0
			Totals	199	174 (87%)	66	104	3	1	0
+	+	-	No	9	8 (89%)	0	6	1	0	1
+	-	-	No	0	0	0	0	0	0	0
-	+	-	No	26	25 (96%)	0	25	0	0	0
-	-	-	No	144	123 (85%)	0	109	11	3	0
			Totals	179	156 (87%)	0	140	12	3	1
			Totals	378	330	66	244	15	4	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2019-09-13: a number still in process

Situation at the time of the February 2019 VCM (viruses with collection dates Sep 2018-Jan 2019)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a1b	Number in clade 3C.2a2	Number in clade 3C.2a3	Number in clade 3C.2a4
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir								
+	+	+	Yes	35	29 (83%)	9	19	1	0	0
-	+	+	Yes	17	3 (18%)	0	1	1	1	0
-	-	+	Yes	1	1 (100%)	0	1	0	0	0
+	-	+	Yes	1	1 (100%)	0	1	0	0	0
			Totals	54	34 (63%)	9	22	2	1	0
+	+	-	No	8	8 (100%)	0	6	1	1	0
+	-	-	No	0	0	0	0	0	0	0
-	+	-	No	24	7 (29%)	0	6	0	1	0
-	-	-	No	143	85 (59%)	0	73	8	3	1
			Totals	175	100 (57%)	0	85	9	5	1
			Totals	229	134	9	107	11	6	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2019-02-12: a number still in process

Table 7a. Summary of influenza A(H3N2) viruses collected after 2019-01-31, analysed by HI

MONTH	H3N2														
	Country	Number received ¹	Number propagated ²	3C.2a: Cell-propagated A/Hong Kong/5738/2014			3C.2a1: Egg-propagated A/Singapore/INFIMH-16-0019/2016			3C.2a2: Egg-propagated A/Switzerland/8060/2017			3C.3a: Cell-propagated A/Kansas/14/2017		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2019															
FEBRUARY															
Algeria	3	0													
Armenia	10	1		1				1			1				1
Austria	1	1	1					1			1				1
Belarus	1	1		1				1			1				1
Bosnia	11	0													
Bulgaria	19	12	2	6	4			12		2	10		2		10
Cyprus	1	0													
Denmark	3	0													
Estonia	2	1	1					1			1				1
Georgia	1	0													
Germany	17	9	8	1			2	7			9				1
Greece	5	1	1					1			1	1			
Iran	2	2	1	1				2			2				2
Iraq	2	0													
Ireland	3	2			2			2			2	2			
Italy	8	3	1	2			1	2			3	2			1
Jordan	7	0													
Kazakhstan	2	2	1	1				2			2				2
Kosovo	1	0													
Latvia	6	5	1	4				5			5	1			4
Macedonia	4	0													
Malta	3	0													
Moldova	2	0													
Morocco	3	0													
Poland	5	0													
Portugal	7	6		6				6			6	6			
Russia	6	5		4	1			5			5				5
Senegal	2	0													
Serbia	4	0													
Slovakia	3	3		2	1			3			3				3
Slovenia	3	1			1			1			1				1
Sweden	2	2	1	1				2			2				2
Switzerland	6	0													
Ukraine	9	8	2		6			8			8		1		7
United Kingdom	10	10	5	5			1	9			10	7	2		1
MARCH															
Armenia	2	0													
Belarus	7	4			4			4			4				4
Bosnia	5	0													
Cameroon	1	1	1				1			1		1			
Cyprus	1	0													
Czech Republic	2	1			1			1			1				1
Denmark	9	1		1				1			1				1
Estonia	2	0													
Finland	5	2		2				2			2	1			1
France	4	3	1	2				3			3	2			1
Georgia	2	1			1			1			1				1
Germany	16	12	5	6	1		1	11			12				
Greece	6	1		1				1			1	1			
Iceland	3	2	1		1		1	1			2		1		1
Iran	3	3	1		2		1	2			3				3
Ireland	11	6		2	4		3	3			6	4	1		1
Italy	9	8	2	5	1			8			8	5			3
Jordan	2	1	1				1			1		1			
Kazakhstan	2	2	2					2		1	2		2		
Kosovo	1	1		1				1			1	1			
Latvia	1	1		1				1			1	1			
Macedonia	7	3		3			1	2			3	2			1
Norway	1	0													
Portugal	2	1			1			1			1				1
Russia	14	2		1	1			2			2				2
Senegal	4	3		2	1		2	1			3	2	1		
Serbia	4	0													
Seychelles	2	0													
Slovakia	2	2			2			2			2				2
Slovenia	3	0													
Sweden	2	0													
Switzerland	6	1			1			1			1				1
Ukraine	1	1			1			1			1				1
United Kingdom	5	0													

Table 7b. Summary of influenza A(H3N2) viruses collected after 2019-01-31, analysed by HI

MONTH	H3N2														
	Country	Number received ¹	Number propagated ²	3C.2a: Cell-propagated A/Hong Kong/5738/2014			3C.2a1: Egg-propagated A/Singapore/INFIMH-16-0019/2016			3C.2a2: Egg-propagated A/Switzerland/8060/2017			3C.3a: Cell-propagated A/Kansas/14/2017		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2019															
APRIL															
Armenia	2	0													
Bosnia	3	2		1	1				2			2			2
Cameroon	1	0													
Czech Republi	2	1	1					1				1	1		
Finland	4	3	2	1				1	2			3	2	1	
France	12	7	3	2	2				7			7	6		1
Georgia	3	0													
Ghana	2	2		1	1			1	1			2			2
Iceland	4	1		1				1				1	1		
Iran	1	1		1					1			1			1
Ireland	2	1		1				1				1			1
Jordan	2	1			1			1				1	1		
Kosovo	2	1	1						1			1	1		
Macedonia	4	1		1					1			1	1		
Madagascar	4	0													
Moldova	1	1		1					1			1			1
Norway	1	0													
Russia	4	3	3						3			3	1	1	1
Senegal	2	1		1				1				1		1	
Seychelles	1	0													
South Africa	2	2	2						2			2			2
United Kingdo	15	0													
MAY															
Armenia	1	1	1					1				1	1		
Finland	1	0													
Georgia	1	0													
Iceland	5	1		1				1				1			1
Madagascar	2	0													
Norway	3	2		1	1				2			2	2		
Russia	1	1	1						1		1		1		
Senegal	1	1		1					1			1	1		
Seychelles	3	0													
South Africa	10	8	4	3	1			2	6			8	4		4
JUNE															
Ghana	2	1	1				1					1	1		
Hong Kong	12	8	3	5				1	7			8			8
Iceland	2	1	1				1			1			1		
Norway	2	1		1					1			1			1
Oman	1	1		1					1			1			1
Seychelles	1	0													
South Africa	6	3		3				1	2			3	3		
JULY															
Norway	3	1		1				1				1		1	
Oman	2	2		1	1				2			2	1		1
AUGUST															
Norway	4	1			1				1			1			1
	458	199	62	91	46	3	27	169	1	5	193	69	14	96	
47 Countries			31.2%	45.7%	23.1%	1.5%	13.6%	84.9%	0.5%	2.5%	97.0%	38.5%	7.8%	53.6%	

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

3. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/5738/2014 (3C.2a)

4. Compared to homologous titres for ferret antisera raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (3C.2a1)

5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/8060/2017 (3C.2a2)

6. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Kansas/14/2017 (3C.3a)

Numbers highlighted in blue show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-02-15 + 2019-02-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock 6/14 SIAT F14/14 ¹ 3C.3a	A/HK 5738/14 MDCK F30/14 ¹ 3C.2a	A/Bretagne 1413/17 SIAT F01/18 ¹ 3C.2a2	A/Singapore 0019/16 Egg 10 ⁻⁴ F46/17 ¹ 3C.2a1	A/La Rioja 2202/18 SIAT F26/18 ¹ 3C.2a1b	A/Switz 8060/17 Egg F27/18 ¹ 3C.2a2	A/Eng 538/18 SIAT F31/18 ¹ 3C.3a	A/Neth 10260/18 Egg F02/19 ¹ 3C.2a1b	A/Neth 10260/18 SIAT F07/19 ¹ 3C.2a1b	A/Norway 3275/18 SIAT F03/19 ¹ 3C.2a1b
				Passage history	Ferret number	Genetic group							
				REFERENCE VIRUSES									
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	160	40	80	160	80	160	160	<	40	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	80	160	160	80	160	160	<	80	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	80	<	640	160	80	640	80	<	80	160
A/Singapore/IN/16-0019/2016	3C.2a1	2016-04-14	E5/E2	<	<	40	640	80	80	40	<	40	<
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E7/E1	<	40	1280	320	80	1280	80	40	80	40
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	40	<	40	80	<	40	640	<	80	<
A/Netherlands/10260/2018	3C.2a1b	2018-02-15	E5/E1	40	<	80	80	320	80	80	1280	320	160
TEST VIRUSES													
A/Abu Dhabi/240/2018	3C.2a1b	2018-01-01	E6/E1	<	<	40	80	160	40	40	640	ND	80
A/Parma/177/2018	3C.2a1b	2018-11-20	SIAT2/SIAT2	160	80	40	160	320	160	80	<	320	160
A/Parma/180/2018	3C.2a1b	2018-11-21	SIAT1/SIAT2	80	40	<	80	80	40	40	<	160	80
A/Parma/179/2018	3C.2a1b	2018-11-22	SIAT1/SIAT1	160	40	<	80	160	40	40	<	160	80
A/Parma/174/2018	3C.2a1b	2018-11-22	SIAT1/SIAT1	160	40	<	40	160	40	40	<	160	80
A/England/630/2018	3C.2a1b	2018-11-23	SIAT1/SIAT1	80	40	160	80	80	160	40	<	80	160
A/Hong Kong/2946/2018	3C.2a1b	2018-11-27	MDCK1/SIAT1	40	<	40	80	80	80	160	<	ND	160
A/England/646/2018	3C.2a1b	2018-11-28	SIAT2/SIAT1	<	<	<	<	80	<	<	<	80	80
A/Constanta/239165/2018	3C.2a1b	2018-12-07	SIAT1/SIAT2	<	<	<	<	80	<	<	<	40	40
A/England/754/2018	3C.2a1b	2018-12-13	MDCK1/SIAT1	<	<	<	<	<	40	80	<	<	<
A/Centre/013/2019	3C.2a1b	2019-01-02	SIAT1	160	<	<	80	160	<	<	<	160	40
A/Pays de Loire/040/2019	3C.2a1b	2019-01-04	SIAT1	80	<	80	40	160	80	40	<	80	160
A/Nord Pas de Calais/054/2019	3C.2a1b	2019-01-07	SIAT1	40	<	<	80	160	40	40	<	80	80
A/England/3/2019	3C.2a1b	2019-01-07	MDCK1/SIAT1	40	<	40	80	80	160	160	<	<	ND
A/Paris/105/2019	3C.2a1b	2019-01-08	SIAT1	<	<	<	<	80	<	<	<	80	40
A/Iasi/239836/2019	3C.2a1b	2019-01-09	SIAT1	80	40	80	80	160	80	40	<	80	160
A/Palermo/327/2018	3C.2a2	2018-11-28	SIAT3/SIAT1	160	40	640	160	160	640	160	<	80	160
A/Ghana/758/2018	3C.2a3	2018-10-29	SIAT1	80	<	<	40	80	80	40	<	80	<
A/Nord Pas de Calais/2726/2018	3C.2a3	2018-12-26	SIAT1	80	<	<	40	80	80	40	<	40	40
A/Bourgogne/074/2019	3C.3a	2019-01-02	SIAT1	40	<	40	<	<	<	320	<	40	<
A/Haute Normandie/085/2019	3C.3a	2019-01-07	SIAT1	<	<	<	<	<	<	320	<	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-03-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Neth 10260/18	A/Neth 10260/18
				SIAT F14/14 ¹	MDCK St J F60/17 ¹	SIAT F01/18 ¹	Egg 10 ⁻⁴ F46/17 ¹	SIAT F26/18 ¹	Egg F27/18 ¹	SIAT F31/18 ¹	Egg F02/19 ¹	SIAT F07/19 ¹
	Passage history			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.2a1b
REFERENCE VIRUSES												
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	160	40	80	160	80	160	160	<	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	80	80	160	80	160	160	<	40
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	80	640	160	80	640	160	<	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	<	40	40	640	160	160	80	<	40
A/Switzerland/8060/2017	clone 57 3C.2a2	2017-12-12	E7/E1	40	80	1280	640	80	1280	80	40	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	80	<	40	80	40	80	640	<	80
A/Netherlands/10260/2018	3C.2a1b	2018-02-15	E5/E1	<	40	80	80	320	80	80	1280	320
TEST VIRUSES												
A/Israel/8135/2018		2018-10-19	C1/SIAT1	<	<	<	40	<	<	320	<	<
A/Israel/178/2018	3C.2a1b	2018-11-13	C1/SIAT1	160	40	80	160	160	160	80	<	160
A/Ukraine/8122/2018	3C.2a1b	2018-12-18	MDCK1/SIAT1	<	<	<	<	80	<	<	<	80
A/Nordrhein-Westfalen/41/2019	3C.2a1b	2019-01-04	C1/SIAT1	80	<	<	80	160	80	80	<	160
A/Israel/8701/2018	3C.2a3	2018-11-06	C1/SIAT1	40	<	<	80	80	80	40	<	40
A/Dakar/29/2018	3C.3a	2018-06-12	P1/SIAT1	40	<	80	80	<	80	640	<	80
A/Dakar/28/2018	3C.3a	2018-06-12	P0/SIAT1	40	<	40	80	<	80	320	<	40
A/Dakar/25/2018	3C.3a	2018-06-12	P0/SIAT1	40	<	40	80	<	40	640	<	80
A/Dakar/10/2018	3C.3a	2018-08-03	P0/SIAT1	80	40	80	160	80	160	640	<	80
A/Israel/144/2018	3C.3a	2018-11-04	C1/SIAT1	40	<	40	80	<	80	640	<	80
A/Israel/133/2018	3C.3a	2018-11-04	C1/SIAT1	40	<	40	80	<	80	640	<	80
A/Israel/177/2018	3C.3a	2018-11-13	C1/SIAT1	40	<	40	80	<	80	640	<	40
A/Israel/8989/2018	3C.3a	2018-11-18	C1/SIAT1	40	<	40	40	<	40	320	<	40
A/Israel/10481/2018	3C.3a	2018-11-23	C1/SIAT1	<	<	<	40	<	40	320	<	40
A/Israel/10480/2018	3C.3a	2018-11-23	C1/SIAT1	<	<	<	<	<	40	320	<	40
A/Israel/590/2018	3C.3a	2018-12-18	C1/SIAT1	40	<	40	40	<	40	320	<	40
A/Israel/10422/2018	3C.3a	2018-12-20	C1/SIAT1	40	<	<	40	<	40	320	<	40
A/Nordrhein-Westfalen/11/2019	3C.3a	2019-01-21	C1/SIAT1	80	<	40	80	<	80	640	<	80
A/Nordrhein-Westfalen/22/2019	3C.3a	2019-01-28	C1/SIAT1	80	<	40	80	<	80	640	<	80
A/Berlin/23/2019	3C.3a	2019-01-31	C1/SIAT1	80	<	40	80	<	80	640	<	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-03-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Neth 10260/18	A/Neth 10260/18	A/Norway 3275/18
				SIAT	MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT	SIAT
	Passage history			F14/14 ¹	St J F60/17 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F02/19 ¹	F07/19 ¹	F03/19 ¹
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.2a1b	3C.2a1b
	Genetic group												
REFERENCE VIRUSES													
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	160	160	80	320	80	320	320	<	80	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	160	160	320	160	320	320	40	80	320
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	160	640	320	80	640	320	40	80	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	40	160	80	640	160	160	80	40	40	40
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E7/E1	40	160	1280	640	160	1280	160	40	80	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	40	40	40	40	<	40	640	10	80	<
A/Netherlands/10260/2018	3C.2a1b	2018-02-15	E5/E1	40	160	160	160	640	160	80	1280	640	160
TEST VIRUSES													
A/Oman/5362/2018	3C.2a1b	2018-10-24	SIAT1	80	40	<	40	160	80	80	<	160	80
A/Guangxi-Babu/1913/2018	3C.2a1b	2018-10-29	MDCK2/SIAT1/SIAT1	<	<	<	<	40	<	40	<	40	40
A/Oman/5838/2018	3C.2a1b	2018-11-07	SIAT2	160	80	40	80	160	80	80	<	160	160
A/Oman/5804/2018	3C.2a1b	2018-11-07	SIAT2	160	160	40	80	160	80	80	<	160	160
A/Oman/6018/2018	3C.2a1b	2018-11-13	SIAT1	80	40	<	40	160	40	40	<	80	80
A/Oman/6106/2018	3C.2a1b	2018-11-15	SIAT1	80	80	<	40	160	40	80	<	160	80
A/Kharkiv/446/2018	3C.2a1b	2018-11-23	SIAT2/SIAT1	40	40	<	40	160	40	40	<	320	160
A/Kharkiv/444/2018	3C.2a1b	2018-11-23	SIAT2/SIAT1	40	40	<	<	160	<	40	<	160	160
A/Kyiv/463/2018	3C.2a1b	2018-11-30	SIAT1/SIAT1	<	<	<	<	80	<	80	<	80	80
A/Khmelnyskyi/544/2018	3C.2a1b	2018-12-06	SIAT2/SIAT1	80	40	40	80	160	80	80	<	160	160
A/Khmelnyskyi/545/2018	3C.2a1b	2018-12-08	SIAT2/SIAT1	160	80	40	80	160	80	80	<	160	160
A/Kyiv/487/2018	3C.2a1b	2018-12-10	SIAT1/SIAT1	<	40	<	<	80	<	40	<	160	160
A/Kyiv/491/2018	3C.2a1b	2018-12-13	SIAT2/SIAT1	<	<	<	<	40	<	40	<	80	40
A/Kyiv/493/2018	3C.2a1b	2018-12-14	SIAT1/SIAT1	40	<	<	<	80	<	40	<	160	160
A/Khmelnyskyi/547/2018	3C.2a1b	2018-12-16	SIAT2/SIAT1	80	<	<	40	80	40	40	<	80	80
A/Kyiv/510/2018	3C.2a1b	2018-12-17	SIAT2/SIAT1	160	80	<	40	80	40	40	<	160	80
A/Luxembourg/341/2019	3C.2a1b	2019-01-03	SIAT1	80	80	<	80	160	40	40	<	80	160
A/Luxembourg/1177/2019	3C.2a1b	2019-01-06	SIAT1	<	40	<	<	80	<	80	<	160	160
A/Luxembourg/2293/2019	3C.2a1b	2019-01-14	SIAT1	160	80	40	80	160	80	80	<	160	320
A/Macedonia/723/2019	3C.2a1b	2019-01-28	SIAT1	<	80	80	<	80	80	80	<	80	160
A/Beijing-Chaoyang/12841/2018	3C.2a2	2018-11-26	MDCK1/SIAT1/SIAT1	160	160	320	160	80	320	160	<	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-03-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock 6/14 SIAT	A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Eqq 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Neth 10260/18 Egg	A/Neth 10260/18 SIAT	A/Norway 3275/18 SIAT
				F14/14 ⁻¹ 3C.3a	St J F60/17 ⁻¹ 3C.2a	F01/18 ⁻¹ 3C.2a2	F46/17 ⁻¹ 3C.2a1	F26/18 ⁻¹ 3C.2a1b	F27/18 ⁻¹ 3C.2a2	F31/18 ⁻¹ 3C.3a	F02/19 ⁻¹ 3C.2a1b	F07/19 ⁻¹ 3C.2a1b	F03/19 ⁻¹ 3C.2a1b
REFERENCE VIRUSES													
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	160	160	80	160	80	160	160	<	40	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	160	160	160	80	160	160	40	80	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	320	1280	320	160	1280	320	40	80	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	<	80	40	640	160	160	80	40	40	40
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E7/E1	<	160	640	320	80	640	80	40	80	40
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	40	<	40	40	40	40	320	<	40	<
A/Netherlands/10260/2018	3C.2a1b	2018-02-15	E5/E1	<	160	80	80	320	80	80	1280	320	160
TEST VIRUSES													
A/Saint-Etienne/1998/2018		2018-10-15	MDCK3/SIAT1	40	40	40	40	160	40	40	40	80	80
A/Lyon/CHU/R19.02.59/2019		2019-01-02	MDCK2/SIAT1	160	80	80	80	160	160	80	<	80	160
A/Greece/1430/2018	3C.2a1b		SIAT1/SIAT1	<	40	80	<	<	40	<	<	40	160
A/Mauritius/I-974/2018	3C.2a1b	2018-10-26	MDCK2/SIAT1	80	40	<	40	80	40	40	40	80	40
A/Lyon/CHU/R18.116.67/2018	3C.2a1b	2018-11-26	MDCK4/SIAT1	40	40	<	40	160	40	40	40	160	160
A/Mauritius/I-1058/2018	3C.2a1b	2018-11-30	MDCK 1/SIAT1	160	80	40	80	160	40	80	<	160	160
A/Mauritius/I-1072/2018	3C.2a1b	2018-12-07	MDCK2/SIAT1	160	160	40	80	160	40	80	<	160	160
A/Lyon/CHU/R18.128.2/2018	3C.2a1b	2018-12-17	MDCK3/SIAT1	80	80	80	80	80	160	40	<	80	320
A/Mauritius/I-1122/2018	3C.2a1b	2018-12-26	MDCK2/SIAT1	40	40	<	40	80	<	40	<	80	80
A/Lyon/EHPAD/108/2019	3C.2a1b	2019-01-09	MDCK2/SIAT1	40	40	80	40	80	80	<	<	80	160
A/Mauritius/I-1020/2018	3C.2a2	2018-11-15	MDCK 1/SIAT1	160	160	320	160	80	640	160	<	40	320
A/Lyon/2296/2018	3C.2a3	2018-12-20	MDCK3/SIAT1	160	160	80	160	160	160	160	<	80	80
A/EHPAD/Montpellier/2320/2018	3C.2a3	2018-12-21	MDCK2/SIAT1	160	160	80	160	160	160	160	<	160	80
A/Lyon/2335/2018	3C.3a	2018-12-27	MDCK2/SIAT1	40	40	40	40	<	40	320	<	40	<
A/Lyon/CHU/R18.133.93/2018	3C.3a	2018-12-28	MDCK2/SIAT1	80	40	40	80	<	40	320	40	40	<
A/Lyon/CHU/R19.03.77/2019	3C.3a	2019-01-04	MDCK2/SIAT1	40	40	40	40	<	40	320	<	40	<
A/La Rochelle/85/2019	3C.3a	2019-01-05	MDCK2/SIAT1	40	40	40	40	<	80	320	<	40	<
A/Lyon/95/2019	3C.3a	2019-01-09	MDCK2/SIAT1	40	40	40	40	<	80	320	<	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-03-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock	A/HK	A/Bretagne	A/Singapore	A/La Rioja	A/Switz	A/Eng	A/Norway	NEW	NEW
				6/14	5738/14	1413/17	0019/16	2202/18	8060/17	538/18	3275/18	A/Kansas	A/Kansas
	Passage history			SIAT	MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	14/17	14/17
	Ferret number			F14/14 ¹	St J F60/17 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F03/19 ¹	F11/19 ¹	F12/19 ¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	320	160	160	160	160	320	160	160	160	160
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	160	160	160	80	320	160	320	160	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	160	640	160	80	640	160	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	160	80	640	320	160	80	80	40	40
A/Switzerland/8060/2017	clone 57 3C.2a2	2017-12-12	E7/E1	80	160	1280	640	160	1280	80	80	40	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	80	80	80	80	40	80	640	40	320	320
A/Kansas/14/2017	3C.3a	2017-12-14	E7/E2	40	<	40	40	<	<	320	<	1280	1280
TEST VIRUSES													
A/Belgium/G0023/2019	3C.3a	2019-01-03	SIAT1	160	80	80	80	40	80	640	40	320	320
A/Belgium/S0275/2019	3C.3a	2019-01-15	SIAT1	160	80	80	80	40	80	640	40	320	320
A/Niedersachsen/81/2019	3C.3a	2019-01-25	C1/MDCK1	160	80	80	80	40	80	640	40	320	320
A/Nordrhein-Westfalen/53/2019	3C.3a	2019-02-11	C1/MDCK1	160	80	80	80	40	80	640	40	160	160
A/Bayern/53/2019	3C.3a	2019-02-11	C1/MDCK1	160	80	80	80	40	80	640	40	320	320
A/Berlin/28/2019	3C.3a	2019-02-11	C1/MDCK1	160	80	80	80	40	80	640	40	320	320
A/Nordrhein-Westfalen/60/2019	3C.3a	2019-02-18	C1/MDCK1	160	80	80	80	40	80	640	40	320	320
A/Bremen/12/2019	3C.3a	2019-02-18	C1/MDCK1	160	80	80	160	80	80	640	80	320	320
A/Hessen/34/2019	3C.3a	2019-02-18	C1/MDCK1	160	80	80	160	80	160	640	80	320	320
A/Berlin/37/2019	3C.3a	2019-02-21	C1/MDCK1	80	80	80	80	40	80	640	40	320	320
A/Baden-Wuerttemberg/87/2019	3C.3a	2019-02-25	C1/MDCK1	160	80	80	80	40	80	640	40	320	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine NH 2019-20

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-03-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock	A/HK	A/Bretagne	A/Singapore	A/La Rioja	A/Switz	A/Eng	A/Norway	A/Kansas	
				6/14	5738/14	1413/17	0019/16	2202/18	8060/17	538/18	3275/18	14/17	
	Passage history			SIAT	MDCK	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	Egg
	Ferret number			F14/14 ⁻¹	St J F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F11/19 ⁻¹	
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	
REFERENCE VIRUSES													
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	320	160	80	160	80	160	160	80	160	
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	160	160	160	160	160	320	160	160	160	
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	160	160	640	320	80	640	160	160	160	
A/Singapore/INFIMH-16-0019/2016	clone 57	2016-04-14	E5/E2	40	80	40	640	320	160	80	40	40	
A/Switzerland/8060/2017		2017-12-12	E7/E1	40	160	1280	640	160	1280	80	80	80	
A/England/538/2018		2018-02-26	MDCK1/SIAT3	80	40	40	80	40	80	640	<	320	
A/Kansas/14/2017		2017-12-14	E7/E2	<	<	<	80	<	<	320	<	1280	
TEST VIRUSES													
A/Valladolid/5/2019		2019-01-04	SIAT1/SIAT1	40	40	40	80	40	80	640	<	320	
A/Valladolid/27/2019		2019-01-14	SIAT1/SIAT1	80	40	40	80	40	40	640	<	160	
A/Soria/11/2019	3C.2a1b	2019-01-09	SIAT1/SIAT1	80	40	40	80	320	80	40	160	80	
A/Rabat/352/2019	3C.2a3	2019-01-07	MDCK1/SIAT1	160	160	80	80	160	160	320	80	160	
A/Valladolid/560/2018	3C.3a	2018-12-27	SIAT1/SIAT1	40	40	40	80	40	80	640	<	160	
A/Valladolid/2/2019	3C.3a	2019-01-02	SIAT1/SIAT1	40	40	40	80	40	80	640	<	320	
A/Avila/3/2019	3C.3a	2019-01-03	SIAT1/SIAT1	40	40	40	80	40	80	640	<	320	
A/Valladolid/9/2019	3C.3a	2019-01-08	SIAT1/SIAT1	80	40	80	80	40	80	640	40	320	
A/Valladolid/13/2019	3C.3a	2019-01-09	SIAT1/SIAT1	40	<	40	80	40	40	640	<	160	
A/Avila/15/2019	3C.3a	2019-01-10	SIAT1/SIAT1	80	40	40	80	40	80	640	<	160	
A/Palencia/20/2019	3C.3a	2019-01-12	SIAT1/SIAT1	80	40	40	80	40	80	640	40	160	
A/Valladolid/18/2019	3C.3a	2019-01-12	SIAT1/SIAT1	80	40	40	80	80	80	640	<	320	
A/Valladolid/17/2019	3C.3a	2019-01-12	SIAT1/SIAT1	40	<	40	80	40	40	320	<	160	
A/Valladolid/26/2019	3C.3a	2019-01-14	SIAT1/SIAT1	80	80	40	80	80	80	640	<	320	
A/Valladolid/25/2019	3C.3a	2019-01-14	SIAT1/SIAT1	80	80	40	80	80	80	640	<	320	
A/Valladolid/24/2019	3C.3a	2019-01-14	SIAT1/SIAT1	40	<	40	80	<	40	320	<	160	
A/Tetouan/671/2019	3C.3a	2019-01-24	MDCK1/SIAT1	160	160	80	80	80	160	640	40	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ⁻¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-04-12

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	A/Kansas 14/17 Egg	A/Kansas 14/17 SIAT
				St Jude's F60/17 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F03/19 ¹	F11/19 ¹	F2019-083 ¹ CDC
	Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
	Ferret number											
	Genetic group											
REFERENCE VIRUSES												
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	160	160	320	160	160	160	160	80	80
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	160	640	160	80	640	160	160	160	80
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	160	40	640	320	160	80	80	<	<
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/18	320	1280	640	160	1280	80	80	40	40
A/England/538/2018		2018-02-26	MDCK1/SIAT3	40	<	<	<	<	320	<	160	80
A/Kansas/14/2017		2017-12-14	E7/E2	<	<	40	<	<	160	<	640	80
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	40	40	40	<	40	320	<	160	80
TEST VIRUSES												
A/Cameroon/9199/2018		2018-10-09	SIAT1	160	40	80	160	160	160	80	80	80
A/Cameroon/8607/2018		2018-10-11	SIAT1	160	40	80	160	160	80	80	80	80
A/Cameroon/8600/2018		2018-10-17	SIAT1	160	40	80	160	160	160	80	80	80
A/Cameroon/9272/2018		2018-11-02	SIAT1	80	40	40	80	80	40	40	40	40
A/Cameroon/9207/2018		2018-11-02	SIAT1	80	40	40	80	80	80	80	40	40
A/Cameroon/9264/2018		2018-11-12	SIAT1	160	40	40	80	80	80	40	40	80
A/Cameroon/9678/2018		2018-11-19	SIAT1	160	40	40	80	80	80	80	40	80
A/Beijing/2019-15554/2018		2018-12-25	MDCK2/SIAT1	80	40	40	40	40	320	40	160	80
A/Beijing/2019-15554/2018		2018-12-25	E2/E1	640	640	320	640	640	320	1280	640	320
A/Extremadura/283/2019		2019-01-06	SIAT1	80	40	40	<	40	320	<	160	160
A/Castilla La Mancha/242/2019		2019-01-17	SIAT1	80	40	40	40	40	320	<	160	160
A/Castilla La Mancha/337/2019		2019-01-21	SIAT1	80	40	40	<	40	320	<	160	80
A/Malta/53539/2019		2019-01-23	SIAT1	40	<	40	<	40	320	<	160	80
A/Greece/357/2019	T135K	2019-02-04	SIAT1	80	<	<	160	40	<	80	40	40
A/Latvia/02-013062/2019	T135K	2019-02-04	MDCK2/SIAT1	40	<	40	160	<	<	80	<	<
A/Latvia/02-053853/2019	T135K	2019-02-18	MDCK2/SIAT2	80	40	80	320	80	40	160	40	<
A/Latvia/02-073849/2019	T135K	2019-02-25	MDCKx/SIAT1	40	<	<	80	<	<	40	<	<
A/Latvia/02-073821/2019	T135K	2019-02-25	MDCK1/SIAT1	40	<	<	80	<	<	40	<	<
A/Latvia/02-033614/2019		2019-02-12	MDCK2/SIAT1	40	40	40	<	40	320	<	80	80
A/Latvia/03-012433/2019		2019-03-04	MDCKx/SIAT1	40	<	40	<	40	320	<	160	80
A/Greece/1451-19/2019		2019-03-15	SIAT1	40	<	40	<	<	320	<	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-04-17

				Haemagglutination inhibition titre										
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				NEW						NEW				
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Eng 538/18	A/Norway 3275/18	A/Kansas 14/17	
				MDCK	SIAT	Egg 10 ⁻⁴	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	SIAT	Egg	
				St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F13/19 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F14/19 ⁻¹	F03/19 ⁻¹	F11/19 ⁻¹	
	Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.3a	3C.2a1b	3C.3a	
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	80	160	160	160	160	160	80	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	320	320	80	640	160	160	160	80
A/Singapore/INFIMH-16-0019/2016	clone 57	3C.2a1	2016-04-14	E5/E2	80	40	320	320	160	160	40	40	40	<
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	640	320	320	80	1280	80	40	40	<
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	<	<	40	40	<	<	320	640	<	160
A/Kansas/14/2017		3C.3a	2017-12-14	E7/E2	<	<	40	40	<	<	320	160	<	640
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	40	<	40	80	<	40	320	640	<	160
TEST VIRUSES														
A/Cameroon/9008/2018		3C.2a3	2018-10-23	SIAT2	160	80	80	80	160	160	80	80	160	80
A/Cameroon/9680/2018		3C.2a3	2018-11-06	SIAT2	160	80	80	80	160	160	160	80	80	80
A/Cameroon/9661/2018		3C.2a3	2018-11-14	SIAT2	320	80	160	160	320	320	160	160	160	160
A/Rheinland-Pfalz/34/2019	T135K	3C.2a1b	2019-03-01	C1/SIAT1	80	40	40	40	160	40	<	<	160	40
A/Baden-Wuerttemberg/132/2019	T135K	3C.2a1b	2019-03-06	C1/SIAT1	80	40	40	80	160	80	<	<	160	40
A/Hessen/56/2019	T135K	3C.2a1b	2019-03-20	C1/SIAT1	160	40	40	40	160	40	<	<	160	40
A/Hessen/42/2019		3C.3a	2019-02-27	C1/SIAT1	40	40	40	40	<	40	320	640	<	80
A/Berlin/48/2019		3C.3a	2019-03-04	C1/SIAT1	80	40	40	40	<	40	320	640	<	80
A/Saarland/15/2019		3C.3a	2019-03-11	C2/SIAT1	<	<	<	40	<	<	320	320	<	80
A/Bremen/17/2019		3C.3a	2019-03-11	C1/SIAT1	40	<	40	80	<	40	320	320	<	80
A/Nordrhein-Westfalen/82/2019		3C.3a	2019-03-11	C1/SIAT1	40	40	40	40	<	40	320	640	<	80
A/Nordrhein-Westfalen/97/2019		3C.3a	2019-03-14	C1/SIAT1	40	40	40	40	<	40	320	320	<	80
A/Nordrhein-Westfalen/89/2019		3C.3a	2019-03-15	C1/SIAT1	80	40	80	80	<	80	320	640	<	160
A/Niedersachsen/151/2019		3C.3a	2019-03-25	C1/SIAT1	40	<	40	40	<	40	320	320	<	80
A/Bremen/20/2019		3C.3a	2019-03-25	C1/SIAT1	40	<	40	40	<	40	320	320	<	80
A/Nordrhein-Westfalen/101/2019		3C.3a	2019-03-25	C1/SIAT1	40	40	40	40	<	40	320	320	<	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-05-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14 MDCK St Jude's F60/17 ^{*1} 3C.2a	A/Bretagne 1413/17 SIAT F01/18 ^{*1} 3C.2a2	A/Singapore 0019/16 Egg 10 ⁻⁴ F46/17 ^{*1} 3C.2a1	A/La Rioja 2202/18 SIAT F26/18 ^{*1} 3C.2a1b	A/Switz 8060/17 Egg F27/18 ^{*1} 3C.2a2	A/Eng 538/18 SIAT F31/18 ^{*1} 3C.3a	A/Norway 3275/18 SIAT F03/19 ^{*1} 3C.2a1b	A/Kansas 14/17 Egg F11/19 ^{*1} 3C.3a	A/Kansas 14/17 SIAT F17/19 ^{*1} 3C.3a
				Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history
				REFERENCE VIRUSES								
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	160	320	160	320	320	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	320	1280	320	80	1280	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	40	640	160	160	80	40	<
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	640	80	1280	80	80	40
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	<	<	40	<	<	640	<	160
A/Kansas/14/2017		3C.3a	2017-12-14	E7/E2	<	<	80	<	<	320	<	640
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	40	40	80	<	80	640	40	160
TEST VIRUSES												
A/Slovenia/1213/2019	T135K	3C.2a1b	2019-02-19	MDCKx/MDCK1	40	<	<	160	40	<	80	40
A/Switzerland/6764/2019	T135K	3C.2a1b	2019-03-26	SIAT1	40	<	<	160	40	40	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-06-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	A/Kansas 14/17 Egg	A/Kansas 14/17 SIAT	NEW NYMC X-327 A/Kans/14/17 Egg
				St Jude's F60/17 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F03/19 ¹	F11/19 ¹	F17/19 ¹	F16/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	320	160	320	160	320	160	320	160	80	320
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	320	1280	320	160	1280	320	320	160	160	320
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	160	80	640	160	160	80	40	<	<	40
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	160	1280	640	160	1280	160	80	<	<	40
A/England/538/2018		2018-02-26	MDCK1/SIAT3	40	40	40	40	40	640	<	160	320	320
A/Kansas/14/2017		2017-12-14	E7/E2	<	40	40	<	<	320	<	1280	320	2560
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	80	80	80	80	80	1280	40	320	640	640
NYMC X-327 (A/Kansas/14/17)		2017-12-14	Ex/E1	<	40	40	<	40	320	<	640	320	2560
TEST VIRUSES													
A/Antalya/4/2018		2018-12-18	Cx/SIAT1	80	<	40	160	80	80	160	80	40	40
A/Sanliurfa/1763/2018		2018-11-16	SIAT1/SIAT1	<	<	40	80	<	<	<	<	<	<
A/Diyarbakir/1208/2018		2018-11-23	SIAT1/SIAT1	40	<	40	80	40	<	80	40	<	<
A/Samsun/811/2018		2018-11-27	Cx/SIAT1	80	<	<	80	40	<	80	40	<	<
A/Samsun/851/2018		2018-12-10	Cx/SIAT1	80	<	40	160	40	40	80	80	40	40
A/Antalya/3/2018		2018-12-13	Cx/SIAT1	160	<	40	160	80	80	160	80	40	40
A/Turkey/66/2018		2018-12-17	SIAT1	80	<	40	160	40	40	80	40	<	<
A/Antalya/7/2018		2018-12-18	Cx/SIAT1	80	<	80	160	40	80	80	80	80	40
A/Izmir/6/2018		2018-12-21	Cx/SIAT1	80	<	40	160	40	40	80	80	40	40
A/Turkey/70/2018		2018-12-29	SIAT1	80	<	40	160	40	40	80	80	40	40
A/Turkey/52/2019		2019-01-10	SIAT1	80	<	40	160	40	80	160	40	40	40
A/Makhachkala/RII-1/2019	T131K	2019-02-10	MDCK1/SIAT1	40	40	40	80	80	<	160	80	<	<
A/Grozny/RII-3/2019	T131K	2019-02-18	MDCK1/SIAT1	80	80	40	80	80	40	160	<	<	40
A/Makhachkala/RII-5/2019	T131K	2019-02-18	MDCK1/SIAT1	80	80	40	160	80	40	160	80	40	40
A/S. Peteresburg/RII-239/2019	T135K	2019-02-22	SIAT1/SIAT1	80	<	40	160	40	40	80	80	<	40
A/S. Peteresburg/RII-242/2019	T135K	2019-02-23	SIAT1/SIAT1	80	<	40	160	40	40	80	40	40	40
				Vaccine SH 2018 NH 2018-19		Vaccine SH 2019		Vaccine NH 2019-20		Vaccine NH 2019-20		Vaccine NH 2019-20	
Sequences in phylogenetic trees													

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Table 8-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-06-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Norway 3275/18	NYMC X-327 A/Kans/14/17	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹
	Ferret number			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
	Genetic group											
REFERENCE VIRUSES												
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	160	160	320	160	160	160	80
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	160	640	160	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	40	640	160	80	40	<	<
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	160	1280	640	1280	80	80	40	<
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	<	40	40	320	<	160	160
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	40	40	320	<	1280	320
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	80	40	80	80	640	40	320	320
TEST VIRUSES												
A/Kyrgyzstan/42/2019		3C.2a1b	2018-12-19	MDCKx/SIAT1	<	<	40	80	40	<	80	<
A/Austria/1113943/2018		3C.2a1b	2018-12-31	SIAT1/SIAT1	80	<	80	160	80	40	160	80
A/Austria/1114682/2019		3C.2a1b	2019-01-02	SIAT1/SIAT1	160	80	40	160	80	80	320	80
A/Austria/1114171/2019		3C.2a1b	2019-01-03	SIAT1/SIAT1	80	160	40	160	160	80	320	80
A/Austria/1115877/2019		3C.2a1b	2019-01-08	SIAT1/SIAT1	80	<	<	80	40	<	80	<
A/Kyrgyzstan/46/2019		3C.2a1b	2019-01-10	MDCKx/SIAT1	80	<	40	80	<	<	80	<
A/Kyrgyzstan/43/2019		3C.2a1b	2019-01-10	MDCKx/SIAT1	<	<	<	80	<	<	40	<
A/Austria/1117700/2019		3C.2a1b	2019-01-16	SIAT1/SIAT1	80	<	40	160	40	<	160	80
A/Austria/1120261/2019		3C.2a1b	2019-01-25	SIAT1/SIAT1	80	<	40	160	80	40	80	<
A/Austria/1121454/2019		3C.2a1b	2019-01-31	SIAT1/SIAT1	80	<	<	80	40	<	40	<
A/Austria/1121450/2019		3C.2a2	2019-01-31	SIAT1/SIAT1	80	160	40	40	160	80	160	80
A/Austria/1117352/2019		3C.2a3	2019-01-14	SIAT1/SIAT1	80	<	40	80	80	40	40	80
A/Austria/1121455/2019		3C.2a4	2019-01-29	SIAT1/SIAT1	40	40	40	40	40	40	80	40
A/Austria/1115342/2019		3C.3a	2019-01-08	SIAT1/SIAT1	160	80	160	80	160	320	160	320
A/Bulgaria/896/2019	T131K	3C.2a1b	2019-02-01	SIAT2/SIAT1	80	80	<	80	80	40	160	80
A/Zaporizhia/156/2019	T131K	3C.2a1b	2019-02-01	SIAT2/SIAT1	160	80	80	80	80	40	160	40
A/Austria/1129536/2019	T131K	3C.2a1b	2019-02-27	SIAT1/SIAT1	160	160	40	80	80	40	160	<
A/Iran/168336/2019	T131K	3C.2a1b	2019-03-07	SIAT1/SIAT1	<	40	40	80	80	40	160	40
A/Iran/168555/2019	T131K	3C.2a1b	2019-03-15	SIAT1/SIAT1	160	160	160	160	80	80	320	80
A/Iran/171909/2019	T131K	3C.2a1b	2019-04-12	SIAT1/SIAT1	80	80	80	160	80	80	160	80
A/Bulgaria/958/2019	T135K	3C.2a1b	2019-02-01	SIAT2/SIAT1	80	40	80	160	80	80	160	80
A/Bulgaria/949/2019	T135K	3C.2a1b	2019-02-01	SIAT2/SIAT1	80	<	40	160	40	40	80	40
A/Ukraine/8210/2019	T135K	3C.2a1b	2019-02-01	MDCK0/SIAT1	40	<	<	80	<	<	80	<
A/Khmelnytskyi/155/2019	T135K	3C.2a1b	2019-02-01	SIAT2/SIAT1	160	40	80	160	80	80	160	80
A/Bulgaria/947/2019	T135K	3C.2a1b	2019-02-03	SIAT2/SIAT1	<	<	<	40	<	<	40	<
A/Bulgaria/909/2019	T135K	3C.2a1b	2019-02-03	SIAT2/SIAT1	80	40	80	160	80	40	160	40
A/Bulgaria/1052/2019	T135K	3C.2a1b	2019-02-05	SIAT2/SIAT1	40	<	40	160	<	<	80	<
A/Ukraine/8211/2019	T135K	3C.2a1b	2019-02-05	MDCK0/SIAT1	<	<	<	80	<	<	80	<
A/Bulgaria/1051/2019	T135K	3C.2a1b	2019-02-06	SIAT2/SIAT1	40	40	40	320	<	<	160	<
A/Bulgaria/1134/2019	T135K	3C.2a1b	2019-02-12	SIAT2/SIAT2	80	<	40	640	40	<	320	<
A/Khmelnytskyi/149/2019	T135K	3C.2a1b	2019-02-15	SIAT2/SIAT1	40	<	40	160	<	<	160	<
A/Bulgaria/1129/2019	T135K	3C.2a1b	2019-02-16	SIAT2/SIAT1	80	<	40	160	<	<	160	<
A/Iran/167623/2019	T135K	3C.2a1b	2019-02-16	SIAT1/SIAT1	80	<	40	160	80	40	160	40
A/Khmelnytskyi/148/2019	T135K	3C.2a1b	2019-02-16	SIAT1/SIAT1	<	<	<	80	<	<	80	<
A/Iran/168206/2019	T135K	3C.2a1b	2019-02-19	SIAT1/SIAT1	160	40	80	160	80	80	160	80
A/Ukraine/8230/2019	T135K	3C.2a1b	2019-02-20	MDCK1/SIAT1	40	<	<	80	<	<	80	<
A/Bulgaria/1153/2019	T135K	3C.2a1b	2019-02-21	SIAT2/SIAT1	40	<	40	80	<	<	<	<
A/Khmelnytskyi/145/2019	T135K	3C.2a1b	2019-02-27	SIAT1/SIAT1	40	<	<	80	<	<	80	<
A/Iran/168359/2019	T135K	3C.2a1b	2019-03-11	SIAT1/SIAT1	40	<	40	160	40	<	160	<
A/Ukraine/8233/2019	T135K	3C.2a1b	2019-03-11	MDCK1/SIAT1	<	<	<	40	<	<	40	<
A/Bulgaria/902/2019		3C.2a2	2019-02-01	SIAT2/SIAT1	160	320	80	80	320	80	160	80
A/Bulgaria/900/2019		3C.2a2	2019-02-01	SIAT2/SIAT1	160	320	80	80	320	80	160	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19Vaccine
SH 2019Vaccine
NH 2019-20

Table 8-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-06-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Norway 3275/18	NYMC X-327 A/Kans/14/17	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT
	Passage history			St Jude F60/17 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F03/19 ¹	F16/19 ¹	F17/19 ¹
	Ferret number											
	Genetic group			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
REFERENCE VIRUSES												
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	160	160	160	80	160	160	160	160	80
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT5	160	640	160	80	640	80	160	80	80
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	160	40	640	160	160	80	40	10	<
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	160	2560	640	160	2560	160	80	80	<
A/England/538/2018		2018-02-26	MDCK1/SIAT4	40	40	40	40	40	640	<	160	320
NYMC X-327 (A/Kansas/14/17)		2017-12-14	Ex/E1	<	<	80	<	<	320	<	1280	320
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	40	40	80	40	40	640	<	320	320
TEST VIRUSES												
A/Moldova/851.16/2019	T135K	2019-04-16	SIAT1	40	<	<	80	<	<	80	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-06-21

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	NYMC X-327 A/Kans/14/17 Egg	A/Kansas 14/17 SIAT
					St Jude's F60/17 ⁻¹ 3C.2a	F01/18 ⁻¹ 3C.2a2	F46/17 ⁻¹ 3C.2a1	F26/18 ⁻¹ 3C.2a1b	F27/18 ⁻¹ 3C.2a2	F31/18 ⁻¹ 3C.3a	F03/19 ⁻¹ 3C.2a1b	F16/19 ⁻¹ 3C.3a	F17/19 ⁻¹ 3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	160	80	160	160	160	160	80
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	1280	160	80	640	160	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	640	160	160	80	40	40	<
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	640	160	1280	80	80	80	<
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	40	80	40	40	640	<	160	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	80	40	40	320	40	1280	320
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	160	80	160	80	160	640	40	320	320
TEST VIRUSES													
A/Zambia/1-390/2018		3C.2a2	2018-09-25	SIAT1	160	160	160	80	320	320	80	160	160
A/Lisboa/100/2019		3C.3a	2019-01-28	SIAT3/SIAT1	40	<	40	<	40	320	<	160	320
A/Eskilstuna/6/2019	T131K	3C.2a1b	2019-02-13	MDCK0/SIAT1	160	160	80	80	160	80	320	40	40
A/Skovde/1/2019	T135K	3C.2a1b	2019-02-11	MDCK1/SIAT1	40	<	40	80	40	40	160	<	40
A/Denmark/2785/2019	T135K	3C.2a1b	2019-03-13	SIAT2/SIAT1	40	<	40	160	<	<	80	<	<
A/Lisboa/93/2019		3C.3a	2019-02-08	SIAT2/SIAT1	40	<	40	<	40	320	<	160	320
A/Lisboa/92/2019		3C.3a	2019-02-15	MDCK2/SIAT1	40	40	80	<	40	320	40	160	320
A/Lisboa/91/2019		3C.3a	2019-02-20	SIAT1/SIAT1	40	40	80	40	40	640	40	160	320
A/Lisboa/97/2019		3C.3a	2019-02-25	SIAT2/SIAT1	40	40	80	40	40	640	40	160	320
A/Lisboa/94/2019		3C.3a	2019-02-26	SIAT1/SIAT1	40	40	80	<	40	320	<	160	320
A/Lisboa/95/2019		3C.3a	2019-02-27	SIAT2/SIAT1	40	<	40	<	40	320	<	160	320
A/Lisboa/98/2019		3C.3a	2019-03-08	SIAT1/SIAT1	<	<	40	<	<	160	<	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-07-05

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT	A/Norway 3275/18 Egg
					St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹	F21/19 ⁻¹
	Genetic group				3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a	3C.2a1b
REFERENCE VIRUSES														
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	320	320	160	320	320	320	320	160	80
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	320	1280	320	160	640	320	320	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	1280	320	320	80	80	40	<	320
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	320	2560	640	320	2560	160	80	80	<	640
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	40	40	<	80	1280	<	320	640	<
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	<	<	80	<	40	640	40	1280	320	40
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	160	80	160	40	160	1280	40	640	640	<
A/Norway/3275/2018		3C.2a1b	2018-10-04	E6(Am3A)3	160	160	640	160	320	40	80	<	<	640
TEST VIRUSES														
A/Mauritania/200-2018/2018		3C.2a1b	2018-11-05	SIAT1	40	<	<	160	<	<	160	<	<	<
A/Mauritania/08-2019/2019		3C.2a1b	2019-01-02	SIAT1	80	<	80	160	40	40	80	40	<	40
A/Mauritania/09-2019/2019		3C.2a1b	2019-01-03	SIAT1	80	<	40	160	40	40	80	40	<	40
A/Mauritania/15-2019/2019		3C.2a1b	2019-01-07	SIAT1	160	<	160	160	80	40	160	40	<	40
A/Mauritania/10-2019/2019		3C.2a1b	2019-01-08	SIAT1	160	<	320	320	160	160	320	160	<	40
A/Tanger/672/2019		3C.3a	2019-01-24	MDCK2/SIAT1	160	160	320	40	160	1280	40	640	640	<
A/West Kazakhstan/461/2019	T131K	3C.2a1b	2019-02-27	MDCKx/SIAT1	160	80	80	160	80	<	320	40	80	160
A/West Kazakhstan/462/2019	T131K	3C.2a1b	2019-03-01	MDCKx/SIAT1	160	320	160	320	320	160	640	160	160	320
A/West Kazakhstan/463/2019	T131K	3C.2a1b	2019-03-05	MDCKx/SIAT1	160	160	160	320	160	160	320	160	160	320
A/Finland/105/2019	T131K	3C.2a1b	2019-04-09	SIAT1/SIAT1	320	640	320	320	320	640	1280	160	160	320
A/North Kazakhstan/363/2019	T135K	3C.2a1b	2019-02-13	MDCKx/SIAT1	80	<	<	160	<	<	160	<	<	40
A/Finland/92/2019	T135K	3C.2a1b	2019-03-06	SIAT1/SIAT1	80	<	<	160	<	<	320	<	<	<
A/Georgia/658/2019	T135K	3C.2a1b	2019-03-13	SIAT2	40	<	<	160	<	<	160	<	<	<
A/Finland/99/2019		3C.3a	2019-03-26	SIAT1/SIAT1	80	80	160	40	320	640	80	320	320	<
A/Finland/104/2019		3C.3a	2019-04-03	SIAT1/SIAT1	80	40	40	<	80	1280	<	320	640	<
A/Finland/114/2019		3C.3a	2019-04-30	SIAT1/SIAT1	160	80	160	40	80	1280	<	320	640	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ⁻¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19Vaccine
SH 2019Vaccine
NH 2019-20

Table 8-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-07-16

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
					St Jude's F60/17 ⁻¹ 3C.2a	F01/18 ⁻¹ 3C.2a2	F46/17 ⁻¹ 3C.2a1	F26/18 ⁻¹ 3C.2a1b	F27/18 ⁻¹ 3C.2a2	F31/18 ⁻¹ 3C.3a	F03/19 ⁻¹ 3C.2a1b	F16/19 ⁻¹ 3C.3a	F17/19 ⁻¹ 3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	160	320	160	160	320	160	320	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	320	80	640	160	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	640	160	320	160	40	40	80
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	640	160	1280	160	80	80	80
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	80	40	40	40	640	40	160	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	80	40	40	320	40	1280	640
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	80	80	80	40	80	640	40	320	320
TEST VIRUSES													
A/Mauritania/255-2018/2018		3C.2a1b	2018-12-10	SIAT1	160	40	160	160	40	80	80	40	40
A/Komarno/58/2018		3C.2a1b	2018-12-28	MDCK1/SIAT1	40	<	40	80	<	40	80	<	40
A/Trencin/74/2019		3C.2a1b	2019-01-16	MDCK1/SIAT1	40	<	<	40	<	40	40	<	40
A/Tmava/70/2019		3C.2a1b	2019-01-16	MDCK1/SIAT1	40	<	40	80	40	40	80	<	40
A/Nove Zamky/81/2019		3C.2a1b	2019-01-21	MDCK2/SIAT1	80	<	40	160	40	40	80	<	40
A/Nitra/106/2019		3C.2a1b	2019-01-24	MDCKx/SIAT1	80	<	160	320	80	80	160	80	40
A/Prievidza/126/2019		3C.2a1b	2019-01-29	MDCKx/SIAT1	80	<	40	160	40	80	160	<	40
A/Mogilev/1547/2019			2019-03-12	MDCK2/SIAT1	40	<	40	80	<	80	80	<	40
A/PJovazska Bystrica/177/2019	T135K	3C.2a1b	2019-02-05	MDCKx/SIAT1	80	<	80	160	80	80	160	40	40
A/Bratislava/328/2019	T135K	3C.2a1b	2019-02-20	MDCKx/SIAT1	<	<	<	80	<	40	80	<	40
A/Skalica/339/2019	T135K	3C.2a1b	2019-02-21	MDCKx/SIAT1	80	<	40	160	40	80	160	<	40
A/Vitebsk/927/2019	T135K	3C.2a1b	2019-02-28	MDCK2/SIAT1	80	<	40	80	80	80	80	<	40
A/Orsha/1450/2019	T135K	3C.2a1b	2019-03-05	MDCK2/SIAT1	40	<	<	80	<	80	80	<	40
A/Bobruisk/1468/2019	T135K	3C.2a1b	2019-03-06	MDCK2/SIAT1	40	<	<	80	<	80	80	<	40
A/Mogilev/1315/2019	T135K	3C.2a1b	2019-03-06	MDCK2/SIAT1	40	<	<	80	<	40	80	<	40
A/Bratislava/446/2019	T135K	3C.2a1b	2019-03-08	MDCKx/SIAT1	40	<	40	160	40	80	80	<	40
A/Macedonia/1283/2019	T135K	3C.2a1b	2019-03-11	SIAT1	80	<	80	160	80	40	80	<	<
A/Galanta/473/2019	T135K	3C.2a1b	2019-03-12	MDCKx/SIAT1	40	<	<	160	40	<	80	<	40
A/Macedonia/1350/2019		3C.3a	2019-03-18	SIAT1	80	40	160	160	160	640	80	160	320
A/Macedonia/1402/2019		3C.3a	2019-03-22	SIAT1	80	40	80	40	40	640	40	160	320
A/Kosovo/1130/2019		3C.3a	2019-03-29	SIAT1	80	80	80	40	40	640	40	160	320
A/Kosovo/1188/2019		3C.3a	2019-04-08	SIAT1	160	80	80	40	40	640	40	160	320
A/Macedonia/1462/2019		3C.3a	2019-04-15	SIAT1	80	80	80	40	40	640	40	160	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-07-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14 MDCK St Jude's F60/17 ⁻¹	A/Bretagne 1413/17 SIAT F01/18 ⁻¹	A/Singapore 0019/16 Egg 10 ⁻⁴ F46/17 ⁻¹	A/La Rioja 2202/18 SIAT F26/18 ⁻¹	A/Switz 8060/17 Egg F27/18 ⁻¹	A/Eng 538/18 SIAT F31/18 ⁻¹	A/Norway 3275/18 SIAT F03/19 ⁻¹	NYMC X-327 A/Kansas/14 Egg F16/19 ⁻¹	A/Kansas 14/17 SIAT F17/19 ⁻¹
				3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
REFERENCE VIRUSES												
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	160	640	160	320	640	320	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	320	1280	320	160	1280	320	320	160
A/Singapore/INFMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	640	320	160	80	40	80
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	640	160	1280	160	80	80
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	80	80	40	80	80	640	80	640
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	40	40	40	320	40	320
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	160	80	80	80	80	640	80	640
TEST VIRUSES												
A/South Africa/R03804/2019	T131K	3C.2a1b	2019-04-12	MDCK1/SIAT1	160	160	80	160	160	40	320	40
A/South Africa/R03985/2019	T131K	3C.2a1b	2019-04-16	MDCK2/SIAT1	160	160	80	160	160	80	320	40
A/South Africa/R05118/2019	T131K	3C.2a1b	2019-05-13	MDCK2/SIAT1	160	160	80	160	160	160	320	40
A/South Africa/R05114/2019	T131K	3C.2a1b	2019-05-13	MDCK2/SIAT1	160	160	80	160	160	80	320	40
A/South Africa/R05652/2019	T131K	3C.2a1b	2019-05-21	MDCK2/SIAT1	160	320	160	320	160	160	640	80
A/South Africa/R05636/2019	T131K	3C.2a1b	2019-05-22	MDCK2/SIAT1	80	80	160	320	80	80	160	80
A/Iceland/12621/2019	T131K	3C.2a1b	2019-06-15	SIAT1	640	640	640	640	640	640	640	320
A/Hong Kong/2672/2019	T131K	3C.2a1b	2019-06-17	MDCK1/SIAT1	80	80	160	320	160	160	320	80
A/Hong Kong/2673/2019	T131K	3C.2a1b	2019-06-18	MDCK1/SIAT1	80	40	80	160	160	160	320	80
A/Jordan/30076/2019	T135K	3C.2a1b	2019-03-02	SIAT2	640	320	640	320	320	640	320	640
A/Iceland/4/2019	T135K	3C.2a1b	2019-03-04	MDCK1/SIAT1	40	<	40	80	<	40	80	<
A/South Africa/R05176/2019	T135K	3C.2a1b	2019-05-14	MDCK2/SIAT1	160	80	80	320	80	80	320	40
A/Iceland/71/2019	T135K	3C.2a1b	2019-05-19	MDCK1/SIAT1	80	40	160	320	40	80	160	80
A/Hong Kong/2671/2019	T135K	3C.2a1b	2019-06-17	MDCK1/SIAT1	160	<	80	320	80	320	320	80
A/Hong Kong/2670/2019	T135K	3C.2a1b	2019-06-17	MDCK1/SIAT1	160	<	80	160	80	320	320	160
A/Hong Kong/2679/2019	T135K	3C.2a1b	2019-06-18	MDCK1/SIAT1	160	<	80	320	80	160	320	80
A/Hong Kong/2675/2019	T135K	3C.2a1b	2019-06-18	MDCK1/SIAT1	80	<	80	320	80	160	160	80
A/Hong Kong/2669/2019	T135K	3C.2a1b	2019-06-18	MDCK1/SIAT1	80	<	40	320	80	160	320	80
A/Hong Kong/2676/2019	T135K	3C.2a1b	2019-06-19	MDCK1/SIAT1	80	<	80	160	80	160	160	80
A/Iceland/59/2019		3C.3a	2019-04-15	MDCK1/SIAT1	80	40	160	80	40	640	40	160
A/Jordan/30121/2019		3C.3a	2019-04-17	SIAT1	40	40	160	80	40	320	40	160
A/South Africa/R05121/2019		3C.3a	2019-05-14	MDCK1/SIAT1	40	80	80	40	80	640	40	160
A/South Africa/R05619/2019		3C.3a	2019-05-20	MDCK2/SIAT1	80	80	80	40	40	640	40	160
A/South Africa/R06404/2019		3C.3a	2019-06-04	SV/SIAT1	80	80	80	40	40	640	40	160
A/South Africa/R06917/2019		3C.3a	2019-06-10	MDCK1/SIAT1	80	160	160	80	80	640	40	160
A/South Africa/R06867/2019		3C.3a	2019-06-10	MDCK1/SIAT1	80	80	80	40	80	160	40	160
A/Iceland/23/2019			2019-03-18	MDCK1/SIAT1	160	40	160	160	80	160	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-08-02

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Singapore 0019/16 Egg 10 ⁻⁴	A/La Rioja 2202/18 SIAT	A/Switz 8060/17 Egg	A/Eng 538/18 SIAT	A/Norway 3275/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
					St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹
					3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	320	160	320	640	320	320	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT5	160	640	160	80	640	320	160	160	80
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	640	320	160	80	40	40	40
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	160	1280	640	160	1280	80	80	40	80
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	<	<	40	<	40	320	<	160	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	80	<	<	320	<	1280	640
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	<	40	320	<	160	320
TEST VIRUSES													
A/Champagne Ardenne/1908/2019	T131K	3C.2a1b	2019-04-10	MDCK2/SIAT1	80	80	80	160	160	80	160	<	<
A/Estonia/119303/2019	T135K	3C.2a1b	2019-02-11	SIAT2/SIAT2	80	<	40	320	40	40	160	40	40
A/Torino/544/2019	T135K	3C.2a1b	2019-02-19	SIAT3/SIAT2	80	40	160	160	80	80	160	40	40
A/Moscow/127/2019	T135K	3C.2a1b	2019-03-01	SIAT1/SIAT1	40	<	<	80	<	<	80	<	<
A/Torino/780/2019	T135K	3C.2a1b	2019-03-06	SIAT3/SIAT2	80	40	80	160	80	80	160	80	40
A/Czech Republic/1031/2019	T135K	3C.2a1b	2019-03-06	MDCK4/SIAT1	<	<	<	80	<	<	40	<	<
A/Palermo/20/2019	T135K	3C.2a1b	2019-03-06	SIAT4/SIAT2	80	40	40	160	40	80	160	40	<
A/Moscow/146/2019	T135K	3C.2a1b	2019-03-10	SIAT1/SIAT1	<	<	<	80	<	<	80	<	<
A/Parma/379/2019	T135K	3C.2a1b	2019-03-18	SIAT3/SIAT2	40	<	80	80	40	40	80	40	40
A/Nord Pas de Calais/1803/2019	T135K	3C.2a1b	2019-03-26	MDCK2/SIAT1	80	40	80	160	40	40	160	<	<
A/South Africa/R06219/2019	T135K	3C.2a1b	2019-05-31	SV/SIAT1	40	<	80	160	40	40	80	<	<
A/Parma/353/2019		3C.3a	2019-02-26	SIAT4/SIAT1	40	40	80	40	80	640	<	160	320
A/Palermo/12/2019		3C.3a	2019-02-27	SIAT3/SIAT1	40	40	40	<	40	320	<	160	320
A/Palermo/13/2019		3C.3a	2019-03-05	SIAT3/SIAT1	40	40	40	<	40	320	<	80	160
A/Torino/771/2019		3C.3a	2019-03-05	SIAT3/SIAT2	40	40	40	<	40	320	<	160	160
A/Parma/361/2019		3C.3a	2019-03-08	SIAT3/SIAT1	40	40	40	<	40	640	<	160	320
A/Parma/373/2019		3C.3a	2019-03-11	SIAT2/SIAT1	<	40	40	<	40	320	<	160	160
A/Parma/377/2019		3C.3a	2019-03-15	SIAT2/SIAT1	40	40	40	<	40	320	<	160	160
A/Basse Normandie/1798/2019		3C.3a	2019-03-26	SIAT1	40	40	40	<	40	320	<	80	160
A/Centre/1846/2019		3C.3a	2019-03-30	MDCK1/SIAT1	40	40	40	<	80	320	<	160	160
A/Brest/1998/2019		3C.3a	2019-04-07	MDCK1/SIAT1	<	<	<	<	<	320	<	80	160
A/Brest/1999/2019		3C.3a	2019-04-08	MDCK1/SIAT1	40	40	80	40	80	320	<	160	320
A/Picardie/1898/2019		3C.3a	2019-04-08	MDCK1/SIAT1	40	40	40	<	40	320	<	160	160
A/Czech Republic/1275/2019		3C.3a	2019-04-10	MDCK3/SIAT1	80	40	80	40	80	640	<	160	320
A/Brest/2002/2019		3C.3a	2019-04-11	MDCK1/SIAT1	80	80	80	40	80	320	<	160	160
A/Brest/2001/2019		3C.3a	2019-04-11	MDCK1/SIAT1	80	40	80	40	80	320	<	160	160
A/Bretagne/1913/2019		3C.3a	2019-04-11	MDCK1/SIAT1	<	<	40	<	40	320	<	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19Vaccine
SH 2019Vaccine
NH 2019-20

Table 8-18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-08-16

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14 MDCK St Jude's F60/17 ⁻¹	A/Bretagne 1413/17 SIAT F01/18 ⁻¹	A/Singapore 0019/16 Egg 10 ⁻⁴ F46/17 ⁻¹	A/La Rioja 2202/18 SIAT F26/18 ⁻¹	A/Switz 8060/17 Egg F27/18 ⁻¹	A/Eng 538/18 SIAT F31/18 ⁻¹	A/Norway 3275/18 SIAT F03/19 ⁻¹	NYMC X-327 A/Kansas/14 Egg F16/19 ⁻¹	A/Kansas 14/17 SIAT F17/19 ⁻¹
				3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a
REFERENCE VIRUSES												
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	320	160	320	320	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT5	320	640	320	80	640	320	320	160
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	1280	320	320	80	80	80
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	1280	160	2560	160	160	160
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	40	80	<	40	640	<	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	160	<	40	320	<	1280
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	40	80	80	40	80	640	<	320
TEST VIRUSES												
A/England/fmhwY2NxZm9pZg/2018		3C.2a3	2018-12-19	SIAT3/SIAT1	160	80	160	80	80	160	80	320
A/England/138/2019		3C.3a	2019-01-28	SIAT3/SIAT1	80	80	80	40	80	640	40	80
A/Bosnia and Herzegovia/119G/2019		3C.2a1b	2019-01-17	SIAT1/SIAT1	80	40	40	160	80	80	80	40
A/England/93/2019		3C.2a1b	2019-01-31	MDCK2/SIAT1/SIAT1	160	80	160	160	160	160	160	40
A/England/95/2019	T131K	3C.2a1b	2019-02-04	SIAT3/SIAT1	40	40	40	80	40	40	160	<
A/Sydney/53/2019	T131K	3C.2a1b	2019-03-08	E3/E1	40	160	40	40	160	40	320	40
A/New Caledonia/59/2019	T131K	3C.2a1b	2019-04-02	E3/E1	80	320	80	80	160	160	320	160
A/St. Petersburg/1092V/2019	T131K	3C.2a1b	2019-05-05	MDCK1/SIAT1	160	320	160	640	640	320	640	160
A/England/99/2019	T135K	3C.2a1b	2019-02-07	MDCK2/SIAT1/SIAT1	80	40	160	160	80	160	160	80
A/England/102/2019	T135K	3C.2a1b	2019-02-12	SIAT2/SIAT1	80	40	160	160	80	80	80	80
A/Sakha/3014V/2019	T135K	3C.2a1b	2019-04-16	MDCK1/SIAT1	80	40	40	160	40	40	160	40
A/Tomsk/5645V/2019	T135K	3C.2a1b	2019-04-20	MDCK1/SIAT1	160	160	160	320	160	160	320	160
A/Omsk/1008V/2019	T135K	3C.2a1b	2019-04-23	MDCK1/SIAT1	80	40	80	320	160	80	160	320
A/England/eWIsYW9mYWxlaQ/2019		3C.2a2	2019-02-08	MDCK1/SIAT1/SIAT1	160	320	320	160	320	320	160	160
A/England/154/2019		3C.3a	2019-02-07	SIAT2/SIAT1	80	80	80	40	80	640	40	320
A/England/185/2019		3C.3a	2019-02-08	SIAT2/SIAT1	160	40	80	40	80	640	<	320
A/England/155/2019		3C.3a	2019-02-08	SIAT2/SIAT1	40	40	80	40	80	640	<	160
A/England/157/2019		3C.3a	2019-02-25	SIAT2/SIAT1	40	80	80	40	80	320	<	320
A/England/156/2019		3C.3a	2019-02-26	SIAT2/SIAT1	40	80	80	40	80	320	<	320
A/England/222/2019			2019-02-20	SIAT1/SIAT1	40	40	80	<	40	320	<	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-09-13

				Haemagglutination inhibition titre												
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera											
					A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Norway 3275/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17	A/C'church 515/18	NEW A/HK 674/18	
					MDCK	SIAT	Eqq 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT	Egg		
					Ferret number F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹	F32/19 ⁻¹	F29/19 ⁻¹	
					Genetic group	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a		
REFERENCE VIRUSES																
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	160	320	160	160	160	160	160	160	160	80	
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT5	160	640	320	160	320	160	160	160	160	80	80	
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	320	80	640	320	160	80	40	40	40	40	80	
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	320	1280	640	320	1280	160	160	80	80	80	160	
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	40	40	40	40	40	640	<	160	320	<	40	
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	80	40	40	320	<	1280	320	40	40	
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	40	40	320	<	160	320	<	40	
A/Christchurch/515/2019			2019-04-03	E3/E1	320	80	320	320	80	40	320	<	40	320	80	
A/Hong Kong/674/2018	clone 33		2018-04-10	E6	160	80	160	320	80	80	320	80	40	80	640	
TEST VIRUSES																
A/Ghana/1041/2019			2019-01-04	Cx/SIAT1	160	80	160	320	160	160	320	160	80	320	160	
A/Dakar/02/2019			2019-01-21	C1/SIAT2	160	40	160	320	80	80	160	80	80	160	160	
A/Dakar/01/2019			2019-01-21	C1/SIAT2	160	40	160	320	80	80	160	80	80	160	160	
A/Dakar/03/2019			2019-01-29	C1/SIAT2	40	<	80	160	40	40	80	40	40	80	40	
A/Armenia/501/2019	T131K	3C.2a1b	2019-02-07	SIAT2	80	80	80	160	80	80	160	80	40	160	40	
A/Armenia/941/2019		3C.2a1b	2019-05-21	SIAT2	160	160	160	320	160	160	320	160	160	320	160	
A/Oman/4262/2019		3C.2a1b	2019-06-25	SIAT1/SIAT1	80	40	40	160	160	80	160	80	40	160	40	
A/Bosnia and Herzegovina/84219/2019		3C.2a1b	2019-04-09	SIAT1	40	<	<	160	<	<	80	<	<	80	40	
A/Bosnia and Herzegovina/87911/2019	T135K	3C.2a1b	2019-04-11	SIAT1	80	<	80	160	40	80	160	80	<	80	80	
A/Oman/4518/2019		3C.2a1b	2019-07-18	SIAT1	40	<	<	40	<	<	80	<	<	80	<	
A/Garoua/19V-1937/2019		3C.2a3	2019-03-22	CS	160	160	160	160	320	320	160	320	160	160	40	
A/Ireland/22716/2019		3C.3a	2019-02-26	SIAT1	40	<	<	<	<	320	<	80	160	<	40	
A/Oman/4471/2019		3C.3a	2019-07-11	SIAT1	80	80	80	40	80	320	40	160	320	40	40	
A/Ireland/23414/2019			2019-02-26	SIAT1/SIAT1	40	40	80	<	40	320	<	160	320	<	40	
A/Ireland/26826/2019			2019-03-13	SIAT3/SAIT1	80	80	160	320	80	160	320	80	80	160	160	
A/Ireland/26207/2019			2019-03-13	SIAT1/SIAT1	80	80	160	40	160	320	40	320	320	40	40	
A/Ireland/28223/2019			2019-03-19	SIAT2/SIAT1	40	80	80	40	40	640	40	160	320	<	40	
A/Ireland/28663/2019			2019-03-20	SIAT2/SIAT1	40	<	40	<	<	640	<	160	160	<	40	
A/Dakar/11/2019			2019-03-21	C1/SIAT1	80	40	160	40	40	320	40	160	320	<	40	
A/Ireland/29431/2019			2019-03-21	SIAT2/SIAT1	40	80	160	40	40	640	40	160	160	<	40	
A/Ireland/30256/2019			2019-03-26	SIAT3/SIAT3	40	40	80	160	40	40	160	<	<	160	80	
A/Dakar/13/2019			2019-03-27	C1/SIAT1	40	40	80	40	80	320	40	160	320	<	40	
A/Dakar/12/2019			2019-03-27	C1/SIAT2	80	40	160	160	40	40	160	40	80	160	80	
A/Ireland/32103/2019			2019-04-02	SIAT3/SIAT1	80	<	160	320	80	80	160	40	40	160	80	
A/Ghana/1050/2019			2019-04-04	Cx/SIAT1	40	<	80	160	40	80	80	80	<	80	40	
A/Ghana/154/2019			2019-04-04	Cx/SIAT1	80	40	160	160	40	80	40	80	40	160	160	
A/Dakar/08/2019			2019-04-23	C1/SIAT1	80	40	160	320	80	80	160	80	80	160	160	
A/Norway/2032/2019			2019-05-14	SIAT1/SIAT1	40	40	80	<	40	320	<	160	160	40	40	
A/Dakar/10/2019			2019-05-24	C1/SIAT1	80	80	80	40	80	320	40	160	320	<	40	
A/Norway/2099/2019			2019-05-24	SIAT1/SIAT1	80	40	80	40	40	320	<	160	320	40	40	
A/Ghana/312/2019			2019-06-06	SIAT3	160	160	640	640	160	160	640	160	160	320	160	
A/Norway/2155/2019			2019-06-18	SIAT1/SIAT1	80	40	80	160	40	80	80	40	40	160	40	
A/Norway/2224/2019			2019-07-31	SIAT1/SIAT1	80	80	160	320	160	80	160	160	80	320	80	
A/Norway/2220/2019			2019-08-05	SIAT1/SIAT1	40	40	40	80	40	<	160	<	<	80	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40
Sequences in phylogenetic trees

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Vaccine
NH 2019-20

Table 8-20. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) - Summary

			Haemagglutination inhibition titre									
Viruses	Other information	Collection date	Post-infection ferret antisera									
			A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/La Rioja 2202/18	A/Switz 8060/17	A/Eng 538/18	A/Norway 3275/18	NYMC X-327 A/Kans/14/17	A/Kansas 14/17	
			MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT	
			Ferret number F60/17 ⁻¹	F01/18 ⁻¹	F46/17 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F03/19 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹	
Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a	
Genetic group			3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.3a	3C.3a	
REFERENCE VIRUSES												
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	160	160	160	80	160	160	160	80	
A/Bretagne/1413/2017		3C.2a2	2017-10-09	160	640	160	80	640	80	160	80	
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	160	40	640	160	160	80	40	<	
A/Switzerland/8060/2017		3C.2a2	2017-12-12	160	2560	640	160	2560	160	80	<	
A/England/538/2018		3C.3a	2018-02-26	40	40	40	40	40	640	<	160	
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	<	<	80	<	<	320	<	1280	
A/Kansas/14/2017		3C.3a	2017-12-14	40	40	80	40	40	640	<	320	
TEST VIRUSES												
Number of viruses tested*			201	201	201	201*	201	201	201*	201	180	
No with titre reduction ≤2-fold			63	7	3	88	1	83	83	2	60	
%			31.3	3.5	1.5		0.5	41.3		0.9	33.3	
No with titre reduction =4-fold			92	10	27		5	28		19	15	
%			45.8	5.0	13.4		2.5	13.9		9.5	8.4	
No with titre reduction ≥8-fold			46	184	171		195	90		180	105	
%			22.9	91.5	85.1		97.0	44.8		89.6	58.3	
Subgroup 3C.2a1b viruses + T135K												
Number of viruses tested*			76	76	76	76*	76	76	76*	76	73	
No with titre reduction ≤2-fold			22		1	53		3	39	1	4	
%			28.9		1.3			3.9		1.3	5.5	
No with titre reduction =4-fold			28	2	2		1	12		1	2	
%			36.8	2.6	2.6		1.3	15.8		1.3	2.7	
No with titre reduction ≥8-fold			26	74	73		75	61		74	67	
%			34.3	97.4	96.1		98.7	80.3		97.4	91.8	
Subgroup 3C.2a1b viruses + T131K												
Number of viruses tested*			31	31	31	31*	31	31	31*	31	31	
No with titre reduction ≤2-fold			18	4	1	21	1	3	31	1	3	
%			58.1	12.9	3.2		3.2	9.7		3.2	9.7	
No with titre reduction =4-fold			11	6	6		1	9			5	
%			35.4	19.4	19.4		3.2	29.0			16.1	
No with titre reduction ≥8-fold			2	21	24		29	19		30	23	
%			6.5	67.7	77.4		93.6	61.3		96.8	74.2	
Clade 3C.3a viruses												
Number of viruses tested*			66	66	66	66*	66	66	66*	66	48	
No with titre reduction ≤2-fold			17			1		64	0		46	
%			25.8					97.0			95.8	
No with titre reduction =4-fold			41		7			2		16	1	
%			62.1		10.6			3.0		24.2	2.1	
No with titre reduction ≥8-fold			8	66	59		66			50	1	
%			12.1	100.0	89.4		100.0			75.8	2.1	
Subclade 3C.2a2 viruses												
Number of viruses tested*			3	3	3	3*	3	3	3*	3	3	
No with titre reduction ≤2-fold			3	3		1		1	3		1	
No with titre reduction =4-fold					1		2	2			2	
No with titre reduction ≥8-fold					2		1			3		
Subclade 3C.2a3 viruses												
Number of viruses tested*			1	1	1	1*	1	1	1*	1	1	
No with titre reduction ≤2-fold			1			1		1	1		1	
No with titre reduction =4-fold				1	1		1			1		
No with titre reduction ≥8-fold												
					Vaccine SH 2018 NH 2018-19						Vaccine SH 2019	Vaccine NH 2019-20

* Homologous HI titres not available - only results for viruses yielding HI titres of ≥160 with the respective antisera are shown

Reference virus results are taken from individual tables as examples. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Table 9-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-07-08

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus	
				A/Norway 3275/2018		A/Norway 3275/2018		A/La Rioja 2202/18		A/Switzerland 8060/17		A/Kansas 14/2017		NYMC X-327 A/Kansas/14			
				SIAT F03/19 3C.2a1b		Egg F21/19 3C.2a1b		SIAT F26/18 3C.2a1b		Egg F27/18 3C.2a2		SIAT F17/19 3C.3a		Egg F16/19 3C.3a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																	
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	80	119	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/Al3)	80	80	1280	1050	160	127	160	141	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO), N171K, L194P, T203I, H311Q	
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	56	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E6(Am2Al4)c57 10 ⁻⁵	160	132	2560	2418	160	221	5120	4177	40	47	80	61	N96S, T131K, R142K, T160K(-CHO), L194P, R261Q	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	40	30	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	40	20	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N, F193S, S219Y, H311Q, K326R	
TEST VIRUSES																	
A/Greece/1073-19/2019	No sequence	2019-02-22	SIAT1	640	800	80	68	160	160	160	134	<	1	<	0	no sequence	
A/Greece/1381-19/2019	3C.2a1b	2019-03-08	SIAT1	160	160	80	69	320	461	40	20	<	10	<	10	G5R, E62G, Q75H, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Greece/1392-19/2019	3C.2a1b	2019-03-12	MDCK2	320	227	40	56	640	501	<	10	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Greece/1419-19/2019	3C.2a1b	2019-03-12	SIAT1	1280	1431	160	126	1280	1244	160	133	80	61	80	114	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
										Vaccine SH 2019				Vaccine NH 2019-20			

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-07-11

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus
				A/Norway 3275/2018		A/La Rioja 2202/18		A/Switzerland 8060/17		A/Kansas 14/2017		NYMC X-327 A/Kansas/14				
				SIAT F03/19 3C.2a1b		SIAT F26/18 3C.2a1b		Egg F27/18 3C.2a2		SIAT F17/19 3C.3a		Egg F16/19 3C.3a				
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	87	80	119	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q		
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	320	393	40	56	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q		
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E6(Am2A14)c57 10 ⁻⁵	160	132	160	221	5120	4177	40	47	80	61	N96S, T131K, R142K, T160K(-CHO), L194P, R261Q		
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	40	33	40	30	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R		
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	80	70	40	20	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N, F193S, S219Y, H311Q, K326R		
TEST VIRUSES																
A/Iran/168206/2019	3C.2a1b	2019-02-19	SIAT1/SIAT1	320	360	640	689	40	47	40	40	40	40	E62G, K92R, N121K, T135K(-CHO), R142G, N171K, H311Q		
A/Latvia/02-053853/2019	3C.2a1b	2019-02-18	MDCK2/SIAT1	80	108	320	295	<	5	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, T160X(-CHO), N171K, K310R, H311Q		
A/Algeria/55/2019	3C.2a1b	2019-02-20	SIAT1	640	480	640	707	40	20	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, A163S, N171K, S198P, V309I, H311Q		
A/Bulgaria/1153/2019	3C.2a1b	2019-02-21	SIAT2/SIAT1	80	99	160	142	40	40	40	20	40	53	E62G, K92R, N121K, T128A(-CHO),T135K(-CHO), R142G, T160K(-CHO), N171K, S198P, H311Q		
A/S. Peteresburg/RII-239/2019	3C.2a1b	2019-02-22	SIAT1/SIAT1	160	208	320	387	<	0	<	0	<	1	E62G, K92R, N121K, T128A(-CHO),T135K(-CHO), R142G, N171K, S198P, H311Q		
A/Latvia/02-073849/2019	3C.2a1b	2019-02-25	MDCKx/SIAT1	160	233	320	462	<	10	<	1	<	10	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q		
A/Latvia/02-073821/2019	3C.2a1b	2019-02-25	MDCK1/SIAT1	160	131	320	340	40	20	<	10	<	3	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, T160X(-CHO), N171K, H311Q		
A/Khmelnyskiy/145/2019	3C.2a1b	2019-02-27	SIAT1/SIAT1	160	160	320	320	<	1	<	1	<	3	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q, L316M		
A/Iran/168359/2019	3C.2a1b	2019-03-11	SIAT1/SIAT1	320	250	640	701	<	0	<	0	<	0	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q		
A/Bulgaria/1175/2019	3C.2a1b	2019-02-25	SIAT2/SIAT1	640	853	160	187	160	122	40	40	40	40	E62G, K92R, N121K, T131K, R142G, N171K, V182X, Q197R, S219F, H311Q		
A/Austria/1129536/2019	3C.2a1b	2019-02-27	SIAT1/SIAT1	640	835	160	144	160	122	<	0	<	0	E62G, K92R, N121K, T131K, R142G, N171K, S219F, G275V, H311Q		
A/Iran/168336/2019	3C.2a1b	2019-03-07	SIAT1/SIAT1	640	671	80	99	160	129	40	40	40	40	E62G, K83E, K92R, N121K, T131K, R142G, N171K, H311Q		
A/S. Peteresburg/RII-250/2019	3C.2a1b	2019-03-11	SIAT1/SIAT1	1280	1040	160	151	160	176	<	1	<	5	V20X, E62G, K92R, N121K, T131K, R142G, N171K, H311Q		
A/S. Peteresburg/RII-252/2019	3C.2a1b	2019-03-12	SIAT1/SIAT1	1280	1156	320	400	160	220	40	50	40	40	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q		
A/Iran/171909/2019	3C.2a1b	2019-04-12	SIAT1/SIAT1	640	830	160	192	160	149	<	40	<	20	E62G, N81D, K92R, N121K, T131K, R142G, N171K, S219F, H311Q		
A/Bulgaria/1206/2019	3C.2a2	2019-02-19	SIAT2/SIAT1	2560	2793	640	566	5120	4004	160	120	80	73	R33Q, T131K, R142K, A212T, R261Q, D291E		
A/Bulgaria/1208/2019	3C.2a2	2019-02-24	SIAT2/SIAT1	1280	1684	160	177	2560	3044	<	0	<	0	R33Q, T131K, R142K, A212T, R261Q, D291E		
A/Latvia/03-012433/2019	3C.3a	2019-03-04	MDCKx/SIAT1	40	40	40	48	40	40	320	251	320	261	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R		
								Vaccine SH 2019				Vaccine NH 2019-20				

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-07-15

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus	
				A/Norway 3275/2018		A/La Rioja 2202/18		A/Switzerland 8060/17		A/Kansas 14/2017		NYMC X-327 A/Kansas/14			
				SIAT F03/19		SIAT F26/18		Egg F27/18		SIAT F17/19		Egg F16/19			
				3C.2a1b		3C.2a1b		3C.2a2		3C.3a		3C.3a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES															
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	87	80	119	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/La Rioja/2202/2018	3C2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	320	393	40	56	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Switzerland/8060/2017	3C2a2	2017-12-12	E6(Am2AI4)c57 10 ⁻⁵	160	132	160	221	5120	4177	40	47	80	61	N96S, T131K, R142K, T160K(-CHO), L194P, R261Q	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	40	33	40	30	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	80	70	40	20	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N, F193S, S219Y, H311Q, K326R	
TEST VIRUSES															
A/Slovenia/1547/2019	No Sequence	2019-03-13	MDCKx/SIAT2	160	176	320	334	<	3	<	3	<	1	no sequence	
A/Niedersachsen/82/2019	3C.2a1b	2019-02-26	C1/SIAT1	160	231	320	389	40	40	<	10	40	40	E62G, K92R, A106V, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, I267V, H311Q	
A/Switzerland/6543/2019	3C.2a1b	2019-02-26	SIAT1	160	147	320	312	40	20	<	10	40	20	E62G, G78S, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Slovenia/1396/2019	3C.2a1b	2019-03-02	Sx/SIAT1	160	207	320	486	<	10	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Baden-Wuerttemberg/132/2019	3C.2a1b	2019-03-06	C1/SIAT1	320	336	640	902	40	40	<	5	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Switzerland/6968/2019	3C.2a1b	2019-03-07	SIAT1	160	168	640	524	<	10	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Slovenia/1480/2019	3C.2a1b	2019-03-09	Sx/SIAT1	160	167	320	337	40	20	<	5	<	10	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Switzerland/6764/2019	3C.2a1b	2019-03-26	SIAT1	80	76	80	76	<	3	40	20	<	10	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q	
A/Switzerland/0424/2019	3C.2a1b	2019-02-21	SIAT1	640	754	160	182	80	117	40	40	40	40	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Serbia/5114/2019	3C.2a1b	2019-03-01	SIAT1	640	607	160	160	160	133	<	20	40	40	K2E, E62G, N63K(-CHO), K92R, N121K, T131K, R142G, N171K, H311Q	
A/Serbia/5270/2019	3C.2a1b	2019-03-12	SIAT1	1280	993	320	251	160	143	40	20	<	0	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, K207R, S219F, H311Q	
A/Serbia/5238/2019	3C.2a2	2019-03-09	SIAT1	640	901	160	129	1280	1457	80	63	80	76	N122T(-CHO), T131K, R142K, R261Q	
A/Berlin/37/2019	3C.3a	2019-02-21	C1/SIAT1	<	3	40	40	40	20	320	407	320	250	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R	
A/Nordrhein-Westfalen/82/2019	3C.3a	2019-03-11	C1/SIAT1	40	20	40	40	40	40	320	331	320	320	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
								Vaccine SH 2019				Vaccine NH 2019-20		Virus with S137F, A138S, F193S substitutions	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-07-18

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b		A/Norway 3275/2018 Egg F21/19 3C.2a1b		A/La Rioja 2202/18 SIAT F26/18 3C.2a1b		A/Kansas 14/2017 SIAT F17/19 3C.3a		NYMC X-327 A/Kansas/14 Egg F16/19 3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/AI3)	80	80	1280	1050	160	127	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO) , N171K, L194P , T203I , H311Q
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V , D190N , F193S, S219Y , H311Q, K326R
TEST VIRUSES														
A/Sachsen/46/2019	No Sequence	2019-02-25	C1/SIAT1	1280	1426	80	113	160	153	40	47	40	47	no sequence
A/Boras/5/2019	No Sequence	2019-03-24	MDCK0/SIAT1	1280	1862	320	280	320	389	80	66	40	58	no sequence
A/Thuringen/39/2019	3C.2a1b	2019-02-18	C1/SIAT1	640	560	80	80	640	800	<	<	<	<	E62G, K92R, N121K, T128A(-CHO), T131S, T135K(-CHO), R142G, N171K, H311Q
A/Georgia/613/2019	3C.2a1b	2019-02-23	SIAT1	320	304	80	74	640	569	40	40	40	20	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Poland/3029/2019	3C.2a1b	2019-02-25	SIAT1	320	301	160	128	320	469	<	<	<	10	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Moldova/617.09/2019	3C.2a1b	2019-02-26	C1/SIAT1	640	594	160	178	640	516	40	54	40	54	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Denmark/2791/2019	3C.2a1b	2019-03-14	SIAT2/SIAT1	640	579	160	139	1280	1404	40	68	40	40	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Georgia/670/2019	3C.2a1b	2019-03-18	SIAT1	160	207	80	69	320	288	<	10	<	10	N8K(-CHO), E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Schleswig-Holstein/3/2019	3C.2a1b	2019-02-21	C1/SIAT1	640	777	80	66	80	90	<	<	<	<	E62G, K83E, K92R, N121K, T131K, R142G, N171K, H311Q
A/West Kazakhstan/461/2019	3C.2a1b	2019-02-27	MDCKx/SIAT1	320	411	80	90	80	103	40	40	40	40	S54R, E62G, K92R, N121K, T131K, R142G, N158X(-CHO) , N171K, S219F, H311Q
A/West Kazakhstan/462/2019	3C.2a1b	2019-03-01	MDCKx/SIAT1	160	216	40	57	80	61	40	20	<	10	S54R, E62G, K92R, N121K, T131K, R142G, T160X(-CHO) , N171K, S219F, H311Q
A/West Kazakhstan/463/2019	3C.2a1b	2019-03-05	MDCKx/SIAT1	160	158	40	20	40	56	<	10	<	5	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Denmark/2592/2019	3C.2a1b	2019-03-07	SIAT2/SIAT1	640	785	160	143	160	174	40	40	<	10	E62G, N121K, T131K, R142G, N171K, S219F, H311Q
A/Denmark/2643/2019	3C.2a1b	2019-03-09	SIAT2/SIAT1	1280	1222	80	77	160	155	40	42	40	20	E62G, K92R, N121K, T131K, R142G, N171K, I214T, S219F, H311Q
A/Denmark/2790/2019	3C.2a1b	2019-03-14	SIAT2/SIAT1	1280	1257	80	84	320	346	40	43	40	20	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, S219F, H311Q
A/Boras/6/2019	3C.2a1b	2019-03-24	MDCK0/SIAT1	320	404	80	66	80	72	40	20	40	20	D53N, E62G, K92R, N121K, T131K, R142G, T160X(-CHO) , N171K, S219F, H311Q
A/Denmark/2743/2019	3C.2a2	2019-03-15	SIAT2/SIAT1	2560	2406	160	201	640	533	<	<	<	<	Q57L, E62K, S91G, N121K, T128A(-CHO), T135K(-CHO), R142K, A212T, R261Q
A/Nordrhein-Westfalen/60/2019	3C.3a	2019-02-18	C1/SIAT1	<	<	<	<	<	5	320	244	160	164	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Lisboa/91/2019	3C.3a	2019-02-20	SIAT1/SIAT1	<	10	<	3	<	5	320	301	320	307	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Baden-Wuerttemberg/87/2019	3C.3a	2019-02-25	C1/SIAT1	40	20	<	10	40	40	320	366	320	293	S91N, N121S, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Lisboa/97/2019	3C.3a	2019-02-25	SIAT2/SIAT1	<	3	<	<	<	10	320	403	160	236	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, R261Q, H311Q
A/Lisboa/94/2019	3C.3a	2019-02-26	SIAT1/SIAT1	<	0	<	3	40	20	320	248	160	237	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Lisboa/95/2019	3C.3a	2019-02-27	SIAT2/SIAT1	<	0	<	0	<	10	320	303	160	155	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Finland/114/2019	3C.3a	2019-04-30	SIAT1/SIAT1	<	1	<	0	40	20	320	400	160	213	N8D(-CHO), S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), P169S, F193S, H311Q, K326R
														Vaccine NH 2019-20

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-07-22

Viruses				Neutralisation Titre ¹										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus
				Post-infection ferret antisera										
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b		A/Norway 3275/2018 Egg F21/19 3C.2a1b		A/La Rioja 2202/18 SIAT F26/18 3C.2a1b		A/Kansas 14/2017 SIAT F17/19 3C.3a		NYMC X-327 A/Kansas/14 Egg F16/19 3C.3a		
Passage history Ferret number Genetic group				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/AI3)	80	80	1280	1050	160	127	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO), N171K, L194P, T203I, H311Q
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N, F193S, S219Y, H311Q, K326R
TEST VIRUSES														
A/Switzerland/0009/2019	3C.2a1b	2019-02-20	SIAT1	160	130	40	59	160	209	<	1	<	1	E62G, K92R, A106V, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, L177X, S198P, H311Q
A/Switzerland/0472/2019	3C.2a1b	2019-02-20	SIAT1	160	151	40	56	320	257	<	5	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Switzerland/6997/2019	3C.2a1b	2019-03-12	SIAT1	160	147	80	75	160	226	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Greece/1456-19/2019	3C.2a1b	2019-03-15	SIAT1	160	195	80	63	160	235	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Hessen/56/2019	3C.2a1b	2019-03-20	C2/SIAT1	160	146	40	57	160	232	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Switzerland/5740/2019	3C.2a1b	2019-03-08	SIAT1	640	570	40	55	160	133	<	3	<	0	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Hessen/55/2019	3C.2a1b	2019-03-17	C1/SIAT1	320	463	40	40	80	61	<	0	<	10	E62G, K83E, K92R, N121K, T131K, R142G, N171K, K276N, H311Q
A/Niedersachsen/135/2019	3C.2a1b	2019-03-18	C1/SIAT1	640	501	80	72	80	114	40	40	40	20	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, S219F, H311Q
A/Bayern/96/2019	3C.2a1b	2019-03-19	C1/SIAT1	640	542	40	59	80	76	<	0	<	0	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Sachsen/92/2019	3C.2a1b	2019-03-19	C1/SIAT1	320	375	80	64	80	65	40	20	40	20	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Switzerland/2512/2019	3C.2a1b	2019-03-27	SIAT1	640	827	40	40	160	136	40	20	<	5	E62G, K92R, N121K, T131K, R142G, A163X, N171K, Q197R, S219F, H311Q
A/Switzerland/0291/2019	3C.2a1b	2019-03-29	SIAT1	640	607	40	45	160	122	<	10	<	0	E62G, K92R, N121K, T131K, R142G, N171K, S219F
A/Bremen/12/2019	3C.3a	2019-02-18	C1/SIAT1	<	10	<	3	40	20	160	126	160	152	P21L, S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R
A/Hessen/34/2019	3C.3a	2019-02-18	C1/SIAT1	40	20	<	5	40	40	320	256	320	244	S91N, A106T, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Nordrhein-Westfalen/97/2019	3C.3a	2019-03-14	C1/SIAT1	<	3	<	0	<	0	160	156	160	203	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R
A/Nordrhein-Westfalen/89/2019	3C.3a	2019-03-15	C1/SIAT1	<	0	<	0	<	1	160	179	160	204	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Niedersachsen/151/2019	3C.3a	2019-03-25	C1/SIAT1	<	10	<	5	<	10	160	134	160	150	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R
A/Bremen/20/2019	3C.3a	2019-03-25	C1/SIAT1	<	10	<	1	<	10	160	138	160	141	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R
A/Nordrhein-Westfalen/101/2019	3C.3a	2019-03-25	C1/SIAT1	40	20	40	20	40	40	160	176	160	141	P21L, S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, I267V, H311Q, K326R

Vaccine
NH 2019-20

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-08-01

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus
				A/Norway 3275/2018		A/Norway 3275/2018		A/La Rioja 2202/18		A/Kansas 14/2017		NYMC X-327 A/Kansas/14		
				SIAT F03/19 3C.2a1b		Egg F21/19 3C.2a1b		SIAT F26/18 3C.2a1b		SIAT F17/19 3C.3a		Egg F16/19 3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/Al3)	80	80	1280	1050	160	127	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO) , N171K, L194P , T203I , H311Q
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V , D190N , F193S, S219Y , H311Q, K326R
TEST VIRUSES														
A/Vitebsk/927/2019	3C.2a1b	2019-02-28	MDCK2/SIAT1	160	145	80	61	160	227	40	20	40	20	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, R261Q, H311Q
A/Macedonia/1167/2019	3C.2a1b	2019-03-01	SIAT1	160	150	80	69	320	320	40	20	<	10	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Borisov/1246/2019	3C.2a1b	2019-03-04	MDCK2/SIAT1	160	146	80	75	160	211	40	40	40	50	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198Q, H311Q
A/Orsha/1450/2019	3C.2a1b	2019-03-05	MDCK2/SIAT1	160	178	80	80	320	284	40	20	40	20	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Macedonia/1365/2019	3C.2a1b	2019-03-19	SIAT1	320	320	80	116	640	576	<	10	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, D188X, H311Q
A/Macedonia/1424/2019	3C.2a1b	2019-04-01	SIAT1	160	198	80	89	320	397	40	20	<	5	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Hong Kong/2679/2019	3C.2a1b	2019-06-18	MDCK1/SIAT1	160	142	80	79	80	118	40	59	40	20	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Hong Kong/2675/2019	3C.2a1b	2019-06-18	MDCK1/SIAT1	80	107	80	61	80	73	40	20	<	3	E62G, S91N, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Hong Kong/2669/2019	3C.2a1b	2019-06-18	MDCK1/SIAT1	80	94	80	72	80	85	40	43	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Macedonia/1222/2019	3C.2a1b	2019-03-06	SIAT1	640	800	40	44	80	112	<	3	<	1	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q
A/Bobruisk/1469/2019	3C.2a1b	2019-03-13	MDCK3/SIAT1	320	247	80	71	80	101	40	40	40	46	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Kosovo/1159/2019	3C.2a1b	2019-04-03	SIAT1	640	631	160	148	160	172	40	47	40	40	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Macedonia/1468/2019	3C.2a1b	2019-04-22	SIAT1	640	711	80	119	320	298	40	40	<	<	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q, T328X
A/Hong Kong/2678/2019	3C.2a1b	2019-06-18	MDCK1/SIAT1	320	356	80	114	320	468	40	41	40	40	E62G, K92R, N121K, T131K, T135K(-CHO), R142G, N171K, H311Q
A/Macedonia/1350/2019	3C.3a	2019-03-18	SIAT1	40	20	40	20	40	47	320	289	160	216	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q
A/Macedonia/1402/2019	3C.3a	2019-03-22	SIAT1	40	40	40	20	40	53	320	314	320	309	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
A/Kosovo/1130/2019	3C.3a	2019-03-29	SIAT1	<	10	<	5	40	40	320	255	160	156	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, I267V, H311Q, K326R
A/Kosovo/1188/2019	3C.3a	2019-04-08	SIAT1	80	65	40	40	80	61	320	304	320	315	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, I267V, H311Q, K326R
A/Macedonia/1462/2019	3C.3a	2019-04-15	SIAT1	<	5	<	0	<	10	320	255	320	249	A69S, S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
												Vaccine NH 2019-20	Virus with S137F, A138S, F193S substitutions	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-7 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-08-08

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus
				A/Norway 3275/2018		A/Norway 3275/2018		A/La Rioja 2202/18		A/Kansas 14/2017		NYMC X-327 A/Kansas/14		
				SIAT F03/19 3C.2a1b		Egg F21/19 3C.2a1b		SIAT F26/18 3C.2a1b		SIAT F17/19 3C.3a		Egg F16/19 3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/AI3)	80	80	1280	1050	160	127	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO) , N171K, L194P, T203I , H311Q
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N , F193S, S219Y , H311Q, K326R
TEST VIRUSES														
A/Iceland/23/2019	No sequence	2019-03-18	MDCK1/SIAT1	160	198	80	75	80	115	40	57	40	58	no sequence
A/Iceland/70/2019	No sequence	2019-05-19	MDCK1/SIAT1	640	485	80	112	160	148	40	40	40	40	no sequence
A/Switzerland/0029/2019	3C.2a1b	2019-02-19	SIAT1	80	83	40	54	80	99	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Jordan/20101/2019	3C.2a1b	2019-02-19	SIAT1	320	381	160	234	320	301	40	59	80	70	G5E, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, K264E, H311Q
A/Jordan/30060/2019	3C.2a1b	2019-02-26	SIAT1	320	320	320	252	640	486	80	78	80	89	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Iceland/4/2019	3C.2a1b	2019-03-04	MDCK1/SIAT1	40	20	<	0	40	40	<	0	<	0	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, T160K(-CHO) , N171K, S198P, H311Q
A/Bratislava/446/2019	3C.2a1b	2019-03-08	MDCKx/SIAT1	80	102	80	70	80	110	40	40	40	59	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N158X(-CHO) , N171K, H311Q
A/Jordan/10624/2019	3C.2a1b	2019-04-04	SIAT1	160	230	160	182	320	243	40	40	80	64	E62G, K92R, N121K, T128A(-CHO), V130I, T135K(-CHO), R142G, N171K, S198P, H311Q
A/Iceland/54/2019	3C.2a1b	2019-04-12	MDCK1/SIAT1	320	294	160	147	320	287	80	74	80	66	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S262N, H311Q
A/Macedonia/1472/2019	3C.2a1b	2019-04-23	SIAT1	320	348	160	143	320	345	<	0	<	0	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Iceland/71/2019	3C.2a1b	2019-05-19	MDCK1/SIAT1	40	40	<	10	40	45	<	10	<	10	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, T160K(-CHO) , N171K, S198P, H311Q
A/Hong Kong/2671/2019	3C.2a1b	2019-06-17	MDCK1/SIAT1	80	78	80	82	160	126	40	59	40	40	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Iceland/61/2019	3C.2a1b	2019-04-20	MDCK1/SIAT1	160	230	40	40	80	93	<	5	<	0	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Hong Kong/2672/2019	3C.2a1b	2019-06-17	MDCK1/SIAT1	640	615	80	118	160	213	80	98	80	99	E62G, N81D, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Hong Kong/2670/2019	3C.2a1b	2019-06-17	MDCK1/SIAT1	160	189	160	136	160	150	80	97	80	87	E62G, K92R, N121K, T131K, R142G, N171K, F193S, I214T, H311Q
A/Hong Kong/2668/2019	3C.2a1b	2019-06-17	MDCK1/SIAT1	1280	1074	80	117	320	244	40	55	40	56	D53N, E62G, N81D, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Hong Kong/2673/2019	3C.2a1b	2019-06-18	MDCK1/SIAT1	640	768	160	207	320	312	80	103	160	126	E62G, G78S, K92R, N121K, T131K, R142G, N171K, S219F, H311Q
A/Lisboa/98/2019	3C.3a	2019-03-08	SIAT1/SIAT1	<	5	<	3	<	10	40	50	40	55	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q
A/Iceland/59/2019	3C.3a	2019-04-15	MDCK1/SIAT1	<	10	<	0	<	3	160	150	160	187	G5E, N8S(-CHO), R33Q, S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Jordan/30121/2019	3C.3a	2019-04-17	SIAT1	40	40	40	20	40	44	160	217	160	130	S91N, T128A(-CHO), T135K(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
												Vaccine NH 2019-20	Virus with S137F, A138S, F193S substitutions	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-8 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-09-05

				Neutralisation Titre ¹										HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus	
				Post-infection ferret antisera											
				A/Norway 3275/2018		A/Norway 3275/2018		A/La Rioja 2202/18		A/Kansas 14/2017		NYMC X-327 A/Kansas/14			
				SIAT F03/19		Egg F21/19		SIAT F26/18		SIAT F17/19		Egg F16/19			
				3C.2a1b		3C.2a1b		3C.2a1b		3C.3a		3C.3a			
Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES															
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/AI3)	80	80	1280	1050	160	127	<	5	<	10	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO) , N171K, L194P, T203I , H311Q	
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V, D190N , F193S, S219Y , H311Q, K326R	
TEST VIRUSES															
A/Dijon/1974/2019	3C.2a1b	2019-04-24	MDCK1/SIAT1	160	160	160	127	160	235	80	80	80	68	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137L, A138S, R142G, N171K, F193S, H311Q	
A/South Africa/R06219/2019	3C.2a1b	2019-05-31	SV/SIAT1	80	102	40	40	80	80	40	20	40	48	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, T160K(-CHO) , N171K, H311Q	
A/Pays de Loire/1864/2019	3C.2a1b	2019-04-01	MDCK2/SIAT1	160	226	160	136	160	156	40	56	80	99	E62G, F79L, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Palermo/20/2019	3C.2a1b	2019-03-06	SIAT4/SIAT1	640	491	160	186	320	455	40	42	80	106	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Torino/780/2019	3C.2a1b	2019-03-06	SIAT3/SIAT1	640	618	320	258	320	366	40	58	80	78	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Tomsk/156/2019	3C.2a1b	2019-03-11	SIAT2/SIAT1	640	623	320	253	640	510	80	111	80	77	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q	
A/Moscow/146/2019	3C.2a1b	2019-03-10	SIAT1/SIAT1	640	567	160	206	320	384	80	92	80	72	D53N, E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q	
A/Finland/61/2019	3C.2a1b	2019-03-06	SIAT1/SIAT1	1280	1138	160	157	320	266	80	66	40	40	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Dijon/2017/2019	3C.2a1b	2019-04-21	MDCK2/SIAT1	640	860	160	127	160	212	<	<	<	<	E62G, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Champagne Ardenne/1908/2019	3C.2a1b	2019-04-10	MDCK2/SIAT1	640	486	80	80	160	209	40	40	40	40	E62G, K92R, N121K, T131K, R142G, R150K, T160A(-CHO) , N171K, Q197R, K207R, S219F, H311Q	
A/South Africa/R06432/2019	3C.2a1b	2019-06-05	MDCK1/SIAT1	1280	1387	160	209	320	305	80	103	80	82	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, K207R, S219F, H311Q	
A/South Africa/R06421/2019	3C.2a1b	2019-06-05	MDCK1/SIAT1	640	947	160	181	320	288	80	72	80	79	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, S219F, H311Q	
A/South Africa/R05937/2019	3C.2a1b	2019-05-28	MDCK1/SIAT1	1280	960	160	187	320	309	80	72	80	62	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, S219F, H311Q	
A/Moscow/152/2019	3C.2a1b	2019-03-30	SIAT1/SIAT1	1280	1189	160	234	320	405	80	73	80	92	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Cheboksary/154/2019	3C.2a1b	2019-03-18	SIAT2/SIAT1	2560	2146	320	384	320	460	160	196	160	195	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Vladivostok/5/2019	3C.2a1b	2019-03-14	SIAT1/SIAT1	1280	1902	320	377	640	610	160	123	160	166	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Brest/2002/2019	3C.3a	2019-04-11	MDCK1/SIAT1	40	40	40	40	<	3	320	199	320	290	S91N, T128A(-CHO), A138S, R142G, S144K, H156Q, L157S, Y159S, T160K(-CHO), F193S, H311Q, S312G, K326R	
A/Bretagne/1913/2019	3C.3a	2019-04-11	MDCK1/SIAT1	40	54	40	40	40	51	160	232	320	320	S91N, T128A(-CHO), A138S, R142G, S144K, H156Q, L157S, Y159S, T160K(-CHO), F193S, H311Q, S312G, K326R	
A/South Africa/R06917/2019	3C.3a	2019-06-10	MDCK1/SIAT1	<	10	<	0	<	0	320	242	160	138	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
A/South Africa/R05121/2019	3C.3a	2019-05-14	MDCK1/SIAT1	<	10	<	5	<	10	320	315	320	240	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
A/Czech Republic/1275/2019	3C.3a	2019-04-10	MDCK3/SIAT1	<	0	<	0	<	3	320	373	160	217	H56Y, K82R, S91N, N121D, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, A304E, H311Q, K326R	
A/Picardie/1898/2019	3C.3a	2019-04-08	MDCK1/SIAT1	160	124	80	71	80	61	320	379	320	331	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), Y161F, F193S, H311Q, K326R	
A/Palermo/13/2019	3C.3a	2019-03-05	SIAT3/SIAT1	40	50	40	20	40	40	320	320	320	248	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R	
												Vaccine NH 2019-20	Virus with S137F, A138S, F193S substitutions		

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-09-09

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹												HA1 substitutions for 3C.2a/3a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus		
				Post-infection ferret antisera														
				A/Norway 3275/2018		A/Norway 3275/2018		A/La Rioja 2202/18		A/Kansas 14/2017		NYMC X-327 A/Kansas/14		A/Christchurch 515/2019			A/Switzerland 8060/17	
				SIAT F03/19	3C.2a1b	Egg F21/19	3C.2a1b	SIAT F26/18	3C.2a1b	SIAT F17/19	3C.3a	Egg F16/19	3C.3a	Egg F32/19	3C.2a1b		Egg F27/18	3C.2a2
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT2 10 ⁻³	640	596	80	61	80	87	<	16	<	13	320	302	80	119	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	E6 (Am3/AI3)	80	80	1280	1050	160	127	<	5	<	10	40	53	160	141	E62G, K92R, A106V, N121K, T131K, R142G, T160K(-CHO) , N171K, L194P , T203I , H311Q
A/La Rioja/2202/2018	3C.2a1b	2018-04-02	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	20	40	28	320	469	40	56	E62G, K92R, N121K, T128A, T135K, R142G, N171K, H311Q
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	320	288	160	160	<	0	40	30	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	46	40	30	80	70	320	285	2560	2008	<	0	40	20	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), G186V , D190N , F193S, S219Y , H311Q, K326R
A/Christchurch/515/2019	3C.2a1b	2019-04-03	E3/E1	640	600	1280	1321	640	564	40	20	<	10	1280	1760	ND	ND	E62G, K92R, N121K, T131K, R142G, N171K, L194P , Q197R, S219F, H311Q
TEST VIRUSES																		
A/Parma/379/2019	3C.2a1b	18/03/2019	SIAT3/SIAT1	640	549	160	135	320	278	40	55	80	87	640	540	80	80	E62G, V88I, K92R, N121K, T128A(-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Hong Kong/2675/2019	3C.2a1b	18/06/2019	MDCK1/SIAT1	80	107	80	61	80	73	40	20	<	3	320	332	ND	ND	E62G, S91N, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Hong Kong/2669/2019	3C.2a1b	18/06/2019	MDCK1/SIAT1	80	94	80	72	80	85	40	43	<	0	320	369	ND	ND	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/Hong Kong/2671/2019	3C.2a1b	17/06/2019	MDCK1/SIAT1	80	78	80	82	160	126	40	59	40	40	320	282	ND	ND	E62G, K92R, N121K, T128A(-CHO), T135K(-CHO), S137F, A138S, R142G, N171K, F193S, H311Q
A/South Africa/R05114/2019	3C.2a1b	13/05/2019	MDCK2/SIAT1	1280	985	160	197	640	480	80	88	40	52	640	560	160	186	V36X, E62G, K92R, N121K, T131K, R142G, N158X(-CHO) , N171K, Q197R, K207X, I217X, S219F, H311Q
A/Bosnia and Herzegovia/351G/2019	3C.2a2	05/03/2019	SIAT1/SIAT1	1280	1187	160	123	160	153	80	66	80	89	640	594	2560	2406	G5X, P21S, K92R, A106V, T131K, R142K, R261Q
A/Bosnia and Herzegovia/335G/2019	3C.2a2	04/03/2019	SIAT1/SIAT1	2560	1920	160	137	160	200	80	118	160	122	1280	968	5120	4332	N122T(-CHO), T131K, R142K, R261Q, P289L
A/South Africa/R06867/2019	3C.3a	10/06/2019	MDCK1/SIAT1	<	0	<	0	<	10	320	255	320	311	<	1	40	40	S91N, T128A(-CHO), A138S, R142G, S144K, Y159S, T160K(-CHO), F193S, H311Q, K326R
A/Brest/1998/2019	3C.3a	07/04/2019	MDCK1/SIAT1	<	3	40	20	40	20	160	221	320	264	40	20	40	20	S91N, T128A(-CHO), A138S, R142G, S144K, H156Q, L157S, Y159S, T160K(-CHO), F193S, P273L, H311Q, S312G, K326R
Repeats												Vaccine		Vaccine				
ND = Not Done												NH 2019-20		SH 2019		Virus with S137F, A138S, F193S substitutions		

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Sequences in phylogenetic trees

Table 9-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Viruses			Passage History		Neutralisation Titre ¹									
					Post-infection ferret antisera									
					A/Norway 3275/2018 SIAT F03/19 3C.2a1b		A/Norway 3275/2018 Egg F21/19 3C.2a1b		A/La Rioja 2202/18 SIAT F26/18 3C.2a1b		A/Switzerland 8060/17 Egg F27/18 3C.2a2		A/Kansas 14/2017 SIAT F17/19 3C.3a	
Passage history Ferret number Genetic group			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read
REFERENCE VIRUSES														
A/Norway/3275/2018	3C.2a1b	SIAT2 10 ⁻³	640	596	80	61	80	87	80	119	<	16	<	13
A/Norway/3275/2018	3C.2a1b	E6 (Am3/AI3)	80	80	1280	1050	160	127	160	141	<	5	<	10
A/La Rioja/2202/2018	3C.2a1b	SIAT1/SIAT1 10 ⁻¹	160	197	40	55	320	393	40	56	40	20	40	28
A/Switzerland/8060/2017	3C2a2	E6(Am2AI4)c57 10 ⁻⁵	160	132	2560	2418	160	221	5120	4177	40	47	80	61
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2 10 ⁻³	<	15	<	10	40	33	40	30	320	288	160	160
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1	40	46	40	30	80	70	40	20	320	285	2560	2008
TEST VIRUSES														
Number of viruses tested			147		115		147		36		147		147	
No with titre reduction ≤2-fold			71				90		2		36			
%			48.3				61.2		5.6		24.5			
No with titre reduction =4-fold			32		6		22				19			
%			21.8		5.2		15.0				12.9			
No with titre reduction ≥8-fold			44		109		35		34		92		147	
%			29.9		94.8		23.8		94.4		62.6		100.0	
Subgroup 3C.2a1b viruses + T135K														
Number of viruses tested			57		41		57		19		57		57	
No with titre reduction ≤2-fold			21				48							
%			36.8				84.2							
No with titre reduction =4-fold			25		3		7				6			
%			43.9		7.3		12.3				10.5			
No with titre reduction ≥8-fold			11		38		2		19		51		57	
%			19.3		92.7		3.5		100.0		89.5		100.0	
Subgroup 3C.2a1b viruses + T131K														
Number of viruses tested			44		35		44		9		44		44	
No with titre reduction ≤2-fold			40				31				2			
%			90.9				70.5				4.5			
No with titre reduction =4-fold			4		2		12				9			
%			9.1		5.7		27.3				20.5			
No with titre reduction ≥8-fold					33		1		9		33		44	
%					94.3		2.2		100.0		75.0		100.0	
Clade 3C.3a viruses														
Number of viruses tested			34		31		34		3		34		34	
No with titre reduction ≤2-fold											33			
%											97.1			
No with titre reduction =4-fold			1				2							
%			2.9				5.9							
No with titre reduction ≥8-fold			33		31		32		3		1		34	
%			97.1		100.0		94.1		100.0		2.9		100.0	
Subclade 3C.2a2 viruses														
Number of viruses tested			6		3		6		3		6		6	
No with titre reduction ≤2-fold			6				6		2		1			
No with titre reduction =4-fold											3			
No with titre reduction ≥8-fold					3				1		2		6	
									Vaccine SH 2019		Vaccine NH 2019-20			

All test viruses had collection dates after 2019-01-31 and 94/98 yielded full-length HA gene sequences

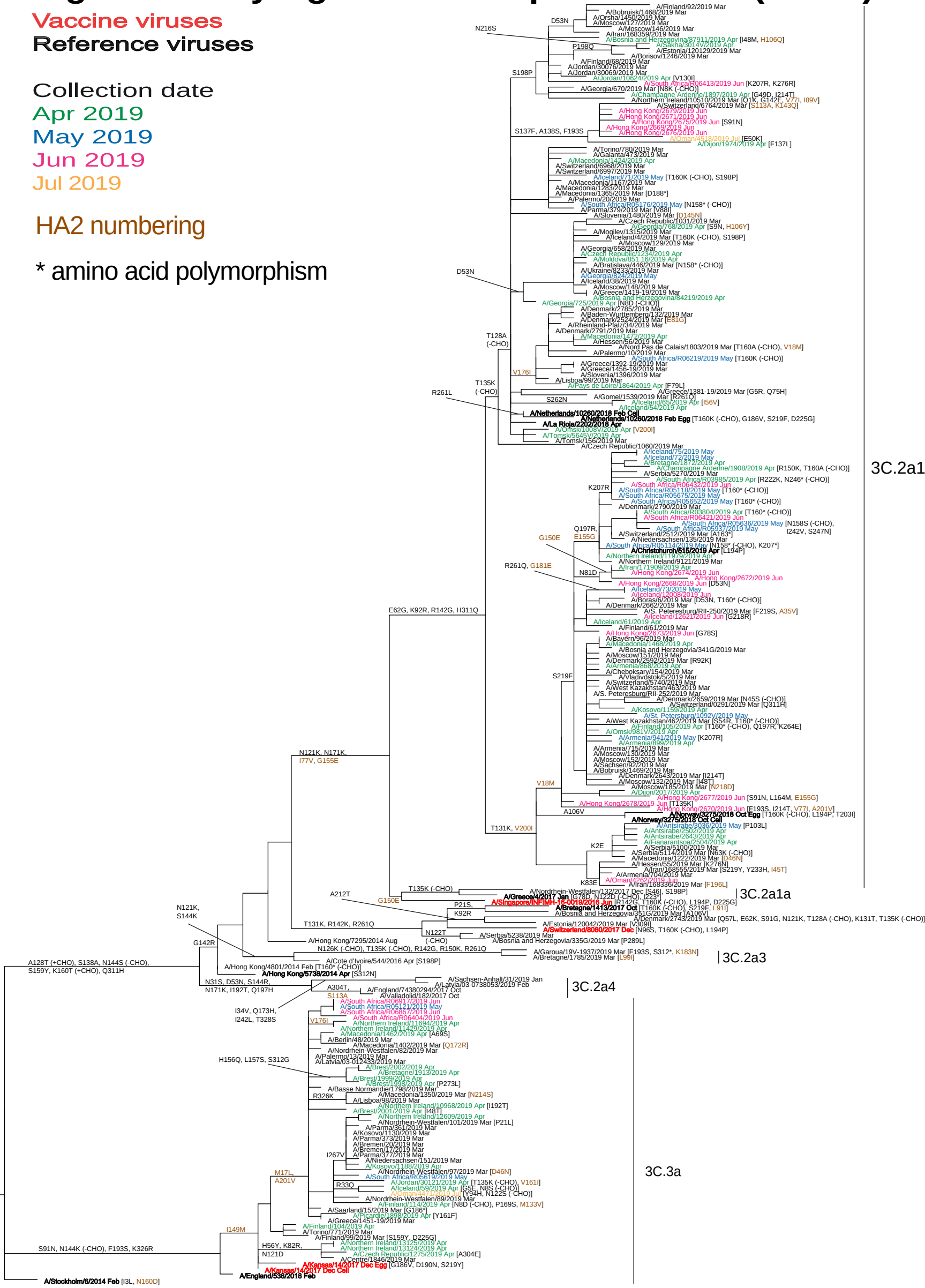
Figure 6. Phylogenetic comparison of A(H3N2) HA genes

Vaccine viruses
Reference viruses

Collection date
Apr 2019
May 2019
Jun 2019
Jul 2019

HA2 numbering

* amino acid polymorphism



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**

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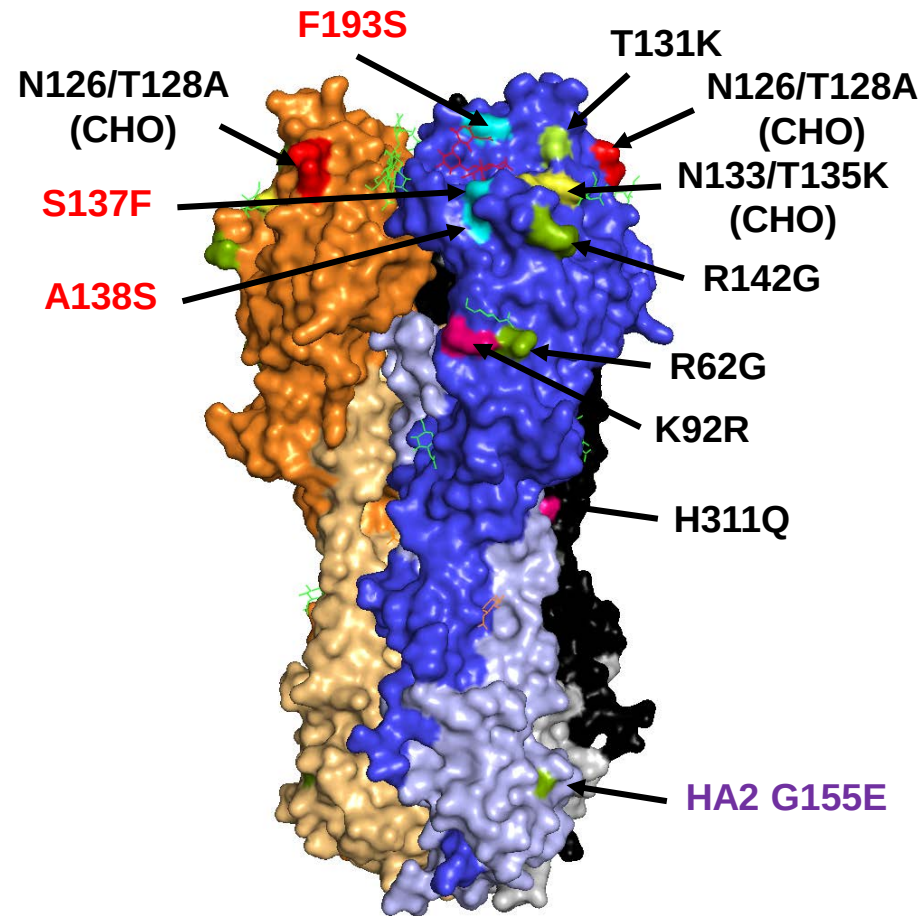
	310	320	330	340	350	360	370	380	390	400	
A/Stockholm/6/2014	TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKFS										3a
A/England/538/2018			R								3a
A/Kansas/14/2017			R								3
A/Northern Ireland/13125/2019			R								3a
A/Oman/4471/2019			R		L						3a
A/Kosovo/1188/2019			R		L						3a
A/Bretagne/1913/2019		G	R		L						3a
A/South Africa/R06867/2019			R		L						3a
A/Hong Kong/5738/2014		HN									2a
A/Switzerland/8060/2017		H									2a2
A/Bretagne/1413/2017		H									2a2
A/Singapore/INFIMH-16-0019/2019		H									2a1a
A/Greece/4/2017		H									2a1a
A/Oman/4262/2019											2a1b
A/Antsirabe/3036/2019											2a1b
A/Norway/3275/2018											2a1b
A/Armenia/941/2019					M						2a1b
A/Iceland/12008/2019					M						2a1b
A/Hong Kong/2672/2019					M						2a1b
A/Christchurch/515/2019					M						2a1b
A/South Africa/R06432/2019					M						2a1b
A/La Rioja/2202/2018											2a1b
A/Iceland/65/2019								V			2a1b
A/South Africa/R06219/2019											2a1b
A/Georgia/824/2019											2a1b
A/Iceland/71/2019											2a1b
A/Dijon/1974/2019											2a1b
A/Oman/4518/2019											2a1b
A/Hong Kong/2675/2019											2a1b
A/South Africa/R06413/2019											2a1b
A/Sakha/3014V/2019											2a1b
A/Finland/92/2019											2a1b

	410	420	430	440	450	460	470	480	490	500	
A/Stockholm/6/2014	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSMNKLFETKTKQLRENAEDMNGCGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF										
A/England/538/2018											
A/Kansas/14/2017								M		N	
A/Northern Ireland/13125/2019								M		N	
A/Oman/4471/2019								M		N	
A/Kosovo/1188/2019								M		N	
A/Bretagne/1913/2019								M		N	
A/South Africa/R06867/2019								M		N	
A/Hong Kong/5738/2014										N	
A/Switzerland/8060/2017										N	
A/Bretagne/1413/2017			I							N	
A/Singapore/INFIMH-16-0019/2019		V						E	E	N	
A/Greece/4/2017		V						E	E	N	
A/Oman/4262/2019		V						E	E	N	
A/Antsirabe/3036/2019		V						E	E	N	
A/Norway/3275/2018		V						E	E	N	
A/Armenia/941/2019		V						E	E	N	
A/Iceland/12008/2019		V						E	E	N	
A/Hong Kong/2672/2019		V						E	E	N	
A/Christchurch/515/2019		V						E	E	N	
A/South Africa/R06432/2019		V								N	
A/La Rioja/2202/2018		V						E	E	N	
A/Iceland/65/2019		V						E	E	N	
A/South Africa/R06219/2019		V						E	E	N	
A/Georgia/824/2019		V						E	E	N	
A/Iceland/71/2019		V						E	E	N	
A/Dijon/1974/2019		V						E	E	N	
A/Oman/4518/2019		V						E	E	N	
A/Hong Kong/2675/2019		V						E	E	N	
A/South Africa/R06413/2019		V						E	E	N	
A/Sakha/3014V/2019		V						E	E	N	
A/Finland/92/2019		V						E	E	N	

	510	520	530	540	550	
A/Stockholm/6/2014	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI					
A/England/538/2018						
A/Kansas/14/2017						
A/Northern Ireland/13125/2019						
A/Oman/4471/2019			V			
A/Kosovo/1188/2019			V			
A/Bretagne/1913/2019			V			
A/South Africa/R06867/2019			V			
A/Hong Kong/5738/2014						
A/Switzerland/8060/2017						
A/Bretagne/1413/2017						
A/Singapore/INFIMH-16-0019/2019						
A/Greece/4/2017						
A/Oman/4262/2019			I			
A/Antsirabe/3036/2019			I			
A/Norway/3275/2018			I			
A/Armenia/941/2019			I			
A/Iceland/12008/2019		E	I			
A/Hong Kong/2672/2019			I			
A/Christchurch/515/2019			I			
A/South Africa/R06432/2019			I			
A/La Rioja/2202/2018						
A/Iceland/65/2019						
A/South Africa/R06219/2019		I				
A/Georgia/824/2019						
A/Iceland/71/2019						
A/Dijon/1974/2019						
A/Oman/4518/2019						
A/Hong Kong/2675/2019						
A/South Africa/R06413/2019						
A/Sakha/3014V/2019						
A/Finland/92/2019						

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

a) Subgroup 3C.2a1b



b) Clade 3C.3a

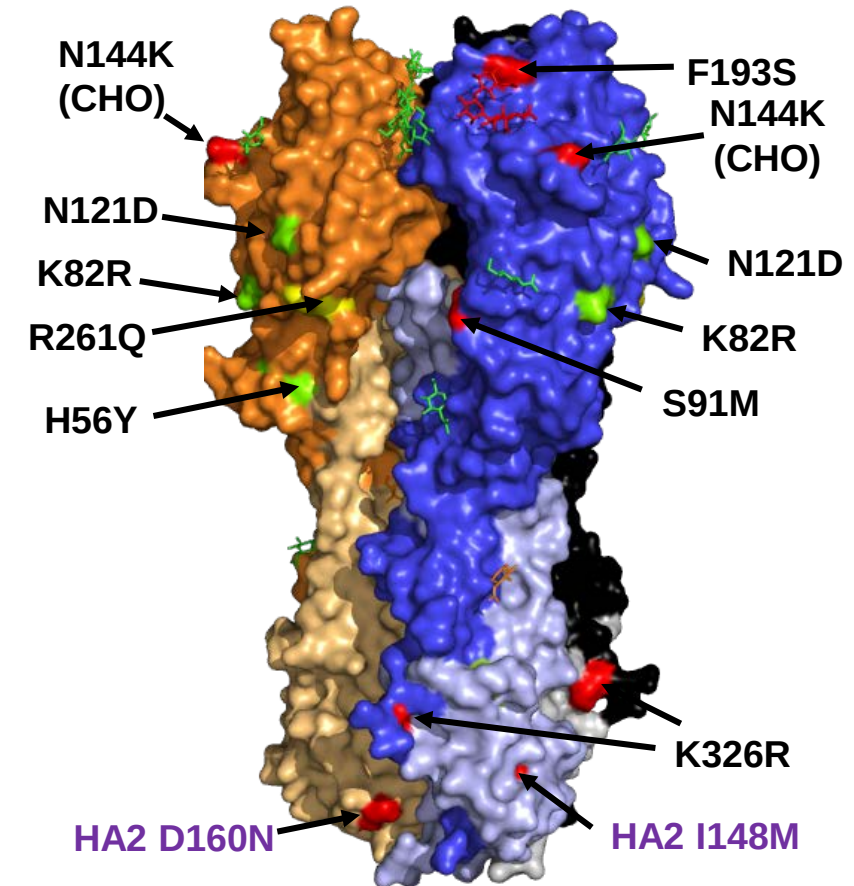


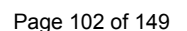
Figure 8. Amino acid substitutions in the HA of (a) recent subgroup 3C.2a1b H3N2 viruses compared with A/Singapore/INFIMH-16-0019/2016 and (b) clade 3C.3a viruses compared with A/Switzerland/9715293/2013

Vaccine viruses

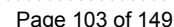
Reference viruses

Apr 2019
May 2019
Jun 2019
Jul 2019

D151*: +CHO if *=N



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400					
A/Stockholm/6/2014	PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNGVWMGRTINET	SR	LG	YET	FK	VIEGWSNPKSKLQ	TNRQ	VI	DR		3a				
A/England/538/2018		T		N		K		D		V	I	3a			
A/Kansas/14/2017		T		N		K		D		V	I	3a			
A/Northern Ireland/13125/2019		T		N		K	V		D		V	M	3a		
A/Oman/4471/2019		T		N		K		E	D		V	S	I	3a	
A/Kosovo/1188/2019		T		N		K		E	D		V	S	I	3a	
A/Bretagne/1913/2019		T		N		K			D		V	S	I	3a	
A/South Africa/R06867/2019		T		N		K			D		V	S	I	3a	
A/Hong Kong/5738/2014														2a	
A/Switzerland/8060/2017		S		N					D		V	S	I	2a2	
A/Bretagne/1413/2017		S	R	N					D		V	S	I	2a2	
A/Singapore/INFIMH-16-0019/2019				N					D		V	S	I	2a1a	
A/Greece/4/2017				N					D		V	S	I	2a1a	
A/Oman/4262/2019		I		R	S	N	K	D			V	S	I	2a1b	
A/Antsirabe/3036/2019		I		R	S	N	K	D			V	S	I	2a1b	
A/Norway/3275/2018		I		R	S	N	K	D			V	S	I	2a1b	
A/Armenia/941/2019		I		R	S	N	K		D		V	S	I	2a1b	
A/Iceland/12008/2019		I		R	S	N	K			D		V	S	I	2a1b
A/Hong Kong/2672/2019		I		R	S	N	K			D		V	S	I	2a1b
A/Christchurch/515/2019		I		R	S	N	K			D		V	S	I	2a1b
A/South Africa/R06432/2019		I		R	S	N	K			D		V	S	I	2a1b
A/La Rioja/2202/2018		I			S	FN				D		V	S	I	2a1b
A/Iceland/65/2019		I			S	N	FN			D		V	S	I	2a1b
A/South Africa/R06219/2019		I			S	N				HD		V	S	I	2a1b
A/Georgia/824/2019		I			S	FN				D		V	S	I	2a1b
A/Iceland/71/2019		I			S	N				D		V	S	I	2a1b
A/Dijon/1974/2019		I			S	N	K			D		V	S	I	2a1b
A/Oman/4518/2019		I			S	N				D		V	S	I	2a1b
A/Hong Kong/2675/2019		I			S	N	K			D		V	S	I	2a1b
A/South Africa/R06413/2019		I			S	N				D		V	S	I	2a1b
A/Sakha/3014V/2019		I			S	N				D		V	S	I	2a1b
A/Bobruisk/1468/2019		I			S	N				D		V	S	I	2a1b
	410	420	430	440	450	460									
A/Stockholm/6/2014	GDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIIVFCGTSCTYGTGSGWPDGADLNLMP	I													
A/England/538/2018															
A/Kansas/14/2017															
A/Northern Ireland/13125/2019															
A/Oman/4471/2019															
A/Kosovo/1188/2019															
A/Bretagne/1913/2019															
A/South Africa/R06867/2019															
A/Hong Kong/5738/2014															
A/Switzerland/8060/2017															
A/Bretagne/1413/2017															
A/Singapore/INFIMH-16-0019/2019															
A/Greece/4/2017															
A/Oman/4262/2019							N								
A/Antsirabe/3036/2019															
A/Norway/3275/2018															
A/Armenia/941/2019															
A/Iceland/12008/2019															
A/Hong Kong/2672/2019			N												
A/Christchurch/515/2019															
A/South Africa/R06432/2019															
A/La Rioja/2202/2018															
A/Iceland/65/2019															
A/South Africa/R06219/2019															
A/Georgia/824/2019			X												
A/Iceland/71/2019															
A/Dijon/1974/2019															
A/Oman/4518/2019											HT				
A/Hong Kong/2675/2019											HT				
A/South Africa/R06413/2019															
A/Sakha/3014V/2019															
A/Bobruisk/1468/2019															

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza B virus analyses.

Influenza B viruses represented only 11.8% of the samples received with collection dates after 2019-01-31 and were received from NICs in 21 countries or administrative regions in four WHO Regions: African (3: Madagascar, Mauritania and Senegal), Eastern Mediterranean (3: Iran, Jordan and Oman), European (14) and Western Pacific (1: Hong Kong SAR) (**Table 1**). B/Victoria lineage viruses outnumbered B/Yamagata lineage viruses at a ratio of ~3:1.

B/Victoria lineage.

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-11**: test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 11-4 to 11-11** are those with collection dates after 2019-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. A summary of the recognition of the test viruses collected after 2019-01-31 is shown in **Table 11-12**. Three distinct patterns of antigenicity were detected.

A total of 72 viruses with collection dates after 2019-01-31 were analysed, of these gene sequencing has been completed for 60. Five viruses collected after 2019-01-31 had no deletion in HA1. These viruses, from Iran, Hong Kong and Turkmenistan, were recognised poorly by antisera raised against viruses with a two amino acid deletion in HA1 (Δ 162-163 viruses: cell culture-propagated B/Norway/2409/2017 and both the cell culture- and egg-propagated cultivars of B/Colorado/06/2017) and a virus with a triple amino acid deletion in HA1 (Δ 162-164 virus: cell culture-propagated B/Cote d'Ivoire/1662/2018) at titres $\geq$ 8-fold lower than the homologous titres of the antisera. However, the antisera raised against cell culture-propagated B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016 recognised these five test viruses at titres within 4-fold of the homologous titres of the antisera. Antisera raised against egg-propagated reference viruses with no deletion in the HA (the former vaccine virus B/Brisbane/60/2008, B/Malta/63671/2011 and A/South Australia/81/2012) recognised these viruses poorly.

Sixteen viruses had HA genes in the Δ 162-163 group and all were recognised by the antiserum raised against the cell culture-propagated cultivar of B/Colorado/06/2017 (Δ 162-163) at titres within 4-fold of the titre of the antiserum with the homologous virus (14 within 2-fold of the homologous titre); the antiserum raised against cell culture-propagated B/Norway/2409/2017 (Δ 162-163) recognised fifteen of the sixteen viruses at titres within 4-fold of the titre of the antiserum with the homologous virus (14 within 2-fold of the homologous titre), and similar results were seen for the antiserum raised against the egg-propagated cultivar of B/Colorado/06/2017 which recognised 14 of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus

(13 within 2-fold of the homologous titre). However, the Δ 162-163 test viruses were recognised poorly by antisera raised against egg-propagated or cell culture-propagated viruses without HA1 amino acid deletions (cell culture-propagated B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016, and egg-propagated B/Brisbane/60/2008, B/Malta/636714/2011 and B/South Australia/81/2012) and by the antiserum raised against a virus with a triple amino acid deletion in HA1 (Δ 162-164 virus: cell culture-propagated B/Cote d'Ivoire/1662/2018).

The majority of viruses (54% of those tested and 65% of those for which gene sequencing had been completed) were identified as having HA genes in the Δ 162-164 group. Approximately 61% of these viruses were recognised by an antiserum raised against B/Cote d'Ivoire/1662/2018 (Δ 162-164) at titres 2-fold lower than, equal to or higher than the low titre of the antiserum with the homologous virus. However, 15 Δ 162-164 viruses, from Hong Kong, Iceland, Sweden and Georgia, also had HA1 K136E amino acid substitution and were recognised less well by this antiserum. The antiserum raised against the cell culture-propagated cultivar of B/Colorado/06/2017 recognised ~75% of the Δ 162-164 group test viruses at titres within 4-fold of the homologous titre of the antiserum but the antiserum raised against cell culture-propagated B/Norway/2409/2017 recognised only five of these test viruses (13%) at titres within 4-fold of the homologous titre of the antiserum. Moreover, the antiserum raised against egg-propagated B/Colorado/06/2017 recognised only nine of these test viruses (23%) at titres within 4-fold of the homologous titre of the antiserum. The majority (>85%) of the Δ 162-164 group test viruses were recognised poorly by the antisera raised against cell culture- or egg-propagated viruses with no deletion in HA1.

An antiserum raised against B/Hong Kong/269/2017 (Δ 162-164, with I180T and K209N) was used in **Table 11-7** and recognised none of the test viruses analysed, many of which were in the Δ 162-164 with K136E group.

Phylogenetic analysis of the HA genes of B/Victoria lineage viruses sequenced at the Crick, and other representatives, is shown in **Figure 11** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 12**. All recently circulating viruses analysed carried HA genes that fell into genetic clade 1A, the B/Brisbane/60/2008 genetic clade and all fell in a genetic group defined by three **HA1** amino acid substitutions compared with B/Brisbane/60/2008: **I117V**, **N129D** and **V146I**. Three virus clusters have emerged within clade 1A with deletions in HA1. Test viruses with HA genes encoding the two amino acid deletion in **HA1** (**Δ 162-163**) also encoded amino acid substitutions **D129G** and **I180V** in **HA1** and **R151K** in **HA2** and were seen in samples received from Ireland (n=1), Macedonia (n=1) and Madagascar. The HA genes of most viruses with a three amino acid deletion in **HA1** (**Δ 162-164**) fall into a recently emerging group encoding **K136E** substitution in **HA1** with most in this group also encoding **G133R** substitution in **HA1**. **Figure 13** shows the locations of amino acid substitutions that define the three groups of deletion variants on a B/Brisbane/60/2008 HA structure.

Figure 14 shows a phylogenetic tree for the NA gene and **Figure 15** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The

NA gene phylogeny is largely congruent with the HA phylogeny and the Δ 162-163 group of viruses are defined by **NA K371Q** amino acid substitution. Viruses in the main recently emerging Δ 162-164 genetic group encode a **A395T** substitution in the NA.

B/Yamagata lineage.

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-7**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 13-3 to 13-7** are viruses with collection dates after 2019-01-31. The clade into which the HA gene falls is shown. A summary of recognition of test viruses by all of the post-infection ferret antisera is shown in **Table 13-8**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Southern Hemisphere 2019 season and the Northern Hemisphere 2019-2020 season, recognised all of the 18 test viruses collected after 2019-01-31 at titres within 4-fold of the titre with the homologous virus, and all but one (94%) within 2-fold. However, a post-infection ferret antiserum raised against a cell culture-propagated cultivar of B/Phuket/3073/2013 recognised only six (33%) of the recently collected test viruses at titres within 4-fold, and only four (9%) within 2-fold, of its titre with the homologous virus.

Antisera raised against other viruses from the B/Wisconsin/1/2010-B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and cell culture-propagated B/Mauritius/1791/2017 and B/Mauritius/I-762/2018, recognised well the test viruses collected after 2019-01-31. Respectively, 100%, 89%, 83% and 100% were recognised at titres within 4-fold of the titres of the antisera with their homologous viruses and 90%, 89%, 83% and 91% were recognised at titres within 2-fold of the homologous titres of the antisera.

An antiserum raised against egg-propagated B/Massachusetts/02/2012 (clade 2) recognised 15 (83%) of the 18 recently collected test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, and 14 (78%) at titres within 2-fold of the homologous titre. However, an antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised only five (28%) of the 18 recently collected test viruses at titres within 4-fold, and 17% within 2-fold of its homologous titre. An antiserum raised against another cell culture-propagated clade 2 virus, B/Estonia/55669/2011, recognised 55% of test viruses at titres within 4-fold of, and 22% at titres within 2-fold of, the titre of the antiserum with the homologous virus.

Figure 16 shows a phylogenetic analysis of the HA genes of B/Yamagata lineage viruses sequenced at the Crick and other representative viruses. An amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 17**. All tested viruses with HA gene sequences determined (13 of 18) fall in clade 3, the B/Phuket/3073/2013 clade, and, like the vast majority of recently circulating viruses, fall

into a genetic group defined by two **HA1** amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**.

Figure 18 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 19** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) with the great majority of viruses falling in a group defined by **NA I49M, I171M, D342K** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 NA sequence. A number of subgroups have emerged within this group, notably those defined by **R65H** and **S402P** amino acid substitutions.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAI) zanamivir and oseltamivir was assessed phenotypically and by full NA gene sequencing, for 82 B/Victoria-lineage and 24 B/Yamagata-lineage viruses with collection dates after 2019-01-31 (**Table 14**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

Conclusion.

Viruses of the B/Victoria and the B/Yamagata lineages were received in small numbers with those of the B/Victoria lineage predominating over those of the B/Yamagata lineage at a ratio of 3:1. Viruses of the B/Victoria lineage fell into three genetic groups. The predominant group, representing 65% of the viruses sequenced, had HAs with a three amino acid deletion in **HA1** (Δ **162-164**) with the additional **HA1** amino acid substitution **K136E**. Viruses with HA genes encoding a **deletion** of two amino acids in **HA1** (Δ **162-163**) represented 27% of those sequenced, and viruses lacking amino acid deletions represented only 8.3% of those sequenced. Viruses with either deletion are antigenically distinct from the other viruses in clade 1A and the Δ **162-163** and Δ **162-164** variants are antigenically distinct from each other.

Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the majority were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2019-01-31

MONTH	B Victoria lineage										
Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against								
			Clade 1A viruses			Clade 1A(Δ2) virus			Clade 1A(Δ3) virus		
			° 2 fold ³	4 fold ³	8 fold ³	° 2 fold ⁴	4 fold ⁴	8 fold ⁴	° 2 fold ⁵	4 fold ⁵	8 fold ⁵
2019											
FEBRUARY											
Georgia	2	2			2		1	1	1		1
Iran	1	1	1					1			1
Ireland	1	1			1	1					1
Senegal	1	1		1		1			1		
Slovenia	1	1			1		1		1		
Sweden	1	1			1			1			1
MARCH											
Georgia	1	1			1			1	1		
Hong Kong	1	1			1			1	1		
Ireland	1	1			1	1			1		
Norway	1	0									
Russia	2	2			2			2	2		
Senegal	2	2	1	1		2			2		
Slovenia	1	1			1		1		1		
Sweden	1	1			1			1			1
APRIL											
Iceland	2	2			2	1		1			2
Macedonia	1	1			1		1				1
Madagascar	1	0									
Mauritania	1	1			1			1	1		
Norway	1	0									
Russia	2	2			2	1		1	2		
Turkmenistan	7	7	7					7			7
MAY											
France	1	1			1			1	1		
Hong Kong	2	2			2		1	1	2		
Madagascar	9	3			3	3					3
Norway	3	0									
Portugal	1	0									
Russia	1	1			1	1			1		
JUNE											
Hong Kong	15	15	1		14	5	9	1	3		12
Madagascar	14	11			11	10	1				11
Norway	3	0									
Oman	8	8			8	3	5		8		
JULY											
Norway	4	0									
Oman	2	2			2		2		2		
	95	72	10	2	60	29	22	21	31	0	41
17 Countries			13.9%	2.8%	83.3%	40.3%	30.6%	29.2%	43.1%	0.0%	56.9%

Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

Propagated to sufficient titre to perform HI assay

Compared to homologous titres for ferret antisera raised against at least one of two tissue-culture propagated B/Brisbane/60/2008-like equivalents

(B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016) - Clade 1A

Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Cote d'Ivoire/1662/2018 - Clade 1A(Δ3) 'African'

Numbers highlighted in blue show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Table 11-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-03-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ¹	F10/18 ²	F37/18 ²
	Passage history													
	Ferret number													
	Genetic group			1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	640	320	40	40	<	40	80	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	320	320	320	320	40	40	<	40	80	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	320	160	640	160	40	40	<	40	80	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	40	40	80	320	80	80	<	20	<	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	20	<	20	80	160	80	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	20	<	20	40	80	80	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	40	160	40	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	<	<	<	<	<	<	40	80	40	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	40	80	20	<	<	40	160	160	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	320	10	<	20	<	<	<	<	20	<	40
TEST VIRUSES														
B/Ghana/AC004018/2018	1A(Δ3)	2018-10-10	C1/MDCK1	40	<	<	<	<	<	<	<	40	<	40
B/Guangdong-Nanshan/36/2018	1A(Δ3)	2018-11-12	MDCK3/MDCK1	80	<	<	<	<	<	<	<	40	<	40
B/Fujian-Yanping/11948/2018	1A	2018-11-21	MDCK3/MDCK1	2560	40	<	20	80	80	40	<	<	<	<
B/Jiangxi-Qingshanhu/314/2018	1A(Δ2)	2018-11-29	MDCK3/MDCK1	320	<	<	<	<	<	<	<	80	20	<
B/Sichuan-Gaoxin/513/2018	1A(Δ3)	2018-12-03	MDCK2/MDCK1	320	<	<	<	<	<	<	20	40	<	40
B/Ghana/4134/2018	1A(Δ3)	2018-12-07	MDCK1	80	<	<	<	<	<	<	<	40	<	40
B/Norway/4183/2018	1A(Δ3)	2018-12-27	MDCK1/MDCK1	320	<	<	<	<	<	<	<	40	<	10

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

Vaccine[#]

Vaccine[§]

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

[§] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Table 11-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-03-19

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F44/17 ^{*4}	F29/13 ^{*4}	F25/16 ^{*4}	NIB F47/16 ^{*4}	F15/16 ^{*2}	F16/16 ^{*2}	F40/17 ^{*4}	F09/18 ^{*4}	F10/18 ^{*4}	F37/18 ^{*2}	
				Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)
				REFERENCE VIRUSES											
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	160	320	160	40	40	<	40	80	<	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	320	320	320	160	40	40	<	20	80	<	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	160	160	640	160	40	40	<	40	80	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	<	<	<	160	80	80	<	20	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	<	<	<	80	80	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	<	<	<	<	80	80	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	40	160	40	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	<	<	40	80	40	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	80	20	40	<	<	<	20	160	160	10	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	160	<	<	<	<	<	<	<	20	<	20	
TEST VIRUSES															
B/Kanagawa/AC1738/2018	1A		MDCK2 /MDCK1	2560	<	<	<	<	80	<	<	<	<	<	
B/Tokushima/19/2018	1A(Δ2)	2018-03-13	MDCK2/MDCK2/MDCK1	2560	40	<	80	<	<	<	40	160	160	10	
B/Dakar/07/2018	1A(Δ3)	2018-08-27	P0/MDCK1	80	<	<	<	<	<	<	<	40	<	20	
B/Dakar/17/2018	1A(Δ3)	2018-10-08	P0/MDCK1	40	<	<	<	<	<	<	<	40	<	40	

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

⁴ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

⁵ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Vaccine[§]

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-04-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08	B/Bris 60/08	B/Malta 636714/11	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16	B/Norway 2409/17	B/Colorado 06/17	B/Colorado 06/17	B/CIV 1662/18
				Passage history	Egg	Egg	Egg	MDCK	MDCK	MDCK	MDCK	MDCK	Egg	MDCK
				Ferret number	Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F10/18 ²
Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)			
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	640	160	160	160	80	20	20	<	20	40	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	160	320	160	80	20	20	<	10	40	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	160	160	320	80	20	20	<	20	40	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	20	80	20	80	40	40	<	<	<	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	10	40	10	20	80	80	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	10	40	10	<	40	40	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	<	<	<	<	<	<	80	80	20	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	10	<	<	80	160	40	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	80	40	40	<	<	<	40	80	80	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	80	10	20	20	<	<	<	<	10	10	40
TEST VIRUSES														
B/Guangdong-Nanshan/2018-33566/2018	1A(Δ3)	2018-11-12	MDCK4/MDCK1	320	<	20	<	<	<	<	<	20	<	20
B/Guangdong-Nanshan/38/2018	1A(Δ3)	2018-11-12	MDCK4/MDCK1	640	40	40	20	<	<	<	<	10	20	10
B/Guangdong-Nanshan/2018-33566/2018	1A(Δ3)	2018-11-12	E2/E1	1280	80	80	80	20	<	<	<	40	80	40
B/Guangdong-Nanshan/38/2018	1A(Δ3)	2018-11-12	E2/E2	320	20	40	10	<	<	<	<	<	10	10
B/Sichuan-Gaoxin/531/2018	1A(Δ3)	2018-12-26	MDCK3/MDCK1	640	20	40	20	<	<	<	<	20	20	20
B/Sichuan-Gaoxin/531/2018	1A(Δ3)	2018-12-26	E4/E1	320	40	40	20	10	<	<	<	10	20	20
B/Nigeria/3352/2018	1A(Δ3)	2018-01-16	C1/MDCK1	80	<	20	40	<	<	<	10	20	<	40
B/Temara/652/2019	1A(Δ3)	2019-01-28	MDCK1/MDCK1	40	<	<	<	<	<	<	<	20	<	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadrivalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Table 11-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-06-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F10/18 ²	F37/18 ²
	Passage history													
	Ferret number													
	Genetic group			1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	160	160	320	80	20	20	<	20	40	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	160	160	160	80	20	20	<	10	20	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	160	160	320	80	20	20	<	20	20	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	20	40	20	80	40	40	<	<	<	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	10	20	10	40	40	40	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	10	20	10	40	40	40	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	40	40	10	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	<	<	10	<	<	<	40	80	20	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	40	40	10	<	<	20	80	80	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	160	10	20	20	<	<	<	<	20	10	20
TEST VIRUSES														
B/Ashgabat/1/2016	1A	2016-11-30	MDCK1/MDCK1	1280	10	20	10	40	40	40	<	<	<	<
B/Ashgabat/2/2016	1A	2016-12-07	MDCK1/MDCK1	1280	<	20	10	20	20	20	<	<	<	<
B/Ashgabat/3/2016		2016-12-14	MDCK1/MDCK1	1280	10	40	20	40	20	20	<	<	<	<
B/Ashgabat/4/2016	1A	2016-12-14	MDCK1/MDCK1	640	<	10	10	<	20	20	<	<	<	<
B/Ashgabat/5/2017	1A	2016-12-14	MDCK1/MDCK1	1280	10	20	20	40	40	40	<	<	<	<
B/Ashgabat/6/2017	1A	2016-12-14	MDCK1/MDCK1	1280	10	20	10	20	40	40	<	<	<	<
B/Slovenia/308/2019	1A(Δ3)	2019-01-15	MDCKx/MDCK1	320	<	20	10	<	<	<	20	20	10	20
B/Slovenia/787/2019	1A(Δ3)	2019-02-01	MDCKx/MDCK1	320	<	20	10	<	<	<	20	20	20	20
B/Slovenia/1654/2019		2019-03-25	MDCKx/MDCK1	320	<	20	10	<	<	<	20	20	10	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadrivalent vaccines SH 2018[§] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]Vaccine[§]

Table 11-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-06-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											B/CIV 1662/18 MDCK
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg		
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F10/18 ²		
				1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	160	320	320	20	40	<	20	40	<	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	640	320	320	320	40	40	<	20	40	<	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	320	160	320	160	20	20	<	20	40	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	80	40	20	80	40	40	<	<	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	80	20	10	40	40	40	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	80	20	20	80	40	80	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	<	<	<	<	<	40	40	40	10	<	
B/Colorado/06/2017	1A(Δ2)		MDCK1/MDCK2	80	<	<	<	<	<	40	40	40	10	<	
B/Colorado/06/2017	1A(Δ2)		E5/E2	1280	320	40	40	20	<	<	40	80	160	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	80	20	10	10	<	<	<	<	20	<	10	
TEST VIRUSES															
B/Zambia/4-374/2018	1A	2018-09-12	MDCK1	2560	80	<	20	80	40	80	<	<	<	<	
B/Zambia/4-409/2018	1A(Δ2)	2018-10-01	MDCK1	80	<	<	10	<	<	<	40	40	20	<	
B/Zambia/4-434/2018	1A(Δ2)	2018-10-15	MDCK1	80	<	<	10	<	<	<	40	40	20	<	
B/Zambia/1-420/2018	1A(Δ2)	2018-10-30	MDCK1	80	<	<	10	<	<	<	40	40	20	<	
B/Zambia/1-428/2018	1A(Δ2)	2018-10-31	MDCK1	320	<	<	10	<	<	<	40	40	20	<	
B/Zambia/2-439/2018	1A(Δ2)	2018-11-16	MDCK2	80	10	<	10	<	<	<	40	40	20	<	
B/Zambia/1-463/2018	1A(Δ2)	2018-11-25	MDCK1	80	20	<	10	<	<	<	40	40	20	<	
B/Zambia/1-462/2018		2018-11-30	MDCK1	80	10	<	10	<	<	<	40	40	20	<	
B/Iran/164310/2019	1A	2019-02-10	SIAT2/MDCK1	2560	40	<	10	80	40	80	<	<	<	<	
B/Stockholm/1/2019		2019-02-24	MDCK1/MDCK1	640	80	<	10	<	<	<	<	<	10	<	
B/Eskilstuna/2/2019	1A(Δ3)	2019-03-06	MDCK1/MDCK1	640	80	<	10	<	<	<	<	<	20	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadrivalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]Vaccine[§]

Table 11-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-07-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁻²	F29/13 ⁻²	F25/16 ⁻²	NIB F47/16 ⁻²	F15/16 ⁻²	F16/16 ⁻²	F40/17 ⁻²	F09/18 ⁻⁴	F10/18 ⁻²	F37/18 ⁻²	
Genetic group				1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	640	320	40	40	<	40	80	<	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	5120	640	640	320	320	40	40	<	20	80	<	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	640	160	320	160	20	40	<	20	40	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	160	80	40	160	80	80	<	<	20	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	<	10	80	80	80	<	<	20	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	2560	80	<	10	80	80	160	<	<	20	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	20	<	<	<	<	<	40	80	20	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	20	<	<	<	<	<	40	80	40	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	320	40	80	<	<	<	40	80	160	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	80	40	<	20	<	<	<	<	<	<	20	
TEST VIRUSES															
B/Mauritania/32-2019/2019	1A(Δ3)	2019-01-16	MDCK1	640	80	<	40	<	<	<	<	20	10	40	
B/Georgia/569/2019	1A(Δ3)	2019-02-22	C2/MDCK1	1280	160	80	80	40	<	<	<	20	40	40	
B/Georgia/568/2019	1A(Δ3)	2019-02-22	C2/MDCK1	320	40	<	<	<	<	<	<	<	<	<	
B/Iceland/7775/2019	1A(Δ3)	2019-04-03	MDCK1	160	20	<	<	<	<	10	10	40	<	<	
B/Macedonia/587/2019	1A(Δ2)	2019-04-04	MDCK1/MDCK1	80	10	<	<	<	<	<	20	20	10	<	
B/Mauritania/145-2019/2019	1A(Δ3)	2019-04-10	MDCK1	40	20	<	<	<	<	<	<	<	<	20	
B/Iceland/60/2019	1A(Δ3)	2019-04-19	MDCK1/MDCK1	160	20	<	<	<	<	10	<	<	<	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Vaccine[§]

Table 11-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-07-17

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre														B/CIV 1662/18 (B/HK/269/17) MDCK	BX-75C (B/HK/269/17) Egg	B/HK 269/17 Egg
				Post-infection ferret antisera																
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 (B/HK/269/17) MDCK	BX-75C (B/HK/269/17) Egg	B/HK 269/17 Egg				
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F10/18 ²	F37/18 ²	F15/19 ²	F50/17 ²				
Genetic group				1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	1A(Δ3)	1A(Δ3)				
REFERENCE VIRUSES																				
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	320	320	320	40	40	<	20	40	<	40	10			
B/Malta/636714/2011		1A	2011-03-07	E4/E1	2560	640	320	320	160	20	40	<	20	40	<	<	10			
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	160	640	320	40	40	<	20	40	10	40	10			
B/Hong Kong/514/2009		1B	2009-10-11	MDCK1/MDCK2	2560	160	80	40	160	80	80	<	<	<	20	<	<			
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	80	20	20	80	80	80	<	<	<	<	<	<			
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	80	40	20	80	80	80	<	<	<	<	<	<			
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	20	<	<	<	<	<	40	40	20	<	<	<			
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	40	<	<	<	<	<	40	80	20	<	<	<			
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	40	20	<	<	20	80	80	<	40	10			
B/Cote D'Ivoire/1662/2018		1A(Δ3)	2018-07-25	P0/MDCK3	80	40	<	10	<	<	<	<	<	<	10	<	<			
NYMC BX-75C		1A(Δ3)		E#6382	1280	320	80	80	40	<	<	<	20	80	10	640	160			
B/Hong Kong/269/2017	Is 2 clone 33	1A(Δ3)	2019-10-10	E6 (Am2AI4)	1280	320	40	80	40	<	<	<	40	80	10	640	80			
TEST VIRUSES																				
B/Kanagawa/AC1867/2019		1A(Δ3)		E3	1280	160	40	40	20	<	<	<	40	40	80	80	<			
B/Kanagawa/IC18169/2019		1A(Δ3)		E3	640	160	40	80	20	<	<	<	40	40	40	40	<			
B/Georgia/675/2019		1A(Δ3)	2019-03-21	MDCK2	80	20	<	<	<	<	<	<	<	<	20	<	<			
B/Hong Kong/60/2019		1A(Δ3)	2019-03-23	MDCK1/MDCK1	<	40	<	<	<	<	<	<	<	<	20	<	<			
B/Hong Kong/218/2019		1A(Δ3)	2019-05-08	MDCK1/MDCK1	<	<	<	<	<	<	<	<	<	<	10	<	<			
B/Hong Kong/308/2019		1A(Δ3)	2019-05-18	MDCK1/MDCK1	80	40	<	<	<	<	<	<	20	<	40	<	<			
B/Hong Kong/584/2019		1A(Δ3)	2019-06-16	MDCK1/MDCK1	40	20	<	<	<	<	<	<	20	<	40	<	<			
B/Hong Kong/574/2019		1A(Δ3)	2019-06-17	MDCK1/MDCK1	80	20	<	<	<	<	<	<	20	<	40	<	<			
B/Hong Kong/587/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	80	40	<	<	<	<	<	<	40	<	<	<	<			
B/Hong Kong/580/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	160	40	<	<	<	<	<	<	20	<	<	<	<			
B/Hong Kong/579/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	160	40	<	<	<	<	<	<	20	<	<	<	<			
B/Hong Kong/586/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	80	40	<	<	<	<	<	<	20	<	<	<	<			
B/Hong Kong/576/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	320	40	<	<	<	<	<	<	20	<	<	<	<			
B/Hong Kong/585/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	160	40	<	<	<	<	<	<	20	<	<	<	<			
B/Hong Kong/575/2019		1A(Δ3)	2019-06-18	MDCK1/MDCK1	80	40	<	<	<	<	<	<	20	<	40	<	<			
B/Hong Kong/588/2019		1A(Δ3)	2019-06-19	MDCK1/MDCK1	160	80	<	20	<	<	<	<	40	<	<	<	<			
B/Hong Kong/583/2019		1A(Δ3)	2019-06-19	MDCK1/MDCK1	80	80	<	<	<	<	<	<	40	<	<	<	<			
B/Hong Kong/582/2019		1A(Δ3)	2019-06-19	MDCK1/MDCK1	80	80	<	20	<	<	<	<	40	<	<	<	<			
B/Hong Kong/581/2019		1A	2019-06-19	MDCK1/MDCK1	1280	80	<	20	40	40	40	<	<	<	<	<	<			
B/Hong Kong/578/2019		1A(Δ3)	2019-06-19	MDCK1/MDCK1	80	40	<	<	<	<	<	<	40	<	<	<	<			
B/Hong Kong/577/2019		1A(Δ3)	2019-06-19	MDCK1/MDCK1	80	40	<	<	<	<	<	<	20	<	<	<	<			

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

⁵ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Table 11-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-07-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F10/18 ²	F37/18 ²	
Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)				
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	640	320	40	40	<	20	80	10	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	1280	640	640	320	40	80	<	20	80	10	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	640	320	640	320	40	80	<	40	80	10	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	160	80	80	160	80	160	<	<	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	<	40	80	80	160	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	2560	80	40	40	160	80	160	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	160	20	<	<	<	<	<	80	80	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	320	40	<	<	<	<	<	80	80	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	160	40	40	20	<	<	40	80	160	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	160	80	<	20	<	<	<	<	<	<	40	
TEST VIRUSES															
B/Zambia/1-405/2018	1A(Δ2)	2018-10-21	SIAT2	320	40	<	40	<	<	<	80	80	40	<	
B/Sicilia/3/2019	1A(Δ3)	2019-01-19	MDCK1	640	40	<	10	<	<	<	<	<	<	80	
B/Moscow/8/2019	1A(Δ3)	2019-03-05	MDCK1/MDCK1	160	40	<	<	<	<	<	<	10	<	80	
B/Moscow/13/2019		2019-03-30	MDCK2/MDCK1	160	40	<	<	<	<	<	<	10	<	80	
B/Moscow/16/2019	1A(Δ3)	2019-04-17	MDCK1/MDCK1	640	80	<	<	<	<	<	<	<	<	160	
B/Paris/2038/2019	1A(Δ3)	2019-05-16	MDCK2/MDCK1	640	80	<	<	<	<	<	<	<	<	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

Vaccine[#]Vaccine[§]¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

* B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Table 11-9. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-08-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ⁴	F25/16 ⁴	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ²
Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)			
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	160	320	160	40	20	<	40	80	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	640	320	320	320	40	40	<	10	160	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	640	160	320	320	40	40	<	10	160	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	80	40	40	160	80	80	<	<	<	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5120	80	<	20	160	80	160	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	<	<	80	10	80	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	20	<	<	<	<	<	40	40	160	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	40	<	<	10	<	<	80	80	160	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	20	20	20	<	<	20	40	320	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	80	40	<	10	<	<	<	<	10	40	40
TEST VIRUSES														
B/Chelyabinsk/963/2018	1A(Δ3)	2018-12-07	MDCK1/MDCK1	320	40	<	<	<	<	<	<	40	80	40
B/Ireland/21508/2019	1A(Δ2)	2019-02-22	MDCK1/MDCK1	160	20	<	<	<	<	<	40	80	160	<
B/Ireland/28115/2019	1A(Δ3)	2019-03-19	MDCK3/MDCK1	1280	80	<	40	20	<	<	<	40	160	40
B/Ashgabat/5/2019	1A	2019-04-15	MDCK2/MDCK1	2560	80	40	<	40	160	80	<	<	<	<
B/Ashgabat/4/2019		2019-04-15	MDCK2/MDCK1	2560	80	40	<	40	160	80	<	<	<	<
B/Ashgabat/3/2019		2019-04-15	MDCK2/MDCK1	2560	80	40	<	40	160	80	<	<	<	<
B/Ashgabat/2/2019		2019-04-15	MDCK1/MDCK1	2560	80	40	20	40	160	80	<	<	<	<
B/Birobidzhan/1023V/2019	1A(Δ3)	2019-04-16	MDCK1/MDCK1	80	20	<	<	<	<	<	<	40	40	40
B/Ashgabat/8/2019	1A	2019-04-22	MDCK2/MDCK1	1280	80	10	<	40	160	80	<	<	<	<
B/Ashgabat/7/2019	1A	2019-04-22	MDCK1/MDCK1	1280	80	20	<	40	80	40	<	<	<	<
B/Ashgabat/6/2019		2019-04-22	MDCK1/MDCK1	1280	80	40	<	40	160	80	<	<	<	<
B/St. Petersburg/1152V/2019	1A(Δ3)	2019-05-14	MDCK1/MDCK1	640	40	<	<	<	<	10	10	80	160	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Vaccine[§]

Table 11-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-08-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ⁴	F25/16 ⁴	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ²	
				1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	1280	320	320	320	40	40	<	40	160	10	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	640	320	320	160	40	40	<	20	160	10	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	640	320	640	320	80	40	<	40	160	10	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	5120	160	160	40	160	80	80	<	20	10	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	80	<	80	80	80	<	20	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	2560	80	80	<	80	80	80	<	10	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	10	<	<	<	<	10	80	80	160	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	10	<	<	<	<	10	80	80	160	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	160	80	40	20	<	<	40	80	320	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	160	40	20	<	<	<	10	<	20	40	40	
TEST VIRUSES															
B/Dakar/02/2019	1A(Δ3)	2019-01-08	C1/MDCK1	160	40	<	<	<	<	20	<	40	40	40	
B/Dakar/06/2019	1A(Δ3)	2019-02-22	C1/MDCK1	160	40	<	10	<	<	20	<	40	40	40	
B/Dakar/07/2019	1A(Δ3)	2019-03-04	C1/MDCK2	40	10	<	<	<	<	20	<	40	40	40	
B/Dakar/08/2019	1A(Δ3)	2019-03-21	C1/MDCK2	1280	40	<	<	<	40	40	<	80	80	80	
B/Antananarivo/2877/2019	1A(Δ2)	2019-05-06	MDCK1	160	10	40	<	<	<	<	40	80	160	<	
B/Antananarivo/3336/2019	1A(Δ2)	2019-05-29	MDCK1	160	20	40	<	<	<	<	80	80	160	<	
B/Antananarivo/3426/2019	1A(Δ2)	2019-05-30	MDCK1/MDCK1	160	20	40	<	<	<	10	80	80	320	<	
B/Antananarivo/3435/2019	1A(Δ2)	2019-06-03	MDCK1/MDCK1	160	10	40	<	<	<	<	40	40	160	<	
B/Antananarivo/3493/2019	1A(Δ2)	2019-06-06	MDCK1/MDCK1	80	10	<	<	<	<	<	20	40	80	<	
B/Antananarivo/3624/2019	1A(Δ2)	2019-06-11	MDCK1	80	10	<	<	<	<	<	40	80	160	<	
B/Antananarivo/3566/2019	1A(Δ2)	2019-06-11	MDCK1	80	10	<	<	<	<	<	40	80	160	<	
B/Antananarivo/3592/2019	1A(Δ2)	2019-06-12	MDCK1	160	10	<	<	<	<	<	40	40	160	<	
B/Antananarivo/3593/2019	1A(Δ2)	2019-06-12	MDCK1	160	10	<	<	<	<	<	40	80	160	<	
B/Antananarivo/3688/2019	1A(Δ2)	2019-06-17	MDCK1	160	10	<	<	<	<	<	40	80	160	<	
B/Antananarivo/3743/2019	1A(Δ2)	2019-06-19	MDCK1	40	<	<	<	<	<	<	<	20	20	<	
B/Antananarivo/3745/2019	1A(Δ2)	2019-06-19	MDCK1	160	10	<	<	<	<	<	40	80	160	<	
B/Antananarivo/3779/2019	1A(Δ2)	2019-06-24	MDCK1	160	10	20	<	<	<	10	80	80	160	<	
B/Antananarivo/3780/2019	1A(Δ2)	2019-06-24	MDCK1	160	10	20	<	<	<	10	80	80	160	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]Vaccine[§]

Table 11-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-09-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ⁴	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ⁴
Passage history	Ferret number	Genetic group		1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	1280	320	320	160	40	20	<	40	160	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	1280	640	320	320	40	40	<	20	160	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	640	160	320	160	40	40	<	40	160	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	160	160	40	160	80	80	<	10	40	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	80	<	80	160	160	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	40	<	40	80	80	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	80	80	160	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	<	<	<	<	<	<	40	80	160	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	40	40	10	<	<	40	80	320	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	160	40	10	<	<	<	<	<	20	80	40
TEST VIRUSES														
B/Brisbane/35/2018		2018-10-19	E4/E1	320	80	40	40	10	<	<	<	20	160	40
B/Victoria/705/2018		2018-10-30	E3/E1	640	80	40	40	20	<	<	<	20	160	40
B/Washington/02/2019		2019-01-19	C2/MDCK1	640	40	<	<	<	<	<	<	20	80	40
B/Washington/02/2019		2019-01-19	E3/E1	320	80	40	20	10	<	<	<	20	160	20
B/Shanghai-Changning/1228/2019		2019-02-25	C4+1/MDCK1	160	<	20	<	<	<	<	<	20	80	40
B/Shanghai-Changning/1228/2019		2019-02-25	E4+1/E1	640	80	80	40	20	<	<	<	20	320	80
B/Kanagawa/IC18169/2019	1A(Δ3)	2019-04-27	hCK2/MDCK1	320	40	<	<	<	<	<	<	40	80	40
B/Oman/4100/2019		2019-06-09	SIAT2/MDCK1	160	40	<	<	<	<	<	<	40	80	80
B/Oman/4116/2019		2019-06-12	SIAT1/MDCK1	160	40	<	<	<	<	<	<	40	80	80
B/Oman/4147/2019	1A(Δ3)	2019-06-15	SIAT1/MDCK1	320	40	<	<	<	<	<	<	40	80	80
B/Oman/4229/2019	1A(Δ3)	2019-06-22	SIAT1/MDCK1	160	40	<	<	<	<	<	<	20	80	40
B/Oman/4234/2019		2019-06-22	SIAT1/MDCK1	320	80	10	40	<	<	<	<	20	40	40
B/Oman/4241/2019	1A(Δ3)	2019-06-23	SIAT1/MDCK1	320	80	10	40	<	<	<	<	20	80	20
B/Oman/4309/2019	1A(Δ3)	2019-06-28	SIAT1/MDCK1	80	40	<	<	<	<	<	<	20	40	40
B/Oman/4332/2019		2019-06-30	SIAT1/MDCK1	320	40	<	<	<	<	<	<	20	80	40
B/Oman/4324/2019	1A(Δ3)	2019-07-02	SIAT1/MDCK1	160	40	<	<	<	<	<	<	20	80	40
B/Oman/4357/2019		2019-07-04	MDCK1	80	<	<	<	<	<	<	<	20	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

\$ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Vaccine[#]Vaccine^{\$}

Sequences in phylogenetic trees

Table 11-12. Antigenic analyses of influenza B viruses (Victoria lineage) with collection dates after 2019-01-31 - Summary

Viruses	Other information	Post-infection ferret antisera										
						*	*	*	*	*		
		B/Bris 60/08	B/Malta 636714/11	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16	B/Norway 2409/17	B/Colorado 06/17	B/Colorado 06/17	B/CIV 1662/18	
		Passage history Egg	Egg	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK		
		F44/17 ^{*2}	F29/13 ^{*2}	F25/16 ^{*2}	NIB F47/16 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	F40/17 ^{*2}	F09/18 ^{*4}	F10/18 ^{*2}	F37/18 ^{*2}	
		Genetic group	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)
REFERENCE VIRUSES												
B/Brisbane/60/2008	1A	640	320	640	320	40	40	<	20	80	10	
B/Malta/636714/2011	1A	1280	640	640	320	40	80	<	20	80	10	
B/South Australia/81/2012	1A	640	320	640	320	40	80	<	40	80	10	
B/Hong Kong/514/2009	1B	160	80	80	160	80	160	<	<	<	<	
B/Ireland/3154/2016	1A	80	<	40	80	80	160	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	80	40	40	160	80	160	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	20	<	<	<	<	<	80	80	<	<	
B/Colorado/06/2017	1A(Δ2)	40	<	<	<	<	<	80	80	<	<	
B/Colorado/06/2017	1A(Δ2)	160	40	40	20	<	<	40	80	160	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	80	<	20	<	<	<	<	<	<	40	
TEST VIRUSES												
Genetic group	No tested	No with										
All viruses	72	Titre reduced ≤2-fold				1	10	11	16	29	15	31
		%				1.4	13.9	15.3	22.2	40.3	20.8	43.1
		Titre reduced =4-fold	1		1	9	2	3	22	22	11	
		%	1.4		1.4	12.5	2.8	4.2	30.6	30.6	15.3	
		Titre reduced ≥8-fold	71	72	71	62	62	59	53	21	46	41
		%	98.6	100.0	98.6	86.1	86.1	81.9	73.6	29.1	63.9	56.9
1A	5	Titre reduced ≤2-fold				2	5	5				
		Titre reduced =4-fold				3						
		Titre reduced ≥8-fold	5	5	5				5	5	5	5
1A-Δ2	16	Titre reduced ≤2-fold							14	14	13	
		Titre reduced =4-fold							1	2	1	
		Titre reduced ≥8-fold	16	16	16	16	16	16	1		2	16
1A-Δ3	39	Titre reduced ≤2-fold					1	1	3	13	2	24
		%					2.6	2.6	7.7	33.3	5.1	61.5
		Titre reduced =4-fold	1		1	1		2	2	16	7	
		%	2.6		2.6	2.6		5.1	5.1	41.0	18.0	
		Titre reduced ≥8-fold	38	39	38	38	38	36	34	10	30	15
		%	97.4	100.0	97.4	97.4	97.4	92.3	87.2	25.7	76.9	38.5

Vaccine[#]

Vaccine^{\$}

[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

^{\$} B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

* In no assay did these antisera give a homologous titre of >80: all < titres for test viruses were placed in the ≥8-fold reduced category.

Reference virus results are taken from individual tables as examples. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 11. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Apr 2019

May 2019

Jun 2019

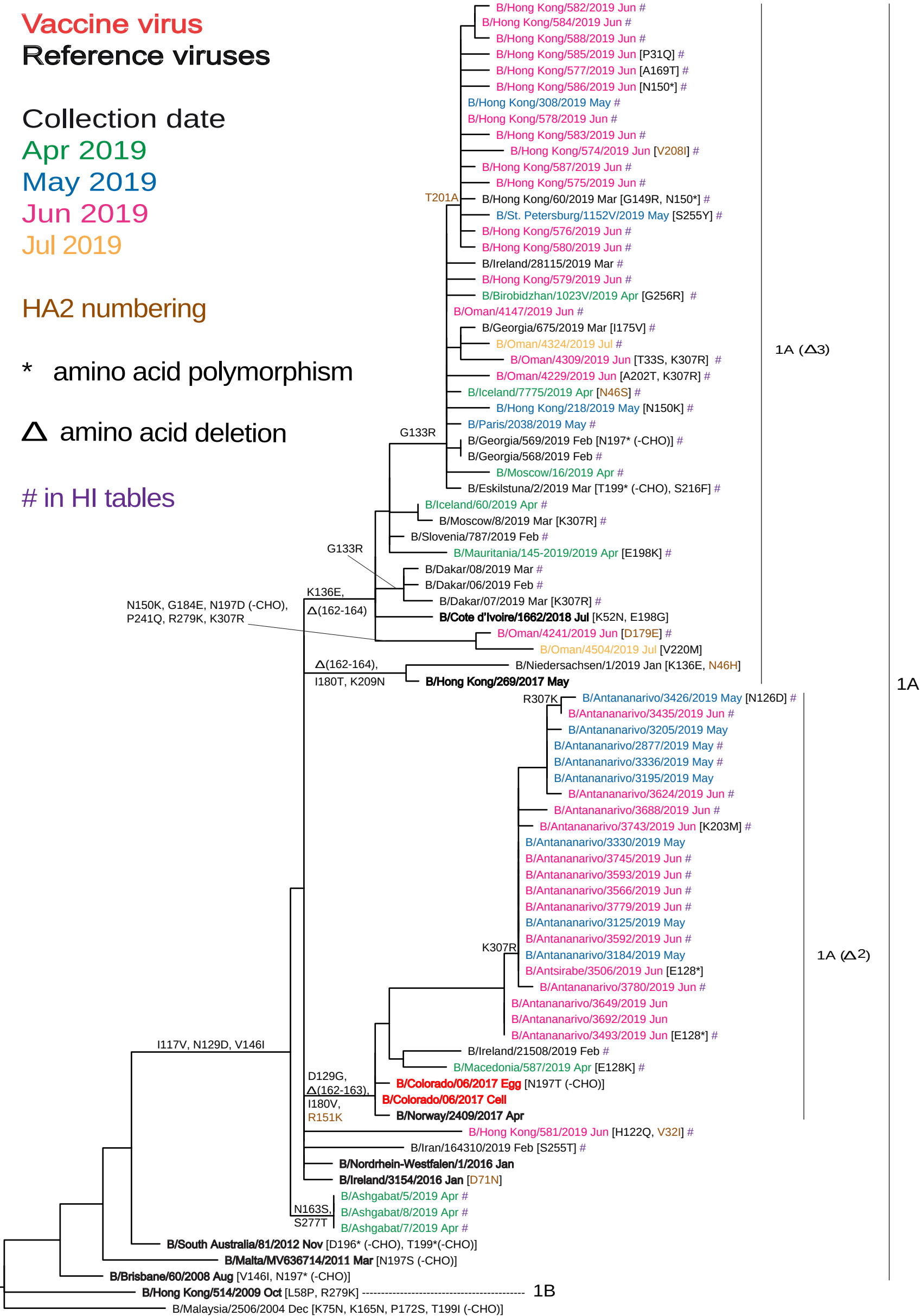
Jul 2019

HA2 numbering

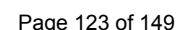
* amino acid polymorphism

Δ amino acid deletion

in HI tables



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400																
B/Hong Kong/514/2009	GG	LN	KSKPY	TTGE	HAKA	IGNC	PIWVK	TPLK	LANG	TKYR	PPAK	LLKE	RG	FFG	AIAG	FLEGG	WEGMI	AGWH	GYTSH	GAGH	VAVAA	DLKSTQ	EAIN	KITK	NLN	1B
B/Brisbane/60/2008	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1A
B/South Australia/81/2012	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1A
B/Ashgabat/8/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1A
B/Nordrhein-Westfalen/1/2016	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1A
B/Norway/2409/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Colorado/06/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Macedonia/587/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Antananarivo/3649/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Antsirabe/3506/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Antananarivo/3743/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Antananarivo/2877/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Antananarivo/3426/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ2
B/Hong Kong/269/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Niedersachsen/1/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	H.	.	.	1AΔ3
B/Oman/4504/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Oman/4241/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Cote d'Ivoire/1662/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Dakar/06/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Mauritania/145-2019/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Iceland/60/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Moscow/16/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Paris/2038/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Hong Kong/218/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Oman/4229/2019	.	.	R.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Birobidzhan/1023V/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/St. Petersburg/1152V/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Hong Kong/574/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Hong Kong/585/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3
B/Hong Kong/582/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1AΔ3

	410	420	430	440	450	460	470	480	490	500		
B/Hong Kong/514/2009	SLSE	LEVKNLQRLSGAM	DELHNEILELDE	KVDDL	LRAD	TISSQIELAV	LLSNEGIINSE	DEHLLALERK	LKMLGPSA	VEIGNGCFT	EKHKCNQTCLDR	IA
B/Brisbane/60/2008												
B/South Australia/81/2012												
B/Ashgabat/8/2019												
B/Nordrhein-Westfalen/1/2016												
B/Norway/2409/2017												
B/Colorado/06/2017												.K.
B/Macedonia/587/2019												.K.
B/Antananarivo/3649/2019												.K.
B/Antsirabe/3506/2019												.K.
B/Antananarivo/3743/2019												.K.
B/Antananarivo/2877/2019												.K.
B/Antananarivo/3426/2019												.K.
B/Hong Kong/269/2017												
B/Niedersachsen/1/2019												
B/Oman/4504/2019												
B/Oman/4241/2019												
B/Cote d'Ivoire/1662/2018												
B/Dakar/06/2019												
B/Mauritania/145-2019/2019												
B/Iceland/60/2019												
B/Moscow/16/2019												
B/Paris/2038/2019												
B/Hong Kong/218/2019												
B/Oman/4229/2019												
B/Birobidzhan/1023V/2019												
B/St. Petersburg/1152V/2019												
B/Hong Kong/574/2019												
B/Hong Kong/585/2019												
B/Hong Kong/582/2019												

	510	520	530	540	550	560	570
B/Hong Kong/514/2009	AGTFDAGEFSLPTFDSLNTTAASLNDDGLNHTIILYYSTAASSLAVTILMIAIFVVYMYSRDNVSCSICL						
B/Brisbane/60/2008							
B/South Australia/81/2012							
B/Ashgabat/8/2019							
B/Nordrhein-Westfalen/1/2016							
B/Norway/2409/2017							
B/Colorado/06/2017							
B/Macedonia/587/2019							
B/Antananarivo/3649/2019							
B/Antsirabe/3506/2019							
B/Antananarivo/3743/2019							
B/Antananarivo/2877/2019							
B/Antananarivo/3426/2019							
B/Hong Kong/269/2017							
B/Niedersachsen/1/2019							
B/Oman/4504/2019							
B/Oman/4241/2019			E				
B/Cote d'Ivoire/1662/2018							
B/Dakar/06/2019							
B/Mauritania/145-2019/2019							
B/Iceland/60/2019							
B/Moscow/16/2019							
B/Paris/2038/2019							
B/Hong Kong/218/2019							
B/Oman/4229/2019							
B/Birobidzhan/1023V/2019							
B/St. Petersburg/1152V/2019					A		
B/Hong Kong/574/2019					A	I	
B/Hong Kong/585/2019					A		
B/Hong Kong/582/2019					A		

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

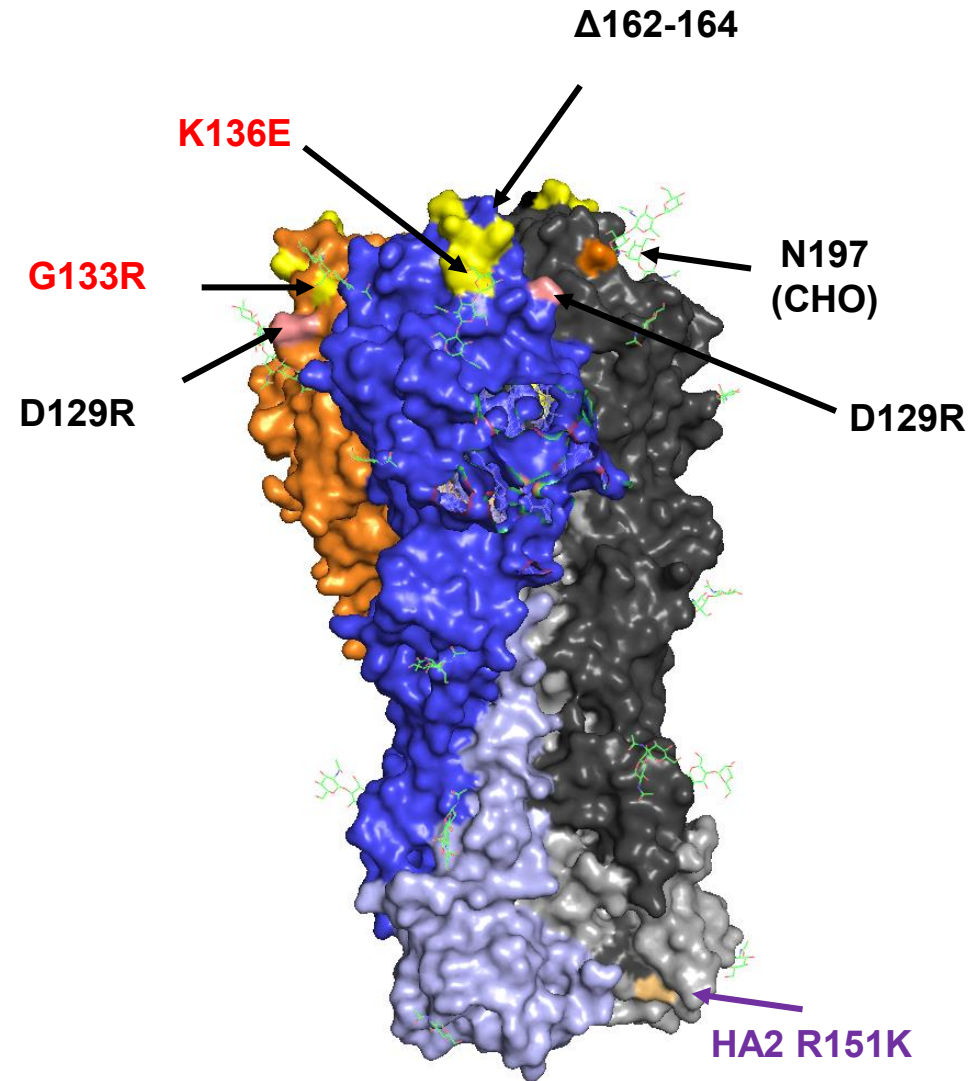


Figure 13. Amino acid substitutions and deletions in the HA of recent B/Victoria lineage viruses mapped onto B/Brisbane/60/2008

Figure 14. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Apr 2019

May 2019

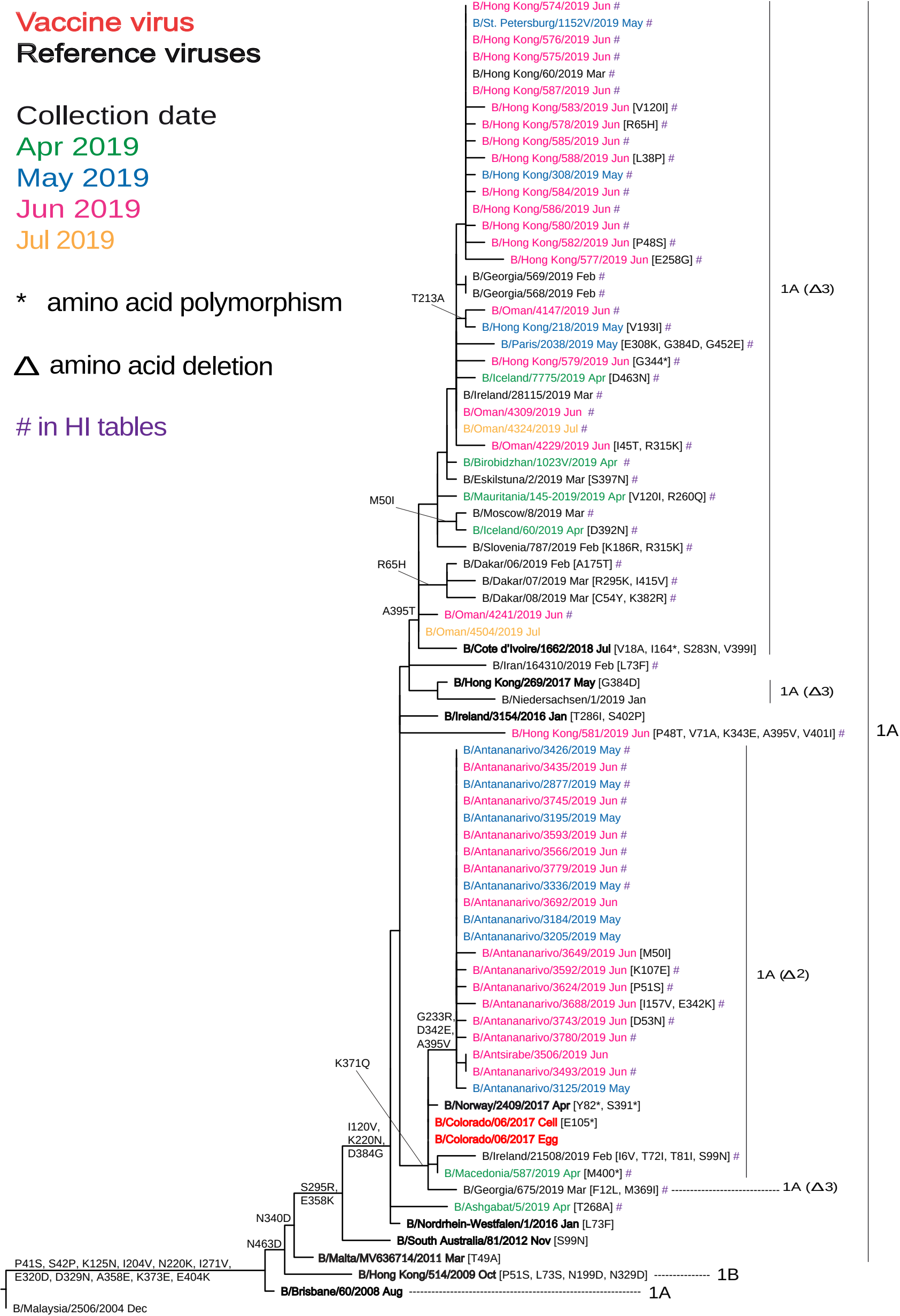
Jun 2019

Jul 2019

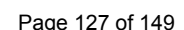
* amino acid polymorphism

Δ amino acid deletion

in HI tables



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2019-01-31

MONTH	B Yamagata lineage				
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³
2019					
FEBRUARY					
Greece	1	1	1		
Russia	2	2	2		
Senegal	1	0			
Sweden	1	1		1	
Turkmenistan	1	1	1		
MARCH					
Austria	2	2	2		
France	1	1	1		
Greece	1	1	1		
Hong Kong	1	1	1		
Iran	1	1	1		
Jordan	2	1	1		
Norway	2	0			
Russia	1	1	1		
United Kingdom	2	2	2		
APRIL					
Hong Kong	1	1	1		
Norway	3	0			
Russia	1	1	1		
MAY					
Jordan	1	1	1		
Norway	3	0			
JUNE					
Norway	1	0			
Oman	1	0			
	30	18	17	1	0
13 Countries			94.4%	5.6%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-03-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK	B/Maur I-762/18 MDCK
				SH614 ^{*1,4}	F39/17 ^{*3}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}
	Passage history												
	Ferret number												
	Genetic Group												
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	80	160	20	40	<	80	10	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	160	640	80	160	10	320	20	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	<	10	320	20	80	<	80	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	<	20	320	80	160	10	320	20	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	<	20	320	40	160	10	160	20	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	5120	320	320	640	160	320	160	320	160	>1280
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	20	160	40	160	<	160	20	80
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	2560	<	40	160	20	80	20	160	40	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK2	2560	40	80	160	40	80	40	160	80	640
TEST VIRUSES													
B/EHPAD/Libourne/1958/2018	3	2018-10-12	MDCK2/MDCK1	1280	20	20	160	20	40	<	80	40	160
B/Toulon/2308/2018	3	2018-12-18	MDCK2/MDCK1	2560	40	40	320	40	80	10	160	40	320
B/Tunis/21/2019	3	2018-12-27	MDCK1	1280	<	20	160	20	40	10	80	20	160
B/Norway/252/2019	3	2019-01-11	MDCK2	1280	<	20	160	20	40	<	80	20	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 < = <20; 4 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-03-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762/18
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK
				Ferret number	SH614 ^{*1,4}	F39/17 ^{*3}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
Genetic Group	3	2	2	2	3	3	3	3	3	3			
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	160	40	160	20	10	<	40	10	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	160	80	640	80	20	<	160	20	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	<	10	640	40	80	<	80	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	<	10	320	40	40	<	160	20	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	<	<	160	40	80	<	80	<	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	5120	160	320	320	80	80	80	320	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	<	160	20	40	<	80	10	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	20	80	20	20	10	40	20	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	<	20	80	20	20	<	40	20	160
TEST VIRUSES													
B/Cote d'Ivoire/2007/2018	3	2018-09-14	MDCK1	5120	80	320	320	160	20	80	320	160	640
B/Dakar/08/2018	3	2018-09-17	P0/MDCK1	2560	160	320	160	80	160	80	160	80	640
B/Dakar/14/2018	3	2018-09-24	P1/MDCK1	2560	<	80	160	20	20	20	160	40	320
B/Dakar/15/2018	3	2018-10-08	P0/MDCK1	1280	<	40	80	20	20	10	80	40	160
B/Dakar/22/2019	3	2018-11-05	P0/MDCK1	2560	20	80	160	40	20	20	80	40	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 < = <20; 4 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-04-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762/18
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK
				Ferret number	SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
Genetic Group	3	2	2	2	3	3	3	3	3	3	3		
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	160	20	20	<	40	<	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	160	160	320	80	80	20	160	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	20	20	320	20	40	<	80	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	10	160	40	80	<	160	10	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	640	<	10	80	20	40	<	80	<	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	320	320	80	160	80	320	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	10	80	20	40	<	80	10	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	10	20	80	20	40	10	40	20	80
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	10	20	80	20	40	<	40	20	160
TEST VIRUSES													
B/Saudi Arabia/465861/2018	3	2018-09-03	MDCK2	1280	20	20	80	10	40	10	40	20	80
B/Wyoming/11/2018	3	2018-09-25	C2/MDCK1	1280	20	40	80	20	40	10	80	20	160
B/Greece/1037-19/2019	3	2019-02-20	MDCK1	1280	20	20	40	20	40	10	40	20	80
B/Greece/1244-19/2019	3	2019-03-01	MDCK1	1280	20	40	80	20	40	10	80	20	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 <= <40; 2 <= <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-06-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762
				Egg SH614 ^{1,3}	MDCK F39/17 ²	MDCK F10/16 ²	Egg F06/17 ²	Egg F36/15 ²	Egg F05/17 ²	MDCK F27/15 ²	Egg F25/17 ²	MDCK F04/18 ²	MDCK F05/19 ²
	Passage history												
	Ferret number												
	Genetic Group												
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	40	160	40	20	<	40	<	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	160	160	640	80	160	20	160	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	20	10	320	20	40	<	40	<	10
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	<	10	160	20	40	<	80	10	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	<	10	80	20	80	<	80	<	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	5120	160	320	320	80	160	80	160	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	<	80	20	40	<	40	<	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	2560	10	20	80	20	40	<	40	20	80
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	10	20	80	20	40	<	40	20	160
TEST VIRUSES													
B/Sudan/287/2018	3	2018-01-31	MDCK1	1280	10	10	80	10	40	<	40	10	40
B/Austria/1118469/2019	3	2019-01-21	MDCK1	640	10	20	80	20	40	10	80	<	160
B/Grozny/RII-15/2019	3	2019-02-18	MDCK1/MDCK1	1280	10	20	80	20	40	<	40	20	80
B/Grozny/RII-9/2019	3	2019-02-22	MDCK1/MDCK1	2560	10	20	80	20	40	<	40	20	80
B/Austria/1130261/2019		2019-03-04	SIAT1/MDCK1	2560	40	80	160	80	160	40	160	80	320
B/Austria/1133679/2019	3	2019-03-14	SIAT1/MDCK1	1280	20	10	80	40	40	10	40	20	80
B/Iran/169602/2019	3	2019-03-30	SIAT2/SIAT1	2560	80	320	320	80	160	40	160	80	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-07-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762
				Egg SH614 ^{*1,3}	MDCK F39/17 ^{*2}	MDCK F10/16 ^{*2}	Egg F06/17 ^{*2}	Egg F36/15 ^{*2}	Egg F05/17 ^{*2}	MDCK F27/15 ^{*2}	Egg F25/17 ^{*2}	MDCK F04/18 ^{*2}	MDCK F05/19 ^{*2}
	Passage history Ferret number Genetic Group			3	2	2	2	3	3	3	3	3	3
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	160	80	80	20	40	<	160	<	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	320	320	320	80	160	80	160	80	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	320	20	20	160	20	40	<	80	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	10	10	160	40	80	<	160	20	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	640	<	10	80	20	80	<	80	<	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	320	320	160	80	320	160	160	80	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	10	10	80	20	40	<	160	<	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	20	40	80	20	40	10	80	20	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	20	40	40	20	40	10	80	20	80
TEST VIRUSES													
B/Egypt/106/2019	3	2019-01-03	MDCK2	640	10	20	40	10	20	<	40	<	80
B/Mauritania/22-2018/2018	3	2018-02-07	MDCK1	1280	20	40	80	20	40	40	40	<	80
B/Mauritania/30-2018/2018	3	2018-02-14	MDCK1	640	10	20	40	10	40	<	40	<	80
B/Mauritania/32-2018/2018	3	2018-02-15	MDCK1	640	20	40	40	10	40	10	80	<	80
B/Macedonia/548/2018	3	2018-03-15	MDCK1/MDCK1	1280	20	20	80	10	40	40	40	<	80
B/Stockholm/2/2019	3	2019-02-14	MDCK0/MDCK1	640	10	20	80	10	40	40	40	<	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-07-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13 Egg SH614 ^{1,3}	B/Estonia 55669/11 MDCK F39/17 ²	B/Mass 02/12 MDCK F10/16 ²	B/Mass 02/12 Egg F06/17 ²	B/Wis 1/10 Egg F36/15 ²	B/Stock 12/11 Egg F05/17 ²	B/Phuket 3073/13 MDCK F27/15 ²	B/Phuket 3073/13 Egg F25/17 ²	B/Maur 1791/17 MDCK F04/18 ²	B/Maur I-762 MDCK F05/19 ²
				Passage history Ferret number Genetic Group	3	2	2	2	3	3	3	3	3
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	80	160	40	40	<	160	<	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	320	640	640	160	160	80	320	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	40	20	320	40	80	<	160	<	40
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	10	20	320	80	160	10	320	40	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	<	20	160	40	160	<	160	<	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	320	320	160	160	160	160	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	10	10	160	40	80	<	320	<	80
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	20	40	160	40	80	20	160	40	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	2560	40	80	80	40	80	20	160	40	320
TEST VIRUSES													
B/Jordan/30070/2019	3	2019-03-07	MDCK1	2560	20	80	320	40	80	20	160	40	320
B/Moscow/11/2019		2019-03-12	MDCK1/MDCK1	2560	40	80	160	40	160	20	320	80	320
B/Brest/1993/2019		2019-03-24	MDCK1/MDCK1	2560	20	40	160	40	80	20	160	40	320
B/Hong Kong/61/2019	3	2019-03-24	MDCK1/MDCK1	2560	40	20	160	20	80	20	320	40	320
B/Hong Kong/80/2019	3	2019-04-09	MDCK1/MDCK1	2560	320	1280	640	320	320	160	640	320	1280
B/Moscow/15/2019		2019-04-17	MDCK1/MDCK1	2560	20	80	160	40	80	20	320	40	320
B/Jordan/30151/2019	3	2019-05-12	MDCK1	2560	40	80	320	40	80	20	160	40	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine[#]

Table 13-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-08-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/Maur	B/Maur
				3073/13	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	1791/17	I-762
	Passage history			Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK
	Ferret number			SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}
	Genetic Group			3	2	2	2	3	3	3	3	3	3
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	320	40	40	160	10	<	<	40	<	40
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	160	640	640	80	80	20	320	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	320	20	20	320	20	40	<	80	<	10
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	640	<	20	320	20	40	<	80	<	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	320	<	20	160	10	40	<	80	<	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	80	320	320	40	160	40	160	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	10	160	20	40	<	80	<	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	640	<	20	160	40	<	<	40	20	80
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	640	<	20	80	40	<	<	40	20	160
TEST VIRUSES													
B/Dakar/01/2019		2019-01-08	C1/MDCK1	1280	40	40	80	20	20	10	10	<	160
B/Vladikavkaz/39V/2019	3	2019-01-16	MDCK1/MDCK1	2560	80	160	160	80	80	40	40	80	640
B/Komi/489V/2019	3	2019-01-28	MDCK1/MDCK1	1280	20	40	80	20	10	10	10	<	160
B/Dakar/03/2019	3	2019-01-30	C1/MDCK1	640	20	40	80	10	10	10	10	<	80
B/Dakar/04/2019		2019-01-31	C1/MDCK1	1280	<	40	40	10	10	10	10	<	80
B/Ashgabat/1/2019	3	2019-02-25	MDCK1/MDCK1	640	<	20	40	10	<	<	40	<	80
B/England/247/2019	3	2019-03-08	SIAT1/MDCK1	640	<	20	40	10	<	10	40	<	80
B/England/152/2019		2019-03-19	SIAT2/MDCK1	1280	80	160	160	40	80	20	160	80	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

Sequences in phylogenetic trees

Vaccine#

Table 13-8. Antigenic analyses of influenza B viruses (Yamagata lineage) with collection dates after 2019-01-31 - Summary

Viruses	Other information	Passage history Ferret number Genetic Group	Haemagglutination inhibition titre								
			Post-infection ferret antisera								
			B/Estonia 55669/11 MDCK F39/17 ²	B/Mass 02/12 MDCK F10/16 ²	B/Mass 02/12 Egg F06/17 ²	B/Wis 1/10 Egg F36/15 ²	B/Stock 12/11 Egg F05/17 ²	B/Phuket 3073/13 MDCK F27/15 ²	B/Phuket 3073/13 Egg F25/17 ²	B/Maur 1791/17 MDCK F04/18 ²	B/Maur I-762 MDCK F05/19 ²
			2	2	2	3	3	3	3	3	3
REFERENCE VIRUSES											
B/Estonia/55669/2011	2	160	80	160	40	40	<	160	<	160	
B/Massachusetts/02/2012	2	320	640	640	160	160	80	320	80	320	
B/Massachusetts/02/2012	2	40	20	320	40	80	<	160	<	40	
B/Wisconsin/1/2010	3	10	20	320	80	160	10	320	40	160	
B/Stockholm/12/2011	3	<	20	160	40	160	<	160	<	80	
B/Phuket/3073/2013	3	160	320	320	160	160	160	160	160	640	
B/Phuket/3073/2013	3	10	10	160	40	80	<	320	<	80	
B/Mauritius/1791/2017	3	20	40	160	40	80	20	160	40	160	
B/Mauritius/I-762/2018	3	40	80	80	40	80	20	160	40	320	
TEST VIRUSES											
Genetic group	No tested	No with									
All viruses	18*	Titre reduced ≤2-fold	4	3	14	16	16	4	17	15	18
		%	22.3	16.7	77.7	88.9	88.9	22.2	94.4	83.3	100.0
		Titre reduced =4-fold	6	2	1	2	2	2	1		
		%	33.3	11.1	5.6	11.1		11.1	5.6		
		Titre reduced ≥8-fold	8	13	3		2	12		3	
		%	44.4	72.2	16.7		11.1	66.7		16.7	
Vaccine [#]											

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19, SH 2019 and NH 2019-20

* Sequences were available for 13 test viruses - all were Clade 3 (B/Phuket/3073/2013-like)

Reference virus results are taken from individual tables as examples. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 16. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus
Reference viruses

Collection date

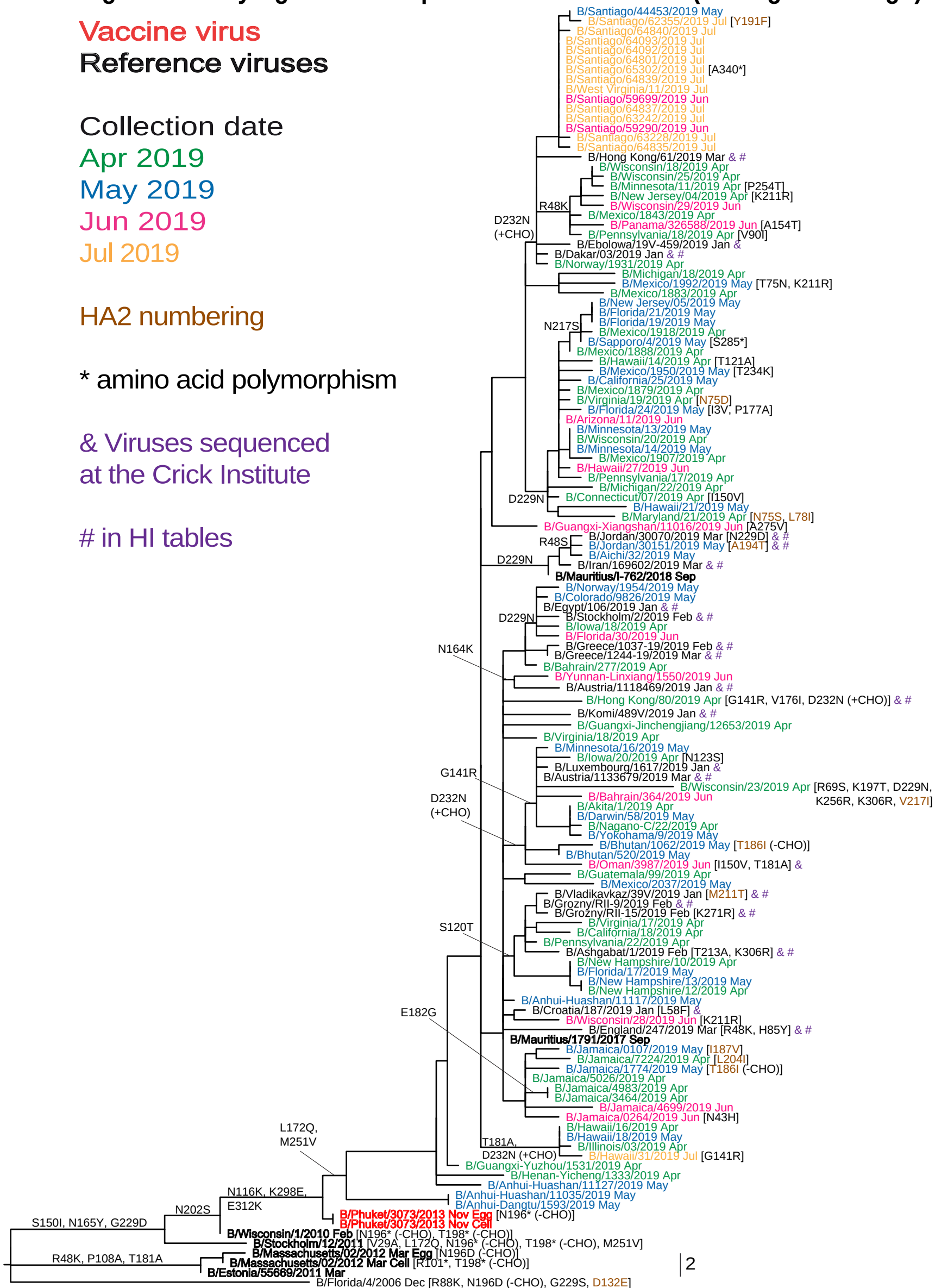
Apr 2019
May 2019
Jun 2019
Jul 2019

HA2 numbering

* amino acid polymorphism

& Viruses sequenced
at the Crick Institute

in HI tables



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**

[illegible][illegible]

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	310	320	330	340	350	360	370	380	390	400
B/Estonia/55669/2011	G L N K S K P Y Y T G E H A K A I G N C P I W V K T P L K L A N G T K Y R P P A K L L K E R G F F G A I A G F L E G G W E G M I A G W H G Y T S H G A H G V A A D L K S T Q E A I N K I T K N L N S									
B/Massachusetts/02/2012	.	.	.	.	.	.	.	.	.	.
B/Wisconsin/1/2010	.	.	.	.	.	.	.	.	.	.
B/Phuket/3073/2013	.	K.	.	.	.	.	.	.	.	.
B/Anhui-Dangtu/1593/2019	.	K.	.	.	.	.	.	.	.	.
B/Anhui-Huashan/11127/2019	.	K.	.	.	.	.	.	.	.	.
B/Hawaii/31/2019	.	K.	.	.	.	.	.	.	.	.
B/Jamaica/1774/2019	.	K.	.	.	.	.	.	.	.	.
B/Mauritius/1791/2017	.	K.	.	.	.	.	.	.	.	.
B/England/247/2019	.	K.	.	.	.	.	.	.	.	.
B/Groznny/RII-15/2019	.	K.	.	.	.	.	.	.	.	.
B/Oman/3987/2019	.	K.	.	.	.	.	.	.	.	.
B/Bahrain/364/2019	.	K.	.	.	.	.	.	.	.	.
B/Hong Kong/80/2019	.	K.	.	.	.	.	.	.	.	.
B/Greece/1037/19/2019	.	K.	.	.	.	.	.	.	.	.
B/Norway/1954/2019	.	K.	.	.	.	.	.	.	.	.
B/Mauritius/I-762/2018	.	K.	.	.	.	.	.	.	.	.
B/Iran/169602/2019	.	K.	.	.	.	.	.	.	.	.
B/Jordan/30070/2019	.	K.	.	.	.	.	.	.	.	.
B/Guangxi-Xiangshan/11016/2019	.	K.	.	.	.	.	.	.	.	.
B/Hawaii/21/2019	.	K.	.	.	.	.	.	.	.	.
B/Arizona/11/2019	.	K.	.	.	.	.	.	.	.	.
B/Sapporo/4/2019	.	K.	.	.	.	.	.	.	.	.
B/New Jersey/05/2019	.	K.	.	.	.	.	.	.	.	.
B/Norway/1931/2019	.	K.	.	.	.	.	.	.	.	.
B/Panama/326588/2019	.	K.	.	.	.	.	.	.	.	.
B/Wisconsin/29/2019	.	K.	.	.	.	.	.	.	.	.
B/Hong Kong/61/2019	.	K.	.	.	.	.	.	.	.	.
B/Santiago/64840/2019	.	K.	.	.	.	.	.	.	.	.
B/Santiago/62355/2019	.	K.	.	.	.	.	.	.	.	.

Figure 18. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus
Reference viruses

Collection date

Apr 2019
May 2019
Jun 2019
Jul 2019

* amino acid polymorphism

in HI tables

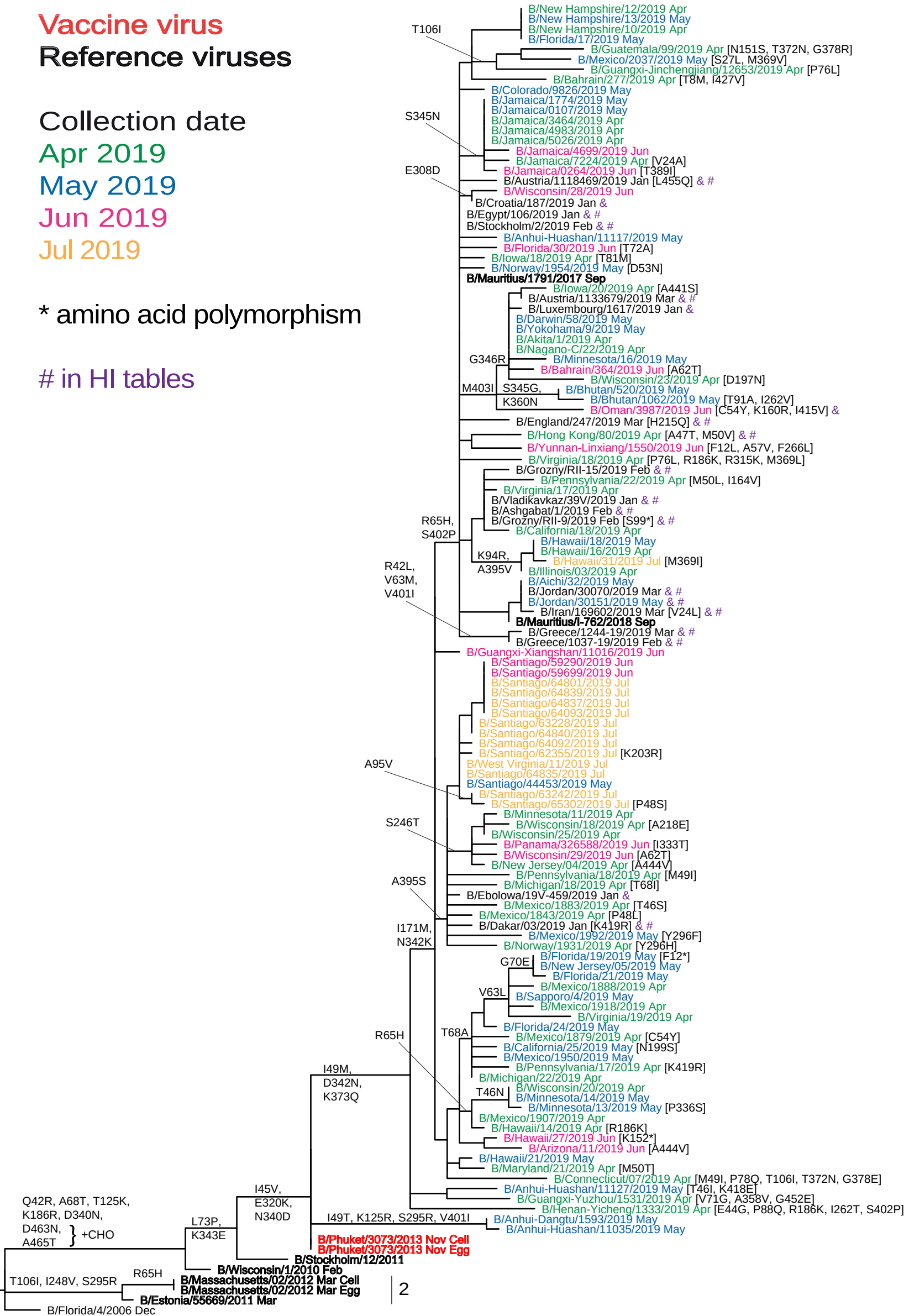


Figure 19. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
B/Estonia/55669/2011	M	L	P	S	T	I	Q	T	L	T	L	F	L	T	S	G	G	V	L	L	S	L	S	L	S	L	S	L	S	L	L	S	D	I	L	L	K	F	S	Q	T	E	I	T	A	P	I	M	P	L	D	C	A	N	A	S	N	V	Q	A	V	N	R	S	A	A	K	G	V	T	L	L	L	P	E	F	E	W	T	Y	P	R	L	S	C	P	G	S	T	F	Q	K	A	L	L	I	S	P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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	310	320	330	340	350	360	370	380	390	400
B/Estonia/55669/2011	PFVKLN	VDTDAE	IRLMCTET	YLDTPR	PNDGSI	TGPCE	SDGDKG	SGGIKGG	FVHQ	RMA
B/Massachusetts/02/2012										
B/Wisconsin/1/2010				N						
B/Phuket/3073/2013		K		E						
B/Anhui-Dangtu/1593/2019		K		E						
B/Anhui-Huashan/11127/2019		K		NE						
B/Hawaii/31/2019		K		KE			I	Q		V
B/Jamaica/1774/2019		K		KE	N			Q		
B/Mauritius/1791/2017		K		KE				Q		
B/England/247/2019		K		KE				Q		
B/Grozy/RII-15/2019		K		KE				Q		
B/Oman/3987/2019		K		KE				Q		
B/Bahrain/364/2019		K		KE	R			Q		
B/Hong Kong/80/2019		K		KE				Q		
B/Greece/1037/19/2019		K		KE				Q		
B/Norway/1954/2019		K		KE				Q		
B/Mauritius/I-762/2018		K		KE				Q		
B/Iran/169602/2019		K		KE				Q		
B/Jordan/30070/2019 Mar		K		KE				Q		
B/Guangxi-Xiangshan/11016/2019		K		KE				Q		
B/Hawaii/21/2019		K		KE				Q		
B/Arizona/11/2019		K		KE				Q		
B/Sapporo/4/2019		K		KE				Q		
B/New Jersey/05/2019		K		KE				Q		
B/Norway/1931/2019		K		KE				Q		S
B/Panama/326588/2019		K		T	KE				S	
B/Wisconsin/29/2019		K		KE				Q		S
B/Ebolowa/19v-459/2019		K		KE				Q		S
B/Santiago/64840/2019		K		KE				Q		S
B/Santiago/62355/2019		K		KE				Q		S
	410	420	430	440	450	460				
B/Estonia/55669/2011	VSME	EPGW	YSFG	FEIK	DKKCD	VPCIG	IEMV	HGGK	TTW	HSAA
B/Massachusetts/02/2012										
B/Wisconsin/1/2010									N	T
B/Phuket/3073/2013									N	T
B/Anhui-Dangtu/1593/2019	I								N	T
B/Anhui-Huashan/11127/2019			E						N	T
B/Hawaii/31/2019		P							N	T
B/Jamaica/1774/2019		P							N	T
B/Mauritius/1791/2017		P							N	T
B/England/247/2019		P							N	T
B/Grozy/RII-15/2019		P							N	T
B/Oman/3987/2019		PI			V				N	T
B/Bahrain/364/2019		PI							N	T
B/Hong Kong/80/2019		P							N	T
B/Greece/1037/19/2019	IP								N	T
B/Norway/1954/2019		P							N	T
B/Mauritius/I-762/2018		P							N	T
B/Iran/169602/2019		P							N	T
B/Jordan/30070/2019 Mar		P							N	T
B/Guangxi-Xiangshan/11016/2019									N	T
B/Hawaii/21/2019									N	T
B/Arizona/11/2019					V				N	T
B/Sapporo/4/2019									N	T
B/New Jersey/05/2019									N	T
B/Norway/1931/2019									N	T
B/Panama/326588/2019									N	T
B/Wisconsin/29/2019									N	T
B/Ebolowa/19v-459/2019									N	T
B/Santiago/64840/2019									N	T
B/Santiago/62355/2019									N	T

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 14. Antiviral (NAI) testing of isolates with collection dates after 2019-01-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2019									
February	220	1	127	1	7		5		361
March	126	1	127		10		10		274
April	22	1	47		14		3		87
May	5		22		8		3		38
June	22		23		37		2		84
July	6		4		6		1		17
August	4		1						
Totals	401	3	350	1	82	0	24	0	861

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2019-09-18

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Cyprus/919/2019	A(H1N1)pdm09	2018-02-16	109 (HRI)	3.96	N295S
A/Trnava/535/2019	A(H1N1)pdm09	2019-04-10	268.7 (HRI)	1.05	H275Y
A/Denmark/2697/2019 [§]	A(H1N1)pdm09	2019-03-14	1.4	34 (RI)	Q136X, D151X
A/Latvia/03-0738053/2019	A(H3N2)	2019-02-26	5	24.6 (RI)	D151X, V165I

[§] The clinical specimen was not available for sequencing

WORLDWIDE INFLUENZA CENTRE
WHO CC for Reference & Research on Influenza
THE FRANCIS CRICK INSTITUTE
LONDON NW1 1AT
United Kingdom

**Report prepared for the WHO annual consultation on the
composition of influenza vaccines for the Southern
Hemisphere 2020**

23rd – 26th September 2019

Annex

**Phylogenetic analyses of all influenza type A
viruses, with collection dates after 2019-01-31,
characterised at the Worldwide influenza
Centre as of 2019-09-16.**

Vaccine viruses

Reference viruses

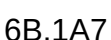
Apr 2019
May 2019
Jun 2019
Jul 2019



Reference viruses

Jul 2019

in HI tables



Reference viruses

Jul 2019

@ in VN tables

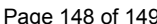


Figure A-4. Influenza A(H3N2) NA genes

Vaccine viruses

Reference viruses

Collection date

Apr 2019

May 2019

Jun 2019

Jul 2019

in HI tables

@ in VN tables

